

SELF STUDY REPORT

Submitted to



NATIONAL ASSESSMENT AND ACCREDITATION COUNCIL (NAAC)

An Autonomous Institution of the University Grants Commission

P.O. Box No.1075, Nagarbhavi

Bangaluru - 560072

by



Chhatrapati Shivaji Institute of Technology



(Approved by AICTE, New Delhi and affiliated to CSVTU, Bhilai)

Shivaji Nagar, Balod Road, Durg - 491 001. Chhattisgarh, India.

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CHHATRAPATI SHIVAJI INSTITUTE OF TECHNOLOGY, DURG

Ref./CSIT/NAAC/2014-15/.....

Date :

Dr. Mahesh P.,
Principal,
CSIT, Durg (C.G.), INDIA.

Sub. : Submission of Self Study Report (SSR) and NAAC Assessment Fee for Accreditation by NAAC (1st Cycle).

Ref. : i) Letter No. NAAC/E&NE/HKA/Cycle-1/SSR/CHCOGN21243/2014 dated 14th Aug., 2014.

ii) Our letter No. CSIT/SSR-NAAC/2014/341 dated 25th Sept., 2014.

To

**The Director,
National Assessment and Accreditation Council,
P.O.Box No. 1075,
NAGARBHAVI, Bangalore – 560072.**

Sir,

Sincere Greetings!

- i) With reference to your letter cited above we are submitting the Self Study Report (SSR). The documentation of the SSR is as per the guidelines given by NAAC. The SSR was uploaded on our website (www.csitdurg.in) on 25th Sept., 2014 and remains uploaded. It is not protected by password and is reader friendly.
Five Hard Copies of SSR and Soft Copy in CD and Pen-Drive are enclosed herewith.
- ii) We are herewith enclosing a Demand Draft on State Bank of India bearing No. 769114 dated 19.11.2014 of Rs.3,37,080/- in favour of "The Director, NAAC" and payable at Bangalore towards the NAAC Assessment Fee.
- iii) The Committee of NAAC may kindly decide to visit our Institute in any one of the following slots. We are specifically suggesting these slots because the climatic conditions at our place are pleasant during Feb. and March. Moreover, according to the calendar of our University, all classes will be in full swing during this period.

A) Feb., 26,27, 28

B) March 12, 13, 14

C) March 19, 20, 21

Kindly let us know the time slot convenient to the Peer Team of NAAC for visiting our Institute.

Thanking you,

Yours faithfully,

Principal
CSIT, Durg

Encl. : As stated above plus one copy of CSIT's Information Brochure with compliments to the Director, NAAC.



Rajesh Verma <registrar@csitdurg.in>

College Trackd

1 message

NAAC <admn@naac.gov.in>
To: registrar@csitdurg.in
Cc: rceastneast@naac.gov.in

Fri, May 16, 2014 at 3:02 PM

DEAR USER,

YOU HAVE SUCCESSFULLY SUBMITTED LOI.**INSTITUTION NAME :**CHHATRAPATI SHIVAJI INSTITUTE OF TECHNOLOGY, DURG (C.G.)**ADDRESS:**SHIVAJI NAGAR, BALOD ROAD KOLIHAPURIPOST PISEGAON, DURG**CITY :**DURG-BHILAI NAGAR**STATE:**CHHATTISGARH**PINCODE :**491001**INSTITUTION EMAILID :** registrar@csitdurg.in**YOUR INSTITUTION TRACKID IS :**CHCOGN21243**PLEASE NOTE DOWN YOUR TRACK ID FOR FUTURE REFERENCE.****NAAC WILL REVERT TO YOU AT THE EARLIEST, IN NO CASE LATER THAN 40 DAYS.**

-----This is an auto generated mail and please do not reply to this mail. -----



Rajesh Verma <registrar@csitdurg.in>

[SPAM] LOI status of CHHATRAPATI SHIVAJI INSTITUTE OF TECHNOLOGY, DURG (C.G.),DURG-BHILAI NAGAR

2 messages

NAAC <admn@naac.gov.in>
To: registrar@csitdurg.in
Cc: rceastneast@naac.gov.in

Thu, Jun 12, 2014 at 3:57 PM

Dear User,

Institution Name :CHHATRAPATI SHIVAJI INSTITUTE OF TECHNOLOGY, DURG (C.G.)

Your Institution TRACKID is :CHCOGN21243

Thank you for your interest in the Assessment and Accreditation process of NAAC.

Your LOI is accepted and you may kindly submit your SSR within six months from receipt of this intimation.

**Please note: No separate intimation will be mailed on the above matter,
also refer to instruction as in the link provided before submission of SSR / RAR.**

For further details: http://www.naac.gov.in/sites/naac.gov.in/files/Duties_and_Responsibilites.pdf

-----This is an auto generated mail and please do not reply to this mail. -----

Rajesh Verma <registrar@csitdurg.in>
To: "Dr. N. T. Khobragade" <deanacademic@csitdurg.in>

Thu, Jun 12, 2014 at 3:58 PM

[Quoted text hidden]



Najesh Verma <registrar@csitdurg.in>

IEQA submissioni message

ieqa@naac.gov.in <ieqa@naac.gov.in>

Tue, Jul 1, 2014 at 10:17 AM

Reply-To: rceastneast@naac.gov.in

To: registrar@csitdurg.in

Cc: rceastneast@naac.gov.in, ieqanaac@gmail.com

Dear Sir/Madam,

You have successfully submitted your IEQA.

Track Id : CHCOGN21243

College Name: CHHatrapati shivaji institute of technology, durg (C.G.)

The IEQA evaluation report is also been sent, please check your email.

Thanks and regardsDirector,
NAAC

 CHCOGN21243.pdf
13K

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Chhatrapati Shivaji Institute of Technology

Approved by AICTE New Delhi | Affiliated to CSVTU, Bhilai | An ISO 9001 : 2008 Certified | NBA Accredited

Ref./CSIT/SSR-NAAC/2014/341.....

Date : 23-09-2014

To
The Director,
National Assessment and Accreditation Council,
Nagarbhavi, Bangalore – 560072.

Sub. : Uploading of SSR on the Website of Our Institute.

Ref. : Letter No. NAAC/E&NE/HKA/Cycle-1/SSR/CHCOGN21243/2014
dated 14th Aug., 2014.

Dear Sir,

Greetings from CSIT!

We have carefully completed the work on SSR. We are now uploading it on the Website of our Institute. We have not protected it by password. It is accessible to all those who are interested in accessing the same.

We assure you that it shall continue to remain online till the completion of Assessment and Accreditation Process by NAAC.

Thanking you,

With regards,

Yours sincerely

Dr. Mahesh P.

Principal-CSIT
PRINCIPAL

Chhatrapati Shivaji Institute
of Technology, Durg



SINCE 1999

Chhatrapati Shivaji Institute of Technology

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Ref./CSIT/SSR-NAAC/2014/...342...

Date : 23-09-2014

Declaration by the Head of the Institute

It is certified that the data included in the "Self Study Report" are true to the best of my knowledge.

The SSR is prepared by us after thorough interaction with our colleagues and no part of it had been outsourced.

We are aware of the fact that the information submitted by us in the SSR is subject to validation by the Peer Team of NAAC.

Dr. Mahesh P.

Principal – CSIT

PRINCIPAL

**Chhatrapati Shivaji Institute
of Technology, Durg**



PREFACE

Realizing the importance of Engineering Education, its contribution to the growth and prosperity of human civilization, Eminent Academicians, Educationists, Administrators and Enthusiastic Social Workers of the state of Chhattisgarh had initiated the formation of a society under the aegis of Shivanath Shikshan & Seva Samiti, registered under the provision of the statute 44, of Madhya Pradesh Vishwavidyalaya Adhiniyam 1973 and Chhatrapati Shivaji Institute of Technology was founded at Durg in the year 1999. The aims and objectives of the society were and continue to remain so are stated below:

- i) To deliver world class education and excellent technical, moral and ethical training to the students admitted in CSIT.
- ii) To maintain stellar performance and high standard while imparting engineering education and appropriate training to the students.
- iii) To contribute to the overall development of the state by supplying appropriate technical man power that would shoulder the responsibility of development with confidence.
- iv) To contribute towards the much needed technology for rural development and conservation of Environment.

CSIT is spread over 29.2 acres of lush green and sprawling campus in picturesque surroundings and provides serene environment required for stimulating intellectual creativity and innovation.

CSIT has now a proud history of training students of seven different branches of Engineering at U.G. level and three branches at P.G. level and does have the satisfaction for the acceptance of its students by different organizations in the country (Growth of CSIT is given in Table no. 1).

In the present scenario of technical education in our country a typical tide is surging across India. Quantitative growth of colleges to the tune of 3500 and sanctioned intake 17lacs + seats has finally given rise to 40% vacancies across the country. Paucity of Qualified HR needed to handle the system has resulted in poor quality of pass outs and hence their non-acceptance for the job. End result of this is UNEMPLOYMENT of Engg. Grads.

Under such a scenario there are still some institutes that have developed excellent infrastructure and are located at people –friendly places and therefore attract good students and good teachers. As a result of these factors, their pass outs are of good quality and find good placements.

With humility, we claim that we have an excellent infrastructure, dedicated faculty much better than any other institute in the state and location–wise we

are located at a very easily approachable place. This has given us added momentum for our comfortable journey during the past 15 years.

Established in 1999, CSIT has approval from AICTE and the Govt. of Chhattisgarh and is affiliated to the “Chhattisgarh Swami Vivekanand Technical University, Bhilai”. We get year to year affiliation.

Three of our U.G. Courses, namely, Mech., CSE and E&TC were accredited for the first time by NBA in 2008. At present, Mech (2nd time), EEE and E&I have received accreditation by NBA in Feb 2013.

It's a matter of pleasure and pride for us to put it on record that the founder Chairman of CSIT Mr. Ajay Prakash Verma has been the recipient of “Edupreneurs Award – 2013” in recognition of his contribution to Engineering Education in India, the award being instituted by Engineering Watch Magazine.

Excellence of CSIT has been recognized by third parties like:-

- i) Outlook-Marketing & Development Research Survey which has included CSIT in top 100 Colleges of the Country at the 98th position in their latest survey (Outlook: June 2014).
- ii) CSIT is ranked 3rd among the top Engineering Colleges of Excellence by Global Human Resource Development Centre in Competition Success Review (June 2014).
- iii) We have been ranked 57th among best private institutes in India by GHRDC-CSR (June 2014).

With the credentials mentioned above, the college is now submitting its “Self Study Report” to the NAAC for getting assessed for the purpose of accreditation.

Table no. 1 Growth of Intake in CSIT

Year \ Dept.	CSE	E&TC	MECH	EEE	E&I	IT	MEX	M.TECH (CSE)	M.TECH (E&TC)	M.TECH (MECH)
1999-2000	60	60	60							
2002-2003	60	60	60	60						
2004-2005	60	120	120	60	60					
2008-2009	90	120	120	60	60	60				
2009-2010	120	120	120	60	60	60	60			
2010-2011	120	120	120	60	60	60	60	18	18	18

Executive Summary

Chhatrapati Shivaji Institute of Technology (CSIT) was established in the year 1999 and was affiliated to Pt. Ravishankar Shukla University, Raipur. The Technical University came into being in the year 2005 and since then CSIT is affiliated to Chhattisgarh Swami Vivekanand Technical University (CSVTU), Bilai. The beginning was humble. It started with 3 U.G. Courses with the sanctioned intake of 180. It started growing gradually but steadily and in the very first decade it assumed the dimensions of seven U.G. Courses and three P.G. Courses with sanctioned intake of 600 at U.G. level and 54 at P.G. level. To meet the requirement of nurturing 7 U.G. Courses and 3 P.G. Courses, the management has raised beautiful infrastructure sufficient to take care of institutional activities.

In a contiguous plot of 29.2 Acres, we have a built up space of 41592.0 sq. m. where we conduct curricular and co-curricular activities, seminars & conferences, workshops & FDPs and mega events like All India Conference & Laqshya.

“Go Green” concept has been implemented in our campus. Apart from maintaining lovely garden and lawns, use of non-conventional energy sources, management of sewage water by installing sewage treatment plant, application of solar water heaters for hostels, solar lighting system for college premises are the steps taken to uphold the concept of “Go Green”.

In the present scenario of technical education a typical tide is surging across India. Year after year, there is addition of New Engineering Colleges and increase in sanctioned seats of established engineering colleges without upper limit. Every college requires qualified staff. In this bewildering context, it has become very difficult, if not impossible to get suitable Ph.Ds to occupy the positions of Professors and Associate Professors in various Departments of Engineering Science. We are desperately battling against acute shortage of qualified staff.

Fortunately, at our Institute, we have 12 Ph.Ds and 17 are pursuing Ph.D who would hopefully get their degrees within next 2 years. A good strength of our faculty lies in 76 Post Graduates. Thus, at CSIT, Human Resource needed for imparting education is of very good quality. It's a fusion of skills & ability, values & virtues, wisdom & character and laughter & seriousness. People working here form a team of committed group of determined teachers ready to give stellar performance in building the technical manpower and intellectual capital of India.

Members of the faculty work on different committees. Thus, the faculty has the freedom of speech and thought. It is empowered to analyze and criticize the process. The net result of the extensive meetings of different committees



gives rise to masterpiece ideas which are born and challenged. Some of them survive and grow and the others are allowed to die. That is how we work in CSIT.

The intake we get is a group of students who belong to rural and semi urban area. Inherently weak in communication in English, the rural boys need longer time for getting fluency in English and mastering the engineering courses. Our faculty therefore has a stupendous task to perform so that our students can elegantly be transformed into acceptable engineering grads both at the national level and international level. And we are proud to submit that our placements range between 65 to 75 percent every year.

Finally, we hereby submit an honest account in full of CSIT to the Esteemed National Assessment and Accreditation Council of India for its consideration.

The SWOC Analysis

Strengths:

- NBA Accredited(Mech, EEE and E&I)
- Good Infrastructure
- Qualified and experienced faculties
- Well equipped laboratories including computer labs with latest and licensed softwares
- Large reservoir of books and online journals in Central library
- Good Connectivity
- Wi-fi enabled campus
- 100 % power backup
- Bank and ATMs in campus
- Industry Institute Partnership Cell(IIPC), Entrepreneurship Development Cell(EDC) and Renewable Energy Cell (REC) which are functional
- FDPs/Workshops/Conferences are organized regularly
- Recipient of grants under MODROB, RPS, FDP from AICTE and some grants from other sponsors for research activities
- Development of scientific temper through projects from IIIrd Semester onwards
- Personality grooming classes from 2nd year onwards.

Weaknesses:

- Research centre needs to be established
- Majority of students is from rural background
- Local recruitment is insignificant

Opportunities:

- To develop into an autonomous institute
- Collaboration with industries
- Vast pool of students in schools
- Requirement of technical man power in the region
- Establishment of major industries in the region

Challenges:

- Competition from nearby colleges
- Less number of PhDs.
- Fast changing technology
- Competition from other universities
- Guests from corporates-less due to distance from metros
- Lack of IT Park of software industries in CG.



Profile of the Institute

**Profile of the Affiliated / Constituent College****1. Name and Address of the College:**

Name :	Chhatrapati Shivaji Institute of Technology		
Address :	Kolihapuri, P.o. Pisegaon, Balod Road Durg		
City : Durg	Pin : 491001	State: Chhattisgarh	
Website :	www.csitdurg.in		

2. For Communication:

Designation	Name	Telephone with STD code	Mobile	Fax	Email
Principal	Dr Mahesh B. Parappagouda	0788-2320884 2229901,		0788- 2334808	Principal @csitdurg.in
Steering Committee Co-ordinator	Dr N. T. Khobragade	0788-2211381		0788- 2334808	deanacademi cs@csitdurg. in

3. Status of the Institution:

Affiliated College	☒
Constituent College	--
Any Other (specify)	--

4. Type of Institution**a. By Gender**

i	For Men	--
ii	For Women	--
iii	Co Education	☒

b. By Shift

i	Regular	☒
ii	Day	--
iii	Evening	--

5. It is a recognized minority institution?

Yes	---
No	☒

6. Sources of funding:

Government	--
Grant-in-aid	--
Self-financing	☒
Any other	--

7. a. Date of establishment of the college: ...06/09/1999 (dd/mm/yyyy)

b. University to which the college is affiliated /or which governs the college (If it is a constituent college)

Chhattisgarh Swami Vivekanand Technical University Bhilai

c. Details of UGC recognition:

Under Section	Date, Month & Year (dd-mm-yyyy)	Remarks (If any)
i. 2 (f)	---	---
ii. 12 (B)	---	---

(Enclose the Certificate of recognition u/s 2 (f) and 12 (B) of the UGC Act)

d. Details of recognition/approval by statutory/regulatory bodies other than UGC (AICTE, NCTE, MCI, DCI, PCI, RCI etc.)

Under Section/ clause	Recognition/Approval details Institution/Department Programme	Day, Month and Year (dd-mm-yyyy)	Validity	Remarks
AICTE	Central/1-2008782228/2014/EOA	04-06-2014	1 year	

(Enclose the recognition/approval letter)

8. Does the affiliating university Act provide for conferment of autonomy (as recognized by the UGC), on its affiliated colleges?

Yes	---
No	☒

9. Is the college recognized

a. by UGC as a College with Potential for Excellence (CPE)?

Yes	---
No	☒

b. for its performance by any other governmental agency?

Yes	---
No	☒

10. Location of the campus and area in sq.mts:

Location *	Rural
Campus area in sq. mts.	118169.48
Built up area in sq. mts.	41592.0

(* Urban, Semi-urban, Rural, Tribal, Hilly Area, Any others specify)

11. Facilities available on the campus (Tick the available facility and provide numbers or other details at appropriate places) or in case the institute has an agreement with other agencies in using any of the listed facilities provide information on the facilities covered under the agreement.

Auditorium	☒	01
seminar complex	☒	06
Sports facilities	☒	Indoors and Outdoor games
play ground	☒	Cricket, football, Volleyball,
swimming pool	No	--
gymnasium	☒	01
Hostel		
Boys' hostel		☒
i.	Number of hostels	02
ii.	Number of inmates	300
iii.	Facilities	Rooms with attached bathroom, TV hall, Water cooler, hygienic food, library and reading room
Girls' hostel		☒
i.	Number of hostels	02
ii.	Number of inmates	300
iii.	Facilities	Rooms with attached bathroom, TV



		hall, Water cooler, hygienic food, security guard, wi-fi., reading room
Working women's hostel		No
i	Number of inmates	--
ii	Facilities(Residential facilities for teaching and non-teaching staff (give numbers available — cadre wise))	Professor Duplex Banglows 4 Associate Professors Flat 8 Non teaching and others Flats 25
Cafeteria	☒	02
Health centre	☒	01
First aid, Inpatient, Outpatient, Emergency care facility, Ambulance	☒	First Aid, Ambulance,
Health centre staff		
Qualified doctor	☒	Part-time
Qualified Nurse	☒	Full time
Facilities like banking, post office, book shops	☒	Bank, Stationery shop.
Transport facilities to cater to the needs of students and staff	☒	07 Buses
Animal house	No	--
Biological waste disposal	No	--
Generator or other facility for management/regulation of electricity and voltage	☒	Power backup by two diesel generators total capacity 375 kva. Solar plant installed 200KWp
Solid waste management facility	No	--
Waste water management	☒	One Recycling Plant installed, one lac liters per day
Water harvesting	☒	Rain water harvesting, 10 locations

12. Details of programmes offered by the college (Give data for current academic year) 2013-14

S. N.	Programme Level	Name of the Programme/ Course	Duration	Entry Qualification	Medium of instruction	Sanctioned /approved Student strength	No. of students admitted
1.	Under-Graduate	BE CSE	4 Years	10+2	English	120	97
2		BE ETC	4 Years	10+2	English	120	86
3		BE MECH	4 Years	10+2	English	120	124
4		BE EEE	4 Years	10+2	English	60	62
5		BE EI	4 Years	10+2	English	60	23
6		BE IT	4 Years	10+2	English	60	15
7		BE MEX	4 Years	10+2	English	60	57
1	Post-Graduate	M Tech. Computer Science & Engineering. (CSE)	2 years	UG	English	18	18
2		M. E. Communication System (ETC)	2 years	UG	English	18	18
3		M Tech Industrial Engineering & Management (Mech)	2 years	UG	English	18	16

13. Does the college offer self-financed Programmes?

Yes	☒
No	
If yes, how many	10

14. New programmes introduced in the college during the last five years if any?

Yes	☒	No	--	Number	03 (M.Tech)
-----	-------------------------------------	----	----	--------	--------------

15. List the departments: (respond if applicable only and do not list facilities like Library, Physical Education as departments, unless they are also offering academic degree awarding programmes. Similarly, do not list the departments offering common compulsory subjects for all the programmes like English, regional languages etc.)

Faculty	Departments (eg. Physics, Botany, History etc.)	UG	PG	Research
Engineering	Computer Science and Engineering	Yes	Yes	
	Electronics and Telecommunication Engineering	Yes	Yes	
	Mechanical Engineering	Yes	Yes	
	Electrical and Electronics Engineering	Yes		
	Electronics and Instrumentation Engineering	Yes		
	Information Technology Engineering	Yes		
	Mechatronics Engineering	Yes		

16. Number of Programmes offered under (Programme means a degree course like BA, BSc, MA, M.Com...)

a.	Annual System	Nil
b.	Semester System	10
c.	Trimester System	Nil

17. Number of Programmes with

a.	Choice Based Credit System	Nil
b.	Inter/Multidisciplinary Approach	Nil
c.	Any other (specify and provide details)	Nil

18. Does the college offer UG and/or PG programmes in Teacher Education?

Yes	
No	☒

19. Does the college offer UG or PG programme in Physical Education?

Yes	--
No	☒

20. Number of teaching and non-teaching positions in the Institution

Positions	Teaching faculty						Non-teaching staff		Technical staff	
	Professor		Associate Professor		Assistant Professor					
	*M	*F	*M	*F	*M	*F	*M	*F	*M	*F
Sanctioned by the UGC / University / State Government <i>Recruited</i>										
<i>Yet to recruit</i>										
Sanctioned by the Management/ society or other authorized bodies <i>Recruited</i>	14	04	10	06	89	67	48	14	43	07
<i>Yet to recruit</i>										

*M-Male *F-Female

21. Qualifications of the teaching staff:

Highest qualification	Professor		Associate Professor		Assistant Professor			
	*M	*F	*M	*F	*M	*F	*M	*F
Permanent teachers								
Ph.D.	06	02	01				07	02
M.Phil.			01	01	02	06	03	07
PG ME/M Tech	09	02	07	04	46	14	62	20
PG MSc, MA, MCom, MBA, MCA	03	02	02	01	08	25	13	28
UG B Tech	01		02		30	27	33	27
Temporary teachers Nil								

*M-Male *F-Female

22. Number of Visiting Faculty /Guest Faculty engaged with the College

➤ Nil

23. Furnish the number of the students admitted to the college during the last four academic years.

Categories	2013-14		2012-13		2011-12		2010-11	
	Male	Female	Male	Female	Male	Female	Male	Female
SC	21	05	12	08	19	05	25	09
ST	31	0	18	10	22	05	21	08
OBC	105	33	129	50	180	67	190	64
General	184	85	201	91	191	105	252	123
Others	-	-	-	-	-	-	-	-

24. Details on students enrollment in the college during the current academic year: 2013-14

Type of students	UG	PG	M. Phil.	Ph.D.	Total
Students from C.G.	452	52	--	--	504
Students from other States	12	00	--	--	12
NRI Students	00	00	--	--	--
Foreign Students	00	00	--	--	--
Total	464	52	--	--	516

25. Dropout rate in UG and PG (average of the last two batches)

UG	1.094%	PG	2.25%
----	--------	----	-------

26. Unit Cost of Education

(Unit cost = total annual recurring expenditure (actual) divided by total number of students enrolled)

(a) Including the salary component

Rs. 69357

(b) Excluding the salary component

Rs 40000

27. Does the college offer any programme/s in distance education mode (DEP)?

Yes	---	No	☒
-----	-----	----	-------------------------------------

28. Provide Teacher-student ratio for each of the programme/course offered

Program	Course	Teacher Student Ratio
BE	Computer Science and Engg	1:15
	Electronics and Telecommunications	1:15
	Mechanical Engineering	1:15
	Electrical and Electronics Engineering	1:15
	Electronics and Instrumentation	1:15
	Information Technology	1:15
	Mechatronics	1:15
M Tech	Computer Science and Engineering	1:12
	Communication System	1:12
	Industrial Engineering & Management	1:12

29. Is the college applying for Accreditation:

Cycle 1	☒	Cycle 2	----	Cycle 3	-----	Cycle 4	-----
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Re-Assessment

(Cycle 1 refers to first accreditation and Cycle 2, Cycle 3 and Cycle 4 refers to re- accreditation)

30. Date of accreditation* (applicable for Cycle 2, Cycle 3, Cycle 4 and re-assessment only)

Not applicable



** Kindly enclose copy of accreditation certificate(s) and peer team report(s) as an annexure*

31. Number of working days during the last academic year. 287

32. Number of teaching days during the last academic year 198

33. Date of establishment of Internal Quality Assurance Cell (IQAC)
IQAC (dd/mm/yyyy)

Not applicable

34. Details regarding submission of Annual Quality Assurance Reports (AQAR) to NAAC

Not applicable

35. Any other relevant data (not covered above) the college would like to include. (Do not include explanatory/descriptive information)

- | | | |
|---|--------------------------|--|
| 1 | Funding by AICTE | Grants received so far Rs...97 lac |
| 2 | Internet | 10 Mbps BSNL Leased line |
| 3 | Wi-Fi enabled campus | Extensive coverage |
| 4 | Digital Valuation Centre | Valuation of answer sheets assigned by Technical University for undergraduate courses of Engineering. |
| 5 | Award | The Chairman of this Institute Mr Ajay Prakash Verma has received a prestigious award “ Engineering Watch Edupreneures Awards 2013 ” for his contribution to Engineering Education in India.” |
| 6 | Solar Power Plants | Two plants installed each of 100 kwp |
| 7 | Power Back Up | Two DG sets 250 + 125 KVA |
| 8 | Sewage Treatment Plant | Second organization in state to install this plant. |
| 9 | Yoga Classes | Yoga classes are held in premises. |



- 10 E-Tap conduction A technical awareness program is conducted for the Students of XII to understand the scope of engineering education.
- 11 Solar dryers are encouraged among the nearby villagers.

Criterion – I

Infrastructure



CRITERION - I**CURRICULAR ASPECTS****1.1 Curriculum Planning and Implementation**

1.1.1 State the vision, mission and objectives of the institution, and describe how these are communicated to the students, teachers, staff and other stakeholders.

Vision:

To nurture world-class intellectual growth by imparting high quality futuristic technical education to emerge as one of the best technical institutes to serve the mankind.

Mission:

- To create inspiring environment that will be most suited for research, creativity and innovation.
- Providing quality technical education at par with global standards.
- To accomplish National Center of Excellence in collaboration with industry.
- To establish international alliances and collaborations with foreign universities to achieve global excellence.

Quality policy:

To produce all rounded skillful and suave engineering graduates with intellect and knowledge along with positive outlook and savvy.

Quality Objectives:

1. Continual improvement in scope with rapid technical growth, in National & International market.
2. Professional broad vision, global perspectives and environmental changes management.
3. Exposure of students to most advanced technologies through frequent Industrial visits, Seminars, Workshops and Group Discussions under guidance of Eminent Personalities of varied respective spheres/fields.
4. To upgrade the existing Lab facilities with latest modern equipments.
5. Teachers' competency mapping & training.

Goals

Short term

1. To establish Quality standard in Education and to implement our quality policy of ISO 9001: 2008 for better Teaching – Learning process.
2. To achieve Quality Assurance and Ensurance through AICTE – NBA Accreditation.
3. To achieve 100% result with academic excellence.
4. To organize at least one faculty development programme (FDP) per year.
5. To frequently depute the faculty for various Faculty development programmes, conferences and workshops.
6. To encourage the students to take up Industry specific projects. Minimum one project/Dept.

Long term

1. To evolve into a Center of Excellence in Engineering and Technology by undertaking acknowledged research and development works.
2. To start Post Graduation and PhD programs in all branches of Engineering.
3. To develop Robotics Lab.

1.1.2 How does the institution develop and deploy action plans for effective implementation of the curriculum? Give details of the process and substantiate through specific example(s).

1. College Academic Committee conducts extensive meetings with the HODs and decides the strategy for effective implementation of the curriculum.
2. We have a college calendar which defines the details of the university dates regarding the schedule of the semester.
3. Lesson plans are an important aspect of every college teaching methodology. This gives a lot of emphasis on well defined lesson plan according to the university syllabus. These lesson plans are prepared by the faculty member prior to the start of the semester.
4. Periodic meetings are held in the respective departments under the respective HODs where strategies for effective planning & implementation of the curriculum are being carried out on regular basis.
5. All the faculty members irrespective of the branches have to develop their course files which are audited and updated regularly with the efforts of the faculty members in order to maintain quality of course file in each semester.
6. Unit tests, Class tests and Assignments are regular features of the assessment procedure of the students. For weak and below average students extra classes are conducted before and after the college hours.
7. Apart from these, we invite people of repute from various industries,

organizations and institutes to share their practical knowledge. This helps in enhancing the technical skills of the students.

8. Industrial visits and vocational training are also arranged by the college for the students to get familiar with industrial processes which they study in the curriculum.
9. In addition, we do have specialized training & placement cell that shoulders the responsibilities of inviting various companies for ultimate purpose of fetching training and job opportunities for the students.
10. We do have fully equipped in-house laboratories, Entrepreneurship Development Cell, Robotic Cell, Renewable Energy Cell, Industry Institute Partnership Cell, original softwares such as Mi-power, Matlab, Lab-view, SIMULIA, MiniTab16, Oracle, Fluidsim etc.

1.1.3 What type of support (procedural and practical) do the teachers receive (from the University and/or institution) for effectively translating the curriculum and improving teaching practices?

1. After receiving the notification from the university about the schedule of the semester, our institute plans its activities.
2. The institute prepares a plan of action for imparting the curriculum and training the students in the best possible manner.
3. The management extends its full support in all activities so that the execution is in consonance with the planning.
4. Lessons and Tutorials of the various subjects irrespective of the branches are prepared by the various departments prior to the commencement of the semester.
5. As per the guidelines for effective and efficient teaching practice, OHP & LCD projectors are made available to each department by the college authority.
6. NPTEL lectures and E-journals of various subjects are available in college digital library for better teaching & learning process.
7. College encourages the departments to conduct workshops, seminars, refresher courses etc. on recent technologies.
8. Faculty members are allowed to have duty leaves and financial reimbursement for expenditure incurred towards attending workshops, seminars and paper presentations.
9. Regular training programs are organized with the help of expert faculty members of premier institutes such as NITs and IITs.
10. Internet facility is available 24 hrs. a day through LAN and Wi-Fi for browsing the latest trends and technological up gradations.
11. Periodically refresher courses and seminars are organized by our university. Staff members are encouraged to participate in such programmes which help them in updating their knowledge and improving teaching skills. The college bears the expenditure on account of participation fee, Registration fee etc.

1.1.4 Specify the initiatives taken up or contribution made by the institution for effective curriculum delivery and transaction on the Curriculum provided by the affiliating University or other statutory agency.

1. Internet connection through Wi-Fi and LAN are provided by the college.
2. Institution encourages its faculties for giving lessons on LCD projectors and optical projectors for which sufficient number of LCDs and OHPs are available.
3. Institution also encourages its faculty members and students to participate in paper presentations, seminars and workshops to enhance their technical skills.
4. Before commencement of every semester, lesson plans are prepared and updated by the faculty member. Academically weak students are encouraged to take extra classes for the subjects they are weak in.
5. Institution has very good library which remains open from 8A.M. to 8P.M. Number of journals, papers, lecture notes, seminar proceedings etc are available in the Library.
6. Lab manuals are provided to students in which all the experiments according to the university syllabus are available. The Lab manuals are regularly updated.
7. Every department has association which takes care of Extracurricular and Co-curricular activities.
8. Gym and musical classes are also available for the students.
9. Bus transportation facility helps the day scholars so that they can commute safely.

1.1.5 How does the institution network and interact with beneficiaries such as industry, research bodies and the university in effective operationalization of the curriculum?

1. Industry Institute Partnership Cell (IIPC) has been established by the college for the purpose of keeping effective and efficient communication with prominent industries.
2. Guest lectures, seminars, industrial visits and recruitment activities are being carried out with an ease through IIPC reference.
3. College encourages and motivates faculty members to take up research projects sponsored by governmental funding agencies which are beneficial for enhancement of technical skills.
4. College has an Alumni Association and Training & placement cell which help to maintain professional relationship with the representatives of various industries.
5. Few faculty members of the college are actively involved in the curriculum design and development whenever the university proposes the revision of curriculum.

6. PEO's (Program Educational Objectives) and PO's (Program Outcomes) are formed by all the departments for effective operationalisation of the curriculum.
7. All the faculty members are members of the professional bodies like ISTE, IETE, CSI, ISOI, NAFEMS and IE and attend the conferences which in turn help them to refine their lectures and deliver them effectively to the students.

1.1.6 What are the contributions of the institution and/or its staff members to the development of the curriculum by the University? (Number of staff members/departments represented on the Board of Studies, student feedback, teacher feedback, stakeholder feedback provided, specific suggestions etc.

Table 1.1 Contributions of staff members to the development of the curriculum by the university:

S.N.	Name of the faculty	Department	Contributions
1	Mr. Rajesh Kumar, Professor & HOD	Mechatronics	Chairman, Board of Studies in Mechatronics.
			Member, Academic Council in University.
			Member, Examination Committee in University.
			Observer in University Examinations.
			Member, Syllabus Revision Committee in University.
2	Mrs. Archana Tiwari, Professor & HOD	EI	Chairman, Board Of Studies in Electronics & Instrumentation.
3	Dr. V. K. Chandra, Professor & HOD	EEE	Member of Board of Studies in University.
			Member of Syllabus Review Committee in University.
4	Mr. A. Jafri	EEE	Member of Expert Committee in University.
			Member of State Innovation Council in University.
5	Mrs. Padmini Sharma	EEE	Member of Syllabus Review Committee in University.

6	Mr. Mangal Singh, Professor & HOD	ETC	Member of Syllabus Review Committee in University.
			Member of Syllabus Review Committee in University (EI).
7	Mr. Deepak Sharma	ETC	Member of Syllabus Review Committee in University.
8	Ms. Sonika Arora, Professor & HOD	IT	Member of Syllabus Review Committee in University (ETC).
9	Mr. Santosh Kumar Sharma, Professor & HOD	Mechanical	Member of Board Of Studies in University.
			Member Syllabus Review Committee in University.
			Member of Examination Committee in University.
10	Mr. Jitendra Chaturvedani	Mechanical	Member of State Innovation Council in University.
11	Mr. Rajesh Gupta	EI	Member of Board Of Studies in University.
12	Mr. Shailendra Dewangan	EI	Member of Board Of Studies in University.
13	Mr. Apoorva Mishra	CSE	Member of State Innovation Council in University.
14	Dr. Anil Kashyap, Professor & HOD	Applied Mathematics	Member of Syllabus Review Committee in University.
15	Mrs. Renu Tripathi, Professor & HOD	Applied Physics	Member of Board of Studies and Exam Committee in University.
			Member of Syllabus Review Committee in University.
16	Mrs. Sumita Sengupta	Applied Physics	Member of Syllabus Review Committee in University.

1.1.7 Does the institution develop curriculum for any of the courses offered (other than those under the purview of the affiliating university) by it? If 'yes', give details on the process ('Needs Assessment', design, development and planning) and the courses for which the curriculum has been developed.

➤ No

1.1.8 How does institution analyze/ensure that the stated objectives of curriculum are achieved in the course of implementation?

College has some pre-determined objectives and the achievement of these objectives is ensured as given below:

1. Proper analysis of the feedback from the outgoing students is carried out. Their suggestions are discussed in a meeting by the Principal. Reformatations in content delivery are made accordingly.
2. College has teacher guardian scheme in the ratio of 1:20 .All TGs personally take care of the students through proper monitoring and counseling.
3. Employers and recruiters feedback is also obtained to check the ability of the students performance at the time of campus interviews.
4. Performance in Class tests, Unit tests, tutorial and semester examination are also a regular feature of the analysis. This analysis helps us in modifying our approach in classroom teaching.
5. Periodic review is taken by the departmental HODs through regular meetings about the coverage of syllabus and students' performance.

1.2 Academic Flexibility

1.2.1 Specifying the goals and objectives give details of the certificate/diploma/skill development courses etc., offered by the institution.

Our college imparts education at Under Graduate and Post Graduate Degree level in the following branches:

Table 1.2 List of courses offered.

	Programs offered	Sanctioned Intake
UG Level	Computer Science & Engineering (CSE)	120
	Mechanical Engineering (ME)	120
	Electronics & Telecommunication Engineering (ET&T)	120
	Electrical & Electronics Engineering (EEE)	60
	Electronics & Instrumentation Engineering (E&I)	60
	Information Technology Engineering (IT)	60
	Mechatronics Engineering (MEX)	60
	Computer Science & Engineering (CSE)	18
PG Level	Communication Engineering (ET&T)	18
	Industrial Engineering & Management (ME)	18

In addition to imparting prescribed curriculum to meet the requirements stipulated by the University, our Institute organizes skill development courses. The details of such programs are given below:

- Personality Development Programme
- Entrepreneurship Development Cell (EDC)
 - Organizing Entrepreneurship Awareness camps for students
 - Expert Talk for Students
 - Skill Development Programme
 - Formation of Solar Energy Cell
- Industry Institute Partnership Cell
- Robotics Cell
- Renewable Energy Club
- Training and Placement
- Software Development Cell
- Sponsored and Funded programmes
 - Various sponsored and funded programmes were organized by the institute. Details of the same are available in respective departments.
- AICON (All India Conference) is organized every year.
- Workshops and Seminars are regular feature

1.2.2 Does the institution offer programmes that facilitate twinning /dual degree? If 'yes', give details.

- **No**, the institute does not offer such programmes.

1.2.3 Give details on the various institutional provisions with reference to academic flexibility and how it has been helpful to students in terms of skills development, academic mobility, progression to higher studies and improved potential for employability.

Range of Core/ Elective options offered by the University and those opted by the college

- **Core options :** Please refer **Table 1.2 (Page no. 18)**
- **Elective options :**

Liberal options to choose elective subjects give good academic flexibility.

Elective options available to students enrolled in the Four Year Degree Program are as follows:

All the students have the choice of selecting their own elective subjects from the list of subjects proposed by the affiliating University.

The Elective subjects offered are based on the latest technological trends and industrial needs so that at the end of the program the student is ready for the industry.

List of UG Electives

S. No.	Name of the Departments	Semester	Electives	No. of choice offered
1.	UG- All Branches	6th	PE 1	6
		7th	PE 2	7
		8th	PE 3	5
			OE 4	19

List of PG Electives

S. No.	Name of the Departments	Semester	Electives	Total
1.	PG- All Branches	1st	Elective - 1	5
		2nd	Elective -2	5
		3rd	Elective - 3	5

In addition to the elective subjects, the students also do project work during the final semester that provides an opportunity for them to prove their technical skills.

Enrichment Courses:

The college offers following enrichment courses:

- Bridge courses.
- Computer literacy courses to students from non computer background.
- Personality Development Programmes.
- Communication Skill Development Programmes.
- Specialized Soft Skill Training to students to enable them to perform well in the campus recruitments.
- The students are offered assistance and support for preparation for GATE, GRE, TOEFL, CAT and GMAT exams by the Faculty Members to ease their passage to higher studies.
- The placement cell periodically hosts mock placement tests and interviews with the help of experts from the industry and gives feedback to the participants.

1.2.4 Does the institution offer self-financed programmes? If ‘yes’, list them and indicate how they differ from other programmes, with reference to admission, curriculum, fee structure, teacher qualification, salary etc.

All the programs offered by the college are self-financed only. All these programs are approved by the AICTE and affiliated to CSVTU, Bhilai.

For details please refer to **Table 1.2. (Page no. 18)**

- **Admissions:** 75% admissions are carried out on the basis of merit in Common Entrance Test (CGPET), 15% seats are allotted under Management Quota based on the merit of CGPET or IIT-JEE and 10% seats are allotted as Outside State Quota.
- **Curriculum:** The curriculum is designed by the Chhattisgarh Swami Vivekanand Technical University (CSVTU), Bhilai (C.G) and implemented by the college through the well prepared patterns of instructions, teaching aids like LCD projectors. For effective planning, implementation and delivery of the curriculum, computer and audio equipments are frequently used.
- **Fee Structure:** As fixed by Admissions and Fee Regulatory Committee (AFRC) of Govt. of Chhattisgarh.

1.2.5 Does the college provide additional skill oriented programmes, relevant to regional and global employment markets? If ‘yes’ provide details of such programme and the beneficiaries.

For regional and global employment markets, the institution offers following skill oriented programmes:

- Soft Skill Development Programs are organized for first year to final year students. Experts are invited to share their ideas.
- For improving practical knowledge, students are allowed to go for Industrial visits after 1st year.
- Students are permitted to go for industrial visits to interact with the outside world. Such visits are organized by the Industry Institute Partnership Cell of our college.
- Quantitative, Logical and Analytical Skill Development Programs are organized periodically.
- Value added courses in core areas are conducted for improving technical knowledge.
- Workshops, Conferences & Seminars are organized in different departments.
- Design projects are given more importance so that the students are adequately skilled and better employable.

- Industry oriented projects for final year students prove to be quite useful.
- Specific add-on programs are organized by the various departments, besides adding additional contents to some of the individual subjects to enhance the skills of the students:
 - Training on “**ABAQUS 6.10**” conducted by Mechanical Department of CSIT Durg by Prof. Pankaj Gupta & Mr. Abhishek Dewangan from 24th February, 2014 to 6th March, 2014.
 - Technical Training on “**Entrepreneurship Awareness**” organized by EDC from 16th-18th Feb, 2012.
 - Technical Training on “**Fluid Power & Hydraulic System**” organized by EDC for Mechanical Department from 15th Sep to 5th October, 2011.
 - Technical Training on “**Six Sigma Yellow Belt**” organized by MEA, CSIT, Durg for students of Mechanical Department from 20th Aug to 21st Aug, 2009.
 - Program on “**MATLAB & Simulink**” conducted by Mechatronics Department by Mr. Nirmal Pandey for pre-final students from 24th-26th Feb, 2014.
 - Program on “**Industrial Automation and Special Purpose Machine Building**” conducted by Mechatronics Engg. Department for the final year students.
 - Program on “**Programming Practice & Computer Ethics**” conducted by Computer Science & Engineering Department for final year students.
 - Program on “**Wireless Sensor Network**” conducted by Computer Science & Engineering Department for final year students.
 - Program on “**Network Programming**” conducted by Computer Science & Engineering Department for pre final year students.
 - Program on “**Cryptography & Network Security**” conducted by Computer Science & Engineering Department for final year students.
 - Program on “**JAVA with Database Connectivity**” conducted by Information Technology Department for final year students in session of 2013-14.
 - Program on “**PLC and SCADA**” conducted by Electrical & Electronics Engineering Department for final year students in session of 2013-14.

1.2.6 Does the University provide for the flexibility of combining the conventional face-to-face and Distance Mode of Education for students to choose the courses/combination of their choice? If ‘yes’, how does the institution take advantage of such provision for the benefit of students?

- **No**, the institute does not offer such programmes.

1.3. Curriculum Enrichment

1.3.1. Describe the efforts made by the institution to supplement the University's Curriculum to ensure that the academic programs and Institutions goals and objectives are integrated?

To supplement the University's curriculum the following initiatives are taken to integrate the academic programs and Institution's goals.

- To reach the goals and objectives, the institute has added some additional inputs in the syllabi to face the current trends in competitive areas.
- All faculty members identify the gaps in the curriculum and include topics to bridge the gaps.
- The institute encourages the faculty members to attend different faculty development programs, conferences, seminars and workshops to know the latest technology and scenario that can help to bridge the gap between University's curriculum and Institution's goals.
- Assignments are given of all subjects in addition to unit wise written/oral tests, quizzes etc.
- Guest lectures, seminars and workshops are organized in the institute to create the awareness about different additional courses, skills and industries.
- Industrial visits and training programs are arranged to develop and enhance the knowledge about the industries.
- The college supplements the University's curriculum by adding special courses like personality development, communication skill development.
- Various competitive exams like mock-GATE, mock-CAT and Aptitude test are conducted in the college to encourage the students to compete and succeed in these exams.
- Campus recruitment training and mock interviews are arranged in the college to succeed in interviews conducted by various companies.
- Additional experiments beyond the University syllabi are designed and conducted in all laboratories.
- Plenty of books are available in the central library beyond the University prescribed books.
- National journals, international journals, magazines, e-journals, e-books are available in the central library.

1.3.2. What are the efforts made by the institution to modify, enrich and organize the curriculum to explicitly reflect the experiences of the students and cater to needs of the dynamic employment market?

- The institute follows the syllabus designed by CSVTU, Bilai but when delivering the lectures to the students; our faculties supplement it with

their own proficiency and experience which is helpful to students in this highly competitive world.

- The Training and Placement Cell of the institute regularly interacts with the HR managers and executives of different industries and organizations and gets the information regarding the demands and expectations of corporate sectors. College conducts some Add-on courses and extra classes under the guidance of various committees to communicate these demands to the students to overcome the deficiencies in the students.
- Special training and orientation programs are conducted in the institute to make the students globally acceptable.
- The final year students are encouraged to make their main projects related to present technology and real time problems. For analysis and collection of the data, students are allowed to visit industries and organizations, if necessary.
- Mini projects for lower semesters (3rd, 4th, 5th, and 6th), seminars and viva-voice are made compulsory so that the students can gain skills and enhance confidence.
- The college is totally computerized and has internet connections and intranet facilities at the administration and academic levels. Campus is completely surrounded with Wi-Fi connection.

1.3.3. Enumerate the efforts made by the institution to integrate the cross cutting issues such as Gender, Climate Change, Environmental Education, Human Rights, ICT etc., into the curriculum?

The college organizes National Level Seminars and Conferences, for which experts are invited to share their experience and knowledge.

Gender:

- A Women Grievance Cell is established in the college.
- The college has given the charge of Convener to one of the senior women faculty & there are members who take care of this cell.
- This cell provides an exclusive platform to women employees and girl students to express their grievances either in person or in writing through complaint box.

Environmental Education:

- The institution has rain water harvesting system, to increase the ground water level.
- College has a Sewage Treatment Plant (S.T.P.) with a capacity of one lac litres per day.
- The NSS unit of college provides awareness to the students regarding the environment by organizing the following activities in the campus:-

1. Plantation
2. Campus Cleaning

Human Rights:

- College organizes awareness program for students on Human Rights.
- College forms an Anti-Ragging committee that takes necessary action to make the campus ragging free.

ICT:

- In today's scenario Information & Communication Technology plays a vital role for providing or sharing any important information.
- The college has the facility of group messaging, so that any information can be passed to the students easily.

1.3.4. What are the various value-added courses/enrichment programmes offered to ensure holistic development of students?**Moral and ethical values:**

- "Health, Hygiene & Yoga" is the separate subject introduced in the curriculum by the affiliating university.
- Special lectures are arranged on "Art of Living" & "Meditation". Eminent person from the religious society Prajapati Brahma kumari is invited to deliver such lectures.

Employable and life skills:

- The Departmental Associations of the college involve students in social service activities by arranging program on 15th August & 26th January at Vridha Ashram & Sneha Sampada.
- Guest Lecture on "Stress Management" is also conducted in the institute to enrich the holistic development of the students.
- The institute also gives importance to the co-curricular & extra-curricular activities by organizing Techno-fest events on Robotics, National level Paper Presentation (for Students).
- The Institute Celebrates Teachers' Day and Engineers' Day to enlighten the importance of teacher-student relationship and the role of engineers in nation building.

Better career options:

- To provide better career options for the students, college organizes personality development classes & Mock Tests so that they can easily get selected in campus drive.
- Training and Placement Cell continuously interacts with the employers to know about new areas of knowledge in demand and new career options available for the students.
- In-house actions are taken to enrich the students accordingly.

Community orientation:

- The college NSS team organizes Blood Donation Camp and Tree Plantation to create awareness among the students on various social and environmental issues.
- The Institute organizes E-Tap (Technical Education Awareness Program) for 12th Class science students of the state to enhance their understanding about technical education.

1.3.5. Citing a few examples enumerate on the extent of use of the feedback from stakeholders in enriching the curriculum?**Students:**

- The College has implemented student feedback system software for all the branches.
- Two different types of feedbacks are taken from the students:-
 - Faculty Feedback
 - Subject-wise Assessment
- In a semester these feedbacks are taken twice, one in the mid of the semester and the second at the end of the semester. Improvement is checked in second feedback.

Alumni:

- Every year college invites the alumni for a get-together. The college takes feedback from the alumni on that occasion.

Parents:

- College organizes parents' teacher meet in which feedback and suggestions are taken from the parents for further improvement.

1.3.6. How does the institution monitor and evaluate the quality of its enrichment programmes?

Public acceptance of any Institute depends upon:

- ✓ Discipline on the campus.
- ✓ Good teaching and good results.
- ✓ Good placement of final year students.

Discipline:

- Punctuality of students is monitored with the help of teacher guardians.
- There is a systematic approach towards class-room interaction, laboratory work, seminars, minor and major projects and industrial visits.

Teaching process:

- We conduct the institute level faculty development program before the start of every semester. Teachers are made ready to systematically deliver the goods.
- Curriculum coverage is as per the stipulations of university. If a teacher finds missing links in the prescribed curriculum, he/she conducts lectures on Add-on topics. Such topics are further strengthened by organizing expert lectures.
- As per the university requirements, we are supposed to deal with minor and major projects in 7th and 8th semester. At the institute level, we have introduced minor projects to students from 3rd semester onwards. This makes them ready to handle useful and developmental problems in 7th and 8th semester.

Result analysis:

- Our emphasis is on giving better results in university examination. Result analysis indicates the subjects in which the performance is poor. Special efforts are made to improve the result of such subjects for the succeeding batches.
- Feedback obtained from the students helps us to considerable extent in removing our weaknesses.

Placement of students:

- Training and Placement department imparts training to the eligible students, conducts institute level tests. Declares result. Students realize their strengths and weaknesses and accordingly use their latent intelligence to improve upon before they appear for actual placement interviews.
- The strategies followed and the programmes thus carried out by the institute are all leading to enrichment of the UG degree program and give us the satisfaction of discharging our duties to meet the expectations of our students and the other stakeholders.

1.4 Feedback System

1.4.1 What are the contributions of the institution in the design and development of the curriculum prepared by the University?

- Our Institute is affiliated to the Chhattisgarh Swami Vivekanand Technical University Bhilai.
- Many senior faculty of the institution are members of the various syllabus committees and also the Board of Studies constituted by CSVTU.
- Faculty members regularly attend workshops and seminars and offer their suggestions to enrich the curriculum.
- The following faculty members have been nominated as members in different Boards of Studies and the syllabus review committee of the University.

For details please refer **Table 1.1 (Page no. 16)**

1.4.2 Is there formal mechanism to obtain feedback from students and stakeholders on curriculum? If yes, how is it communicated to the University and made use internally for curriculum enrichment and introducing changes/new programs?

The college collects feedback from the students, alumni, parents, employers/industries and experts from academic field.

Students Feedback:

- Feedback about class room teaching is taken online twice in a semester and corrective steps are taken to improve the teaching.
- Students' Council meetings are organized by TGs almost every week to get feedback from the students.
- Teacher Guardians periodically conduct meetings which provide platform for the students to discuss all academic issues. Difficulties expressed by the students are noted and are communicated to the university either through the senior faculty who are members in the various boards of studies/syllabus committees etc. or forwarded to the university when suggestions are invited during syllabus revision.

Exit Feedback:

- The institution has defined Program Educational Objectives (PEOs) and Program Outcomes (POs) for each department. A PEO & PO Review Committee functions in each department comprising faculty members and Head of the Department.
- Feedback is obtained from the outgoing students on curriculum and worthy suggestions are communicated for improvement.

Subject wise Feedback:

- Subject wise feedback is used internally to check the fulfilment of objectives of the subjects. Additional contents are delivered to bridge the gap between prescribed curriculum and the actual requirement of the topic or topics.
- Additional laboratory hours are provided to conduct experiments beyond the syllabus to enrich the practical knowledge of the students.

Alumni:

- Alumni Meets are organized every year to obtain their views on curriculum and training.
- Alumni Meets create an opportunity for interaction with the students who are currently studying in the college. The current trends of engineering fields are discussed in these meets.
- The alumni of the college who have moved on to industry or for higher studies also give a feedback on how their years in the institution have helped them to perform in their places of work/study. The alumni give constructive suggestions to the students and advise them to remain focused towards present studies.

Parents:

- Interactive sessions between the college officials and parents are regularly conducted to invite their suggestions.
- The constructive suggestions are considered for the development of students.

Employers/Industries:

- Representatives of various industries give extremely useful feedback regarding the employability of the students and also their expectations from the students. Hence, any additional requirement that the industries may appreciate is provided by the college.
- The suggestions / comments are communicated to the affiliating University through the members of the Board of Studies.

1.4.3 How many new programmes/courses were introduced by the institution during the last four years? What was the rationale for introducing new courses/programmes?

- Please refer to **Criteria-3 Table 3.9 (Page no. 98-99)**

Criterion – II

Facilities @ CSIT



Bus Facility



Staff Quarters



Bank



Hostel



Cafe Corner



ATM

CRITERION II

TEACHING-LEARNING AND EVALUATION

2.1. Student Enrolment and Profile

2.1.1. How does the institute ensure publicity and transparency in the admission process?

Publicity:

- The institute is located in beautiful, green and serene ambience in the city of Durg, Chhattisgarh. The institute enjoys good reputation on account of its continued and visible commitment to the cause of technical education in the region.
- The institute maintains publicity in a planned manner and goes on throughout the year and not just during the time of admission. The plan for the admission process is advertised by the Directorate of Technical Education in the local newspapers and on their website.
- The institute advertises through the leading regional and English daily news papers and hoardings placed prominently at different parts of the state and in institution website.
- The notification advertisement contains detailed information about the UG and PG courses, Intake of students, eligibility criteria, process of admission and the facilities provided by the institute management and year wise campus placement records.
- The brochure gives updated information about the course curriculum, fee structure, facilities available and the rules and the regulations of the university and the Institute.
- The institute pamphlets are distributed to the nearby schools in order to motivate the students to join in our institute.
- All information is also available in the institute website : www.csitdurg.in

Transparency:

- The institute follows the academic calendar, provided by CSVTU and also prepares the institutional and the departmental academic calendar based upon it.
- The institute admission is done through the online counseling done by the Directorate of Technical Education for the students who have appeared in CGPET.
- The students are also admitted through the JEE on the basis of the preference given by the candidates in the online counseling.
- The students are also admitted through the Management Quota seats.
- Therefore, the equity and social justice are guaranteed through transparency and adherence to the rules.

2.1.2. Explain in detail the criteria adopted and process of admission (Ex. (i) merit (ii) common admission test conducted by state agencies and national agencies (iii) combination of merit and entrance test or merit, entrance test and interview (iv) any other) to various programmes of the Institution.

- The institute offers only undergraduate and post graduate courses in Engineering
- As per the directives of the university the admission of the students into the institute commences after the allocation of the Branch and the Institute through the online counseling.
- The admission selection process is based purely on the online counseling
- The admission procedure strictly adheres to rules which are laid down by the university and state government.
- Our institute also follows the reservation policy of the government.

2.1.3. Give the minimum and maximum percentage of marks for admission at entry level for each of the programmes offered by the institute and provide a comparison with other institutes of the affiliating university within the city/district.

- The institute offers four year Under Graduate courses in various departments like (1) Electronics and Telecommunication Engineering, (2) Computer Science and Engineering (3) Mechanical Engineering (4) Electrical and Electronics Engineering (5) Electronics and Instrumentation (6) Information Technology and (7) Mechatronics Engineering.
- The institute also offers Post Graduate courses in (1) Mechanical Engineering (Industrial Engineering and Management) (2) Electronics and Telecommunication Engineering and (3) Computer Science and Engineering
- Admission to particular department is made after the counseling only.
- Students seeking admission through management must also have to satisfy the requirements and should have good percentage in their academics.
- The institute strictly follows the Government policies in admission. The minimum percentage of eligibility for entrance test is 45% and the maximum percentage varies from year to year depending on general response to engineering programmes.
- The admission procedure for PG & UG courses is done strictly according to the conditions laid by the CSVTU, DTE and AICTE.

Table 2.1 Courses-wise lowest & highest ranks for UG - B.E. Programmes

C.	D.	CGPET Rank																											
		MECH		ET&T		CSE		EEE		E&I		IT		MEX															
		Lowest	Highest	Lowest	Highest	Lowest	Highest	Lowest	Highest	Lowest	Highest	Lowest	Highest	Lowest	Highest														
2013-14																													
	GEN		875		9097		3773		28957		1545		27271		2577		15339		17549		29541		14956		29158		1893		26823
	OBC		2192		8992		2639		25609		7437		31563		3512		21321		9359		30837		11130		25015		4532		28486
	SC		9536		24821		27622		0		24877		0		3746		23761		15061		28325		0		0		9762		27096
	ST		6133		31647		15209		0		0		0		29723		0		0		0		0		0		3940		0
2012-13																													
			792		5909		2202		29422		2897		21266		2978		11912		8902		25384		7977		32157		4614		17106
2011-12																													
			1504		5861		1255		19295		3049		23740		1550		7787		7417		28853		8157		29564		1717		19426
2010-11																													
			1305		28210		2411		28012		3110		20935		2301		28432		5985		30156		8238		28420		1612		27250

(Note: - C. = Category and D. = Department)

Table 2.2 Course-wise lowest & highest ranks for PG-M.Tech Programmes

Category	GATE Rank					
	Communication Engg.		Comp. Science. Engg.		Industrial Engg. and Management	
	Lowest	Highest	Lowest	Highest	Lowest	Highest
2013-14						

GATE	91	801	171	856	34	297
2012-13						
GATE	15	580	16	777	36	739
2011-12						
GATE	35	347	75	209	3	255
2010-11						
GATE	209	642	202	512	104	104

2.1.4. Is there a mechanism in the institution to review the admission process and student profiles annually? If 'yes' what is the outcome of such an effort and how has it contributed to the improvement of the process?

- The institute verifies and reviews all the profiles of the students admitted annually.
- The institution has well planned, well defined and well designed mechanisms as far as reviewing the profiles and merits of the students.
- The academic section regularly monitors and provides information to the departmental heads for proper corrective measures.
- The institute has strong admission team which works round the year to get the right kind of students through advertisement in print and electronic media, seminars in schools ,technical education awareness programmes (e-tap) and visits of school students to the institute .
- `Teacher Guardian Report` of the students is maintained which consists of the parameters like, academics, attendance , marks scored, behaviour ,sports and extra-curricular activities and over all development .
- The students are motivated by the faculty members through regular counseling sessions so that they transform into positive personality and students feel more confident to excel.
- As a result of these activities in the last Fourteen years, the institute has noticed a continuous growth of students in both academics and discipline. The students have learnt to energize their positive attitude to increase their potential into many constructive activities and got placed into renowned industries.

2.1.5. Reflecting on the strategies adopted to increase/improve access for following categories of students, enumerate on how the admission policy of the institution and its student profiles demonstrate/reflect the National commitment to diversity and inclusion

- The reservation policy of the government for the admission of the students belonging to the SC/ST/OBC category was not applicable to private engineering institutes till 2012-2013. The reservation policy has come into existence for private engineering institutes from 2013-2014.

- The category wise breakup of the students is given below

Table 2.3 UG

Categories	2013-14		2012-13		2011-12		2010-11	
	Male	Female	Male	Female	Male	Female	Male	Female
SC	21	5	12	8	19	5	25	9
ST	31	0	18	10	22	5	21	8
OBC	105	33	129	50	180	67	190	64
General	184	85	201	91	191	105	252	123
Others	-	-	-	-	-	-	-	-
Total	341	123	360	159	412	182	488	204

Table 2.4 PG

Categories	2013-14		2012-13		2011-12		2010-11	
	Male	Female	Male	Female	Male	Female	Male	Female
SC	1	2	-	-	3	-	-	1
ST	-	-	-	-	-	-	-	-
OBC	12	13	8	2	10	4	10	4
General	15	9	16	13	13	16	17	13
Others	-	-	-	-	-	-	-	-
Total	28	24	24	15	26	20	27	18

SC/ST/OBC:

- Upon the implementation of the reservation policy of the government the students from SC/ST/OBC are allotted seats as per state government policies. The benefits are extended to these students in our institution as per the state government rules.
- The institution continuously bestows its efforts to create awareness on the importance of higher studies i.e. education as a means of empowerment which brings a change in the society.
- The faculty members of our institution visit the neighboring villages in order to promote the awareness of higher studies by counseling the students belonging to the non creamy layers.

- The faculty members also explain the benefits of the reservation policies in the academic institutes.

Table 2.5 the no. of girl students admitted during the last four years

Program	Courses	2013-14	2012-13	2011-12	2010-11
B.E.	ET&T, EEE, E&I, MECH, MEX, CSE, IT	123	159	182	204
M.E.	Comm. Engg., CSE, Industrial Engg.	CSE=13 ETC=11 MECH=1 Total=25	CSE=9 ETC=6 MECH=0 Total=15	CSE=9 ETC=10 MECH=1 Total=20	CSE=8 ETC=10 MECH=0 Total=18
Percentage of all students		148/654= 22.62%	174/654= 26.60%	202/654= 30.89%	222/774= 28.68%

Table 2.6 Minority students admitted during last four years

Program	Courses	2013-14	2012-13	2011-12	2010-11
B.E.	ET&T, EEE, E&I, MECH, MEX, CSE, IT	ET&T=3 EEE=3 E&I=1 MECH=5 MEX=0 CSE=3 IT=0 Total=15	ET&T=2 EEE=0 E&I=2 MECH=1 MEX=3 CSE=7 IT=1 Total=16	ET&T=4 EEE=2 E&I=1 MECH=2 MEX=5 CSE=4 IT=2 Total=20	ET&T=4 EEE=3 E&I=3 MECH=3 MEX=5 CSE=5 IT=0 Total=23
M.E.	Comm. Engg., CSE, Industrial Engg.	CSE=00 ETC=00 MECH=00	CSE=00 ETC=00 MECH=00	CSE=01 ETC=01 MECH=01	CSE=01 ETC=00 MECH=00
Percentage of all students		2.29%	2.45%	3.51%	3.10%

2.1.6 Provide the following details for various programmes offered by the institution during the last four years and comment on the trends. i.e. reasons for increase / decrease and actions initiated for improvement.

Table 2.7

Programmes	Branch	Number of students admitted	Demand Ratio
UG 1.B.E. (2013-14) 2.B.E. (2012-13) 3.B.E. (2011-12) 4.B.E. (2010-11)	B.E.	464 519 594 692*(Including second shift)	77.33 86.5 99.0 96.11
PG 1.ME/M.Tech (2013-14)	Mech	17	94
	E & TC	18	100
	CSE	18	100
2. ME/M.Tech(2012-13)	Mech	7	39
	E & TC	17	94
	CSE	16	89
3. ME/M.Tech (2011-12)	Mech	12	67
	E & TC	18	100
	CSE	16	89
4. ME/M.Tech. (2010-11)	Mech	9	50
	E & TC	18	100
	CSE	18	100

S. N.	Program	Course	Student's Intake	Students Admitted	Demand Ratio
2013-14					
1	UG	Electronics and Telecommunication Engineering	120	86	71.67
2		Electrical and Electronics Engg.	60	60+2(Fee Waiver)	100
3		Electrical and Instrumentation Engg.	60	23	38.33
4		Computer Science Engg.	120	97	80.83
5		Mechanical Engg.	120	120+3(Fee	100

				Waiver)	
6		Mechatronics Engg.	60	57	95.0
7		Information Technology	60	15	25.0
1	PG	Communication Engg.	18	18	100.0
2		Computer Science Engg.	18	18	100.0
3		Industrial Engg.and Management	18	17	94.44
2012-13					
1	UG	Electronics and Telecommunication Engineering	120	104	86.67
2		Electrical and Electronics Engg.	60	56	93.34
3		Electrical and Instrumentation Engg.	60	48	80.00
4		Computer Science Engg.	120	111	92.5
5		Mechanical Engg.	120	120	100.00
6		Mechatronics Engg.	60	53	88.34
7		Information Technology	60	28	46.67
1	PG	Communication Engg.	18	16	88.89
2		Computer Science Engg.	18	18	100.00
3		Industrial Engg.and Management	18	7	38.89
2011-12					
1	UG	Electronics and Telecommunication Engineering	120	116	96.67
2		Electrical and Electronics Engg.	60	62	103.34
3		Electrical and Instrumentation Engg.	60	57	95.00
4		Computer Science Engg.	120	122	101.60

5		Mechanical Engg.	120	126	105.00
6		Mechatronics Engg.	60	64	106.67
7		Information Technology	60	50	83.34
1	PG	Communication Engg.	18	18	100.00
2		Computer Science Engg.	18	16	88.89
3		Industrial Engg.and Management	18	12	66.67
2010-11					
1	UG	Electronics and Telecommunication Engineering	120	124	103.34
2		Electrical and Electronics Engg.	120	120	100.00
3		Electrical and Instrumentation Engg.	60	57	95.00
4		Computer Science Engg.	180	175	97.22
5		Mechanical Engg.	120	129	107.5
6		Mechatronics Engg.	60	61	101.67
7		Information Technology	60	26	43.34
1	PG	Communication Engg.	18	18	100.00
2		Computer Science Engg.	18	16	88.89
3		Industrial Engg.and Management	18	14	77.78

2.1.7. Was there an instance of the Institute discontinuing a programme during last four years? If yes, indicate the reasons.

➤ No

2.2. Catering to Student Diversity

2.2.1. How does the institution cater to the needs of differently-abled students and ensure adherence to government policies in this regard?

- The institution follows the government policies regarding the needs of differently-abled students.
- The differently-abled students use the lift installed specifically for the purpose.

2.2.2. Does the institution assess the students' needs in terms of knowledge and skills before the commencement of the programme? If 'yes', give details on the process.

- The students are admitted to different programs on the basis of their ranks and the option exercised during counseling.
- The students hail from diverse academic background.
- The degree of command over English as the language of communication also varies, thus, necessitating understanding and categorization of the students within a program for the effective dissemination of thoughts and ideas.
- It is achieved by assessing the students in terms of knowledge and skills through different means.
- The Teaching scheme and the significance and the application of the courses are also emphasized upon the students.
- The faculty then teaches in the class depending upon the information obtained about the knowledge and the skills of the class.
- The institute takes care that every section has an appropriate mix of students from different programmes thus necessitating understanding and categorization of the students within a program for effective dissemination of thoughts and ideas.
- The academically weak students are identified by the faculty and it takes care of them by arranging special classes.

2.2.3 What are the strategies drawn and deployed by the institution to bridge the knowledge gap of the enrolled students to enable them to cope with the programme of their choice? (Bridge/ Remedial/ Add-on/ Enrichment Courses, etc.)

- The institute understands that the knowledge gap of the students can exist in two ways

- The gap at the entry level to the program of their choice on account of insufficiency of the syllabi of the qualifying examination, and
- The insufficiency of the course content prescribed by the University on account of the developments in science and technology.
- The Departments conduct doubt clearing, tutorial and remedial classes.
- Add –on programmes on latest topics are organized by the departments.
- Bridge courses are developed to link the course with the developments in technology.
- Guest Lectures are organized regularly.
- Robotic competitions, training programmes, workshops, and seminars are conducted to expose the students to the technological world.

2.2.4 How does the institute sensitize its staff and students on issues such as gender, inclusion, environment etc.?

- The faculty members are considered as the foundation stones on which the success stories of the students are built upon.
- The institute focuses upon the metamorphosis of its students into responsible global citizens with a strong national character imbibed in them.
- The institute underlines the equality of opportunity and facilities for both the genders.
- An atmosphere of mutual respect and co-existence is fostered.
- The institute has a Women`s Grievance Redressal Cell (SWAR) to take up problems and issues related to female faculty members and students.
- Women`s Day is celebrated regularly.
- Women empowerment can be observed as several key positions are headed by Female Faculty.
- The departmental student associations also have a fair representation of the girl students.
- The institute understands the significance of the ecological balance and has made sincere efforts since its inception by developing a green ambience.
- The layout of the college emphasizes the importance given by the institute towards establishing and maintaining an eco-friendly environment for studies.
- The institute has developed an efficient Rainwater Harvesting System.
- The institute organized an All India Conference (AICON) with the theme `Go Green` in 2012 to showcase the significance of environmental awareness.
- The institute promotes tree plantation and environment awareness programmes.

2.2.5 How does the institution identify and respond to special educational/learning needs of advanced learners?

- The advanced learners and the intelligent students are identified and picked up by the faculty members.
- Such students are exposed to a higher level of application oriented knowledge beyond the scope of the syllabus.
- Such students are encouraged to prepare for the examinations like GATE, CAT, GRE, Public Sectors etc.
- These students are guided and given financial assistance by the institute to carry out projects.
- The students are encouraged to participate in workshops and seminars by giving financial assistance.
- The faculty members devote additional time, advanced learning materials and assistance to the individual students continuously.
- These students are continuously exposed to motivational sessions.

2.2.6 How does the institute collect, analyze and use the data and information on the academic performance (through the programme duration) of the students at risk of drop out (students from the disadvantaged sections of society, physically challenged, slow learners, economically weaker sections etc.)?

- The academic performance of the students from the disadvantaged sections of society, physically challenged, slow learners, economically weaker sections etc.. is detected by the concerned faculty members during their lectures in classrooms.
- Such students are also identified on account of their non-participation in the academic, co-curricular and extra - curricular activities.
- Low marks, incomplete and wrong answers become a matter of concern.
- Such students are counseled by the individual and also by the team of faculty members and efforts are made to arrive at the cause of the problem and to find a solution to it.
- Poor and irregular attendance is also observed and analysed critically.
- The reasons for less attendance are found out by communicating with the students and the parents over telephone and also by post.
- The Public Relation Cell keeps a track of the communications made with the parents.
- The institute has a very strong Teacher-Guardian system .The appointed Teacher-Guardian remains with the allocated batch of the students till the batch passes out.
- The teacher-guardian maintains the data pertaining to the batch and acts as a mentor.
- Additional academic resources are provided to such students.

2.3 Teaching-Learning Process.

2.3.1. How does the institute plan and organize the teaching, learning and evaluation Schedules? (Academic calendar, teaching plan, evaluation blue print etc.)

The Affiliating University i.e. Chhattisgarh Swami Vivekanand Technical University, Bhilai brings out its own academic calendar every year.

CSIT not only follows the academic calendar as prescribed by the University but it also incorporates its own activities within the same framework at the institutional level. Moreover, the departments are also given the flexibility to include the departmental activities within the same framework.

So, the academic calendar of the university is further tuned to the activities of the individual departments.

Upon the allotment of the courses to be taught in the forthcoming academic session, the teaching plan is prepared by the faculty members keeping the teaching scheme in view and strict adherence to it which is monitored by the Head of the Department.

2.3.2. How does IQAC contribute to improve the teaching –learning process?

We have not constituted IQAC.

Instead we have College Academic Committee.

Realising the necessity of remaining competitive with other institutes in the region, our Management remains committed to its objective of imparting sound technical education with ethical values and hence has set up Academic Committee to translate its objective into reality. Our main objectives are –

- (a) Discipline on the campus
- (b) Good Results and
- (c) Good Placement of students.

Functions of this committee are:

- 1) Close watch on all activities related to academics.
- 2) Analysis of Class Test Results. Depending upon the performance, necessary corrective actions are proposed and followed.
- 3) Academic committee conducts Analysis of the University Results. If the result of any subject / subjects is less than 70%, the committee recommends a shift in the previous approach of teaching such subject/subjects.
 - Our experience indicates the improvement in the result at subsequent Exams.
- 4) Placement of Students:

- Placement of students from a particular Institute has become a slogan, which is quite deceptive.
 - Unfortunately, the parents who admit their children in Engg. College are not aware about the process of placement.
 - Companies that visit our campus for recruitment of final year students insist upon the strong and consistent Academic profile of the student which ranges from 65% onwards score at all levels from X SSC to VI Semester having no tolerance for a single back log at any stage. This criteria excludes many students.
 - Students fulfilling expected academic profile have to prove their genius in the tests conducted by the Company Representatives. This profile finally ends up with limited placement.
 - The situation of placement is further aggravated by the overall recession of the Indian Economy.
- 5) In spite of the barriers cited above our Academic committee has made it a point to impart proper training to the eligible students which helps them in the selection process. The pleasing result is 75% placement of CSIT students.
- 6) The same Academic Committee shall shoulder the responsibility of preparing the Annual Quality Assurance Report after getting accreditation by NAAC.
- 7) Thus, the College Academic Committee takes care of the Institutional Quality Management.

2.3.3. How is learning made more student-centric? Give details on the support structures and systems available for teachers to develop skills like interactive learning, Collaborative learning and independent learning among the students?

The learning has to be student centric because the student is the main stake holder and all the policies are designed keeping him in mind.

- The faculty gets all the resources from the institute which is essential for the overall development of the student.
- Dedicated Audio-visual hall facilitates video lectures.
- The students have access to NPTEL video lectures.
- The Central Library is powerhouse of knowledge .It contains thousands of books, periodicals, journals, magazines, newspapers, cd`s etc. The students can access the library whenever they feel like.
- The reference section provides a very good collection of reading material.
- The reference section also has online access to journal publications like IEEE .
- Internet access in internet laboratory and the wi-fi facility in the campus are available.

2.3.4. How does the institution nurture critical thinking, creativity and scientific temper among the students to transform them into life-long learners and innovators?

The institute understands that the Engineering programmes should be taught in manner in which the curiosity and interest are generated in the students. Once it is done then the student will himself foray into the technological sphere and try to find solution to his own queries and that posed by others.

The understanding of the programme should be in terms of Why? When? and How ?

The institution motivates the students to participate in activities like seminars, workshops, technical competitions, guest lectures etc which provide avenues for the thought process to ignite.

Moreover, since the day a student enters into the fold of the department, he is made to do project works of appropriate level. This may be an individual activity or in a group, depending upon the nature and the magnitude of the project. The Project work requires the application of the knowledge acquired through different courses which makes the students think of their own and convert their thought into physical reality.

The students are also made to perform additional experiments, beyond the scope of the syllabus, on the same setup.

The students are also encouraged to take up teaching and research as their profession thus transforming them into life long learners.

2.3.5. What are the technologies and facilities available and used by the faculty for effective teaching? Eg: Virtual laboratories, e-learning - resources from National Programme on Technology Enhanced Learning (NPTEL) and National Mission on Education through Information and Communication Technology (NME-ICT), open educational resources, mobile education, etc.

The technology and the facilities available and used by the faculty for effective teaching are

- Dedicated Audio-visual halls with LCD Projectors and Computer
- Powerpoint Presentations
- Video Presentations
- Video Lectures (NPTEL), 8000 hours of NPTEL video courses on 200 subjects are provided in the library.
- Webinars
- E-learning(Resource materials from different NITs and IITs)
- Digitalised lecture notes

2.3.6. How are the students and faculty exposed to advanced level of knowledge and skills (blended learning, expert lectures, seminars, workshops etc.)?

The institute has the wholesome development of its faculty and the students uppermost in the mind. It encourages development processes through different means.

Faculty:

- The faculty is sponsored for higher studies like M.Tech/Ph.D.
- The faculty is encouraged for research publications.
- Expenses of participation in Conferences/Seminars/Workshops are reimbursed.
- The institute regularly organizes Faculty Development Programmes
- Participations in Faculty Development Programmes are encouraged and sometimes made compulsory.
- All India Conference (AICON) is being organized since 2009.

Students:

- The students are encouraged to participate in conferences and seminars
- The expenses incurred are reimbursed
- Guest lectures are organized regularly
- Add-on Programs are conducted on relevant topics
- The students visit the industries every semester.

2.3.7 Detail (process and the number of students benefitted) on the academic, personal and psycho-social support and guidance services (professional counseling/mentoring/academic advise) provided to students?

Apart from teaching the students it is also necessary that they be made strong in terms of mind and body to cope up with the stressful life. The students should also know the opportunities which they will have after graduating.

- Counseling is done consistently by the faculty members.
- The teacher guardians constantly look after the students and provide them with all kinds of support and guidance.
- Information regarding job opportunities is provided by the TPO
- Yoga sessions are conducted regularly
- Industrial Experts interact with the students and motivate them to join the industries
- The institute has tied up with youngistaani.com, experts in aptitude development, for the personality development of the students.

2.3.8. Provide details of innovative teaching approaches/methods adopted by the faculty during the last four years? What are the efforts made by the institution to encourage the faculty to adopt new and innovative approaches and the impact of such innovative practices on student learning?

The institute feels that the conventional teaching approach should be supplemented with the latest technology to make the teaching-learning process more effective and interesting.

During the last four years the faculty members of the institute have adopted several innovative methods.

- Video lectures
- Powerpoint presentations
- Add-on programmes
- Webinars
- Internet facilities
- Workshops on latest softwares
- Participations in online faculty development programmes conducted by IIT`s.
- Sharing of experience by the participants of conferences/ seminars/ workshops etc. with the faculty/students

There has been a visible impact on the students on account of innovative teaching approaches and the students have begun to realize the significance and the potential of Engineers.

2.3.9. How are library resources used to augment the teaching-learning process?

The institute has a very well equipped Central Library. It contains books, journals, thesis, CD`s etc. for acquiring knowledge on subjects related to curriculum and beyond it. Each of the department has its own Departmental library as well

Central Library

- The library is accessed by the students and the faculty members alike.
- Library cards are issued to the students and the faculty
- An additional library card is provided to the students who come within Top 20 ranks in the class
- The library software provides information on the status of the stock
- The books are frequently recommended by the departments and upon the approval of the library committee the purchase is made.
- The Central Library also has a digital library where the online journals can be accessed.

The college has a central library with plinth area 1912 sqm with all modern facilities. The library has the following resources:

Table 2.8 Library Resources

Titles	6842
Total No. of Volumes	58,761
Reference Books	7,215
Library Automation	All the data relating to the library is computerized and bar coding has been introduced.
E-Books	20
E-Journals	194 (IEEE-145, SPRINGER-49)
Print Journals	80
Project Reports	2,500
Digital Library	
Number of Computers	40
Multimedia and Internet facility Broad Band internet Connectivity	10 Mbps
Library Server	HP Prolient ML 350, 2 GB Ram 146 GB Hard disk, HP Motherboard, and Intel Processor
Library Networks Membership	DELNET
Discussion Rooms	02 (for start-stop mode of e-learning)
Video Projection Room	01
Satellite Enabled Sources	Nil
E-Learning Material	8000 Hours of NPTEL Video Courses on 200 subjects
Media Resource Centre	Reprographic, Printing and Document
Scanning facility Number of users per day	400 (on average)

Departmental Library:

The departmental library caters to the requirement of the books by the faculty for ready reference. It contains books pertaining to the Academic Programme.

2.3.10. Does the institution face any challenges in completing the curriculum within the planned time frame and calendar? If 'yes', elaborate on the challenges encountered and the institutional approaches to overcome these.

No such situation has ever arisen because the lesson plans are prepared with the academic calendar and the number of classes are made available for the completion of the course under consideration.

However, if the situation necessitates the taking of the extra classes then it can be done after planning under the guidance of the Head of the Department and seeking the approval of the Principal.

2.3.11. How does the institute monitor and evaluate the quality of teaching learning?

The core essence of any academic institute is effective teaching learning and its monitoring. CSIT makes all out efforts to see that quality education is imparted to the students.

- Continuous monitoring of the Teaching Plan and its effective implementation is done by the Head of the Department
- Feedback is taken from the students twice in a semester, which gives detailed information on different attributes of the teacher.
- The feedback obtained from the students is discussed with the faculty by the Head of The Department.
- The marks obtained in the class-tests are also an indication of the effectiveness of the teaching learning process.

2.4. Teacher Quality

2.4.1 Provide the following details and elaborate on the strategies adopted by the institute in planning and management (recruitment and retention) of its human resource (qualified and competent teachers) to meet the changing requirements of the curriculum

CSIT has a well defined recruitment policy for the teaching and the non-teaching staff.

The Head of the Department sends a requirement with a justification to the Principal for the recruitment of teaching staff based on a specific course or fulfillment of Teacher-Student ratio or Cadre Ratio much before the commencement of the semester.

Upon the approval of the requirement, the institute floats advertisement in national and regional newspapers. The applications received are scrutinized, a merit list is prepared and the eligible candidates are called for interview. A Selection Committee comprising of the concerned Head of Department, Senior Faculty Members, Subject Expert, Dean (Academics), Principal and Director conducts the interview and the demonstration of the subject knowledge and

the teaching skills of the candidates and submits the report of selected candidates to the Chairman. The Principal in consultation with the management takes the final decision and issues appointment order. Direct appointment may be given in case of exceptionally good candidate. Same procedure is adopted in case of non-teaching staff.

2.4.2. How does the institution cope with the growing demand/ scarcity of qualified senior faculty to teach new programmes/ modern areas (emerging areas) of study being introduced (Biotechnology, IT, Bioinformatics etc.)? Provide details on the efforts made by the institution in this direction and the outcome during the last three years.

The scarcity of qualified senior faculty looms countrywide not only for modern areas or emerging areas of study but also in the traditional programmes.

In order to fulfill the requirement of such kind of faculty the institute floats advertisements in leading national newspapers.

2.4.3. Providing details on staff development programmes during the last four years elaborate on the strategies adopted by the institution in enhancing the teacher quality

(a) Nomination to staff development programmes

Academic Staff Development Programmes							
Number of faculty nominated							
		Refresher courses	HRD programmes	Orientation programmes	Staff training conducted by the university	Staff training conducted by other institutions	Summer / winter schools, workshops, etc.
CSE	2013-14					20	6
	2012-13					23	6
	2011-12					19	
	2010-11					17	
ME	2013-14		4		2	4	26
	2012-13				2	4	6
	2011-12				4		4
	2010-11						3
EEE	2013-14	14		15		3	33
	2012-13	12	14	14			14
	2011-12	10			2	5	8

	2010-11	10				1	7
ETC	2013-14					4	8
	2012-13					3	11
	2011-12				1	11	35
	2010-11						36
IT	2013-14						1
	2012-13						4
	2011-12						7
	2010-11						1
E & I	2013-14					2	5
	2012-13						4
	2011-12						5
	2010-11						3
MEX	2013-14		1		1		5
	2012-13		1			1	2
	2011-12					1	3
	2010-11						3
Physics	2013-14					1	
	2012-13						
	2011-12						3
	2010-11					2	
Chemistry	2013-14					1	
	2012-13						
	2011-12		1				1
	2010-11		1				1
Maths	2013-14						
	2012-13	1			1		1
	2011-12						2
	2010-11						
Communication Skill	2013-14						
	2012-13						
	2011-12				2	1	
	2010-11						

(b) Faculty Training programmes organized by the institution to empower and enable the use of various tools and technology for improved teaching-learning

- Faculty Development Programmes have been conducted by departments on varied topics.

(c) Faculty Involvement in Workshops/seminars/conferences

- Invited as resource persons in Workshops / Seminars / Conferences organized by external professional agencies
- Participated in external Workshops / Seminars / Conferences recognized by national/ international professional bodies
- Presented papers in Workshops / Seminars / Conferences conducted or recognized by professional agencies
- Participated in Summer Schools/Winter Schools

(d) The institution has left no stone unturned to see that the faculty members are provided with the resources which can enhance their quality.

- The access to the online journals provides a cutting edge in imparting the latest information in technology.
- The Central library being a large reservoir of books is an important source of getting knowledge.
- Interaction with industrial experts and the visits to the industries add to the exposure of the faculty.
- Interaction with subject experts paves the way for a better understanding of the course.

2.4.4. What policies/systems are in place to recharge teachers? (e.g.: providing research grants, study leave, support for research and academic publications teaching experience in other national institutions and specialized programmes industrial engagement etc.)

The institution extends full support for the professional development of the faculty.

- Faculty members are encouraged to pursue M.E./M.Tech. and Ph.D. through Faculty Development Schemes.
- Research Paper publications in reputed national and international journals are encouraged.
- Applications for research grants are encouraged.
- Participations in refresher courses, faculty development programmes, national and international conferences, workshops, seminars, summer schools and winter schools are motivated by reimbursing the expenditure incurred.
- The institute also conducts conference every year for the benefit of the faculty.

The Institute also motivates the faculty by following means

- Availing grants/ funds,
- sanctioning training program and leaves to scholars,
- Supporting in terms of technology and information needs, adequate infrastructure & human resource.
- Providing duty leave to faculty for participating and presenting papers at national and international conferences.

2.4.5. Give the number of faculty who received awards / recognition at the state, national and international level for excellence in teaching during the last four years. Enunciate how the institutional culture and environment contributed to such performance/ achievement of the faculty.

The institutional culture and environment have played a very significant role in bringing laurels to the institute which is a testimony to the devotion, diligence, strong character, discipline and the aspiration for achieving academic excellence. The list of the award winners will be shown to the peer team during visit.

2.4.6. Has the institution introduced evaluation of teachers by the students and external Peers? If yes, how is the evaluation used for improving the quality of the teaching-learning process?

Yes, the evaluation of teachers is done twice by the students in a semester. The format is transparent and easy to understand and use. The format includes the essential attributes of an effective teacher. The purpose of the feedback is to gauge the acceptance of a teacher by the students in terms of Teaching-Learning.

The Principal takes cognizance of the feedback and calls for explanation from the faculty members with less than 70 % marks to their credit and also gives directions and suggestions for improvement.

No, the external peers do not evaluate the teachers.

2.5. Evaluation Process and Reforms.

2.5.1. How does the institution ensure that the stakeholders of the institution especially students and faculty are aware of the evaluation processes?

The institution ensures that the stakeholders of the institution especially students and faculty are aware of the evaluation processes.

- The students are evaluated by the faculty through the class-tests, viva-voce, question-answer sessions, quizzes etc.
- The assignment copies are evaluated with due promptness.

- The completion of the laboratory records also becomes a parameter for evaluation.
- The model answers of the class-tests are displayed so that the students can understand the strengths and the weaknesses of their answers.
- The students evaluate the teachers twice a semester.
- The feedback format encompasses all the desirable attributes of a teacher.
- The Principal takes a note of the feedback result and instructs the faculty members whose total marks obtained fall below a certain level to improve upon the deficiencies.
- The Head of Department seeks explanation and the line of action to overcome the deficiencies.

2.5.2. What are the major evaluation reforms of the university that the institution has adopted and what are the reforms initiated by the institution on its own?

The major evaluation reforms in the University are as follows

- Digitalisation of the Answer copies
- Evaluation of the answer copies on computers
- Setting up of Digital Valuation Centres at Different institutes.
- Filling up of practical marks in OMR Sheets
- Filling up of Teacher's Assessment and Sessional Marks in the University software.
- Use of ICR sheets and CDs for the submission of Teacher's Assessment and Sessional Marks.
- The evaluation reforms initiated by the institute on its own.
- The institute is one of the pioneers in setting up of the Digital Valuation Centre in its premises.
- Internal assessment for the students is conducted as per the University criteria.
- Class tests are conducted to evaluate the performance of students.
- Assignments, projects, seminars, practicals are considered as evaluation parameters

2.5.3. How does the institution ensure effective implementation of the evaluation reforms of the university and those initiated by the institution on its own?

- The evaluation reforms of the university are followed in the best of the spirit.
- The institute follows the evaluation guidelines for class tests and practicals.
- The institute also follows the evaluation guidelines as laid down for the mandatory summer training.

- The students are briefed about the various evaluation processes.
- All the records are maintained i.e. answer sheets, class test marks, assignment evaluation details, attendance etc.

The institution has adopted the followings steps for evaluation reforms i.e.

- The institute maintains the University pattern of the question paper for the class tests.
- The questions given for assignments are important from the University examination point of view.
- An unitwise question bank is prepared for all the courses and given to the students.
- Internal assessment is conducted as per the university criteria.
- The evaluation is also done through assignments, projects, seminars , practicals etc.

2.5.4. Provide details on the formative and summative evaluation approaches adopted to measure student achievement. Cite a few examples which have positively impacted the system.

Formative approach:

- The University is the sole authority for the formative evaluation. The faculty members follow the formative assessment guidelines while evaluating an answer copy or while awarding marks in practical examinations.
- The evaluation through courses like health, hygiene and yoga, value education, personality development, managerial skills etc. gives substantial information about the students development.

Summative approach:

- Classroom teaching by using various methods makes a student capable of appearing in any examination. Tutorial classes comprise of 30 students in a batch for better understanding of the problems.

2.5.5. Enumerate on how the institution monitors and communicates the progress and performance of students through the duration of the course/programme? Provide an analysis of the students results / achievements (Programme/course wise for last four years) and explain the differences if any and patterns of achievement across the program /courses offered

The class-tests are conducted as per the academic calendar prepared by the institute .The marks are displayed on the notice board and the answer copies are shown in the classrooms.

The parents are informed by post and also telephonically regarding the performance of their ward.

The University results are available on the website of the university i.e. www.csvtu.ac.in

Table 2.9 Result of last 4 batches

Batch	Branch	Overall pass	1st Division	2nd Division
2010-14	Mech	75.18	92	8
	ETC	52.38	58	8
	CSE	75	45	30
	EEE	74.29	66	12
	EI	75	24	18
	IT	50	12	0
	MEX	28.8	12	0
2009-13	Mech	74.43	85	14
	ETC	63	70	13
	CSE	77	76	1
	EEE	76.19	44	4
	EI	63.07	38	3
	IT	71.74	21	8
	MEX	21.31	8	3
2008-12	Mech	65.21	66	24
	ETC	70.37	77	18
	CSE	53	52	1
	EEE	60.32	33	5
	EI	61.9	31	7
	IT	44.73	11	6
	MEX	NA	NA	NA
2007-11	Mech	67.39	72	21
	ETC	69.67	71	78
	CSE	48	48	NA
	EEE	65.67	39	5
	EI	65.21	30	15
	IT	NA	NA	NA
	MEX	NA	NA	NA

2.5.6. Detail on the significant improvements made in ensuring rigor and transparency in the internal assessment during the last four years and weightages assigned for the overall development of students (weightage for behavioral aspects, independent learning, communication skills etc.

The process of internal assessment is transparent and the students are aware of the system.

The average of two best marks obtained in the class-tests is considered for Third to Sixth semester and in Seventh and Eighth Semester it is the best

marks obtained in the class-test which are recommended as class-test marks to the University.

The internal assessment is made by the Faculty Members keeping in mind the following aspects / factors of student's performance during the academic year:

- Class attendance
- Class assignments
- Marks obtained in class-tests

2.5.7. What are the mechanisms for redressal of grievances with reference to evaluation both at the institute and University level?

- **At the Institute level**

The students contact the teacher concerned and the teacher points out the mistakes and deficiencies in the answers and suggests the corrections, by showing the answer sheet, attendance record or practical performance records.

- **At the University Level**

There is a provision of revaluation and re-revaluation after paying the appropriate fees. The grievance of the students, if it affects a larger number of students or is of general nature, is communicated by the Principal to the University.

2.6. Student Performance and Learning Outcomes

2.6.1. Does the institute have clearly stated learning outcomes? If 'yes' give details on how the students and staff are made aware of these?

- The institute has categorically stated its Vision and Mission which is prominently displayed at different places.
- The attainment of the Vision and the Mission requires a well laid programme coupled with the determination to achieve it.
- The Vision and the Mission are in tune with the expectations of the society and are in the best interest of the youth and the nation.
- The institute aims towards inculcating a scientific thinking and reasoning process in the aspirants of Engineering programmes.
- The institute realizes that a well laid down policy and its proper implementation only can lead to the achievement of the Vision and the Mission.
- Each of the Engineering program has defined its Programme Objectives and Outcomes.
- The Programme Objectives and the Outcomes are prominently displayed near the Departments.
- The Course objectives and its outcomes are also stated and made known to the faculty and the students alike.

2.6.2. How are the teaching, learning and assessment strategies of the institution structured to facilitate the achievement of the intended learning outcomes?

- The institute considers its students as willful partners in the learning process.
- They require consistent inputs of tested, proven and rational knowledge.
- A closed cycle of Teaching-Learning-Assessment-Teaching is adopted
- The teaching is imparted with the course objective and the outcome in the mind.
- The attentiveness, inquisitiveness and the attitude of the students in the class indicate the Learning atmosphere in qualitative terms.
- The assessment of the students is done through the Class-tests, question-answer sessions, quizzes, surprise tests etc.
- The assessment indicates the degree of Learning by the students in quantitative terms.
- Model answers of the Class-test question papers are displayed on the notice board and discussed in the class room.
- The results of the class-test serve as the barometer for the re-teaching or re-orientation of the teaching required for more fruitful result.
- The Department decides the modus operandi for each of the course depending upon the result of the assessment.
- The teachers are taken into confidence and are made aware of the need of the situation.
- The subject teaching is then done accordingly.
- Subsequently, a second class test confirms the effectiveness of the strategies adopted.
- Then, if required, extra classes take care of the additional inputs required.

2.6.3. What are the measures/initiatives taken up by the institution to enhance the social and economic relevance (quality Jobs, entrepreneurship, innovation and research aptitude) of the courses offered?

- The effectiveness of the Engineering Programmes is driven by the need of the industry.
- The technical manpower requirement of the industry acts as an indicator for the aspiring students of Engineering to exercise their option for specific Programme.
- The industry is driven by the economic effectiveness of a Programme which in turn decides its economic relevance.
- The quantity and the quality of the job offered by the industry decide the inclination of the students towards a specific programme.
- The society also looks for relevant programme in terms of utility and job opportunity.

- The society attaches a tag of priority to the Engineering programmes and the institutes that act as an unwritten guideline for the prospective entrants.
- In order to increase the acceptance of a programme, the institution tries to make the programme lucrative by emphasizing upon its real time application, planning and executing live projects, encouraging the students to participate in competitions, fostering a research environment and arranging for appropriate jobs.
- Guest lectures on topics of human relevance provide information beyond the Engineering subjects.
- The institute tries to enhance the employability of the students by means of finishing schools which focuses on appropriate attitude and etiquettes.
- The efforts are made to chisel the students through interaction with the industry experts.
- When a student excels in the chosen field of Engineering then the social and economic relevance of the programme increases by itself.

2.6.4. How does the institution collect and analyze data on student learning outcomes and use it for planning and overcoming barriers of learning?

- The Class tests are conducted as per the academic calendar
- Atleast two class tests are conducted every semester
- The class-tests have the same pattern as that of the University question papers.
- The model answers are prominently displayed for the students to arrive at the best answer and also to understand the lacuna and the mistakes committed.
- The marks obtained in the class-tests are an indication of the degree of the subject knowledge acquired.
- Poor or less marks indicate ineffective teaching-learning process.
- The students also express their opinion on different student centric parameters in the two feedbacks conducted in a semester.
- The feedback reports are analyzed by the Principal and then forwarded to the Head of the Department to interact with the faculty and the students on the basis of the marks given to the faculty for a specific trait by the students.
- The Head of the Department also interacts with the students for the redressal of the grievance of the students expressed through the feedbacks.

Barriers of Learning

- Diverse educational, social and economical background of the students
- Inexperience of the teachers
- Poor communication skill of the students
- Lack of interest in the subject or course

Methods of Overcoming the Barriers of Learning

- The students are made to understand the significance of engineering education and how it helps in the development of mind, widening of the thinking process, rationalization of thoughts, development of self, upliftment in the standard of living, increased acceptance by the society and to contribute towards the development of mankind.
- Aptitude Development classes are conducted from the Third Semester onwards.
- Personality Development classes are also conducted.
- Inexperienced faculty members are encouraged to upgrade their educational qualification, participate in Faculty Development Programmes, conferences, seminars, workshops, summer and winter schools, teacher-training programmes etc.
- The students are encouraged to communicate in English.
- Special English Language classes are conducted for the students.
- The subjects are made more interesting to the students by stressing upon the significance and the real time application and the contribution they would make when they graduate.
- The students are made to visualize and understand the role of the Engineers for the development of the nation and the society.

2.6.5. How does the institution monitor and ensure the achievement of learning outcomes.

The institute has a well defined, transparent and stable academic and administrative system.

- It is mandatory to attend the theory and the laboratory classes
- The time –table is prominently displayed
- The classes are of fixed duration
- Tutorials are a part of the teaching scheme
- The assignments are given on a regular basis
- The assignments are corrected promptly and appropriate markings are made by the Faculty
- Attendance is taken in all the theory and the laboratory periods.
- The attendance is displayed fortnightly and the students with less attendance are counseled
- The parents of the students with short attendance are contacted telephonically, by e-mails and also the information is sent by post by the Public Relations Cell.
- Classroom attendance is encouraged by giving cash incentives to students with 95% or more attendance
- Semester toppers and University toppers are given cash incentives.
- The classroom performance is also considered during the finalization of sessional marks.
- Regular counseling is given to the students for proper orientation of efforts.

- Additional practice sessions are given in the laboratories to the students who need it.
- Faculty members are encouraged to conduct surprise tests, quizzes, question-answer sessions etc.

2.6.6. What are the graduate attributes specified by the institute/affiliating university? How does the institute ensure the attainment of these by the students?

The institute aspires towards the metamorphosis of the raw and energetic students into bundles of technical powerhouse that will have the potential to guide the world with competence and confidence towards better life.

- The institute aims towards making the students academically sound.
- The institute endeavours to enhance the communication skills of the students.
- Personality development of the students is the key focus of the institute.
- The institute aims towards making its students better employable
- The institute strives to mould the students into socially responsible global citizens.
- The institute believes that the students will be able to handle complex engineering problems and apply the knowledge and the skills acquired during the learning phase.
- The faculty sensitizes the students towards the environment, social concerns, gender bias, discrimination between rich and poor, human rights so that the students can act appropriately whenever required.

The graduate attributes are:

GA1 : Engineering Knowledge

GA2 : Problem Analysis

GA3 : Design/ Development

GA4 : Conduct Investigations of Complex Problems

GA5 : Modern Tool Usage

GA6 : The Engineer and Society

GA7 : Environment and Sustainability

GA8 : Ethics

GA9 : Individual and Teamwork

GA10 : Communication

GA11 : Project Management and Finance

GA12 : Life-long Learning

- The programme outcomes are developed to attain the Graduate Attributes and to meet the programme educational objectives.

Programme Outcomes (POs) :

Programme Outcomes are narrower statements that describe what students are expected to know and be able to do by the time of graduation. These relate to the skills, knowledge, and behavior that students acquire in their matriculation through the programme.

Engineering programmes must demonstrate that their students attain the following outcomes:

PEO1: An Ability to apply knowledge of mathematics, science, and engineering,

PEO2: An Ability to design and conduct experiments, as well as to analyze and interpret data

PEO3: An Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability

PEO4: An Ability to function on multidisciplinary teams

PEO5: An Ability to identify, formulate, and solve engineering problems

PEO6: An Understanding of professional and ethical responsibility

PEO7: An Ability to communicate effectively

PEO8: The Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental and societal context

PEO9: A Recognition of the need for, and an ability to engage in life-long learning

PEO10: A Knowledge of contemporary issues, and

PEO11: An Ability to use the techniques, skills, and modern engineering tools necessary for engineering practice.

- Programme educational objectives are formulated as given below based on stake holders feedback.

Programme Educational Objectives (PEOs):

PEO1: To prepare graduates with solid foundation in mathematics, science and related engineering subjects so as to comprehend, analyze, design and apply the Knowledge to real life problems.

PEO2: To prepare graduates who can achieve a high level of technical expertise so that they are able to succeed in their chosen profession.

PEO3: To inculcate in students the spirit of lifelong learning, professional attitude, ethics, and teamwork and to develop effective communication skills, multidisciplinary approach for a successful professional career.

PEO4: To enhance industry-institute interaction with reputed academic and industrial organizations in order to impart knowledge and hands-on experience to students thus making them engineers of excellence.

Criterion – III

Laboratories



Mechanical Lab



Electronics & Instrumentation Lab



Mechatronics lab



Mechanical - Workshop



Computer Lab



Electronics & Telecom. Lab

Criterion III

Research, Consultancy and Extension

3.1 Promotion of Research

3.1.1 Does the institution have recognized research center/s of the affiliating University or any other agency/organization?

This institution is fifteen years old. There is approved research centre for CSE and we have applied for Mechanical Engg. Permission is awaited.

3.1.2 Does the Institution have a research committee to monitor and address the issues of research? If so, what is its composition? Mention a few recommendations made by the committee for implementation and their impact.

Yes. The research committee consists of Director, Principal, Dean, HODs and PG Co-ordinators.

Role of research committee:

- To encourage faculties to participate in national / international conferences, workshops, short-term courses, faculty development programmes etc.
- To encourage students to participate in national / international conferences, workshops, short-term courses, faculty development programmes etc.
- To encourage faculties for organizing conferences, workshops, short-term training programmes, faculty development programmes etc.
- To improve the level of projects.
- To guide faculties and students in their dissertation work.
- To encourage students for interdisciplinary research.
- To encourage staff/students for higher education.
- To upgrade the experiments/laboratories

Impact:

- Number of participants is exponentially increasing since last four years. Some of the staffs are reviewers /members of conferences and journals.
- We are annually organizing national level conference (All India Conference) in our institute. We have also conducted workshops, FDPs etc.
- Students have started projects from IIIrd semester. All the PG students are doing research oriented projects. Each department is having a project

committee consisting of senior faculties. Time to time they interact with students and help/guide them in their dissertation work.

- Students (UG, PG) have started doing interdisciplinary projects.
- Some of our staffs are doing MTech and PhD in reputed universities and organizations.
- Have received funds from AICTE, DRDO and CGCOST.

Table 3.1 Funds Received

S. No.	Title	Chief coordinator	Amount (Rs)	Ref no
1	System Lab (MODROBS)	Dr Anurag Verma	1800000	9-38/RIFD/MODROB/Policy-1/2013-14(PVT) 15/07/2013
2	Modernization of Power System	Dr.Vivek Chandra	1220000	9-37/RIFD/MODROB/Policy-1/2013-14(PVT) on 12/07/2013
3	FDP	Dr.Vivek Chandra	245000	1-7/AICTE/RIFD/FDP/P(1)-4/2012-2013,28/02/2013
4	Dev.of Centre of Excellence PRD.DEV	Mr.Santosh Ku. Sharma	920000	Refno:- 8024/MOD/29/2011/12 on 20/07/2012
5	Advanced Mechanism Design centre	Dr Anurag Verma	700000	Ref no:- 8023/RID/RPS/21/11/12 on 09 /05/12
6	MODROBS(C SE)	Dr.Sipi Dubey	1500000	Ref no:-200-21/FIN/2001-2002/1016 Date 05/02/10
7	MODROBS(E TC)	Mr Mangal Singh	1300000	Ref:- 200-21/FIN/2001-2002/1015 on 05/022010
8	Staff Development Program	Dr. Sipi Dubey	350000	Ref:-35/FIN/2001-2002/1066 on 08/02/10

9	R&D Project	Mr.N H kela	300000	Ref no:-NBC(C.G.) DPD- NFS/5438/RIF_innovati ons/RPR10010/CSIT200 9-10 Duration 24/02/2010 to sept/2014
10	R&D Project	Mr.N H kela	74000	Ref:-DSIR/TePP/TMC- CSV TU-1/09/96 sanctioned:-74000& received:37000 Dated on 20/10/2009 to 13/08/2013
11	R&D Project	Mr.Sanjay Singh	75000	Project No:RDUG2012010 on 20/04/2012
12	Review of drawing of SEMI BED TRUCK TRAILER	Mr. A.P.Kumbhare	20000	REG. NO.: CSIT/996 on 25/6/2011
13	R&D Project	Mr.Sanjay Singh	30000	Project no :- RDUG2014045
14	R&D Project	Mr.Sanjay Singh	50000	Project no :- RDUG2013018
15	R&D Project	Mr.Sanjay Singh	50000	Project no :- RDUG2013096

- Design, advanced experiments are also performed in laboratories.

3.1.3 What are the measures taken by the institution to facilitate smooth progress and implementation of research schemes/projects?

In order to facilitate smooth progress and implementation of the research projects, the institution implements following schemes:

- **Autonomy to the principal investigator**

Full autonomy is given to the principal investigator.

- **Timely availability or release of resources**

The resources which are required for the implementation of research projects are made available in short time. Some of the resources which are commonly required for the purpose of research are made readily available like: books

from central / departmental library, Journals, facility of e-library which provides access to various journals.

- **Adequate infrastructure and human resources**

The institution provides the facilities in terms of infrastructure and human resources required for the research projects. There are various labs which are being provided for the purpose of research and supporting staffs are made available to assist the researchers.

- **Time-off, reduced teaching load, special leave etc. To teachers**

The institution provides relaxation to teachers involved in research in terms of reduced teaching load, special leaves etc-, so that they can dedicate more time on their research projects.

30 extra leaves are provided per year to PhD scholars, DLs are being sanctioned for conferences, faculty development programs, seminars etc-.

- **Support in terms of technology and information needs**

The institute provides various facilities to meet the technology and information needs related to the research.

- The institute provides the licensed software to the researchers to conduct their research.
- The institute provides the facility of NPTEL video lectures, e-library etc.

3.1.4 What are the efforts made by the institution in developing scientific temper and research culture and aptitude among students?

College regularly conducts seminars, guest lectures, conferences & workshops etc., for the benefit of both students and faculty members.

Faculty members are motivated to acquire higher qualifications with full support by providing necessary leave and the required sponsorship.

E-journal and hard copy of reputed journals, NPTEL video, advanced library, latest softwares etc are provided by the institute so that staff and students are benefitted.

Add-on classes, guest lectures, GATE Classes, aptitude classes are regularly organized by various departments.

Industrial visits are also organized to bridge the gap between industry and institute.

3.1.5 Give details of the faculty involvement in active research (Guiding student research, leading Research Projects, engaged in individual / collaborative research activity, etc.

Various faculties of the institution have been actively involved in research. The institute encourages research, tries to imbibe a culture of research in the college and provides the facilities and resources required for that.

- **Faculty involvement in Guiding Research students / Higher Qualification / Research Projects during last 4 years.**

a) Faculty involvement in Guiding Research students:

The details of different faculties who have been involved in - guiding PhD projects are as follows:

S. No.	Activity	Name of the Faculty with designation	Dept.	Title of Research	Remark/ Status
1	Ph. D. Guide	Dr. Mahesh P.	Mechanical	Modeling of Pressure Die Casting Process with Conventional Regression and Neural Network Based Approach	Completed (2013)
2	Ph. D. Guide	Dr. Anil Kashyap, Professor and HOD-Maths	Mathematics	A Study of Fuzzyset and fuzzy logic with some engineering application	Registered December 2011

b) No. of Faculty acquired Higher Qualifications :

1. No. of Faculty awarded Ph.D. : 02
2. No. of Faculty submitted thesis for the award of Ph.D. : 01

c) Faculty undertaken Research Projects :

The details of different faculties who have been involved in leading research projects are as follows:

Table 3.2 Faculty Research Project

S. No	Name of the faculty with designation	Dept.	Title of Research	Status-Progress/Completed
1	Dr. Anurag Verma	Mechanical	Advanced Mechanism design centre	Progress
2	Dr. Vivek Chandra	Electrical & Electronics Engg	Modernization of power system	Progress
3	Mr. Santosh Kumar Sharma	Mechanical	Development of centre of excellence PRD. DEV	Completed
4	Mr. N H Kela	Mechanical	R & D Project	Completed
5	Mr. N H Kela	Mechanical	R & D Project	Progress
6	Mr. Sanjay Singh	Mechanical	R & D Project	Completed
7	Mr A.P.Kumbare	Mechanical	Review of drawing of SEMI BED TRUCK TRAILER	Completed
8	Mr.Sanjay Singh	Mechanical	R & D Project	Progress
9	Mr.Sanjay Singh	Mechanical	R & D Project	Completed
10	Mr.Sanjay Singh	Mechanical	R & D Project	Completed

3.1.6 Give details of workshops/ training programmes/ sensitization programmes conducted/organized by the institution with focus on capacity building in terms of research and imbibing research culture among the staff and students.

The institution focuses on imbibing research culture among the staff and students of the institute and in order to promote the culture of research, the institution has organized several workshops, training programs, seminars, guest lectures and conferences. The details of these programs are as follows:

No. of Workshops/ training programmes/ sensitization programmes conducted/organized by the institution

2013-14	2012-13	2011-12	2010-11
11	10	12	8

3.1.7 Provide details of prioritised research areas and the expertise available with the institution.

The institution has many experienced and well qualified faculties, who have one or more areas of research. The research areas have been represented as follows:

Department of Mechanical Engineering: Scheduling in Manufacturing System, System Simulation, Six Sigma, Quality Engineering, Design Engineering, Production Management etc.

Department of Mechatronics Engineering: Industrial Engineering & Control System, Computer Aided Design, Artificial Intelligence, Image Processing etc.

Department of Electronics & Telecommunication: Embedded Systems, Optical Comm, Digital Signal Processing, Neural Network/Speech Signal Processing.

Department of Information Technology: Wireless Communication, 4G, LTE, OFDM, Wireless Adhoc Network etc.

Department of Computer Science & Engineering: S/W engineering, Image and signal processing, web application, Bio informatics, CN, Web application, C, C++, Java, Robotics, DBMS etc.

Department of Electrical & Electronics Engineering: Basic Electrical machines, Electrical Power System, Power System and Generation, Signal Processing, Microcontroller etc.

Department of Electronics & Instrumentation: Instrumentation, PLC, Process Control, Control Systems, Signals & Systems, Power Electronics & Robotics etc.

Department of Physics: Solid State Physics, Electronics, luminescence

Department of Mathematics: Algebra, Graph Theory, Operation Research, Discrete Mathematics, Non Linear Functional Analysis, Coding Theory etc.

Department of Chemistry: Environmental & Analytical Chemistry, Water Pollution, Surfactant.

3.1.8 Enumerate the efforts of the institution in attracting researchers of eminence to visit the campus and interact with teachers and students?

The institution has constantly put up efforts to attract the eminent researchers to visit the college and interact with the faculties and students. The different efforts put up by the college to attract the eminent researchers and corresponding outcome are described below:

List of Some Guests Who Visited the Institute

S. No.	Name of expert	Title of Lecture	Year of Visit
1	Mr. Jayant Chaturvedani, (Manager – Operation; Ericsson)	Telecommunication Word, Mobile Network	2014
2	Dr. K. B. Subramaniam (Retd. Principal; Regional Institute of Education, Bhopal)	Nature of Mathematics and Mathematics in Nature	2014
3	Mr. Apoorv Tiwari, (Associate Software Engineer; Accenture)	Cracking Campus Interviews	2014
4	Dr. Ashok Bhansali	Teaching Methodology	2014
5	Dr. Rabindra Ku. Jena	Green Computing	2014

The complete list is available in the department, which can be shown at the time of visit.

3.1.9 What percentage of the faculty has utilized Sabbatical Leave for research activities? How has the provision contributed to improve the quality of research and imbibe research culture on the campus?

About 6% of the faculty members are utilizing sabbatical/ study leave for their research and development activity. Yes, it has contributed to imbibing a culture of research.

List of the faculties who are deputed on sabbatical/academic leave is available with the institute.

Please refer to **Table 3.6 (Page no. 75)**

3.1.10 Provide details of the initiatives taken up by the institution in creating awareness/advocating/transfer of relative findings of research of the institution and elsewhere to students and community (lab to land)

In order to create awareness in the society regarding the findings of research of the institution, some awareness campaigns have been undertaken by the institution:

- The renewable energy club has developed a low cost solar cooker, the project sponsored by National Bank for Agriculture and Rural Development [NABARD]. The college has organized an awareness program about its use in which Mr. N. H. Kela (Professor, Dept. of Mechanical Engineering), demonstrated the working of the solar cooker to the people from the nearby places.

- A Cost Effective Fully Automatic Power Wheel Chair for old age and Disabled people has been fabricated.

3.2 Resource Mobilization for Research

3.2.1 What percentage of the total budget is earmarked for research? Give details of major heads of expenditure, financial allocation and actual utilization.

- Institute financially supports research activity every year.
- Two to three percent of the total budget is earmarked for research.
- The institute provides budget every year for in house R&D, faculty development program, student development program.
- The institute also provides budget for advanced software packages & tools necessary to carryout research projects.
- Text books, reference books and E-Journals related to advanced topics are also added to the institute library for the use to the faculty and students to carry out their research activities.

3.2.2 Is there a provision in the institution to provide seed money to the faculty for research? If so, specify the amount disbursed and the percentage of the faculty that has availed the facility in the last four years?

- Yes, Provision for required amount as seed money.
- Financial assistance to staff for acquiring higher education like M.Tech. / PhD.
- Institute provides registration fee, TA/DA, duty leave to staff for attending FDPs, workshops, conferences and symposia.
- The institute also finances cost of training in advanced software / tools to staff.

Table 3.3 Amount spent towards staff research during last 4 years

Type of support	Financial provision to support staff (Rs. In Lakhs)				
	2010-11	2011-12	2012-13	2013-14	Total
Ph.D	--	11,90,128	13,60,314	96,000	26,46,442
FDP/SDP	--	3,15,288	78,109	--	3,93,397
Paper Presentation	91,544	3,15,288	1,30,014	54,936	5,91,782
Total					36,31,621

3.2.3 What are the financial provisions made available to support student research projects by students?

Table 3.4 Amounts spent towards Students Research

Year	Amount Disbursed to the Students for research (Rs.)
2013-14	42,428
2012-13	91,465
2011-12	--
2010-11	--
Total	1,33,893

3.2.4 How does the various departments/units/staff of the institute interact in undertaking inter-disciplinary research? Cite examples of successful endeavors and challenges faced in organizing interdisciplinary research.

The institute encourages the departments to undertake inter-disciplinary research and to organize workshops and seminars.

Examples

Example 1: SDP on “Image processing and its Advantages” by CSE department of the Institute.

Example 2: Modernization of power system instrumentation and switch gear lab by EEE department.

Example 3: Development of center of excellence by Mechanical department of the institute.

3.2.5 How does the institution ensure optimal use of various equipment and research facilities of the institution by its staff and students?

- Research facilities are utilised to the reasonable extent by P.G. students and staff members.
- The institute library/e-library, NPTEL Center and Computer Centre facilitate a conducive research environment by accessing video lectures, subscribing the necessary national & international online journals worth Rs. 70 lakhs.
- The institute has developed some laboratories as research labs by procuring research specific equipment/instruments and software.
- Using these lab facilities, the staffs carry out experimental work on research projects and publish the results.

3.2.6 Has the institution received any special grants or finances from the industry or other beneficiary agency for developing research facility? If 'yes' give details.

➤ Yes

The details of special grants is given in **Criteria 3.2.7., Table 3.5 (Page no. 72)**

3.2.7 Enumerate the support provided to the faculty in securing research funds from various funding agencies, industry and other organisations. Provide details of ongoing and completed projects and grants received during the last four years.

- The college encourages the faculty to get research projects and provides motivational incentives by awarding cash prizes.
- The college also provides the necessary infrastructure, space, internet facility along with printers for carrying out research activities.
- The college deputed senior faculty to faculty development programs, skill enhancement programs related to research projects, provides travel grant to attend workshops at conferences in national and international level.
- The details of ongoing and completed projects and grants received during last four years are presented in table 3.5 below.

Table 3.5 Research funds received from various funding agencies during last 4 years

S. No.	Duration	Title of the project	Name of the funding agency	Total Grant	Total grant received till date	Project Status	
				Sanctioned	Received		
1	24-2-10	Development and popularisation of low cost solar cooker and low cost solar driers in rural areas	NABARD (National Bank for Agriculture and Rural Development)	3,00,000			Complete
	23-3-10	--	--	--	1,45,200	1,45,200	
	12-2-13 up to Sept 14	--	--	--	1,45,800	3,00,000	

2	20-10-09 up to 13-8-13	Indirect type solar driers for processing food products under TePP	Ministry of Science and Technology Dept of Scientific and Industrial Research	74000	37000	37000	Complete
3	2013-14	Cost effective fully Automatic Power wheel chair for old age & disabled(RDU G2014045)	Institution of Engineers	30000	30000	30000	Complete
4	2012-13	Hybrid Limbed Exploration Vehicle(RDU G2013018)	Institution of Engineers	50000	50000	50000	Complete
5	2012-13	Cost effective Micro Propulsion through indigenous Development of Valveless Pulsejet Systems(RDU G2013096)	Institution of Engineers	50000	50000	50000	Complete
6	2011-12	Vibration analysis of Cracked rod with Circumferential crack(RDUG2012010)	Institution of Engineers	75000	75000	75000	Complete

3.3 Research Facilities

3.3.1 What are the research facilities available to the students and research scholars within the campus?

The college is providing following facilities to encourage research.

i) Library facilities.

The details of library facilities are described in **criteria 2.3.9. (Page no. 46)**

ii) Laboratory facilities:

Each department in our institute provides following laboratory facility with advanced software, which is used for research and development.

Electrical & Electronics Engineering	: Mipower
Mechanical Engineering	: Pro-e, ANSYS, SINUTRAIN, MS-project.
Electronics and Telecommunication Engineering	: Multi Sims, Matlab10
Computer Science and Engineering	: Visual Prolog Advanced operating system lab
Mechatronics Engineering	: Cad lab, M&I lab, Dynamics of machine, lab kinematics of machine lab.
Electronics and Instrumentation Engineering	: Lab view

The complete details are available with the institute.

iii) Licensed Software:

Apart from above mentioned softwares each department in our institute has licensed software for research and development. For instance-

Electrical and Electronics Engineering	Mipower
Mechanical Engineering	Minitab, Cut viewer, Fluidsim, ABACUS.
Electronics and Telecommunication Engineering	Xilinx, Spartan6,

iv) Facility and Financial Assistance:

- Grant of books/equipments/ Conference/workshops etc. is given on merit, subject to the max. Rs. 50,000/- for Ph. D. and Rs. 10,000/- for M. Tech. for total duration of the completion of course.
- Special leave limited to 1 month in a year is granted on the recommendation of guide for Ph. D. candidate.
- Duty leave is granted for Conferences/ Workshops, FDPs & STTPs.

3.3.2 What are the institutional strategies for planning, upgrading and creating infrastructural facilities to meet the needs of researchers especially in the new and emerging areas of research?

1. Necessary software, Lab equipment are promptly purchased.
 2. Reference books, journals are made available.
- Institute encourages students and faculties to promote research environment and provides incentives to those who publish journal papers.
 - Institute conducts AICON i.e. All India National Conference for promoting and propagating research culture among students and faculties.

3.3.3 Has the institution received any special grants or finances from the industry or other beneficiary agency for developing research facilities? If 'yes', what are the instruments/ facilities created during the last four years.

➤ Yes

The details are mentioned in **Table 3.5 (Page no. 72)**

3.3.4 What are the research facilities made available to the students and research scholars outside the campus / other research laboratories?

- The following faculties are doing their research work and using laboratories, library, hard copy journals, e-journals outside the campus and are getting research facility from institute for doing research work outside the campus.

Table 3.6 Faculties persuing PhD

S. No	Name of the faculty	Name of the institute where research is conducted
1	Mr. Mangal Singh	NIT ,Rourkela
2	Mr. Santosh Sharma	NIT ,Rourkela
3	Mr.Sanjay Singh	K.L university ,Vijayawada
4	Mr. R.K. Sahu	Jodhpur National University, jodhpur

5	Mr. Deepak Sharma	Jodhpur National University ,jodhpur
6	Mr. Deepak Singh	NIT ,Rourkela
7	Mrs.Urmila Soni	CSVТУ ,Bhilai
8	Mrs. ArchanaTiwari	CSVТУ ,Bhilai
9	Mr. Rajesh Gupta	Dr. C.V Raman University, Bilaspur
10	Mr. Rajesh Kumar	CSVТУ ,Bhilai
11	Mr.Om Prakash Yadav	CSVТУ ,Bhilai
12	Mrs. Padmini Sharma	Dr. C.V Raman University, Bilaspur
13	Mr.Sanjay Parganiha	CSVТУ ,Bhilai
14	Mrs.Pratibha S Kurup	CSVТУ ,Bhilai
15	Mr.Shailendra Kushwaha	CSVТУ ,Bhilai
16	Mrs. Sumita Sengupta	CSVТУ ,Bhilai

3.3.5 Provide details on the library/ information resource centre or any other facilities available specifically for the researchers?

Library & Information Centre:

Please refer to **Table 2.8** of **criterion 2.3.9** (Page no. 47)

3.3.6 What are the collaborative researches facilities developed / created by the research institutes in the college? For ex. Laboratories, library, instruments, computers, new technology etc.

Table 3.7 Collaborative programs taken up with industry

S. No.	Industry/Institute Name	Outcome
1.	Escorts Construction Equipment LTD.	A guest lecture given by Mr. Ashish Mathur, chief general manager, Escorts Construction Equipment LTD on the topic “Functioning of an organization” for BE final year student (Mech., MEX) on 21 st Sep. 2013.
2.	Talent Acquisition group Espire info labs, Gurgoan.	• A guest lecture was organized on 11th sep. 2013 and the resource person is Mr. Rahul Krisna. • BE 7th Semester Students (CSE, IT, ETC) participated in training program on topic “Fresher Development-A test of Corporate World ”
3	Mobitives IT Pvt.banglore,india	• A guest lecture on mobile application development on android platform for beginners was conducted for 6th semester students (Mechatronics) on 11 th March 2014, resource person is Mr.Rajive Singh, Mobile application engineer. • Total 100 students were benefited with whole

		program of various engineering branches. • They all understood about the introduction to android ,architecture setup, fundamental block, UI development etc.
--	--	---

3.4 Research Publications and Awards

3.4.1 Highlight the major research achievements of the staff and students in terms of

(a) Patents obtained and filed (process and product) :

➤ Nil

(b) Original research contributing to product improvement:

Original research work is carried out at the Institute level in manufacturing science leading to product improvement. The research work is focused on modeling, simulation, analysis and optimization of parameters in metal casting, one of the primary and important manufacturing processes. Research work is carried out on the following casting processes:

- Manufacture of structural steels
- Pressure Die casting process
- Squeeze casting process
- Moulding sand system

(c) Research studies or surveys benefiting the community or improving the services:

Title of Project “To fabricate a Cost Effective Fully Automatic Power Wheel Chair for old age and Disabled.”

(d) Research inputs contributing to new initiatives and social development

Title of Project “Low Cost Solar Cooker and Low Cost Solar Drier in Rural Areas. The details are available at the institute”.

3.4.2 Does the Institute publish or partner in publication of research Journal? If ‘yes’, indicate the composition of the editorial board, Publication policies and whether such publication is listed in any International database?

No. Barring In-House Publication of “Sancharika”, we do not publish any other Research Journal. Moreover, we are not in partnership with other publishers.

3.4.3 Give details of publications by the faculty and students:

Table 3.8 Details of publications by the faculty

S. No.	Name of Faculty with Designation	No of Publication	Remarks
Department of Computer Science & Engineering			
1	Mr Om Prakash Yadav	9	International/national Journal /conferences Publications
2	Mr. Punyaban Patel	15	
3	Mrs Deepty Dubey	13	
4	Mr Khom Lal Sinha	5	
5	Mr Bhupesh K. Dewangan	15	
6	Mrs Manjusha Singh	8	
7	Mr. Ram Kishan Dewangan	7	
8	Ms Usha Kiran	8	
9	Mr Praveen Shende	13	
10	Mr Manojee Roy	4	
11	Mrs Shikha Agrawal	12	
12	Ms. Aradhana Singh	4	
13	Ms. Sachi Choudhary	4	
14	Mr. Vinay Bhardwaj	2	
15	Mr. Bhupesh Kumar Gupta	2	
16	Ms. Rajeshri Lanjewar	2	
17	Mr. Apoorva Mishra	8	
18	Mr. Kamlesh Kumar	2	
19	Ms. Prateeksha Pandey	2	
20	Mr Pushpendra Verma	1	
Mechanical Engineering			
1	Dr. Anurag Verma	1	International/national Journal /conferences Publications
2	Dr. P. Mahesh	26	
3	Mr. Santosh Kumar Sharma	4	
4	Mr. H. R. Chandrakar	1	
5	Mr. Sanjay Singh	3	
6	Mr. Jitendra Chaturvedani	2	
7	Mr. Dehal Ram Nishad	4	
8	Prof. S. R. Paul	1	
9	Mr. N. H. Kela	4	
10	Mr. R. A. Mishra	1	
11	Mr. Rajesh Kumar Ojha	2	
12	Mr. Abhishek Jha	2	

13	Mr. Priyank Shrivastava	3	
14	Mr. Rituraj Chandrakar	17	
15	Mr. Abhilash Trivedi	5	
16	Mr. Sanjay Kumar Sharma	1	
17	Mr. Gargi Sharan Vinod	1	
18	Mr. Manish Pandey	1	
19	Mr. Ram Krishna Ratore	1	
20	Ms. Shoma Mitra	1	
21	Mr. Nilmani sahu	8	
22	Mr. Abhinay Bhatt	3	
23	Ms. Chandrakala Sao	1	
Department of Electronics and Telecommunication			
1	Mr. Mangal Singh	16	International/national Journal /conferences Publications
2	Mrs. Yojana Yadav	3	
3	Mr. Deepak Sharma	1	
4	Mr. Deepak Singh	1	
5	Mr. Rajkumar Sahu	5	
6	Mrs. Chhabikiran Sao	3	
7	Mr. Abhishek Misal	15	
8	Mr. Rahul Sinha	8	
9	Mr. Navneet Kr. Sahu	2	
10	Mr.Pushpendra Singh	8	
12	Mrs. VijetaVerma	1	
13	Mr. TomeshVerma	4	
14	Mr. ChandrakantBadgaiyan	1	
15	Mr. MaheedharDubey	3	
16	Mr. Ayush Swarnkar	1	
17	Mr. Prateek Verma	1	
18	Mrs. Poonam Mishra	2	
19	Ms. Aparna Alone	2	
20	Mr. Ramesh Yadav	1	
21	Ms. Anita Devi Tiwari	8	
22	Ms. Priyanka Khatwani	2	
23	Mrs. Smriti Kumar	6	
24	Mr.Ashish Dewangan	3	
25	Mrs. Suchita Sharma	8	
26	Mr. Ajay Roy	2	
27	Mr. Ayush Swarnkar	1	

Department of Electronics and Instrumentation			
1	Mrs. Archana Tiwari	11	International/national Journal /conferences Publications
2	Mr. Rajesh Kumar Gupta	5	
3	Mr. Nirmal Kumar Pandey	4	
4	Mrs. Urmila Soni	3	
5	Ms. Gomti verma	1	
6	K.Ganpati Shrinivas Sharma	4	
7	Mr. Prabhas Kuma r Gupta	10	
8	Mr. Shailendra Kr.Dewangan	21	
9	Mr. Akash Kumar Mandal	7	
10	Mr. Umesh Sahu	14	
11	Ms. Roma Singhal	1	
12	Mrs. Remya Ani	6	
13	Mr. Indresh Kumar Verma	1	
14	Ms. Suchi Sirmour	4	
15	Mr. Divesh Kumar Gajpal	3	
Department of Electrical and Electronics			
1	Dr. Vivek Chandra	30	International/national Journal /conferences Publications
2	Mrs. Padmini Sharma	4	
3	Mr. Ashraf Jafri	5	
4	Mrs. Manjusha Silas	2	
5	Mr. Naheshwar Pr. Chandra	6	
6	Mr. Somesh Yadav	7	
7	Mr. Rajnish Prasad Pandey	6	
8	Mr. Kushal Tiwari	6	
9	Mr. Homendra Sahu	2	
10	Mr. Vivek Bargate	2	
11	Mrs. Richa Tiwari	4	
12	Ms. Soma Poddar	2	
13	Mr. Amit Pandey	2	
14	Ms. Priyanka Sharma	1	
15	Ms. Rahila Parveen	1	
16	Mr. Chaman Yadav	2	
Department of Mechatronics Engineering			
1	Mr. Rajesh Kumar	16	International/national Journal /conferences Publications
2	Mr. S K Mandelia	10	
3	Mr. Chandrakant Badgaiyan	17	

4	Mr. Pradeep Sahu	6	
5	Mr. Pradeep Agrawal	3	
6	Ms. Indu Sahu	2	
7	Mr. Apurva Chouksey	2	
Department of Information Technology			
1	Ms. Sonika Arora	13	International/national Journal /conferences Publications
2	Mrs. Neelam Dewangan	11	
Department of Mathematics			
1	Dr. Anil Kumar Kashyap	7	International/national Journal /conferences Publications
2	Mrs. Mamta Patel	3	
Department of Chemistry			
1	Dr. Yasmeen Pervez	6	International/national Journal /conferences Publications
2	Mr. Pratibha S. Kurup	5	
3	Mr. Shailendra Kumar Kushawah	7	
Department of Physics			
1	Dr. Renu Tripathi	17	International/national Journal /conferences Publications
2	Mrs. Sumita Sengupta	8	
3	Mr. Sachin H Dhawankar	23	
4	Mr. Sanjay Parganiha	8	

Details of publications by the students:

S. No.	Name of Student	No of Publication	Remarks
1	Abha Tamrakar	01	<i>International/national Journal /conferences Publications</i>
2	Ashish Ku. Tamrakar	02	
3	Gurudutta Verma	02	
5	Sarang Pitale	01	
6	Rubeena Mirza	02	
8	Kranti Kumar Jain	01	
9	Aparna Soni	02	
10	Raj Kumar Singh	01	
11	Shishir Sarkar	02	
12	Aakanksha Dixit	05	
13	Hemlata Joshi	02	
14	Neha Choubey	02	
15	Sanjay kumar Baghel	09	
16	Aman Kumar Chandrakar	01	
18	Princy Diwan	02	

19	Priyanka Daga	03	
20	Yogesh Graham	02	
21	Deepti Sahu	01	
22	Geetanjali Sahu	02	
24	Bhawesh Rahulkar	02	
25	Manoj Pandey	02	
Department of Electronic & Telecommunication			
1	Ramesh Yadav	01	International/national Journal /conferences Publications
2	Ms. Aparna Alone	05	
3	Manjuri Biswas	02	
4	Ajay Kumar Roy	03	
5	Ranjana Chaturvedi	02	

Details of Publications / Paper presentations / Book publications by the faculty during last 4 years.

S. No.	Activity	During			
		2013-14	2012-13	2011-12	2010-11
1	Publication by faculty	2	2	2	2
2	Papers published by faculty & students	318	284	270	217
3	No. of publications listed in International Data Base	34	46	35	12
4	Monographs	-	-	-	-
5	Chapter in Books (Manuals)	-	-	-	-
6	Books edited		2		
7	Books with ISBN / ISSN	2	13	7	4
8	Citation Index*	4	10	16	9
9	SNIP / SJR	-	-	-	-
10	Impact Factor*	5.7971	5.2189	4.8245	5.27
11	H-Index*	1	2	2	1

(* indicates the highest value)

3.4.4 Provide details (if any) of

1. Research Awards received by the faculty: **Nil**
2. Recognition received by the faculty from reputed professional bodies and agencies, nationally and internationally : **Nil**

3.5 Consultancy

3.5.1 Give details of the systems and strategies for establishing institute industry interface?

The institute has formed Industry Institute Partnership Cell (IIPC) for establishing relationship between industries and institute on various fronts.

Each department has nominated one IIPC Coordinator who arranges various activities of the cell. The activities of cell include:

- Arranging Industrial visits, guest lectures and expert talks on technical topics.
- Industrial visits are arranged for the students to the nearby industries of steel, power and other allied industries.
- The IIPC arranges workshops and seminars on Automation, Ethical Hacking, Cloud Computing and Programming.
- The IIPC arranged guest lectures by well known companies like SIEMENS Ltd. Mumbai, and B&R automation, Pune.
- The students are exposed to and are given hands on training on various softwares like Pro-E, CAD/CAM, ANSYS, ABACUS, MATLAB, ORACLE, LABVIEW, FluidSim.

3.5.2 What is the stated policy of the institute to promote consultancy? How is the available expertise advocated and publicized?

1. To promote consultancy the institute has formed Entrepreneur Development Cell and Renewable Energy cell. The cells work for those whose works include a mixture of teaching, research, consultancy and knowledge transfer activities.
2. The institute has conducted various workshops and awareness camps to promote consultancy.



3.. Policy objectives are:

- Enhancing efficient and effective management, planning, implementation, co-ordination and accountability of consultancy activities at the college.
- Ensuring development and periodical updating of consultancy activities.
- Facilitating, sharing and exchange of consultancy information.
- Establishing an adequate outlay of facilities and equipment for consultancy activities at the college.

3.5.3 How does the institute encourage the staff to utilize their expertise and available facilities for consultancy services?

- The institute sponsors and deputed the faculties to attend various seminars, workshops, by making an alternative arrangement of their workload at the college.
- The institute encourages online workshop/training conducted by experts for lifetime members of Indian Society for Technical Education (ISTE).
- The institute has Institutional memberships of IEEE, Springer, ASTM, J-GATE through which faculties/researchers can download research papers by experts.

3.5.4 List the broad areas and major consultancy services provided by the institute and the revenue generated during the last four years?

The broad areas where the consultancy services provided by the institute are like:

- Career counseling
- Financial planning
- Competitive examinations
- Information technologies

Since the consultancy services provided are on honorary basis revenue is not generated through such services.

3.5.5 What is the policy of the institute in sharing the income generated through consultancy (staff involved: institute) and its use for institutional development?

The consultancy offered by college staff is voluntary and hence non – remunerative.

3.6. Extension Activities and Institutional Social Responsibility

3.6.1 How does the institution promote institution-neighborhood-community network and student engagement, contributing to good citizenship, service orientation and holistic development of students?

In order to promote institution-neighborhood-community network and student engagement, the institute has established different cells and organizes various activities/ events throughout the year. The institute has following cells:

- EDC i.e. Entrepreneurship Development Cell, which focuses on developing self-employment activities.
- IIPC i.e. Industry Institute Partnership Cell, which focuses on having collaborations with various industries, so that the students will get exposure and get benefitted.
- The college has a NSS unit with a total strength of 100 members that participates in various social service and community development programs.

The college organizes various events that promote institution-neighborhood-community network and student engagement; such events are as follows:

- The college conducts E-TAP i.e. 'Technical Education Awareness Programme' for the XII standard Science students of the schools, in which they are made aware of the different branches of engineering and their future scope.



- The institute has conducted 'Entrepreneurship Awareness Program' from 13th March to 15th March 2014.



➤ The institute has organized the visit to old-age home on 30th July 2011 and 28th July, 2012.



3.6.2 What is the Institutional mechanism to track student's involvement in various social movements / activities which promote citizenship roles?

The institute uses the following mechanism to track and encourage the involvement of students in social activities:

- The respective teacher guardians of the students participating in social events, keep a record of their activities and they encourage the students for the same.
- The associations in charge of the department also track the participation of their students in social activities.
- Extra lectures and doubt clearing classes are arranged for students who miss their lectures (with prior intimation) because of participation in social events.
- Teacher guardians keep track of their students' involvement in social activities on a weekly as well as on semester basis.

3.6.3 How does the institution solicit stakeholder perception on the overall performance and quality of the institution?

In order to know the perception of the stakeholders (including students, parents, alumni etc) about the overall performance of the institution, a set of activities are being carried out at different levels, some of which are:

- In order to know the opinion of the parents, TG-parent meetings / interactions are being done on a regular basis.
- In order to know the perception of the students regarding the teaching learning mechanism, computer based feedbacks are taken twice per semester.
- In order to know the overall perception of the students, weekly feedback is taken during the counseling sessions.
- The alumni association gives feedback to the institute. Its suggestions are discussed in a meeting by the principal. Some of the suggestions are implemented.



3.6.10 How does the institution plan and organize its extension and outreach programmes? Providing the budgetary details for last four years, list the major extension and outreach programmes and their impact on the overall development of students.

The institution plans and organizes several extension activities and outreach programs for the welfare of the society. The details of the programs and their budget are as follows:

- **Visit to orphanage ‘Saraswati Bal Kalyan Ashram’:**

On the occasion of Independence Day, a visit to orphanage ‘Saraswati Bal Kalyan Ashram’ was organized. Dr. N.T. Khobragade (Dean, CSIT) addressed the students. 110 students and various faculties of the college visited the orphanage. The faculties also told the students about their fundamental rights and duties as responsible citizens.

Budget details:

- The institution donated Rs. 10000 to 'Saraswati Anath Bal Kalyan Samiti', residential school for orphans.
- The association of Computer Science and Engineering department, known by the name of 'Association of Computer Engineers' (ACE), donated Rs. 5000 to 'Saraswati Anath Bal Kalyan Samiti'.
- Gifts worth Rs. 4297 were distributed in the 'Saraswati Anath Bal Kalyan Samiti'.
- Total budget of the orphanage visit = **Rs. 19,287.**



• Visit to old-age home:

Visit to Old-age Home was organized on 06-10-2012. The number of students who participated in this activity was 120. The total budget of the visit was Rs. 6000.



Impact of the outreach activities on students:

- Students get an opportunity to know about their surroundings, the social and economic structure of the society and get ready to make their contribution for the society.
- Students understand their duties as responsible citizens.
- Students get an exposure and opportunity to organize events.
- Such activities contribute to the overall development of the students and complement their technical knowledge.

3.6.5 How does the institution promote the participation of students and faculty in extension activities including participation in NSS, NCC, YRC and other National/ International agencies?

- The institution promotes the participation of students and faculty in extension activities including participation in NSS. The institute has a NSS unit with a total strength of 100 members, affiliated to C.S.V.T.U, Bhilai.
- Tree plantation activity has been organized by the NSS unit in the college.
- In order to encourage students to participate in the events organized by NSS, extra lectures and doubt clearing classes are also arranged for students who miss their lectures (with prior intimation) because of participation in events organized by NSS and duty leaves are provided to employees for participation in the events of the NSS.

3.6.10 Give details on social surveys, research or extension work (if any) undertaken by the college to ensure social justice and empower students from under-privileged and vulnerable sections of society?

There are several extension activities which have been undertaken by the college to ensure social justice and empower the under-privileged students like:

- The college organizes free computer classes for the under-privileged students/youths so that they can learn basics of computer and use it for their livelihood or use it in their daily life.
- Visits to Old Age Home are made by the students and the faculties and the residents are provided help by virtue of arrangement of lunch and providing gifts.



- Every year on 15th August, a visit to deaf and dumb school (Prayas) is being organized, in which many students and some faculties of the institute visit the deaf & dumb school and motivate them, provide them financial assistance and appreciate their creativity.



3.6.10 Reflecting on objectives and expected outcomes of the extension activities organized by the institution, comment on how they complement students' academic learning experience and specify the values and skills inculcated.

Extension activities play a vital role in inculcating values and skills in students. Some of such activities are mentioned below:

- Apart from the technical knowledge students get an opportunity to know about their surroundings, the social and economic structure of the society and to make their contribution for the society.
- Students understand their duties as responsible citizens.
- Students get a sense of social responsibility.
- Students get an exposure and opportunity to organize events.
- Students learn to work effectively in a team.

3.6.8 How does the institution ensure the involvement of the community in its reach out activities and contribute to the community development? Detail on the initiatives of the institution that encourage community participation in its activities?

The institution conducts various reach-out activities and in order to ensure the involvement of the community in those reach-out activities; the institution also conducts awareness programs about the importance of those activities. The following activities have been organized by the institute:

- The institute conducts free computer classes for the unemployed persons of the near-by area.



- The institute conducts various eco-friendly activities like: tree plantation, campus cleaning etc



- The college has conducted 'Entrepreneurship Awareness Program' from 13th March to 15th March 2014.
- The renewable energy club, working on a project sponsored by National Bank for Agriculture and Rural Development [NABARD], has developed a low cost solar cooker. The institute has organized a program regarding the awareness about the use of renewable energy and the working of the low cost solar cooker, in which Mr. N. H. Kela (Professor, Dept. of Mechanical Engineering), demonstrated the working of the solar cooker.



3.6.9 Give details on the constructive relationships forged (if any) with other institutions of the locality for working on various outreach and extension activities.

To develop constructive relationship with the other institutions of the locality for working on various outreach and extension activities the college organizes various events in consultation with the other institutes of the locality. Some of the major events that are organized are as follows:

- The institute organizes 'LAQSHYA' every year, which is a 'techno-management' fest, in which students from various colleges participate in different events and get a chance to implement their ideas.
- The NSS unit of the college participates in various social events like blood donation camps, tree plantation activities etc- in co-ordination with the NSS unit of the other institutes.



3.6.10 Give details of awards received by the institution for extension activities and/contributions to the social/community development during the last four years.

Mr. Ajay Prakash Verma (Chairman, CSIT) was honored with the 'Edupreneurs-2013' award in recognition of his contribution to engineering education in India by "Engineering Watch", a reputed national magazine.



Please refer to Annexure – II

3.7 Collaboration

3.7.1 How does the institution collaborate and interact with research laboratories, institutes and industry for research activities. Cite examples and benefits accrued of the initiatives - collaborative research, staff exchange, sharing facilities and equipment, research scholarships etc.

Collaboration with industries and training institutes gives privileges for students industrial visits, Vocational training and U.G. projects for all semester students.

Major Highlights are:

- Weekly Lecture for Personality Development and frequent assessment tests in Aptitude, Logical, Verbal and Non-Verbal Reasoning, Group discussion, HR interview, Technical interview and Body language for second year to final year of students.
- A full day computer networking workshop is organized containing theoretical aspects and practical knowledge of networking by using special purpose LAN trainer kits.
- A computer hardware awareness program is conducted once in a year for students of all branches that includes hardware components identification, hardware assembly, Operating system installation etc.



Benefits:-

- Provides Industry oriented live projects for the students.
- Develops technical skills and corporate-culture skills.
- By industrial visit students are benefitted and become aware of the functionality & production of any product.
- Students learn about current technologies and their use in the industry.
- More focus on practical knowledge along with theoretical knowledge.

3.7.2 Provide details on the MoUs / Collaborative arrangements (if any) with institutions of national importance/ other Universities/ industries / Corporate (Corporate entities) etc. and how they have contributed to the development of the institution.

➤ Nil

3.7.3 Give details (if any) on the industry-institution-community interactions that have contributed to the establishment / creation/up-gradation of academic facilities, student and staff support, infrastructure facilities of the institution viz. laboratories / library/ new technology /placement services etc.

- Institute invites guest lectures on regular basis from the reputed institutes like GECs, NITs and Central Universities.
- All the engineering departments of our institute organize workshops every year. Since last four years the students and staff of our institute and neighboring institutes have been actively participating.
- The laboratories of our institute are upgraded through MODROBS funds which are sponsored by AICTE.

3.7.4 Highlighting the names of eminent scientists/participants who contributed to the events, provide details of national and international conferences organized by the college during the last four years.

The faculty members of our institute actively participate in staff development program, workshops and seminars organized by this institute and outside institutes to enhance their technical knowledge & skills. The following are the details of workshop/ faculty development programmers / conferences / seminars organized by various departments during the last four years.



Workshop/ Faculty Development Programmes / Conferences / Seminar organized during the last four years:

Dept	2013-14	2012-13	2011-12	2010-11
CSE	1	3	3	2
ETC	2	1	3	1
MECH	2	1	1	1
EEE	1	1	1	1
EI	2	1	1	1
IT	1	1	2	1
MEX	2	2	1	1

Several faculty members of this institute delivered guest lectures in various other institutes:

Year	2013-14	2012-13	2011-12	2010-2011
No. of Guest Lectures	04	02	04	02

3.7.5 How many of the linkages/collaborations have actually resulted in formal MoUs and agreements? List out the activities and beneficiaries and cite examples (if any) of the established linkages that enhanced and/or facilitated –

(a) Curriculum development/enrichment

- Some of our faculties are Members and Chairman of BOS of the university. The BOS suggests the modifications and up gradation in the syllabus in the context which fulfills the needs of industry and current technological changes.
- Some of our senior faculty members are peer reviewers and editors of various reputed national/ international journals for evaluation and approval of research papers.

(b) Internship/ On-the-job training

- The institute collaborates with various industries which provide opportunity to the students to know current trends in industry. This helps our students in acquiring technical skills which prove helpful for placements.

(c) Summer placement

- The institute collaborates with various industries / organizations which help the students to undergo vocational training programs during summer vacation period.

(d) Faculty exchange and professional development

- The various departments organize several workshops/ faculty development programs / conferences / seminars on the current trends and technologies for the benefit of faculties and invite neighboring institutes to exchange the ongoing research activity.
- The following programs are conducted during last 4 years :
 - i) Enhancement in Teaching Learning process.
 - ii) Counseling for Teachers for their teaching methodology.
 - iii) Student feedback
 - iv) Faculty performance appraisal.

(e) Research

Institute encourages the faculty members to take up research projects of AICTE, DST, MHRD and other state and Central Government organizations.

(f) Consultancy

- The institute has established the consultancy cells such as IIPC and EDC that encourage the faculties and students to take consultancy works towards the fulfillment of needs of the society and industry.
- The institute has formed Industry Institute Partnership Cell (IIPC) for establishing relationship between industries and institute.
- To promote consultancy, the institute has formed Entrepreneur Development Cell (EDC) and Renewable Energy cell which include a mixture of teaching, research, consultancy and knowledge transfer activities.
- The institute encourages online workshop/training conducted by experts for lifetime members of Indian Society for Technical Education (ISTE).
- Details are presented in **Criteria 3.5.4. (Page no. 84)**

(g) Extension

The following workshops are conducted by institute to enhance the skill and knowledge of our students:

- Cloud computing workshop
- Ethical hacking
- C programming

(h) Publication

- The institute encourages the faculty to publish their research outcome results in journals and conferences. Details are presented in **Criteria 3.1.7 (Page no. 67)**
- The institute provides support to faculties and students by bearing expenses of research, travel cost and accommodation.
- The institute also motivates faculties and students in the form of giving awards for attending conferences/workshops and publishing research papers in journals

(i) Student Placement

- Providing information of current job requirement skills and market technologies to students.
- Pre-placement training from third year onwards.
- Weekly Lecture for Personality Development and frequent Assessment Tests in Aptitude, Reasoning and Verbal for second year to final year of students.
- Institute has a team of faculties for continuous evaluation and assessment of students by Vocational training, Projects etc.

The Placement details during last 4 years

Year	2013-14	2012-13	2011-12	2010-2011
No. of Students Placed	414	365	254	248

(j) Twinning programs

- We do not have a Degree Course with twinning programme.

(k) Introduction of new courses

Table 3.9 Introduction of new courses

Year	New Courses introduced	Increase in Intake
2013-14	----	----
2012-13	----	----
2011-12	----	----

2010-11	M. Tech. in ETC – 18 M. Tech. in CSE – 18 M. Tech in ME - 18	B.E. CSE - 60 (2 nd Shift) B.E. EEE - 60 (2 nd Shift)
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(l) Student exchange

- The institute collaborates with various industries. Such industries provide opportunity to the students to know the current trends in industry. This helps the students in getting placements.
- The eminent experts from industries and various organizations are invited for the guest lectures for the students for exchange of knowledge and views.
- The institute organizes state level and intercollege fests, conferences, seminars and technical paper contests in which students from various institutions exchange their ideas in the area of engineering.
- The institute conducts industrial tours for students every year for getting exposure to the practical aspects of industry.

The following are eminent faculties of our country who have visited the college for delivering guest lectures and conducting workshops:

1. Ankit Fadia – Ethical Hacking
2. Yashwant Kanetkar – C programming

(m) Any other

- The institute arranges add-on programs for students as well as faculties and lab-staff for enrichment of their knowledge on technical topics that are beyond syllabus.

Engineering departments regularly conduct special classes for GATE aspirants.

3.7.6 Detail on the systematic efforts of the institution in planning, establishing and implementing the initiatives of the linkages / collaborations.

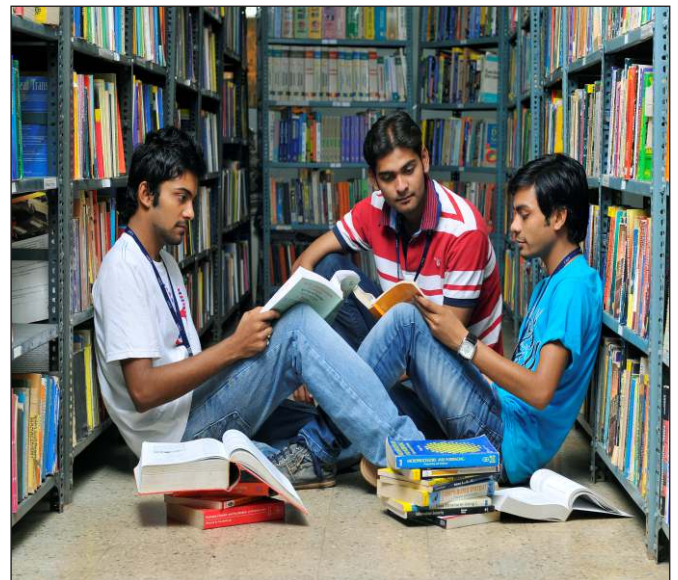
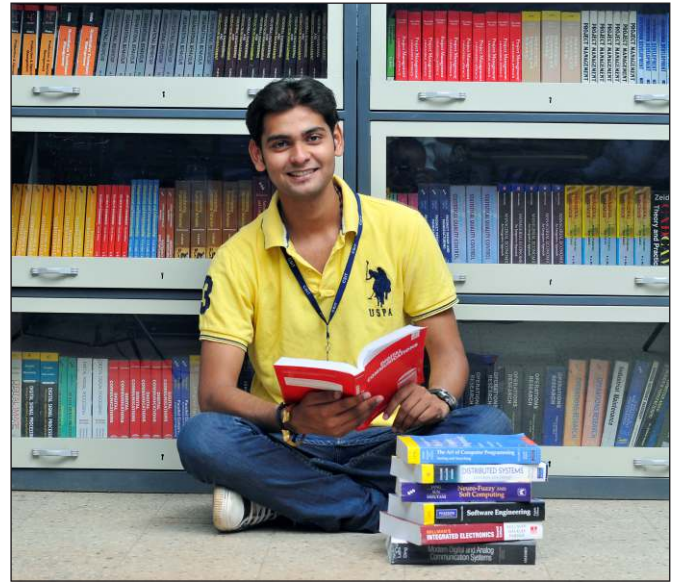
- The institute has an Internal Research Committee.
- The college provides budget for in-house R & D every year.
- Institution provides seed money to undertake industry oriented student projects and in-house R&D projects by faculty.
- Workload of senior faculty involved in major R&D projects and consultancy is reduced.
- Faculty is sponsored regularly to attend important workshops and seminars arranged by leading research organizations.
- Senior faculty of the departments get research projects, MODROBS, FDP and Seminar grants regularly which promote the research culture.



Motivational incentives in the form of cash awards along with commendation certificates are provided to the faculty for publishing papers in reputed journals.

Criterion – IV

Library Facilities



CRITERION - IV

INFRASTRUCTURE AND LEARNING RESOURCES

4.1 Physical Facilities

4.1.1 What is the policy of the Institution for creation and enhancement of infrastructure that facilitate effective teaching and learning?

- The Institute regularly updates physical infrastructure by constructing new buildings and replacing obsolete equipment with new equipments. Every year as per the new courses sanctioned, new buildings are constructed and additional floor space is created.
- Every year budget proposals are submitted by all the departments for purchase of new equipment for laboratories to enhance the technical facilities and to meet the curriculum requirements proposed by the university.
- In addition to the existing central library building, a new library building is coming up with 4 floors with carpet area 658 sq.m. This building can provide 200 seating capacity with reference sections and digital library facilities.
- Laboratories in all the departments are provided with required equipments as per the curriculum specified by the university and also provided with add-on equipment to facilitate the students with advanced experiments in the laboratories which will enable them to carry out research work.

4.1.2 Detail the facilities available for

(a) Curricular and Co-curricular Activities –

- The institute has well furnished ventilated lecture halls for conducting theory and tutorial classes. All the lecture halls are provided with seating capacity of seventy and provided with glass-board, Wi-Fi and LAN enabled internet connectivity.

Details of lecture halls, Seminar halls, library and laboratories:

Particulars	Number of Rooms
Lecture Halls	44
Tutorials	6
Drawing Halls	2
Conference Hall	1
Computer Centres	5
Central Library	1
Department Library	6

Dept. Laboratories	38
Research Centre	1
Central Workshop	1
Seminar Halls	6
Exam Control Office	1
Digital valuation Centre (DVC)	1
Waiting hall (Boys)	5
Waiting hall (Girls)	6



Dynamics of Machine Lab



Central Workshop



EEE Lab



Electrical Machine Lab



Computer Center



- The institute has six air conditioned seminar halls. They are provided with audio systems, LCD projector, Wi-Fi and LAN enabled internet connectivity.
- Apart from central library each department has departmental library to enable the students to use it for reference purpose.
- The institute has well furnished, separate English Language Communication Skills Lab provided with sixty six computers and licensed software named GLOBARENA.
- The institute has 8000 hrs of NPTEL video lectures prepared by IIT Professors on 200 subjects for the use of students and faculty.
- The institute publishes monthly newsletter “Sancharika” for in-house circulation and to keep informed about the happenings and events of the college.
- Entrepreneurship Development Cell (EDC) conducts training programmes to develop the students in the institute, so as to motivate them to become entrepreneurs.
- The Industry-Institute Partnership Cell (IIPC) of the institute is promoting the close interaction of industry and various departments of the institute.

(b) Extra-curricular activities:

- Sports, outdoor and indoor games, gymnasium, cultural activities, yoga classes are being organised by the institute.
- A permanent basketball court and volleyball court are provided with flood light arrangements.
- A separate indoor games facility for playing shuttle badminton, caroms and chess, is provided and it is being used by the faculties & students regularly.
- Qualified & experienced Physical Instructor looks after the day to day games and sport activities of the institute.
- In addition to organising the annual sports regularly the institute organises intra-college league matches in basketball, cricket, football, volleyball, badminton, table-tennis etc.
- The college has an open amphitheatre in 335 sq. m. area.
- Cultural events are organized every year by student associations which give a platform to students to showcase their talent in cultural field. Apart from these events the students participate in various state and college level programs and bring laurels to the college.
- The Institute organizes “Laqshya” program every year which is an Annual National Level Techno Management fest for the under graduate students across the country which aims to inculcate the technical spirit in all its participants.
- Yoga practice is followed in the institute regularly.
- The institute has an NSS unit with total strength of 100 students and affiliated to CSVTU.

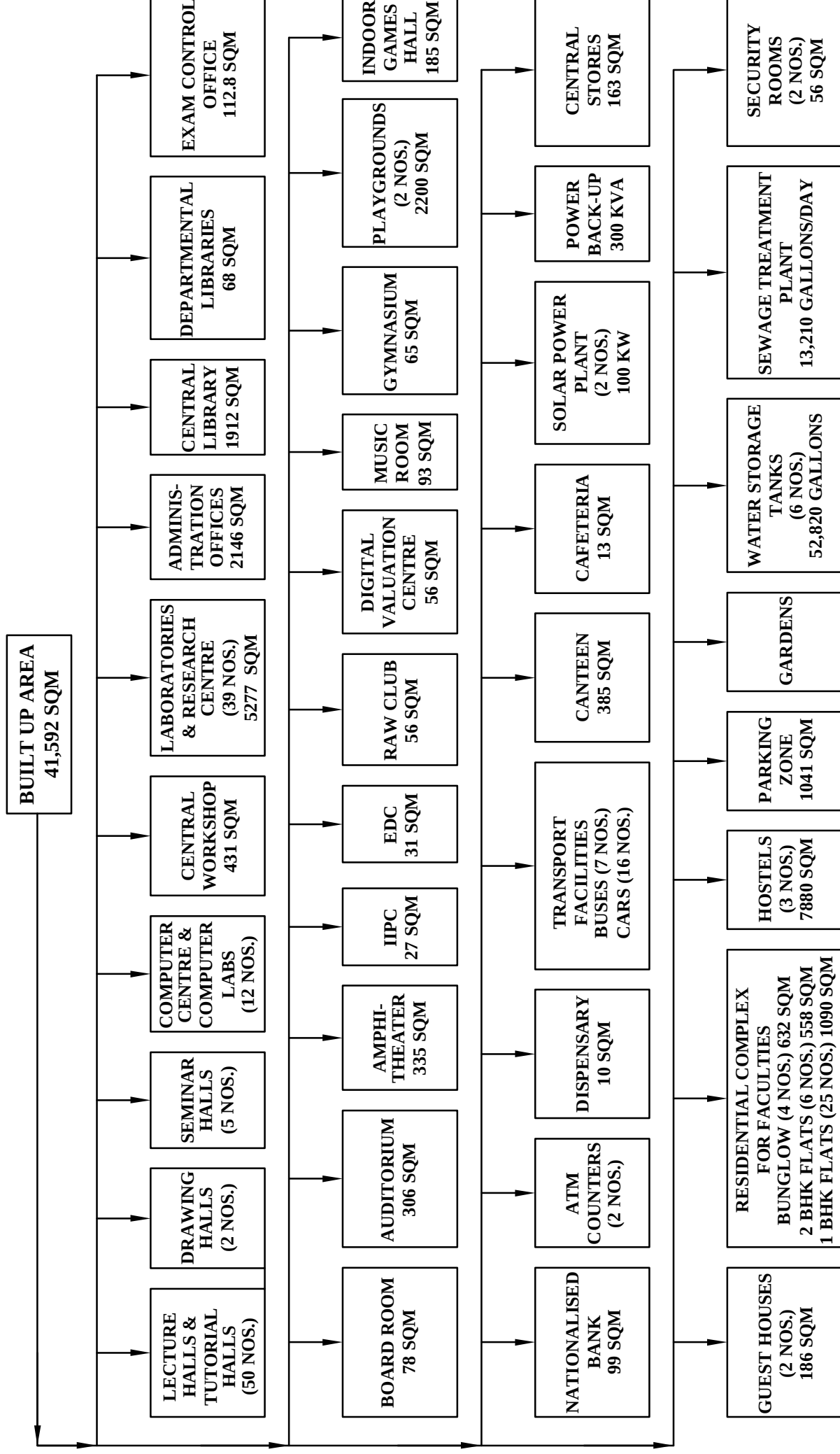


Creating Difference
Since 1999

Chhatrapati Shivaji Institute of Technology

Affiliated to CSVTU, Bhilai | An ISO 9001 : 2008 Certified

PHYSICAL FACILITIES



Facilities	Number available
Volleyball court	1
Basketball	1
Handball Court	1
Cricket ground	1
Football ground	1
Shuttle Badminton Court	1
Kabbadi Court	1
Chess Boards	10
Carom Boards	4
Table Tennis Boards	1
Kho-Kho ground	1

- The institute maintains dispensary with a doctor visiting regularly supported by a full time nurse. Ambulance service is available.



Gymnasium



Volleyball Court



NSS Activities



Women's Cell



Laqshya- A National Level Techno Management Fest

4.1.3 How does the institution plan and ensure that the available infrastructure is in line with its academic growth and is optimally utilized?

- The institute has developed infrastructure as per the norms of AICTE, Technical University and Directorate of Technical Education (DTE).

S. No.	Building Name	Constructed Built-up Area(Sq. m)
1	Block A	3604.7
2	Block B	4616.5
3	Block C	6235.9
4	Block D	3450
5	Block E	9696
6	Old build.	377.9
7	CNC Build.	265
8	R&D Build.	771
9	Central Workshop	431
10	Cafe/Canteen	422
11	Staff Qtr.	2,466
12	Hostels	7,880
13	Parking place	1,041
14	Amphitheatre	335
Total Area		41,592

Facilities developed in the last four years and the amount spent:

S. No.	Items	Expenses in (Rs.) 2013 - 14	Expenses in (Rs.) 2012 - 13	Expenses in (Rs.) 2011 - 12	Expenses in (Rs.) 2010 - 11
1	Acquisition of land; & new buildings and infrastructural built-up	1,03,70,391	1,57,51,253	2,15,74,978	1,34,72,729
2	Library Books & Equipment	25,04,931	25,08,976	16,57,990	41,92,165
3	Library E-Journal, Print Journals & Magazines	4,68,290	10,49,303	6,22,478	2,18,171
4	Laboratory Equipment & Software	55,75,329	99,09,510	39,22,293	55,36,158
5	Games and Sports	1,65,726	22,888	88,780	87,435
6	Buses & Cars Purchase	28,29,088	40,000	14,25,000	35,49,303
7	Furniture	13,54,971	48,20,436	31,83,080	33,03,206
8	Administrative and other Expenditure	12,39,48,119	12,82,53,290	11,20,21,879	8,58,95,961
9	Generator Purchase (D.G.sets)	-	-	-	14,31,218
10	Solar Power Plant (100 KW) (2 no.s)	-	2,25,12,647	-	-

4.1.4 How does the institution ensure that the infrastructure facilities meet the requirements of students with physical disabilities?

- The institute has taken care of physically disabled students. Lift has been provided for the disabled students to attend the class work on first and second floors. Wheel chair facilities are also available. Seating arrangement of physically challenged students is made on ground floor during examinations.

4.1.5 Give details on the residential facility and various provisions available within them:

- Hostel facilities are available for boys and girls in the campus of the institute. Required amenities, Internet, Wi-Fi facilities, recreation facilities, emergency medical facilities, indoor and outdoor game facilities are provided.
- A common music hall facility is provided for the students in the campus for practice and music training classes are conducted for the students.
- Limited residential facilities are available for teaching and non-teaching staff with continuous supply of safe drinking water.
- In addition to the existing residential facilities available for teaching staff, 8 nos. of 1.5BHK Flats with area 618 sq.m. are under construction.
- Security guards look after security requirements for the hostels and college premises.

Residential provision for faculties:

S. No.	Details	Nos.	Area (Sq.m.)
1	Duplex Bungalows	4	632
2	2BHK Flats	6	558
3	Guest House	2	186
4	1BHK Flats	25	1090
5	Security Room	1	28

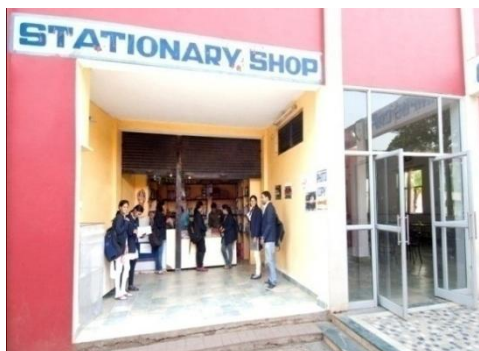
S. No.	Details	Girls Hostel	Boys Hostel 1	Boys Hostel 1
1	Area	4080 sq.m.	1850 sq.m.	1850 sq.m.
2	Number of floors	4	4	4
3	Rooms	95	51	45
4	Capacity (no of students)	300	164	120
5	Occupancy(no of students)	160	106	93
6	Internet	Yes	Yes	Yes
7	Wi-Fi	Yes	Yes	Yes
8	Library facility	Yes		
9	Recreation facility	Yes	Yes	Yes
10	Security	Yes	Yes	Yes
11	Games Room	Yes	Yes	Yes
12	Reading Hall	Yes	Yes	Yes
13	Ambulance	Yes	Yes	Yes
14	Shuttle court	Yes	Yes	Yes
15	Water Purifier	2 Aquaguard	1 Aguaguard	1 Aguaguard



Café



Staff Accommodations



Stationery Shop



Canteen

4.1.6 What are the provisions made available to students and staff in terms of health care on the campus and off the campus?

- There is a dispensary in the institute campus. Ambulance service is provided.
- Employees State Insurance (ESI) scheme is applied to all the eligible employees to get their required off campus medical facilities.

4.1.7 Give details of the Common Facilities available on the campus – Strengthening Women Awareness and Rights (SWAR)

SWAR: Women's Grievance handling Cell is a growing community of women who share a common purpose to learn, educate and collaborate on ideas that have the power to effect meaningful and positive change while fighting the odds they face.

- **Counselling and Career Guidance cell**

A dedicated unit performs comprehensive analysis of students' skills, aptitude, attitude and guides them with structured career plan.

- **Placement Unit:**

- The institute has a Training and Placement Cell responsible for training students in various aspects to compete in Campus Drive. The training module covers Aptitude, Personality Development, GD, Mock Interviews. This helps students to improve their personality and make them employable. The department arranges for both On and Off campus drive.
- The cell is headed by Training and Placement Officer and is assisted by various team members who coordinate the activities needed for the campus drives.

- **Canteen** – Hygienically maintained canteen facility with 428 sq.m. area and seating capacity of 100 is available. Daily 1000 members are using the canteen facility.
- **Cafe** – Cafe is located near the library to get refreshments like snacks, soft drinks, tea & coffee etc.
- **Stationary stores-** It is available near the canteen which has Xerox machine, spiral binding machine and students can procure engineering instruments, aprons and stationary materials from the stores.

4.2 Library as a Learning Resource

4.2.1 Does the library have an Advisory Committee? Specify the composition of such a committee. What significant initiatives have been implemented by the committee to render the library, student/user friendly?

The library has an Advisory Committee. The committee is intended to make plans and coordinate the functioning of the library.

Composition of the Advisory Committee:

The library advisory committee comprises of

1. Director
2. Dean Academic
3. Faculty in charge of library of various departments.
4. Librarian
5. Asst. Librarian
6. Student members from all the departments.

Significant Initiatives of the Committee:

- Preparation of annual library budget to meet the requirements generated from the departments
- The committee looks after the procurement criteria, rules and regulations of the library management including cataloguing and access-administration
- Digital library established
- New library block coming with 4 floors to accommodate reference and digital sections
- Computers are kept for the enquiry of the books according to the title of the book/ author using OPAC.

- Library is stacked with a number of volumes, project reports, e-journals and gives the institute a professional ambience.

4.2.2 Provide details of the following :

- Total area of the library (in Sq. Mts.) : 1912 Sq. m.
- Total seating capacity : 400
- Working hours (On working days, on holidays, before examination days, during examination days, during vacation)

On working days	8.00 AM to 8.00 PM
On Holidays	8.00 AM to 4.00 PM
On Examination days	8.00 AM to 10.00PM
During Preparation Leave	8.00 AM to 7.00 PM

Details of Physical arrangements-

Library is stacked with a number of volumes, back volumes, project reports, e-journals and gives the institute a professional ambience.

Ground Floor: Text Book Section, Circulation Section, Book Bank for All Students, Project Reports.

First Floor: Reference Section, Journals Section, News Papers & Periodicals, reading room, E-Library



E-Library



Issue section



Reference Section



Central Library

4.2.3 How does the library ensure purchase and use of current titles, print and e-journals and other reading materials?

The Library advisory committee reviews the requirements received from all the departments and recommends the same to the college management committee for approval. Prompt approval enables us to get all the requirements of books well before the start of new academic year.

S. NO.	DETAILS		NUMBER
1.	Total number of volumes		58,761
2.	Total number of titles		6,842
3.	New volumes added during 2013-2014		6,273
4.	Expenditure during 2013-2014		Rs. 18,81,900
5.	Number of magazines		10
6.	Number of news papers		15
7.	Journals	International	10
		National	70
8.	e- Journals		194
9.	Expenditure for Journals during 2013-2014		Rs.1,02,114

Details of the amount spent during the last four years

Library	2013-14		2012 – 2013		2011 - 2012		2010 - 2011	
Holdings	Number	Total cost (Rs.)	Number	Total cost (Rs.)	Number	Total cost (Rs.)	Number	Total cost (Rs.)
Text books & Reference books	5295	15,88,500	3342	10,02,600	3166	9,49,800	7366	22,09,800
Journals & Periodicals	80	1,45,315	10	7635	150	244000	115	173360
E - resources	1.IEE 2.Springer	4,68,290	1.IEEE 2.Springer ME 2.Springer ETC &CSE 3.ASTM: DL 4.McGraw-Hill 5.J-Gate	795168	1.IEEE 2.Springer ME 3.ASTM:DL 4.DELNET	467,080	J-Gate	56180
Standards	45	6000/-	-	-	-	-	-	-

4.2.4 Provide details on the ICT and other tools deployed to provide maximum access to the library collection?

- **OPAC-** An OPAC (Online Public Access Catalogue) has been created and two terminals are provided to facilitate its access. It is also made available Via LAN in the campus.
- **Electronic Resource Management Package For e – Journals-** is available and four computers are connected for accessing the Journals in the library.
- **Library Automation**

- All the data relating to the library is computerized and bar coding has been introduced.
- One printer and two Xerox machines are available.
- Internet band width in the library is 10mbps

4.2.5 Provide details on the following items:

• Average number of walk-ins	400
• Average number of books issued/ returned	450
• Ratio of library books to students enrolled	15:1
• Average number of books added during last three years	13874
• Average number of login to OPAC	100
• Average number of login to e-resources	50
• Average number of e-resources downloaded/printed	50
• Number of information literacy trainings organized	Nil
• Details of “ weeding out” of books and other materials	50

4.2.6 Give details of the specialized services provided by the library

Reference	Reference section with 200 seating capacity is available
Reprography	The library has 2 canon photo copy machines
Information deployment and notification	The library displays the latest acquisitions for browsing and content brochure of new arrivals.
ILL (Inter library loan service)	DELNET provides ILL
Download	Yes (E-books etc.)
Printing	Yes
Reading list/Bibliography compilation	Yes
In – house/remote access to e-resources	Yes
User Orientation and awareness	Orientation and awareness seminars are organized for library staff, students and faculty members

Assistance in searching
Databases

Yes

Details of annual library budget and amount spent for purchasing new books and Journals

Year	Amount spent for books (Rs/-)	Amount spent for Journals (Rs/-)
2010-11	22,09,800	1,73,360
2011-12	9,49,800	2,44,000
2012-13	10,02,600	7,95,168
2013-14	15,88,500	6,13,605

4.2.7 Enumerate on the support provided by the Library staff to the students and teachers of the college.

The Library staff always extends services to the students and teachers of the college. Services rendered by the library staff are summarized below:

- Library staff issues books, journals and other reading materials to the students and faculty.
- Call facility for demand on book is available in the library.
- Library is open beyond institute hours for students.
- Library has well furnished students study center with a capacity of 280.
- Book bank facility is available for the students.
- Library staff displays the important media notifications.
- Duplication/ Reprographic facility is available in library.
- Issue of books to the departmental library.
- Faculty/students can access the e-Journals, periodicals and reference books.
- Discussion rooms are planned.

4.2.8 What are the special facilities offered by the library to the visually/physically challenged persons? Give details.

- Issue section is in ground floor.
- Elevator facility is available which helps them to utilize reference section situated in first floor.
- There are no cases of visually challenged students.

4.2.9 Does the library get the feedback from its users? If yes, how is it analyzed and used for improving the library services. (What strategies are deployed by the Library to collect feedback from users? How is the feedback analyzed and used for further improvement of the library services?)

- A suggestion box is available in the library to get feedback.
- Exit feedback is collected from the outgoing students regarding their suggestions for improvement.

4.2.10 List of infrastructural Development

New library building is coming up with four floors and area of 658.0 Sq. m.

4.2.11 Did the library organize workshop for students to facilitate better usage?

Every year an orientation program is conducted for newly admitted students to make them aware of all facilities, rules -regulations and its usages.

4.3 IT Infrastructure

4.3.1. Give details on the computing facility available (hardware and software) at the institution.

1. Number of computers with Configuration (provide actual number with exact configuration of each available system)

S. No.	Configuration	No. of Systems
1	Pentium I-3 processor, INTEL motherboard, 6 GB RAM, 80 GB SATA HDD LED Monitor, keyboard, mouse (optical), ATX cabinet, USB port	100
2	P-IV Dual Core INTEL motherboard, 4GB RAM, 250 GB SATA HDD, LED monitor, keyboard, MOUSE (SCROLL & OPTICAL), ATX cabinet, USB port	145
3	P-IV Dual Core INTEL motherboard, 4 GB RAM, 80 GB SATA HDD color monitor and tft monitor, keyboard, mouse (scroll & optical), ATX cabinet, USB port	439
4	AMD 2000 & 2400 + XP 1 GB RAM, 40 GB HDD, color monitor, keyboard, optical mouse	60

	Total Number of System	744
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2. Computer-student ratio

UG - 1: 4 PG - 1: 2

3. Stand alone facility:

50 systems are provided with stand alone facility

4. LAN facility

All computing labs available in the Institute are provided with LAN facility for easy access.

5. Wi-Fi facility:

Wi – Fi facility is available in the institute with 150 Mbps speed upto 1 Km range and this facility is extended to different locations in and around the campus.

6. Licensed software

Department	Licensed software
Computer Science and Engineering / Information Technology	Visual Prolog, Oracle 11g, RAS studio XE C, C++, Campus License Microsoft, Windows 8.2, Subscription For Cyberoam, Windows 2003, Windows 2003 Client, MATLAB, Oracle 9.0/10.0, Turbo Assembler, Borland builder, AUTOCAD LT 2002, Norton Antivirus 2002, Place Mentor English SW,
Mechanical Engineering	CREO-2.0, ANSYS14, SINUTRAIN 802D, MS-PROJECT 2010, CUTVIEWER Simulation Software, MINITAB 16 , Fluidsim, ABAQUS 6.10, ADAMS
Electrical & Electronics Engineering	MiPOWER (Version 6.0) (10 Users)
Electronics and Telecommunication Engineering	MULTISIM,ULTIBOARD,MULTI MCU (V11), Xilinx ISE System (Edition 13.1), SPARTAN 6, MATLAB 7
Electronics and Instrumentation Engineering	LABVIEW Core + Control & Embedded Software (1 Set of 50 user), RS Logix Micro-500(8.40.00)

English Dept.	GLOBARENA (Placementor)	English	Language	Software
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7. Number of nodes/ computers with Internet facility

- All the systems are provided with internet connectivity

4.3.2 Detail on the computer and internet facility made available to the faculty and students on the campus and off-campus?

- Internet service is available for both faculty and students in the campus.
- Central computing centre is having 60 systems with internet facility. It is kept open for students and staff of all the Departments, even after college working hours.
- Computer labs of all the departments are provided with internet facility.
- All the administrative offices are provided with computer system and internet facility.
- Wi – Fi connectivity is available in and around the campus
- Boys and Girls Hostels are facilitated with Wi-Fi connectivity to access internet.
- All the computers are provided with UPS for power backup. Moreover, un-interrupted power supply is made available in the campus to manage power supply.

Specifications:

- Internet Provider :** Bharat Sanchar Nigam Ltd.,
- Total Band width:** 10 Mbps broad band connections.

4.3.3 What are the institutional plans and strategies for deploying and upgrading the IT infrastructure and associated facilities?

- As and when required, the up-gradation of computer systems is done with latest configuration.
- Individual up gradation of the computers is taken up as per the need and requirements of the various departments or individuals.
- All the computer systems in the campus are regularly monitored by the system administrator and maintenance staff.
- The trouble/problems experienced with the computers in various laboratories are reported by the concerned person through online complaint system via intranet. The computer hardware cell responds to the request and keeps records of the same. The maintenance staff then visits the lab and resolves the problems.

- Computer server room is provided with eight servers of different capacities to cater to the computer requirements.

IT Service management:

- **E-mail system:** Webmail operative through college server is the primary email service offered to institute employees. They can use this communication facility for data exchange and information sharing. All the notifications and circulars are shared through e-mail also.
- **Network security:**
 - The college is having required intranet and internet facility
 - In order to access the internet facility of the institute, user has to get registered. Cyberoam security system is used to allow the access. After registration user name and password is allocated. Unauthorized users are not allowed to use the internet.
 - Unwanted websites are blocked through the server to restrict unwanted message or information transfer or its access.
- **Risk Management:** The risk management of IT assets is associated with the user ownership, operation, involvement and adoption of IT within the institute. The institute has two fold policies to manage the risk.
 - The institutional IT assets such as college website, learning services and campus automation system are protected against threats.
 - Implementing the standards set by the professionals to mitigate the risk. The institute takes no responsibilities to the user's data.
- **Software Assets Management :**
 - It is the policy of the institute to respect all computer software, copy rights and adhere to the terms and conditions of any licence to which institute is a party.
 - All computer softwares acquired by the institute are purchased through the approval of established procurement procedure.
 - Once the software has been received from the suppliers and installed, staff of the concerned departments is responsible for ensuring that the licence document manuals and other associated material are securely and appropriately stored as institute managed assets.

4.3.4 Provide details on the provision made in the annual budget for procurement, upgradation, deployment and maintenance of the computers and their accessories in the institution (Year wise for last four years)

Amount spent for procurement, up gradation, deployment and maintenance of the computers and their accessories in the institution during last four years:

	2013 - 14	2012 - 13	2011 - 12	2010 - 11
Amount Spent	2328882	3098501	3061357	3185180

4.3.5 How does the institution facilitate extensive use of ICT resources including development and use of computer-aided teaching/ learning materials by its staff and students?

- The Institution has adequate computer facility for its faculty. Faculty members can easily access computers with internet facility for learning and preparation of teaching materials in their respective departments.
- Each department has its own computer labs and softwares to fulfill the curriculum demand.
- E-resources available with the Institute can be easily accessed from central computer centre and library.
- NPTEL lectures (video+ notes) are available in computer centre which help students and staff to get ready resources as study material. Moreover, the departments are provided with specific lectures as per the requirement given by faculties.
- Each department is provided with a LCD projector for the faculty to use for teaching-learning purpose.
- Each department in the institute has individual seminar hall provided with LCD projector, PA system and internet facility.

4.3.6 Elaborate giving suitable examples on how the learning activities and technologies deployed (access to on-line teaching - learning resources, independent learning, ICT enabled classrooms/learning spaces etc.) by the institution place the student at the centre of teaching-learning process and render the role of a facilitator for the teacher.

- The institution acts as a facilitator for e-learning material (NPTEL video lectures) and the accession of online lectures of various experts from IITs, NITs and reputed institutes.
- The Institute has subscribed for various e-journals to facilitate research facility for faculty and students.

4.3.7 Does the Institution avail of the National Knowledge Network connectivity directly or through the affiliating university? If so, what are the services availed of?

➤ No.

4.4. Maintenance of Campus Facilities

4.4.1 How does the institution ensure optimal allocation and utilization of the available financial resources for maintenance and upkeep of the following facilities

Every year financial resources are allocated to maintain and upkeep the facilities like building, equipment, computers, vehicles etc.

Amount spent for maintenance of facilities during the last four years:

Items	Expenses in 2013 – 14 (Rs)	Expenses in 2012 – 13 (Rs)	Expenses in 2011 – 12 (Rs)	Expenses in 2010-11 (Rs)
a) Building	60,31,185	57,40,766	58,05,426	41,16,899
b) Equipment & Computers	1,06,609	3,32,680	37,955	34,000
c) Vehicles	11,71,501	8,77,226	11,25,246	6,78,353
d) Any other (Administrative and other Expenditure	40,00,088	33,33,499	22,09,864	15,75,683

4.4.2 What are the institutional mechanisms for maintenance and upkeep of the infrastructure, facilities and equipment of the college?

- Civil Engineer looks after the maintenance of infrastructure and facilities along with two supervisors.
- Electrical supervisor with three electricians takes care of maintenance of electrical equipments and other devices, power generators etc.
- Before the start of the academic session all the departments submit their maintenance requirement to purchase committee. After review and approval by the committee and then by management, departments are allotted fund to ensure workability of the lab equipment.
- Store manager collects requirements of laboratories and furniture from all the departments. After getting approval from the management, store manager ensures timely procurement and supply of items to various departments.
- Preventive maintenance procedure is followed by the lab Incharges to keep the equipments in the laboratories in working conditions.
- **Building Hygiene:** The institute has outsourced the task of hygiene. An agency is performing the duties. The duties are daily cleaning the classrooms, corridors, floors, cleaning of laboratories, staff rooms, dusting of furniture and cleaning of toilets etc.

4.4.3 How and with what frequency does the institute take up calibration and other precision measures for the equipment/instruments?

- Lab in-charges conduct experiments in the laboratories at the beginning of each semester to ensure proper working of the equipments in the laboratories. Sample readings taken by them enable them to update the lab manual.
- Individual department takes care of calibration of precision instruments available in their department.

4.4.4 What are the major steps taken for location, upkeep and maintenance of sensitive equipment (voltage fluctuations, constant supply of water etc.)?

- The college has 100% power backup through, 125 KVA and 250 KVA generator sets in case of power failure.
- All the sensitive equipments and computers are protected from voltage fluctuations and power supply interruptions through the use of UPS devices.
- A 315 KVA transformer was erected in the institute with the following details:

Particulars	Capacity	Nos.
Transformer	315 KVA	1
Main Circuit Breaker	800 amps	1
Bus Bar chamber	3 phase (300x3) amps	1
Distribution Panel	250 amp	4
	160 amp	5
Change over switch	400 amps	2
Total capacity of power backup	375 KVA	

An exclusive electrical maintenance section works for the college with one person as Sub-station in charge, two technicians and two generator operators to look after the electrical equipments.

Supply of water:

- Pure drinking water is available in institute blocks, hostels, canteen during 24 hours with Storage capacity of 2, 00, 000 litres.
- Usage water Storage tanks : 6 Nos.
- Drinking water Storage tanks : 2 Nos
- Water availability in hostel : 24 hours
- Number of aquaguard water purifiers : Academic Premises - 9
- Hostels : 4

Criterion – V

Association Activities



Make your Move



Farewell Party



Blood Donation camp



Welcome Party



Corporate Social Responsibility



Chess

CRITERION V

STUDENT SUPPORT AND PROGRESSION

5.1 Student Mentoring and Support

5.1.1 Does the institution publish its updated prospectus/handbook annually? If 'yes', what is the information provided to students through these documents and how does the institution ensure its commitment and accountability?

➤ **Yes**

- Every Year Institute publishes its **Brochure and Prospectus** and makes it available to student community, society and faculties. These two documents are rich in academic and administrative information. The information Brochure includes the vision and mission of the Institute, programs offered by the Institute, Teaching and Non-teaching staff, Membership of Technical Societies; admission procedure. It provides information about university (CSVTU) rules and regulations and facilities provided by the Institute.
- A monthly News Letter **Sancharika** is published. It contains articles and information about student events such as seminars, Workshops conducted, Technical Awareness Programmes, Departmental association activities and achievements of Students and Faculty members. It is distributed free of cost to all students, faculties and others.
- The institute publishes a Souvenir every year containing information about outgoing students. It is given to the students at the time of their departure from the Institute.

5.1.2 Specify the type, number and amount of institutional scholarships / freeships given to the students during the last four years and whether the financial aid was available and disbursed on time?

- **Yes**, some concessions are granted to students. Fees concession to brother sister studying in the institute.

Details / Session	2013-14	2012-13	2011-12	2010-11
Institutional Scholarships / freeships	Brother Sister Concession was given to 61 students	Brother Sister Concession was given to 58 students	Brother Sister Concession was given to 52 students	Brother Sister Concession was given to 53 students
Total amount given	Rs. 3,66,000	Rs. 3,60,000	Rs. 3,24,000	Rs. 3,18,000

5.1.3 What percentage of students receive financial assistance from state government, central government and other national agencies?

The college follows the admission rules and regulations of Department of Technical Education Chhattisgarh Government. The students are getting admitted through the State CG Pre Entrance Exam (CGPET) and IIT-JEE Mains. They are benefitted by the scholarship scheme provided by the State Government. Provision of Reservation SC =12%, ST=32%, OBC=14% and Un-reserved =42%.

Tuition Fee Waiver Scheme (TFW) – Under this scheme an additional 5% students can be admitted as per the guidelines of the Admission Policy framed by the State Government during the concerned year of admission. This intake is over and above the approved intake. However, TFW is applicable only if the number of students admitted exceeds the approved intake.

Details of financial Assistance given to the students by the State Govt.

Year: 2010-2011

S. No.	Course	Categories	No of Student	Amount
1	BE/ME/M.Tech	OBC	228	4103475
2	BE/ME/M.Tech	ST	15	306795
3	BE/ME/M.Tech	SC	36	761795
4	BE/ME/M.Tech	MINORITY	3	75000
5	BE	TFW	16	Tuition fee

Details of financial Assistance given to the students by the State Govt.

Year: 2011-2012

S. No.	Course	Categories	No of Student	Amount
1	BE/ME/M.Tech	OBC	387	8704935
2	BE/ME/M.Tech	ST	21	582595
3	BE/ME/M.Tech	SC	45	1232854
4	BE/ME/M.Tech	MINORITY	4	99500
5	BE	TFW	11	Tuition fee

Details of financial Assistance given to the students by the State Govt.

Year 2012-2013

S. No.	Course	Categories	No of Student	Amount
1	BE/ME/M.Tech	OBC	387	7605130
2	BE/ME/M.Tech	ST	19	446660
3	BE/ME/M.Tech	SC	39	916510
4	BE/ME/M.Tech	MINORITY	6	148500
5	BE	TFW	Nil	Nil

Details of financial Assistance given to the students by the State Govt.
Year 2013-2014

S. No.	Course	Categories	No of Student	Amount
1	BE/ME/M.Tech	OBC	444	9523160
2	BE/ME/M.Tech	ST	33	858165
3	BE/ME/M.Tech	SC	47	1223775
4	BE/ME/M.Tech	MINORITY	14	349500
5	BE	TFW	6	Tuition fee

5.1.4 What are the specific support services/facilities available for

➤ Students from SC/ST, OBC and economically weaker sections

- As per the Government regulations, all the students belonging to SC /ST / OBC categories receive financial assistance in the form of reimbursement of tuition fee from the Government
- Through Book Bank facility every student gets 6 Books for each semester and additional 3 books can be issued from the Central Library. Reference section of library is kept open for 12 hours for all students.
- Remedial classes are arranged by the college for the academically weak students.
- Teacher Guardian takes special care of weak students, identifying their academic deficiencies.

➤ Students with physical disabilities

- As per the State Government Norms Physically challenged students receive scholarship.
- Wheelchair, Ramp and Lift facility are available for physically challenged students.
- There is provision for making suitable seating arrangement for physically challenged students during examination.

➤ Students to participate in various competitions/conferences in India and abroad.

- Students are encouraged to participate in extra-curricular and co-curricular activities such as technical fests, cultural events, and sports events etc. which are organized by reputed institutions across India.
- The college extends financial support to students participating in various competitions/conferences in India. Attendance is also granted to students who are going out for workshops and conferences.
- Faculty members are deputed to assist and guide the students in respective area. Furthermore, students who secure medals and trophies in such events are given due recognition and appreciation on

Foundation day celebration of the institute.

➤ **Medical assistance to students: health centre.**

- The college has a medical care centre with primary medical facilities.
- The centre also has a full time Nurse.
- The centre has its own round-the-clock ambulance facility.

➤ **Organizing coaching classes for competitive exams**

- College organizes classes for GATE/CAT preparation in college campus. Personality Development programs are conducted for students regularly which enhance the EQ level and communication skills of the students. The college invites Guest speakers from the industry. This helps in getting regional and global employment opportunities for the students.
- Students self assessment test is organized by college through Nicco Ventures Ltd , Cocubes.com, Step Up careers Builders Pvt. Ltd, Monsterscollege, and Rangrut Services India. which helps the students for campus preparation
- College organizes MOCKCAT for the students for preparation of the exams.
- The college has well stocked library books for CAT/GRE/TOEFL/GATE and other Public Service Examinations.
- The college has the laboratories with necessary softwares to enhance students communication and their skills in Spoken English. Extra classes for improvement in communication skill are also organized.
- IIPC cell has been established in the college to promote and extend industrial consultancy and R&D services. IIPC provides a platform for exposure to recent trends in engineering and organizes industrial visits for students.

➤ **Support for slow learners**

- Weak students are identified by the respective Teacher Guardian. Personal counseling is done by the respective Teacher Guardian on regular basis.

➤ **Exposure of students to other institutions of higher learning /corporate /business houses, etc**

- Students are motivated and properly guided to participate in events organized by other institutions. Various Industrial visits are arranged for the students for industrial exposure.
- ISTE & IE. Student chapters have been established for Under-Graduate students to provide global technical exposure.
- Online access to IEEE and SPRINGER journals is available in central E-library for the students of UG and PG.

➤ **Publication of student magazines.**

- College magazine SANCHARIKA is published regularly in which the articles of students are published. Special issue of Technical Research papers is printed after AICON conference.

5.1.5 Describe the efforts made by the institution to facilitate entrepreneurial skills, among the students and the impact of the efforts.

Institute has its Entrepreneurial Development Cell which takes care of conducting various programs for entrepreneurial skill development.

S. No.	Name of Program	Date of the Program	Resource Persons	Industry
2010-2011				
1	3-days Entrepreneurship Awareness Camp	17/02/2011 to 19/02/2011	Mr.G.S.Dey (Ex-Officer of SBI) Mr.P.K.Nimonkar (Project Head, CITCON Raipur) Mr.K.C.Thomas (Project Officer,CITCON,Raipur)	National Science & Technical Entrepreneurship Development Board of Science & Technology, Govt of India, New Delhi & conducted by faculty members of CITCON (Chhattisgarh Industrial & Technical Consultancy Center), Raipur
2011-2012				
2	3-days Entrepreneurship Awareness Camp	16-2-12 to 18-2-12	1.Mr.G.S.Dey (Ex. Bank District Coordinator) 2. Mr. A.S. Kalale (Expert Communication Skill) 3. Mr. K.C. Thomas (Project Officer, CITCON, Raipur)	National Science & Technical Entrepreneurship Development Board of Science & Technology, Govt of India, New Delhi & conducted by faculty members of CITCON (Chhattisgarh Industrial & Technical Consultancy Center), Raipur
2012-2013				
Nil				
2013-2014				

3	2-day Entrepreneurship Awareness Camp	14/03/2014 to 15/03/2014	1.Mr.G.S.Dey (Ex. Bank District Coordinator)	National Science & Technical Entrepreneurship Development Board of Science & Technology, Govt of India, New Delhi & conducted by faculty members of CITCON (Chhattisgarh Industrial & Technical Consultancy Center), Raipur
			2. Mr. A.S. Kalale (Expert Communication Skill)	
			3. Mr. K.C. Thomas (Project Officer, CITCON, Raipur)	

Impact of the Efforts

The training provided has encouraged students and enhanced their entrepreneur skills. The experts gave important ideas and covered the necessary information to setup enterprise. Such type of training inspires the students to become future entrepreneurs.

5.1.6 Enumerate the policies and strategies of the institution which promote participation of students in extracurricular and co- curricular activities such as sports, games, Quiz competitions, debate and discussions, cultural activities etc.

- Sports Track Suits are provided to students for participating in major tournaments.
- Institute has Volleyball, Basketball, Tennis, Table-Tennis and Athletics grounds for enhancing the capabilities of students in field activities and sports. **(Details are already provided in section 4.1.2.b Page no. 102).**
- Institute hosts university level competitions for boys & girls. Cash Awards and Certificates are given in the tournaments conducted in the college.
- Students associations organize various co-curricular activities for the students.
- Institute provides TA, DA to students participating in extracurricular and co-curricular activities such as sports, games, quiz competitions, debates, cultural activities etc at national level.
- Attendance is granted to students participating in technical, cultural or sports events outside the college.
- Institute conducts national level Robotic competition fest known as “LAQSHYA” every year.
- Well equipped Gymnasium facility is available for students.



The institution motivates the students for participating in various extracurricular activities by ensuring proper encouragement. All the facilities are provided and sufficient funds are allotted for the same. The sports and cultural committees monitor the extracurricular activities. Proper academic guidance is ensured for those students who participate in sports activities or other extracurricular activities.

5.1.7. Enumerating on the support and guidance provided to the students in preparing for the competitive exams, give details on the number of students appeared and qualified in various competitive exams such as UGC-CSIR- NET, UGC-NET, SLET, ATE / CAT / GRE / TOEFL / GMAT / Central /State services, Defense, Civil Services, etc.

- Proper guidance for competitive examinations like GATE/CAT/GRE/TOEFL is provided by the respective departments. Preparatory classes and mock tests are conducted for the students.
- NPTEL interactive video lectures are available for technical exam like GATE/IES/PSUs.
- Several volumes of books on career guidance and competitive examinations are available in Library.
- Students who secure good ranks in GATE/GRE/ TOEFL / CAT etc. are given opportunity to deliver talks to their colleagues and juniors so that they can be motivated and benefitted.

No. of Students qualified in GATE

Academic Year	CSE	ETC	EEE	E&I	MECH	MEX	IT
2013-2014	1	2	0	1	05	0	-
2012-2013	1	7	13	2	02	1	-
2011-2012	5	1	2	11	07	0	-
2010-2011	2	1	1	1	10	0	-

No. of Students qualified in GRE/TOFEL/CAT/MAT/PGECET etc

Academic Year	CSE	ETC	EEE	E&I	ME	MEX	IT
2013-2014	2	1	0	0	02	0	0
2012-2013	0	0	0	0	03	2	0
2011-2012	0	2	0	0	-	0	0
2010-2011	0	2	1	0	-	0	0

5.1.8 What type of counseling services are made available to the students (academic, personal, career, psycho-social etc.)

Academic Counseling:

The College has an Academic counseling and guidance cell. Students are guided and counseled by Teacher Guardians and HOD on academic and career prospects. The teacher guardians are available round the clock for the students. Teacher guardians of first year students extend adequate guidance required at initial level. First year students are informed about the scope and nature of various subjects that form the syllabus. From second year onwards, when students come to their respective branches new TGs for a group of 20 students are assigned. These TGs take care of them up to final year. Personalized counseling and parental care is made possible by them. The students are not pressurized in choosing the elective subjects. They are given right kind of counseling which helps them shape their career. The students who need psychological counseling or any type of social counseling are also attended to very carefully.

Personal Counseling:

- College has constituted a dedicated team of Faculty that takes care of academically weak or irregular students.
- Women's Grievance cell is constituted to look into the grievances of female students and female employees to avoid gender specific discrimination and uphold their dignity.

Career Counseling:

- The college has constituted a dedicated team of English Faculty for developing soft skills of the students. Slots are allotted in the time table for conduction of classes throughout the session from 3rd semester to 6th semester students with their regular studies. In these classes, aptitude, reasoning, orientation program, group discussion and mock interviews are conducted to identify and improve the skill set of the students.
- The college is having campus connect initiative with the Nicco Ventures Ltd , **Cocubes.com**, **Step Up careers Builders Pvt. Ltd**, **Monsterscollege**, and **Rangrut Services India** which enhance the employability skills of our students.

Psycho-Social Counseling:

The students during the course of their studies in the college come across various issues. They are, at times, too immature to handle the problems. The college provides them personal counseling. They can share their problems with the teachers. The teacher concerned is very supportive in guiding them to solve their problems. The candidates at times come face to face with certain social issues or problems which tend to bring inferiority of complex in them. The teachers make it sure that no such deterioration happens with the psychosocial understanding of the students. They are counseled to become better human beings and advised to stand tall for the social cause.

5.1.9 Does the institution have a structured mechanism for career guidance and placement of its students? If 'yes', detail on the services provided to help students identify job opportunities and prepare themselves for interview and the percentage of students selected during campus interviews by different employers (list the employers and the programmes).

➤ Yes

Campus Recruitment

Various industries are invited for campus recruitment. Proper efforts are being made by the college to improve the employability skills of the students. In last four years many students are absorbed by the industry before completion of the course. Efforts made by the college are given below:

Training and Placement cell

Training & Placement Cell of the institution comprises of T&P Officer, Office Assistant and Departmental Coordinators. It monitors and organizes continuously the training and placement activities. The cell regularly contacts all relevant industries / companies and conducts placement drives.

Special Training on English Language and Communication Skills

Language classes and Communication Skill development training are arranged to improve the communication skill of students from 1st year onwards. It also helps the students to improve their technical presentation and performance in interview.

Regular Training on Aptitude & Other Soft Skills

Aptitude and soft skills classes are conducted by resource persons from **BookShelves** Bhilai. These classes are started from 2nd year onwards and are compulsory for all students.

Special customized training based on Industry requirement

Each department conducts various training programs in the department to enhance the skills of students in the recent technologies. In addition, guest lectures are arranged on areas relevant to the current industrial scenario.

List of Companies visited for placement drives

Companies Visited	No. of Students selected
2013-14	
On Campus	
Persistent System Ltd, Innovative Design Tech (P) Ltd, Wt It Solutions Pvt Ltd, Merce Technologies Pvt. Ltd., Flour Daniel India Pvt Ltd, Genpact Limited, Jagbros Group, Hcl Infosystems Ltd., United Spirits Limited, Celia Infotech Software Solutions, Fixnix Infosec Solution Pvt Ltd.Hyd, Alert Knowledge Services, Allerin Tech Private Limited, Navi Mumbai, Kma Power India Pvt Ltd, Oil Field Warehouse & Services, Mumbai, Silverton Structures & Projects Pvt Ltd, Jindal Electrical & Machinery Corporation Ludhiana, Anupam Electrical & Engg Industry Mumbai, Tega Industries Limited Kolkata, Tata Consultancy Services, Cognizant Technologies Solutions, Concierge Technologies Pvt Ltd Lucknow, Must Garments, Intelligent Traffic & Embedded Surveillance Systems, Shriram Transport Finance Company Ltd, Mather & Platt Pumps Pvt Ltd, Shriram Industries, Apex Technologies , Vadodara, Narmada Drinks Pvt Limited	414

2012-13	
On Campus	
Fluor Daniel India Pvt. Ltd., Cummins India Ltd., Globussoft Pvt. Ltd., Sobha Softech Pvt Ltd, Stellant Suse Technotics Pvt. Ltd., Paramatrix Technologies (P) Ltd, Poornam Info Vision Pvt. Ltd., Executive Ship Mgmt Pvt Ltd., Inter Globe Technologies, Shriram Transport Finance Co. Ltd., Tega Industries, Vasantha Tool Crafts Pvt. Ltd. Hyd, Aanvi Web Services, Jayaswal Neco Industries Limited, Bhushan Steel Ltd., Bosch Engineering, Wt It Solution, Nexus Engineering, Universal Hunt,	365
2011-12	
On Campus	
A.K. Erectors, Ambuja Cement, Apex Technology, Aspiring Minds, Aster, Bhusan Steel, Capegemini, Cura Healthcare Pvt. Ltd., Fluor Daniel Pvt. Ltd. Jayaswal Neco Industries Ltd., Nextgen Mobile Tech. Ltd., Nexus Consultants, Oilco Services Pvt. Ltd., Rolon Seals, Sasken Communication, Sears Holding Pvt. Ltd., Shriram Tran Fin. Comp. Ltd, Thomson Reuters, Wipro Bpo	254
2010-11	
On Campus	
A K Erectors, Accenture, Cognitel Training Services Pvt. Ltd., Neco Jayaswal Ltd. Nr Switch N Radio Services Pvt. Ltd. Shriram Transport Finance Company Ltd., Wipro, Bpo, Zeppelin Mobile Systems India Ltd.	248

5.1.10 Does the institution have a student grievance redressal cell? If yes, list (if any) the grievances reported and redressed during the last four years.

➤ **Yes**

The student grievance redressal cell is headed by the Dean Academic. It comprises of senior faculty members from each department. The list is given below

S. No	Name	Position Held
1	Dr. N. T. Khobragade , Dean Academic	Convener
2	Mr. N. H. Kela , Professor	Member

3	Mr. R. A. Mishra, Advisor & Assistant Professor, Advisor & Assistant , HoD of EEE	Member
4	Dr. K. Sridhar, Professor	Member
5	Dr. (Mrs.) Renu Tripathi, HoD of Engg physics	Member
6	Dr. (Mrs.) Y. F. Pervez, HoD of Engg Chemistry	Member
7	Mrs. Archana Tiwari, HoD of EI	Member
8	Mr. Rajesh Kumar, HoD of MEX	Member
9	Dr. Vivek Kumar Chandra, HoD of EEE	Member
10	Mr. Santosh Kumar Sharma, HoD of MECH	Member
11	Mr. Mangal Singh, HoD of ETC	Member
12	Ms. Sonika Arora, HoD of IT	Member
13	Dr. Anil Kashyap, HoD of MATHS	Member

The functions of the cell are:

1. To conduct a thorough enquiry on the complaints received from any student, including instances of ragging.
2. To submit the complete enquiry report to the Principal, with its remarks of recommendations on suitable action or punishment to be awarded.
3. On receipt of the above report, the Principal may give an opportunity to the student(s) against whom the complaint was lodged to explain his / her case. The Principal convenes the College Academic Committee meeting in this regard to decide on the measures to be taken. In special cases the opinion of the College Management Committee is also taken before imposing the punishment.

PROFORMA :

Chhatrapati Shivaji Institute of Technology, Durg Dallirajhara Road at Shivaji Nagar near Kolihapuri, Dist. Durg (C.G.) GRIEVANCE REDRESS CELL – STUDENTS	
1. Name of the Student :	
Roll No.....Branch & Year :	
Grievance:	
Date :.....	Signature of the Student
Action Taken (Proposed / Implemented)	
Signature of the Convener	
INFORMATION OF ACTION TAKEN-INTIMATION TO THE STUDENT I was informed of the action taken (Proposed).	
Date:	Signature of the Student
ACKNOWLEDGEMENT – GRIEVANCES REDRESS CELL – STUDENTS	
Received the complaint from.....,	
Roll No..... Class & Year of Study	
Date:.....	Signature of the Student

5.1.11 What are the institutional provisions for resolving issues pertaining to sexual harassment?

Women's Grievance Handling Cell is a growing community of women in CSIT who share a common purpose to learn, educate and collaborate on ideas that have the power to effect meaningful and positive change while fighting the odds they face. It is an expanding network that brings women in CSIT together to learn, laugh, and celebrate their authentic self, their unique gifts and their personal power. The college has a Women Grievance Cell which addresses the grievances of girl students regarding sexual harassment. The cell

comprises of one senior lady faculty member as chairman and one lady faculty member from each department, if available as members. The Dean Academics remains as an ex-officio observer. The composition of the cell is as follows:

S. No.	Name	Position Held
1	Mrs. Archana Tiwari	Chairman
2	Dr. Renu Tripathi	Member
3	Mrs. Sonika	Member
4	Dr. Yasmeen F. Parvez	Member
5	Ms. Priyanka Khatwani	Member
6	Mrs. Neelam Dewangan	Member

Objective

- To uphold the dignity of women at CSIT.
- To facilitate a gender-sensitive and congenial working environment at CSIT, Durg so that women, whether female employee or female student, are not subjected to gender specific discrimination.

Function

- To take action against complaints by any female employee or student as specified in objective of the cell.
- To provide information regarding counseling and support services on CSIT campus.

Activities

Interaction with women to give them motivation and information regarding law and rules which protect them from harassment.

5.1.12 Is there an anti-ragging committee? How many instances (if any) have been reported during the last four years and what action has been taken on these?

Yes, the college has Anti Ragging Committee, Anti Ragging Squad as well as other Student Counseling Committee as provisioned in UGC/AICTE regulation 2009. Constitution of the committee and the progress report is sent to the University.

Name	Designation	Responsibility
Prof. R. A. Mishra	Assoc. Professor (MECH)	Chairman: Anti-Ragging Committee
Mrs. Archana Tiwari	Prof. & Head (E&I)	Vigilance in girls Hostel and its campus

Dr. K. Sridhar	Professor (MECH)	Vigilance in Boys Hostel and its campus
Mr. Rajesh Kumar	Prof & Head (MEX)	FLYING SQUAD
Mr. Santosh Kuma Sharma	Prof & Head (MECH)	INSIDE VIGILANCE

No. of ragging cases in last four years: **Nil**

5.1.13 Enumerate the welfare schemes made available to students by the institution.

Scholarships & Freeships:

Chhattisgarh State Government Scholarships and free-ships available for the students are notified on the notice board of the institution. The teacher Guardian informs the students of the various welfare schemes. The college provides them cash rewards on the basis of their performance in the academics, sports or extracurricular activities.

1. For a paper presented in National Journal Rs. 250/- & for a paper published in International Journal a reward of Rs. 3,000/-.
2. 100% attendance in a semester a reward of Rs. 500/- and for 95% above attendance a reward of Rs. 300/- per semester.
3. Semester topper of the institute in each branch gets Rs. 3000/- cash award.
4. University rank holders for first three positions in each branch will get Rs. 10,000/- and Rs. 7,000/- for rank from 4 to 10.

Bank Services:

In collaboration with Chhattisgarh Rajya Gramin Bank, the institution assists all the students in opening an account with a zero deposit. It empowers students to perform transactions through the bank in the globalized world. It is helpful in availing educational loans. ATM facility is available in the campus.

Counseling & Placement Service:

Regular counseling centre comprises of more than 10 counselors from the faculty. The counselors reach out to the students formally and informally. The placement cell extends its service to the students in career guidance, organizes lectures concerning career planning and invites companies for campus recruitment.

Health Services:

A health centre in the institution takes care of the basic health problems of the

students. The college has made arrangements of a part time doctor and a full time nurse. The college is also having liaison with the nearby hospitals in times of any emergency.

Canteen:

The college Canteen provides hygienic food to the students at affordable rates. The College Management monitors the working of the canteen. As and when any change or reform is required, the committee immediately takes action.

Grievance Redressal Cell:

Grievance Redressal Cell actively interacts with the students to help them to sort out their grievances.

5.1.14 Does the institution have a registered Alumni Association? If 'yes', what are its activities and major contributions for institutional, academic and infrastructure development?

➤ We have Alumni Association

The CSIT Alumni Association will enable the former students to stay in touch with one another and continue their CSIT connections. The Association Staff will engage alumni with the Institute and with one another through events held at different places. CSIT organized alumni meet at Bengaluru as Samagam.

5.2 STUDENT PROGRESSION

5.2.1 Providing the percentage of students progressing to higher education or employment (for last four batches) highlights the trends observed.

Year	Student's Progression % session	No. of students						
		E & TC	MECH	CSE	E&I	IT	EEE	MEX
2013-2014	UG to PG	2	1	1	1	Nil	Nil	Nil
	Employed-Campus Selection	83	94	87	25	8	74	37
2012-2013	UG to PG	7	5	1	1	0	0	3
	Employed-Campus Selection	84	98	64	37	18	39	25

2011-2012	UG to PG	2	8	5	Nil	1	5	Nil
	Employed-Campus Selection	69	77	39	26	3	40	NA
2010-2011	UG to PG	1	11	2	11	1	13	Nil
	Employed-Campus Selection	88	46	31	37	NA	46	NA

5.2.2 Provide details of the programme wise pass percentage and completion rate for the last four years (cohort wise/batch wise as stipulated by the university)? Furnish programme-wise details in comparison with that of the previous performance of the same institution and that of the Colleges of the affiliating university within the city/district. (Batchwise Passout result)

Batch	B.E.						
	ETC	MECH	CSE	EEE	E&I	IT	MEX
2006-10	63.70	61.2	58	56.25	65	--	--
2007-11	69.71	67.39	73	65.67	65.21	--	--
2008-12	71.94	65.21	46	60.32	61.9	44.73	53.54
2009-13	63.84	74.43	48	76.19	63.07	71.79	58.73
2010-14	52.38	75.18	48.71(1shift), 29.82(2 shift)	72.88(A) 62.50(B)	73.86	50	21.31

Batch	M.Tech		
	ETC (Communication)	CSE	MECH
2010-12	83.3	87.5	85.7

5.2.3 How does the institution facilitate student progression to higher level of education and/or towards employment?

Academic Year 2013-14 (Till Date)

Placement Training classes were conducted for B.E. students by Mr. Shantanu Modak & his team from BookShelves Career Planning Centre, Bhilai during session July-Nov 2013 & Jan-Apr 2014.

Academic Year 2012-13

Training and Placement classes were conducted for B.E. students by Mr. Shantanu Modak & his team from Youngistaani Career Planning Centre, Bhilai during session July-Nov 2012 & Jan-Apr 2013.

Mock Test series was specifically done for the campus drive of HCL Technologies, Noida. And we are happy to put on record that our efforts yielded good results.

Academic Year 2011-12

Placement Training classes were conducted for B.E. students by Shantanu Modak & his team from Youngistaani Career Planning Centre, Bhilai during session July-Nov 2011 & Jan-Apr 2012.

One Week Pre-Placement Training classes for campus drive of Accenture, Bangalore were conducted by Faculties of the Institutes.

5.2.4 Enumerate the special support provided to students who are at risk of failure and drop out?

- Special Classes are arranged for the students who are at risk of failure and drop out.
- Regular Counseling conducted by the respective teacher guardian.
- Proper communication is made with parents on their performance by the PR Cell
- Classes for particular subjects, which students feel tough are organized in the form of extra class hours.

5.3 Student Participation And Activities

5.3.1. List the range of sports, games, cultural and other extracurricular activities available to students. Provide details of participation and program calendar.

(a) Sports & Games Facilities:

Sports department looks after the Games and Sports activities inside and outside the campus. The department is headed by the qualified Sports officer .

The college has different sections for indoor games and also has well developed playgrounds for Basketball, Volleyball, Shuttle Badminton, Kho-Kho , Kabaddi, Table Tennis etc.

The participation of students in Games and Sports activities are fundamentally of two types.

1. Inter college competition (University Level).
2. Intramural competition.

Major Activities:

Organizing intramural tournaments in Tennis, Basketball, Table Tennis and Cricket competitions and providing facilities for conducting University team selections for the above events.

Inter college competition:

Inter college competitions are conducted in months of February and March every year and the prizes are distributed by the university.

Intramural competition:

Events like dance, singing, instrument playing, and cricket are organized by different departments of college.

Each department has its students association which is solely run and organized by the students themselves. One faculty member of the Department is nominated as In-charge of the Association and he/she extends guidance.

The Associations normally organize events like indoor and outdoor games and cultural activities after getting due approval from the principal.

Gymnasium: Well equipped Gymnasium facility is available for Boys.

Facilities for Girl Students:

Girl students are encouraged to participate in intramural and inter college sports & games competitions and cultural activities. The following are different events organized by the college;

Indoor: Badminton, Carrom, Chess, Table Tennis, etc.

Outdoor: Track and Field events, Cricket, Football, Volleyball, Basketball, Kho-Kho etc.

- Separate selection trials are conducted for girl students for picking up talented players for participation in Inter Collegiate tournaments.
- Practice sessions are conducted in athletic events.

(b) Cultural activities:

- Cultural Events are conducted by the student associations department wise. Students take permission from their respective HODs. The faculty supervisor of the association guides the students.
- Annual function is organized by the college in the month of Feb.-Mar. and during the festival all the departments work together. A committee of HODs and senior professors is made to supervise all the activities. All faculty members coordinate with the students and advise the students for various issues they face during the preparation of the festival and preparation of cultural events.

The following tables provide the information of the Sports and cultural competitions.

Year	Names of the Events	No. of students Participated	No. of students Won prizes
2010-11	Rhythm (Song And Instrument Playing), Kabaddi, Khokho, Table Tennis, Carrom, Chess, Volleyball, Basketball, Farewell, Teacher's Day, Cricket	90	10
2011-12	Kabaddi, Chess, Rhythm (Song And Instrument Playing), Khokho, Table Tennis, Carrom, Volleyball, Basketball, Teacher's Day, Cricket, Farewell.	96	12

2012-13	Volleyball, Basketball, Rhythm (Song And Instrument Playing), Kabaddi, Table Tennis, Khokho, Carrom, Chess, Farewell, Teacher's Day, Cricket	200	15
2013-14	Relay Race, Kabaddi, Rhythm (Song And Instrument Playing), Khokho, Volleyball, Basketball, Farewell, Teacher's Day, Cricket	223	25



5.3.2 Furnish the details of major student achievements in co-curricular, extracurricular and activities at different levels: University / State / Zonal / National / International, etc. for the previous four years

Co – curricular Activities

Participation of students in paper presentation.

Year	Branch	Activity	UG Students	PG Students
2010-2011	ETC	Paper Presentation	78	2
	ME		2	-
	CSE		19	-
	EEE		1	Na
	IT		Na	Na
	MEX		Na	Na
	E&I		4	Na
2011-2012	ETC	Paper Presentation	30	28
	ME		0	0
	CSE		4	7
	EEE		0	Na
	IT		1	Na
	MEX		1	Na
	E&I		2	Na
2012-2013	ETC	Paper Presentation	8	47
	ME		0	6
	CSE		3	7
	EEE		2	Na
	IT		0	Na
	MEX		0	Na
	E&I		1	Na
2013-2014	ETC	Paper Presentation	3	81
	ME		0	17
	CSE		2	32
	EEE		0	Na
	IT		0	Na
	MEX		1	Na
	E&I		0	Na

Extra Curricular Activities

Student Participation in intercollegiate tournaments.

Year	Event	No of students participated	Prizes won
2010-11	Chess	3	Winner
	Football	16	Winner
2011-12	Chess	2	Winner
	Football	16	Runner
	Athletics	9	Medal
	Volley-Ball	12	Runner
	Basket- Ball	12	Runner
2012-13	Football	16	Runner
	Volley –Ball	12	Runner
	Table Tennis	4	Winner
	Athletics	9	07 Medals
2013-14	Volley Ball	12	Runner
	Foot Ball	16	Runner
	Table Tennis	4	Winner
	Kho-kho	12	Winner
	Kabbaddi	12	Winner
	Basketball (Girls)	12	Winner
	Basketball (Boys)	12	Runner
	Athletics`	7	Medal
	HandBall (Boys)	12	Runner
	Hand Ball (Girls)	12	Winner

5.3.3 How does the college seek and use data and feedback from i) Ex-students and ii) employers is expected to improve the performance and quality of the institutional provisions?

- Exit feedback is taken from outgoing students of all branches.
- Their suggestions are discussed in the meeting of HODs by the Principal and suggestions about enrichment of library are promptly taken care of.
- Suggestions about changes or revision of syllabus of some courses need more time to attend to.
- Majority of the students suggests making efforts for better placement. Training & Placement office of the Institute does its best to improve upon.
- TPO communicates with the Alumni. Some of them do respond and convey their feelings about CSIT. They also make some suggestions for strengthening CSIT further.
- TPO normally makes telephonic talk with some employers. But we do not have compiled data in this respect.

Format of Exit Feedback will be shown at the time of visit of Peer Team.

5.3.4 How does the college involve and encourage students to publish materials like catalogues, wall magazines, college magazine, and other material? List the publications / materials brought out by the students during the previous four academic sessions.

- The college publishes **SANCHARIKA** newsletter every month. It provides information with regards to the professional achievements of students and faculties. Other activities of the departments and opportunities to students and faculty in science & technology are also included in the publication.
- Various departments organize National level paper presentation and Poster contests like **TECHPATRA** (Electronics and Telecommunication Dept.), **FUTECH** (Mechanical Dept.) and **EUREKA** (Electronics and Instrumentation Dept.) which motivate students to prepare technical papers.
- National Level TechFest **LAQSHYA** is organized in which college encourages students to display their models, robots, technical skills through various technical events.

5.3.5 Does the college have a Student Council or any similar body? Give details on its selection, constitution, activities and funding.

➤ **Yes**

The following are the Student Associations of the different departments of the college.

1. ME : **Mechanical Engineering Association (MEA)**
 2. ECE : **Council of Motivated Electronic Talents (COMET)**
 3. CSE : **Association of Computer Engineers (ACE)**
 4. EEE : **Students Association of Electrical & Electronics Engineering (SAEEE)**
 5. EI : **Electronics League of Instrumentation Techno Engineer (ELITE)**
 6. IT : **Building Route of Information Technology (BRITE)**
 7. MEX : **Futuristic Association of Mechatronics Engineers (FAME)**
- The committees of these associations comprise of President from final year, Vice-President from final year, Secretary from pre- final year and

Treasurer from pre-final year. The associations conduct the activities under the guidance of the faculty representatives.

- The departmental association registration fee is Rs.200 for all newly admitted students in their tenure of four years of their study. The college also makes budgetary allocations for student activities.
- In addition to the departmental associations, the student chapters of ISTE, IE(I) also exist. Rs.150/- per student is collected by the departmental associations towards professional membership fee from all the newly admitted students and every student has the choice to become the member of any of the above chapters.

5.3.6 Give details of various academic and administrative bodies that have student representatives on them.

To develop managerial skills in our students, they are involved in different academic and administrative activities; students are nominated as members of various committees.

Academic Bodies

- Career Guidance Cell
- Library Committee

Administrative Bodies

- Hostel Committee
- Robotics Cell
- IIPC and EDC cell
- Members of Teachers day, Engineers day, Technical Fest, Annual Sports and National festival celebration committees.
- Student chapters of ISTE, IE(I).

Student members involved give their opinion on various issues and present their ideas at the respective committee meetings. Their suggestions are given due importance in arriving at decisions. The involvement of students in these committees creates better understanding between the administration and students.

5.3.7 How does the institution network and collaborate with the Alumni and Former faculty of the Institution.

The college maintains the record of the alumni to keep contacts with them and understand the latest demand of industries. The alumni meets are organized regularly at the place suitable for alumni. The departments seek the opinion of

alumni on various developmental activities and their current working area by sending mails to them and requesting them to fill the feedback-form supplied in this regard. Some of them do respond. The institute honors some of their suggestions.



The Format of Feedback form is attached as Annexure - I

Criterion – VI

Sports @ CSIT



Indoor Games



Indoor Games



Outdoor Games



Outdoor Games



Outdoor Games



Outdoor Games

CRITERION – VI

GOVERNANCE, LEADERSHIP AND MANAGEMENT

6.1 Institutional Vision and Leadership

6.1.1 State the vision and mission of the Institution and enumerate on how the mission statement defines the institution's distinctive characteristics in terms of addressing the needs of the society, the students it seeks to serve, institution's traditions and value orientations, vision for the future, etc.?

VISION and MISSION statements are given in Criterion 1.1.1.

Distinctive Characteristics of the College:

- Management is always ready to provide best possible infrastructural facilities and learning resources for faculty, students and research scholars to evolve into a centre of academic and research excellence in the country.
- Motivating working conditions for staff and healthy learning environment for students so that they are prepared for their chosen careers in a global society.
- 3rd in the list of the top emerging colleges of excellence, as per the survey of GHRDC-CSR ranking.
- CSIT is ranked 98th among best private institutes as per OUTLOOK survey.
- Under Corporate Social Responsibility some teachers teach students from rural areas, creating awareness/ guidance to school students regarding engineering and its branches.
- Organization of Techfest (Laqshya) and Foundation Day (6 Sept.), in addition to several extracurricular/co-curricular activities are organized by students under the banner of Departmental Associations.
- Digital library with NPTEL and e- journals is available.
- The Institute is having institutional membership of ISTE, IETE, ISOI, IIIE, NAFEM, and CSI.

Vision for the Future: (Long Term Objectives)

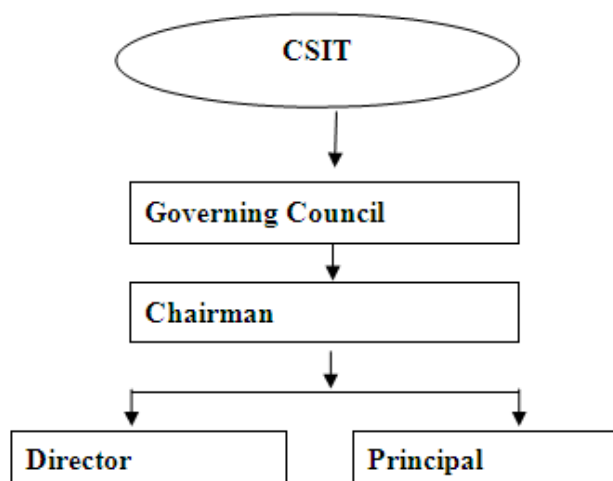
- To maintain the quality of teaching – learning process, developing adequate infrastructure, improving grooming process and increasing the placement of students beyond 75% so as to remain sustainable for accreditation by Agencies Like NAAC, NBA and IEL.
- To become centre of research in the area of ROBOTICS.
- To start P.G. programs in Power System Engg. and Control & Instrumentation.

- To increase the production of solar dryers which are useful to the farmers. At present we have customized production.
- To increase our activity of imparting computer literacy to the house-wives in neighboring villages. At present our focus is on Durg city only.

6.1.2. What is the role of top management, Principal and Faculty in design and regarding implementation of its quality policy and plans?

- CSIT was started by Shivnath Shikshan Seva Samiti in the year 1999 with honest approach of providing value based Technical Education to produce globally acceptable Engineering Graduates. Accordingly it started marching ahead and has completed 15 years of its existence with excellence. It has a Governing Body with Mr. Ajay Prakash Verma as a Chairman who is Mech. Engg. Graduate.
- Governing Body is the Policy Making Body.
- Governing Body ensures excellent infrastructural facilities and conducive learning ambience to the faculty and students.
- It has formed fifteen committees, which shoulder the responsibility of achieving the goal of effective teaching- learning process and other related aspects of the Educational System.
- Leadership given by the Chairman is motivating in nature, establishes harmony in working environment and encourages all to give their best to the noble cause of nourishing the budding Engineering Graduates.
- CSIT follows the QMS (Quality Management System) ISO 9001:2008. QMS has prescribed various measures for different activities of the individual departments and the institute.
- CSIT takes care of execution of the plans with the help of different committees.

Administrative Set-up:



- The responsibility flows from Management to the Director and the Principal.
- The Principal monitors overall development through the Dean, HODs and staff members.
- Frequent meetings are conducted with all HODs by the Principal.
- Minutes of the meetings are conveyed to the Chairman.
- Finally, the Management intimates its acceptance through the Chairman.

Responsibility of the College Academic Committee:

- Academic and other related activities are planned prior to the commencement of the semester. We call it Academic Calendar. It is intimated to all HODs after discussion.
- All departments execute the activities as per the Academic Calendar.

Features of Academic Calendar and approach of CSIT towards imparting education:

- Date –wise planning of activities during a semester is done. Following activities are integral part of our Academic Calendar:
 - i. Class-room instructions.
 - ii. Laboratory classes which help in understanding the application of basic principles.
 - iii. Assignments which help in developing the habit of analyzing the problems and finding their solution.
 - iv. Course- ware :
 - a. Preparation of course file is mandatory for every teacher for the subjects he/she is teaching.
 - b. In addition, text books, important sites, NPTEL series are recommended.
 - v. Independent study culture: Minor projects are to be done by all the students. This is over and above the syllabus. This helps students to become familiar with technology and its application.
 - vi. Information technology Training :

This makes the students proficient in computer application. They become familiar with latest software and I. T. Solutions.
 - vii. Soft Skill Training
It includes training for personality development, smooth communication skill, proper inter- personal relations, team building, leadership traits. These ingredients finally result into all round personality which helps in ‘Career Building.’
 - viii. Industry – Institute – Interface

Industrial training is beneficial for the students to understand the complexities of producing the goods without damaging the environment

ix. Continuous Evaluation:

To give due weightage, as prescribed by CSVTU, class tests are regularly conducted in planned way. This compels majority of students to be punctual and regular in classes.

x. Co-curricular Activities

Participation in various competitions such as Debate, Elocution, Extempore Speech, Essay competition, Dancing, Singing, Art and Crafts etc are mentally exhilarating. These activities encourage students to look beyond books.

xi. Extra-Curricular Activities

- Other than degree, everyone should have something extra with his/her personality.
 - Participation in extra-curricular activities gives added dimension to one's personality.
 - CSIT and the state of C. G. offer ample opportunities to the students in this aspect.
 - Faculty members of CSIT are intimately involved in executing the activities mentioned above.
 - Close monitoring of all activities is done by Director, Principal, Dean and HODs.
 - A teacher is assigned a group of 20 students in a class. He is mentor/guide to that group of students; he looks after their academic and personal activities and conducts meetings from time to time.

xii. Feedback collection and its analysis

- CSIT has Alumni Association.
- Alumni meet is conducted every year.
- Our Chairman and TPO conduct such meets at places where most of the CSIT pass-outs are available.
- Standardized feed-back form is mailed to our pass-outs
- Feedback is received through e-mail or through direct interaction.
- Employers convey their impressions.
- Parents express their opinions through parents' feedback.
- Every student provides feedback twice a semester through online feedback system.

6.1.3 What is the involvement of the leadership in ensuring:

- **The Policy statement:**

- To provide opportunities to the budding engineering graduates to evolve their core-competencies so that they can build up their career as world class professionals.
- The pass-out of CSIT should have adequate domain knowledge.
- Should be persons with good communication skills.
- Should be willing to work in a team.
- Should be promptly employable, and
- Should have willingness for lifelong learning as professionals.

- **Formulation of Action Plan:**

- Curriculum is prescribed by CSVTU.
- Based on the expectations of CSVTU, CSIT makes proper planning to cater to the needs of students so that they are intellectually nourished.
- Academic calendar is designed accordingly.
- Feedback received from Alumni, suggestions made by parents and the ideas received from well-wishers who are working in IITs, NITs and other well known institutes are used while preparing the policy.
- Different committees are constituted. These committees are vigilant throughout the Academic Year and ensure appropriate functioning of the college.
- The Chairman, Director, Principal and the Dean periodically interact with the faculty, students, parents of the students, industrialists and renowned professors. Information gathered through interaction helps us in implementing our policy and effectively running the college.
- CSIT is one of the most disciplined Institutes and is well known for excellent academic culture.
- Apart from coverage of prescribed syllabus, extra efforts are put by the faculty for preparing the students for campus interviews and other competitive examinations.
- The role of Director and Principal is to take care of the Administrative and Academic activities of the institute.
- The Chairman extends his helping hand as and when needed.

Interaction with stakeholders:

- **Students :**

- Regular interaction by the teacher guardian, HOD and Principal.
- Chairman interacts with each class once in a month.

- **Parents:**

Parents meet is organized once in a semester. Suggestions of parents are invited. Suggestions are discussed by the Principal with the HODs. Some

suggestions are accepted for implementation. e. g. Some parents expect frequent feedback from the HOD about the progress of their ward. HOD collects the progress report from TG. It is conveyed to the parents. The parents are happy and satisfied.

- **Employers :**

Representatives of various companies who come for conduction of campus interview, give some suggestions. We accept their suggestions and reform ourselves accordingly.

- **Alumni :**

Suggestions received from Alumni are bit more specific to the organization where they are working. This helps us in preparing the students to face industry- specific interviews.

- **Proper support:**

- i. HODs and Dean review the progress. Analysis of Class-Test Result is presented before the Chairman, Director and Principal. Suggestions for improvement of the score of the students are accepted.
- ii. Being a small town, consultancy in Durg is on a very small scale, but Dept. of Mech. Engg. gets reasonable scope and handles some projects.

Research:

- **Reinforcing the culture of excellence:**

Three Departments are having Post Graduate Courses. Quantum of research work is reasonable in these three departments. They are having their fixed targets. Hence good number of research papers is published every year.

- **Champion organization change:**

- i. Changes in policy are smoothly accepted and implemented.
- ii. Placement of students, publication of research papers by faculties, participation of teachers & students in conferences and workshops are the indicators of institute's progress.

6.1.4 What are the procedures adopted by the institution to monitor and evaluate policies and plans of the institution for effective implementation and improvement from time to time?

- HODs conduct the weekly departmental meetings.
- HODs allocate the responsibilities to the teaching staff and duties to the lab staff.

- Teacher guardians take care of a group of students. One teacher looks after a group of 20 students.
- Every department has students' association. Activities such as co-curricular, extracurricular and extension are carried out by the Associations under guidance of teachers.
- Review meetings are conducted by the Principal.
- The Chairman interacts separately with the HODs and Teaching staff once in a month.
- Interaction of the Chairman with the Dean, Principal and Director is more frequent. Hence any changes suggested by the Chairman are implemented without any time lag.
- Number of theory classes engaged by the teacher and the lab work completed are the indicator of proper academic culture. We monitor this.
- Attendance of students, their performance in class-tests, and punctuality in labs are the parameters used for internal assessment. In case of those who perform poorly we take corrective action with the help of their parents.
- Teachers are sponsored for attending FDPs, Training Programs, Workshops and conferences.
- Students are also sponsored for paper presentation.
- Some of the projects of VIII Sem. students are fully financed by the Management.

All the above actions are helpful for effective implementation of policy and improvement in performance.

6.1.5 Give details of the academic leadership provided to the faculty by the top management?

- The HR policy includes motivation to the faculty for qualification enhancement in the form of study leave, book grant, financial assistance.
- Academic freedom is given to faculty.
- Faculty is encouraged and sponsored for improvement of qualifications.
- Incentives are given to those who have drive and take initiative.
- Overall ambience of college is quite pleasant since there are no instances of favoritism.
- Research work is encouraged and appreciated.
- No compromise is made while investing for Research labs, may be in terms of purchase of equipment or purchase of latest softwares.
- Annual evaluation of faculty is communicated which helps in improvement in deficient areas.

6.1.6 How does the college form leadership at various levels?

- Experienced and qualified people have been appointed as Director, Principal, Dean and Professors. They impart direct and indirect training to other colleagues. Relentless working of the seniors for the betterment of Institute is observed by the juniors. Juniors observe this and try to emulate. This has helped us in creating the second line.
- Our retentivity is reasonably good. Some of our staff members had joined in 1999-2000, 2000-2001 and are still continuing with us. By virtue of their seniority, they are now elevated to the posts of Professors though they had begun their career as Lecturers.

6.1.7 How does the college delegate authority and provide operational autonomy to the departments/units of the institution and work towards decentralized governance system.

The institute believes in the principle of mutual trust. Accordingly it creates an ambience by adopting:

- Participation by all faculty members in developmental issues.
- Maintaining transparency.
- Empowering HODs to formulate the planning, and giving them the freedom to execute the schemes they have planned for the development of the institute.
- Financial requirements of HODs are sanctioned by the Finance Committee of the college.
- P G coordinator monitors day to day activity of P G course in consultation with departmental HOD and Dean academic.
- First year coordinator along with HODs of applied science and humanity look after monthly conduction of classes and labs of first year.

6.1.8 Does the college promote a culture of participative management? If, 'yes', indicate the levels of participative management.

➤ Yes

- Weekly Departmental Meetings are organized by respective HODs.
- Course coverage, Lab work, Add-on courses, Industrial visits etc. are discussed. Deficient Areas are then properly taken care of.
- Minutes of meeting are conveyed to the Chairman, Director, Principal and Dean Academics. Suggestions received from them are accepted and implemented.
- Quarterly meeting of Principal along with all HODs is conducted to discuss academic and administrative activities. Minutes of meeting are

conveyed to Chairman, Director and Dean Academics. Suggestions received from them are accepted and implemented.

- Meetings of Institute Advisory Committee (IAC) and College Academic Committee (CAC) are conducted once in a semester.
- Chairman of the college along with Director, Principal, Dean Academics and all HODs conduct meeting once in a semester to discuss policy, planning, academic progress, student related activities, organization of different events. These matters are further discussed in supreme body i.e. governing body. Governing body meets twice in a year to discuss various issues and aspects related to the development of college.
- Governing body considers the recommendations of Institute Advisory Committee (IAC) and College Academic Committee (CAC) and it chalks out a roadmap to achieve the goals of the institute.

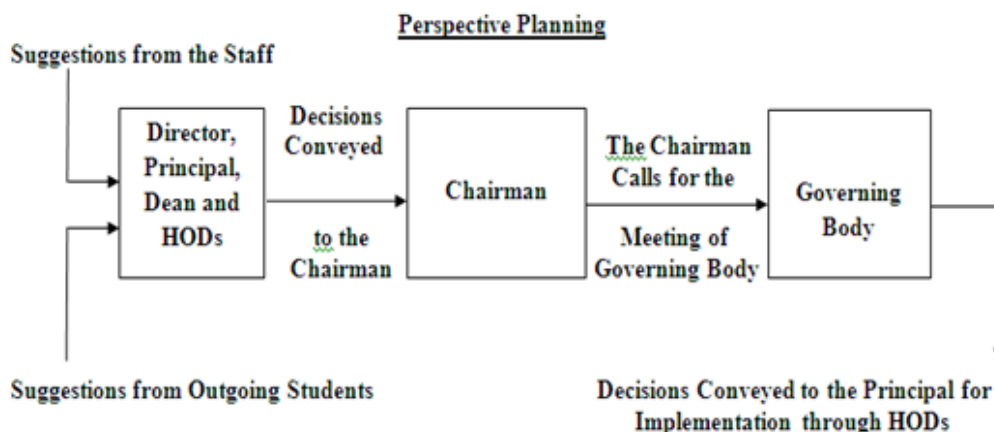
6.2 Strategy Development and Deployment

6.2.1 Does the Institution have a formally stated quality policy? How is it developed, driven, deployed and reviewed?

Yes, College has formally stated quality policy which has been formulated on the basis of the Vision and Mission of the college and is the guiding force that helps departments to plan their activities.

6.2.2 Does the Institute have a perspective plan for development? If so, give the aspects considered for inclusion in the plan.

➤ Yes.



Implementation through HODs

Considering the role of the institute in imparting education and training to the budding engineering graduates a perspective plan is prepared before the start of the Academic Year.

- The Principal conducts a meeting with the HODs. They discuss various issues related with the requirement of the institute for next year and come out with certain decisions.

Useful suggestions made by the outgoing students are taken into consideration at the time of arriving at the decisions.

- The decisions so taken are conveyed to the Chairman of the Governing Body.
- The Chairman of the Governing Body organizes the meeting of the Governing Body, normally in the month of June every year.
- The members of the Governing Body do have their own suggestions.
- Suggestions conveyed to the Chairman and suggestions of the members of the council form the basis of discussion by the council.
- The Governing Body through its deliberations takes decisions in respect of:
 - Expansion of infrastructural facilities,
 - Addition of equipment in various departments,
 - Setting up of new laboratories,
 - Facilities for students and staff,
 - Recruitment of teaching staff depending upon the vacancies,
 - Adequate enrichment of library,
 - Suggestions to establish liaison with some industries.
- The minutes of the meeting are conveyed to the Principal for implementation.

6.2.3 Describe the internal organizational structure and decision making processes.

- Organizational structure is depicted in a separate page

Functions carried out by the officials are summarized in brief.

- Requirements for equipments and consumables are submitted by the HODs to the Principal.
- Purchase Committee takes decision to fulfill the requirements and the purchase-process is carried out. Needed equipment and consumables are thus procured.

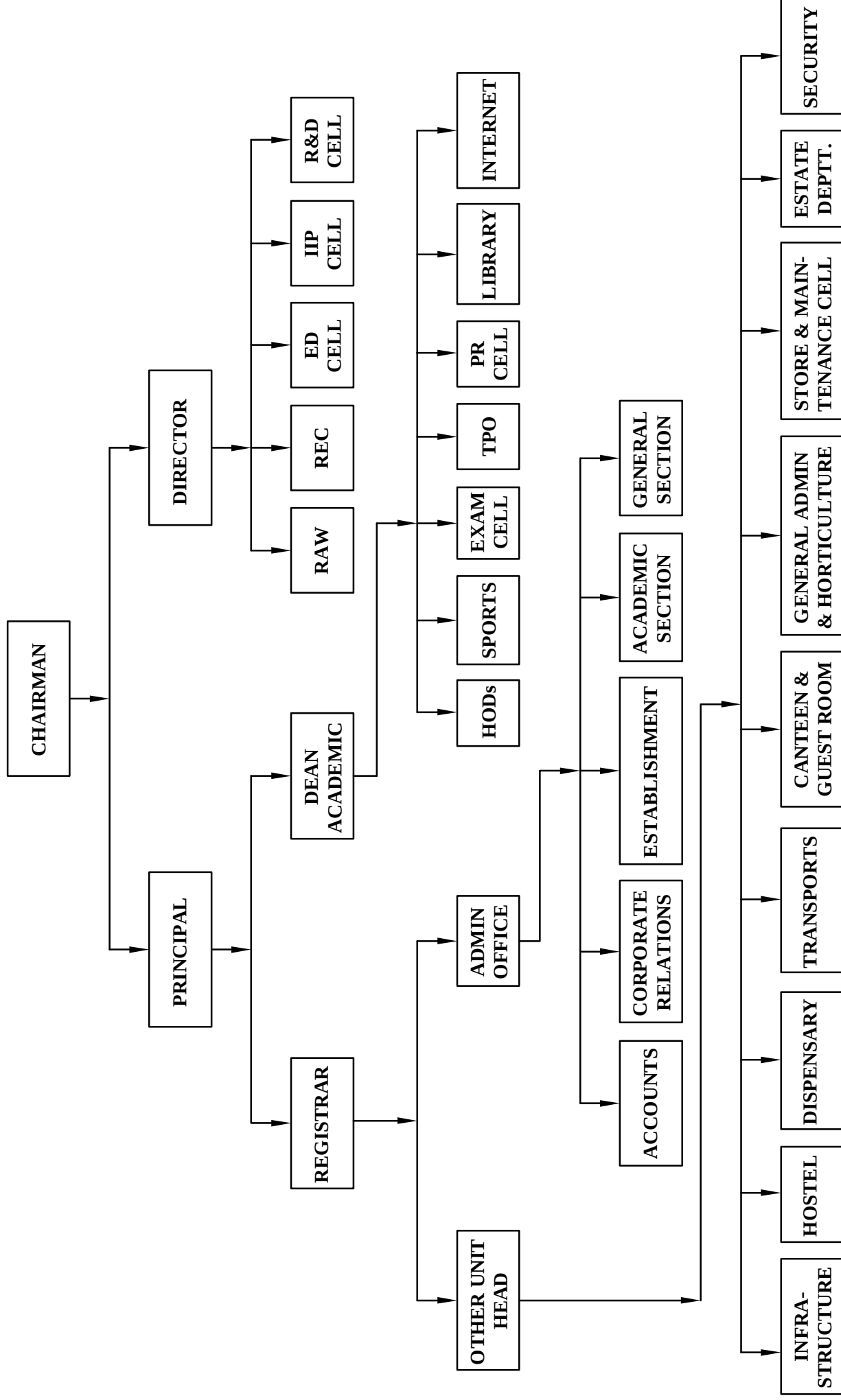


Creating Difference
Since 1999

Chhatrapati Shivaji Institute of Technology

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ORGANIZATION CHART



- First year activities are supervised by a senior staff member.
- All UG Academic and other related activities are taken care of by the respective HODs with the assistance of fellow colleagues.
- R & D activities are supervised by the Director.
- P. G. activities are monitored by the Dean Academics and P. G. Co-ordinators.
- Entire Administrative work is handled by the Principal with the help of Registrar and other office-staff.
- Training and Placement Dept. remains active throughout the year and organizes interviews for placement. Normally 25 to 30 companies visit our campus. T.P.O. and Corporate Relation Manager look after this activity.
- There is a separate Personal Relation Cell for carrying out the Recruitment Process of teaching and non-teaching staff.
- IIP – Cell shoulders the responsibility of organizing industrial visits, to derive the benefits of current practices of various industries.
- Guest Lectures are normally organized by the respective HODs. Occasionally, our Chairman makes some suggestions to invite some Leader of industry to address the students.
- Annual Events like AICON, LAQSHYA etc. are organized by forming need-based committees.
- Computer centers /Labs are properly maintained by a separate Maintenance unit. Purchase of software and timely up gradation is done by this unit.
- Every Deptt. has a Departmental Library. One or two teaching staff members are the custodians of it.
- Senior Librarian takes care of the Central library. A special advisory committee makes suggestions to enrich the Library.
- Maintenance of Laboratories is assigned to the Lab technicians and they discharge their duties under the supervision of the teaching staff who is I/C of the lab.
- Maintenance of premises is outsourced. Supervision is Institutional.

6.2.4 Give a broad description of the quality improvement strategies of the institution for each of the following

- **Teaching & Learning**
- **Research & Development**
- **Community engagement**
- **Human resource management**
- **Industry interaction**

Teaching & Learning:

- Provision of State-of-the art learning resources in Central Library & Departmental libraries.
- Development of student support material.
- Arrangement of compensatory classes for students having low attendance.
- Delivery of Add-on-courses.
- Conduction of pre-placement training classes and campus connect programs.
- Structured course files and lab manuals on all courses.
- Continuous improvement of study resources.
- Provision of lectures in ppts and revision of every unit through Power Point Presentations.

Research & Development

- Formulation of Research Committee
- Allocation of budget for in house R & D
- Incentives and rewards for publications/ research
- Financial assistance to students and staff for developing a prototype or model.

Community Engagement

- Conduction of various career guidance programs for the 12th standard students.
- Blood Donation Camps
- E-Tap (Technical Education Awareness Program) is organized to create awareness among 12th standard science students about various engineering branches.
- Plantation.
- Spreading awareness among people regarding safety measures while driving.
- Financial help to Old Age Home as a part of responsibility towards society.
- Financial help to Orphanage. Students visit Orphanage and perform various activities and distribute gifts every year on 15th August.
- Organizing programs to create awareness among women of rural area about self employment.
- Maintenance of exterior premises of Vriddhashram.
- Providing classes to school students of weaker section of society throughout the year by faculty member.

Human Resource Management

- Transparent policy document.
- Transparent and scientific way of selections.
- Imparting training to the staff & faculty.
- Formulation and communication of policies of the college.
- Support for academic advancement.
- Systematic performance appraisal system and guidance to the people concerned.

Industry Interaction

- Industrial visits are organized to make students aware of various industrial processes.
- Delegates from various industries and organizations are invited to deliver Guest lectures.
- The Industry Institute Partnership Cell (IIPC) promotes interaction of industry and various departments of the institute.
- Entrepreneurship Development Cell (EDC) aims to enable our students to generate income through business activities and empower them by skill development training.

6.2.5 How does the Head of the Institution ensure that adequate information (from feedback and personal contacts etc.) is available for the top management and the stakeholders, to review the activities of the institution?

- The Head of Institution and Chairman of Management Committee get the feedback from parents, alumni, industry, teachers, students and the public with regards to the teaching quality, curriculum, extra-curricular activities and infrastructural demands.
- The students regularly interact with the teacher guardians. HODs and the Principal do discuss various aspects of improvements.
- Chairman interacts with the students of each class once a month for solving their problems not only related with academics but also to groom their personality by taking their personal interviews.
- The various representatives of companies who come for the college placements also provide their suggestions. We accept their views and reform accordingly.
- The training and placement cell keeps record of the students who would be appearing for placement interviews by various companies and interact with them and to prepare them for the placements according to the expectations of Employers.

6.2.6 How does the management encourage and support involvement of the staff in improving the effectiveness and efficiency of the Institutional processes?

Internal Quality Assurance System (IQAS)

- The Training and Placement Cell keeps record of the students who would be appearing for placement interviews of various companies and interacts with them to help them prepare for the placements according to the expectations of Employers.
- Chairman conducts meeting with the staff members. Listens to the suggestions made by the staff.
- Suggestions of the staff members are discussed in the meeting with the HODs.
- Important, Progressive and quality boosting suggestions are accepted for implementation.
- Organization of Annual Events like (i) LAQSHYA and (ii) AICON are the glaring examples of the suggestions given by the students and the staff members.

6.2.7 Enumerate the resolutions made by the Management Council in the last year and the status of implementation of such resolutions.

(a) Governing Body Meeting Resolutions and Actions Taken for the last year

Governing Body

Governing Body of the Institute meets twice in a year to discuss various issues and aspects related to the development of the Institute. It includes the institute's strategic plan which sets the academic aims and objectives of the institution and identifies the financial, physical and staffing needs. Governing body considers the recommendations of Institute Advisory Committee (IAC) & College Academic Committee (CAC) and it chalks out a roadmap to achieve the goals of the institute.

Members of Governing Body:

S. No.	Name & Designation	Institution	Position in Governing Body
1	Mr. Ajay Prakash Verma	CSIT	Chairman
2	Mr. D. K. Verma	CSIT	Member
3	Mrs. S. Deshmukh	CSIT	Member
4	Mrs. N. Verma	CSIT	Member
5	Member		Member, RO, AICTE, Bhopal (Ex-officio)

6	Dr. J. K. Nandi		Member, nominee of CSVTU, Bhilai.
7	Dr. M.L. Jeswani		Member, nominee of CSVTU, Bhilai
8	Member		Member, nominee of DTE, Raipur
9	One Industrialist / Technologist / Education list		Nominated by Govt. of C.G
10	Dr. Anurag Verma, Director	CSIT	Member
11	Dr. Mahesh B. Parappagoudar, Principal	CSIT	Member- Secretary (Ex-Officio)
12	Mr. Mangal Singh, Prof. ETC	CSIT	Member

Sample Minutes of Governing Body held.

- The college should initiate steps required for getting renewal and add two new branches for the NBA Accreditation.
- Teachers should be sponsored for Training in advanced courses that are being conducted by NIT's /IITs/Universities.
- Steps should be taken to start student chapters of the professional bodies like: IETE, IEEE, IE etc., and subscribe reputed technical journals.
- Procedures must be evolved to identify good teachers who may be rewarded suitably. This may be extended to Non-teaching staff also for their contribution to the college.
- The college should plan to undertake "Community service" to the neighboring villages by training artisans in trades like welding, plumbing, house wiring etc.,
- Analysis of the results of students admitted under different categories like social group (caste-wise) and Rural/ Urban may be taken up.

Institute Advisory Committee

Members of the committee:

S. No.	Name and Designation	Organization	Designation in the Committee
1	Mr. Ajay Prakash Verma	CSIT	Chairman
2	Dr. Anurag Verma	CSIT	Member
3	Dr. Mahesh B. Parappagoudar	CSIT	Ex- Officio Secretary
4	Mr. Lokesh Verma	SIMER	Member

5	Mr. Ajay Sinha	O-space Soln.	Member
6	Mrs. Manisha Aditya	SIMER	Member
7	Mr. Mahendra Kashyap	CA	Member
8	Dr. Pramod Gupta	CIMNAMS , DURG	Member
9	Mr. Vinod Jain	Core tub, Bhilai	Member

Frequency and duration of meetings: The committee meets once in an Academic session and conducts a meeting for 1 to 2 hours.

Role of the committee members:

- The members give suggestions as to how to proceed on the “promotion of academic and industry interaction”.
- To give ”Ideas and Directions” for the development of the college.
- Initiate innovative programs so that the career opportunities of students are enhanced.
- To give advice on the desirability of taking up community development programs for the neighboring villages.
- Any other programs which the enlightened members may wish to propose.

College Academic Committee (CAC):

- College Academic Committee is formed to discuss the academic matters of the college.
- Members of the committee give their suggestions and directions for the smooth running of the college in academic aspect.

Members of the Committee:

The listing of the members of the College Academic Committee (CAC) will be shown to the Peer Team at the time of the visit.

6.2.8 Does the affiliating university make a provision for according the status of autonomy to an affiliated institution? If ‘yes’, what are the efforts made by the institution in obtaining autonomy?

➤ No, Not applicable

6.2.9 How does the Institution ensure that grievances / complaints are promptly attended to and resolved effectively? Is there a mechanism to analyze the nature of grievances for promoting better stakeholder relationship?

College constituted committees for:

- Grievance redressal
- Students counseling / mentoring
- Class monitoring
- Women empowerment
- Anti –ragging.

Women Grievance Handling Cell (for details please refer 5.1.11. page no 134)

6.2.10 During the last four years, had there been any instances of court cases filed by and against the institute? Provide details on the issues and decisions of the courts on these?

➤ Nil

6.2.11 Does the Institution have a mechanism for analyzing student feedback on institutional performance? If ‘yes’, what was the outcome and response of the institution to such an effort?

Student feedback is obtained in several ways.

- Feedback from all students is taken twice a semester from students having more than 60% attendance at the time of giving feedback.
- Regular feedback from a group of students is taken by teacher guardian, and HOD.
- From time to time Principal/Director, Chairman receive feedback from class representatives and a group of students.
- Feedback is analyzed by the Principal and his opinion based on feedback is informed to the faculty.
- Exit feedback is taken from final year students. Outcome of feedback is used to map with program objectives and institutional objectives which is helpful in policy making.
- Alumni feedback is received through alumni meet, emails from alumni and by their visit to the campus.
- All these inputs are used to improve overall performance of the institute including policy modification, students employability, training etc.
- Improvement in faculty is further monitored by HOD and Principal. Support and guidance is provided to them by senior faculties.

6.3 Faculty Empowerment Strategies

6.3.1 What are the efforts made by the institution to enhance the professional development of its teaching and non teaching staff?

To enhance the professional development of teaching and non teaching staff:

The Institute

- Organizes the Faculty Development Programs within the Departments
- Organizes the Lab. Staff Training Programs within the Departments
- Sponsors Faculty and Staff to attend Conferences and Workshops outside
- Organizes Conferences and Workshops at the Institute level
- Sponsors Faculty and Staff for higher education
- Organizes the Guest Lecture Programs for the Staff in the Institute
- Organizes Industrial Training Programs through EDC/IIPC Cell for the Faculty
- Motivates the Faculty to deliver guest lectures outside
- Provides opportunity to Staff to develop their managerial skills by organizing National Level Techno Management Fest like LAQSHYA.

6.3.2 What are the strategies adopted by the institution for faculty empowerment through training, retraining and motivating the employees for the roles and responsibility they perform?

- Orientation program is conducted for new faculty members to make them aware about working culture and procedures of the institute.
- Presentation by faculties under FDP on their research areas.
- Facilitating summer vacation courses for faculties.
- Faculties are involved in various events as Event Incharge thus guiding the students for teamwork.
- Faculty development programs are organized in college and at departmental level.
- Faculties are encouraged to participate in workshops and conferences.
- Providing opportunities and appreciation for their achievement in academic/research achievement.
- Special incentives are given to faculties for workshops and conferences.

6.3.3 Provide details on the performance appraisal system of the staff to evaluate and ensure that information on multiple activities is appropriately captured and considered for better appraisal.

Performance Appraisal of the faculty is done by the Principal and Management. Self Appraisal form is given to the faculty, which is filled by him/her. It is submitted to the Principal through HOD. The following points are considered for the Performance Appraisal,

- Academic Performance.
- Examination Results.
- Students Feedbacks.

- Research Publication.
- Books Publication.
- Faculties Commitment towards their duties.
- Achievements of faculty.
- Overall Contribution towards the Institute.

6.3.4 What is the outcome of the review of the performance appraisal reports by the management and the major decisions taken? How are they communicated to the appropriate stakeholders?

- Review of performance appraisal is done on the basis of 100 points and if the feedback is lower than 70% then the faculty is asked to improve.
- Faculties give their plan for improvement which is further monitored by Head and Principal.
- Direct interaction of Principal / Chairman with faculties having unsatisfactory feedback for suitable suggestions.
- Providing support and guidance to faculties for further improvement.

6.3.5 What are the welfare schemes available for teaching and non teaching staff? What percentage of staff have availed the benefit of such schemes in the last four years?

In addition to normal salary and yearly increments, there are provisions to take care of employees' health through Mediclaim Policy. Additionally there are many schemes available for teaching and non teaching staff, such as:

- All faculties and staff members are covered under **Mediclaim** (Insurance Policy) and **Accidental Insurance Policy**.
- All staff members whose salary is less than Rs. 15000 per month are covered by the **Employee State Insurance Corporation (ESIC)**.
- The college contributes 12% of the pay subject to the ceiling (Rs. 780/-pm) per person, towards the Employer's contribution to the **EPF Scheme**.
- **Maternity Leave** granted to a woman employee/Faculty with full pay for a period of three months from the date of its commencement. Maternity leave can be availed only for two issues.
- **Study Leave** may be granted to an employee/Faculty of the institute to enable him to undergo a professional course of study in a Professional or Technical subject connected with the sphere of his duty to the other institute on the conditions agreed upon between the institute and employee/Faculty.
- **Earned Leave (E.L)** of 10 days provided to teaching and non-teaching staff.
- **Encouragement to the Children of Employee of Institute:**

The son or daughter of Employee if admitted in CSIT for U.G. Course in Engineering, he/she will be entitled for 50% discount in tuition fee of the Institute.

Other Benefits:

- Insurance cover for Health (Rs.50, 000/- Mediclaim) and Personal Accidents (Rs.1, 00,000/-). 100% of faculty and staff are availing the benefit.
- Faculty members with five years of continuous service in the college are eligible for Gratuity as per rules.
- EPF: Rs.780/- per month. College pays its contribution of Rs.780/- month if the faculty member pays his contribution.
- Concessional Transport Facilities for Faculties and Staff.

Yearly Increments:

- An annual increment shall ordinarily be drawn as a matter of routine, unless it is withheld.

Incentives and Rewards:

The staff members are granted the following incentives and rewards based on their performance as well as merit:

- For Paper Presentation in National Conference/Seminar:
 - a. To and fro fare (from Durg to the destination station & back) by Train as follows:
 - i) 2nd AC – Director, Principal, Dean, Professor, Advisor & HOD
 - ii) 3rd AC – Associate Professor,
 - iii) 2nd SL – Sr. Assistant Professors, Assistant Professors
 - b. Boarding, Lodging and incidental charges at the destination limited to Rs. 600/- per day (for actual participation date(s) only) if the facility is not provided by the host institution. A certificate from the host institution for not providing this facility will be required for reimbursement.
 - c. Registration charges, limited to Rs. 1000/- only for conferences outside of C.G. State.
- The faculties are permitted to attend Seminar/ Conference/Workshop connected with their field (once in a semester) and are treated on duty.
- The faculties undergoing full time higher Education (M. E./M. Tech) are granted leave for entire duration and they are eligible to purchase books up to the limit of Rs. 10,000/- and this amount is reimbursed. The books will be the property of College Library and can be issued for their use.
- For bringing sponsored project from any Government Agency , a reward of 5% of the total cost of project or Rs. 1,00,000/- whichever is less.

- Incentive in the form of additional increment is granted to those teachers who exhibit outstanding performance. Following factors will be considered for this award:

- i) Students' feedback
- ii) Percentage result of the subjects taught
- iii) Performance appraisal and
- iv) Overall behavior and inter personal relations

6.3.6 What are the measures taken by the Institution for attracting and retaining eminent faculty?

- Higher start is given by the Management in case of eminent faculty.
- Rent free accommodation is provided to them.
- Cars have been given to some faculty members in recognition of their services. This has worked as a good measure for retention.

6.4 Financial Management and Resource Mobilization

6.4.1 What is the institutional mechanism to monitor effective and efficient use of available financial resources?

Financial Management

- Governing body of the institute monitors financial resources. The income is mobilized by way of, **i) Tuition Fees, ii) Hostel Fees and iii) Bus Fees.**
- Hostel fees and Bus fees are utilized for meeting the expenditure of Hostel and Transportation.
- Tuition fee is utilized for meeting the expenditure on account of teaching-learning process and development of infrastructure.
- After assessing the projected income for an academic year, college allocates budget to every department.
- College obtains proposals from various departments which include cost of lab development and other purchases. Duly constituted purchase committee grants the sanction to it.
- Library development: A separate fund is provided for library development which is monitored by library committee.
- Other support: Contingency amount is given to each deptt. for maintenance of laboratories.
- Management takes utmost care to use every Rupee properly so that it is not required to face Audit Objection.

Resource Mobilization

Resource of the Institute is = Fees paid by the students.

6.4.2 What are the institutional mechanisms for internal and external audit? When was the last audit done and what are the major audit objections? Provide the details on compliance.

Every year, internal and external audit are conducted by Chartered Accountant. The last audit was done in July 2013 and there were no objections.

6.4.3 What are the major sources of institutional receipts/funding and how is the deficit managed? Provide audited income and expenditure statement of academic and administrative activities of the previous four years and the reserve fund / corpus available with Institutions, if any.

Major Sources of Income:

Tuition Fee, Hostel Fee and Bus Fee.

Details	2013-14	2012-13	2011-12	2010-11	2009-10
Income	16,88,36,931	18,68,65,220	15,13,77,338	10,89,17,936	10,56,57,047
Academic Expenses	12,25,00,000	11,99,04,154	8,19,00,481	6,24,23,390	5,01,01,791
Administrative Expenses	4,63,36,931	6,25,19,075	4,19,49,411	2,72,28,689	2,82,44,871

6.4.4 Give details on the efforts made by the institution in securing additional funding and the utilization of the same (if any).

College applied for seminar grants and MODROBS schemes of AICTE. College received Rs.1, 30,000/- seminar grants, Faculty Development Programs Rs.1,50,000/- and Rs.93,35,000/- MODROBS grants. The college received Rs.2,67,000/- for organizing remote centre activities of IIT, Mumbai. Mechanical Engineering Department of the college is offering consultancy services and earned Rs.10,66,550/- during the last 4 years.

6.5 Internal Quality Assurance System (IQAS)

6.5.1 Internal Quality Assurance Cell (IQAC)

(a) Has the institution established an Internal Quality Assurance Cell (IQAC)? If 'yes', what is the institutional policy with regard to quality assurance and how has it contributed in institutionalizing the quality assurance processes?

NO. Instead we have College Academic Committee (CAC) and other committees which take care of the Quality Assurance.

(b) How many decisions of the IQAC have been approved by the management/ authorities for implementation and how many of them were actually implemented?

Applicable to 6.5.1 (a) and 6.5.1 (b)

The institute monitors the quality through different committees.

- Committee for maintaining discipline ensures smooth working.
- Being a Co-Ed institute, we have to make it sure that our girl students get proper ambience, so that they are not subjected to gender specific discrimination. Women Grievance Cell is vigilant about it.
- Attendance is carefully monitored. Teacher –Guardians alert the irregular students. In extreme cases parents are contacted.
- Credibility and public acceptance of the institute is judged by the final year results and placements of the students. Results depend upon proper delivery of syllabus which is ensured by conducting weekly meetings to ensure proper coverage of syllabus. Every HOD does conduct weekly meeting.
- Periodic assessment is mandatory. Students, who do not perform well in class-tests, are advised to study. Additional inputs in the form of special classes are given in such cases.
- Lectures on Personality Development, Communication Skill Development and preparing students for facing interview has become an integral part of our institutional time-table. The activity has been outsourced. A group of skilled professionals takes care of this aspect.
- Our ISO 9001:2008 certification, Accreditation by NBA to three of our courses and placement of our students to the extent of 75% are the indicator of assured quality of our Institute.

(c) Does the IQAC have external members on its committee? If so, mention any significant contribution made by them.

We do have external members in some of our committees. Some of their suggestion are accepted.

(d) How do students and alumni contribute to the effective functioning of the IQAC?

Some of the ex-students send their suggestions and share their experiences through mail. Our in-house Associations of students do impressive work in the area of discipline and organization of co-curricular and extracurricular activities.

(e) How does the IQAC communicate and engage staff from different constituents of the institution?

Minutes of the meetings are communicated to the faculty by HODs. Implementation of suggestions is carried out by the faculty.

6.5.2 Does the institution have an integrated framework for Quality assurance of the academic and administrative activities? If 'yes', give details on its operationalization.

➤ **Yes**

Director, Principal and Dean Academic along with Heads of the Departments monitor academic activities. HODs call departmental meeting to assess the syllabus coverage. Heads of different departments have a meeting with the Principal on quarterly basis to give an insight of the departmental working and progress in academic activities.

6.5.3 Does the institution provide training to its staff for effective implementation of the Quality assurance procedures? If 'yes', give details enumerating its impact.

➤ **Yes**

Faculties are motivated to attend different workshops being organized in the leading institutes such as IITs and NITs. Various Faculty Development Programs (FDP) are organized at institute level. Faculty members are motivated to refer to NPTEL lecture series to improve the quality of their lectures delivered. Various journals and reference books are available in the library which meet the requirements of the teachers and students.

6.5.4 Does the institution undertake Academic Audit or other external review of the academic provisions? If 'yes', how are the outcomes used to improve the institutional activities?

➤ **Yes**

The Institute's academic audit is conducted by College Academic Committee (CAC). The committee reviews the following responsibilities:

- Teacher Guardian scheme and its implementation.
- Association Incharges and their activities.
- Subject and Course files of faculties.
- Execution of Departmental Responsibilities.

Based on the outcome of the review, the members of the committee give their

suggestions and directions for the smooth running of the activities in the college.

6.5.5 How are the internal quality assurance mechanisms aligned with the requirements of the relevant external quality assurance agencies/regulatory authorities?

Internal Quality Assurance Mechanism:

The institute functions under the principles, directives and regulations of AICTE, Directorate of Technical Education (C.G.) and Chhattisgarh Swami Vivekanand Technical University (CSVТУ), Bhilai.

- The institute is following the norms of AICTE and has been getting year to year approval from AICTE since its inception.
- The committee appointed by the Directorate of Technical Education (C.G.) visits once in a year whenever there is proposal of the institute for variation of intake and some expansion in terms of addition of a new branch or branches.

The committee normally verifies, i) Infrastructural Facilities, ii) Human Resource, iii) Laboratories and iv) Library.

In the past we had no such deficiencies and all inspections were smooth and in favour of the institute.

- The regulatory body in our case is CSVТУ, Bhilai. The committee appointed by the Vice-Chancellor of the university pays one one-day visit in a year. The committee carries out its inspection with reference to vital parameters including the Results and Placement of students. After being satisfied the committee recommends affiliation. During the past 15 years we have been regularly getting the affiliation. Our 12 batches of UG courses have already been out for market consumption.
- We are ISO 9001:2008 certified institute. Moreover, three of our UG degree programmes are duly accredited by NBA. We are accustomed to regular maintenance of quality in all areas. Hence, we can get ready for inspection very fast. Visitors to our campus express their happiness after their excursion.
- Representatives of the Recruitment Companies normally convey their expectations in terms of cut off marks and other qualities before they come for interview. We get five to six days time to discuss their expectations with our students. Our team of people attached to "Training and Placement Office" interacts with the eligible students and prepares them for facing the interview.

6.5.6 What institutional mechanisms are in place to continuously review the teaching learning process? Give details of its structure, methodologies

of operations and outcome?

The following mechanisms are used to review the teaching learning process:

- Two feedbacks on all the subjects are taken from the students in every semester.
- The feedback is analyzed and evaluated on the scale of 100 and each teacher is provided with a copy of feedback from HODs. Further, teachers are counseled by the Head of Department (HOD), Principal and Director. They convey their expectations to those teachers whose feedback is less than 70%.
- The management also receives the feedback by interacting with Top- 20 students and Class Representatives (CRs) from each class.
- Monitoring is also done through Teacher Guardians (TGs) to assess the uniformity in syllabus coverage, and also the quality of teaching. Weekly feedback is prepared by Teacher Guardians (TGs) and forwarded to HOD.
- The Question bank of individual subject is prepared by using previous five years papers of the university. Assignments are prepared by the faculty. Students are required to complete the assignments in stipulated time period.
- Minor projects are given to individual students to enhance their knowledge in particular field.

Methodology of Operation and Outcomes

Online feedback is taken twice in a semester for evaluating the teachers performance.

This feedback is analyzed and appropriate suggestions are given to the faculty by the HODs. The feedback of the faculty is evaluated on the various teaching skills: punctuality, language, voice, utilization of time, attention to class, topic background, use of source material, systematic communication, capacity to explain, doubt clearing, student attention, assignments, teacher attitude, extra-curricular and opinion from students. HOD interacts with the Class Representative (CR's) of each class and takes the feedback on the teachers once in every month. Based on the analysis of feedback,

- i) Teachers with more than 70% of acceptance get letters of appreciation and
- ii) Teachers with less than 70% of acceptance are advised to improve upon.

6.5.7 How does the institution communicate its quality assurance policies, mechanisms and outcomes to the various internal and external stakeholders?

For communicating the Quality Assurance Policies, the institute:



- Conducts Parents – Teachers meetings.
- Uses college website.
- Achievements of students and faculties are published in the college magazine named “SANCHARIKA”.
- Excellent events and achievements are propagated through News Papers.

Criterion – VII

Energy Conservation & Environment Protection



Power Backup



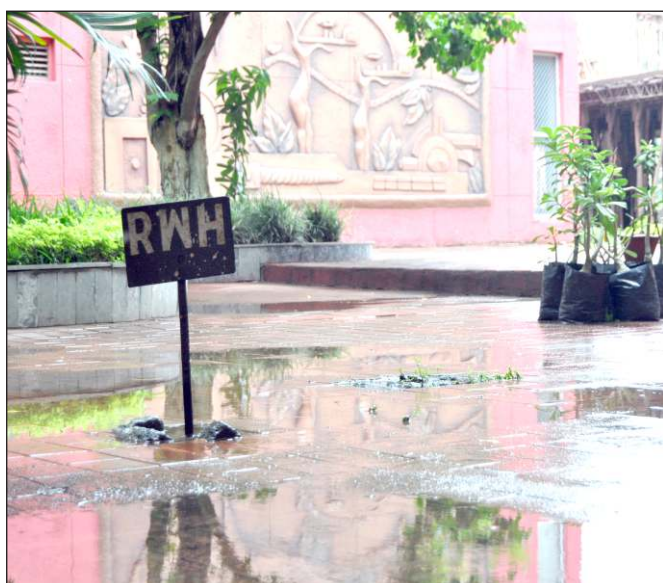
Control Room



Sewerage Treatment Plant



Solar Dryer



Rain water harvesting



Solar Panels

CRITERION – VII

INNOVATIONS AND BEST PRACTICES

The Institute attests importance to environment. It pays due attention to environmental concerns, natural resources and greenery. The Institute is very sensitive towards the use of natural resources. It looks forward, always, to imbibe eco-centric values in students and faculty members. It has Renewable Energy Cell which not only creates awareness among the students but also dedicates itself in making the society aware about environment and energy consumption through camps, workshops and demonstrations. It has about six acres of green cover. The Institute has adopted innovative measures and good practices to achieve its objective of academic excellence.

7.1 Environment Consciousness

CSIT takes keen interest in maintaining the campus highly eco-friendly and energy conservative. Our strategic plans for development have ever been driven by love for nature and its sustenance at all cost. The institution is conscious of its responsibilities to the environment and conducts Green Audit of its campus and facilities. The Institute has been appreciated for its verdure.

7.1.1 Does the Institute conduct a Green Audit of its campus and facilities?

The Institution maintains a checklist of the following in connection with Green Audit.

- Electricity Consumption in the Campus, judicious use of electrical facilities.
- Information about users (staff, students, visitors).
- Water usage and conservation strategies.
- Assessment of treated sewerage water.
- Assessment and comparison of electricity consumption in the Campus.
- Purchase of Energy Star rated gadgets.

7.1.2 What are the initiatives taken by the college to make the campus eco-friendly?

a) Energy Conservation

- The Institute has two Solar Power generation plants with the generation capacity of the 200 KiloWatt. The electricity generated through Solar Power Plants of the Institute is used and supplied to the Grid. The Solar Power Plants of the Institute contribute in reducing the consumption of fossil fuel in generating electricity.

- The Institute has a Renewable Energy Cell to make its students and society aware about judicious use of electricity and other traditional energy resources.
- All the electrical appliances are required to be switched off when not in use.
- Provision of master switches to switch off all lights/fans at the end of work.
- The UPS Batteries in all labs are maintained to reduce charging current of battery.
- All the equipments are maintained regularly in order to avoid the excess energy usage.
- All the classrooms and labs are constructed with better natural light and ventilation facility.
- To save electricity, energy conserving CFL bulbs/tubes are installed in institute campus.
- The procurement principle of the institute is to purchase energy star rated electrical appliances.

(b) Use of Renewable Energy

- The Institute has Sewerage Treatment Plant (which is second in the state) with a processing capacity of one Lakh litre water per day. The treated water is utilized daily in gardening and construction activities, thus, saving fresh water.
- Solar power utilization is available in the campus (two Solar Power Plants of generation capacity 100 KW each)
- Solar power based lighting System is available in the campus.
- Solar power heaters are installed for minimizing the usage of electricity. It provides hot water to the students in the hostel.
- Solar Power projects are done by Mechanical Engineering students.

(c) Rain Water Harvesting

- The buildings are provided with rain water harvesting system.
- The waste water is recycled and used for gardening.
- There are ten rainwater harvesting spots in the campus.

(d) Check Dam Construction

- At present there is no check dam in the Institute premises.

(e) Efforts for Carbon Neutrality.

- Specific parking area for vehicles is allotted for faculty and students.
- Use of plastic bags in the campus is discouraged and by 2015 the campus may be plastic-bag free.
- Alternative methods of garbage disposal are sought other than burning.

- CRT monitors are replaced by LCD Monitors
- Cigarettes and tobacco products are strictly banned in and around the Institute as per the Govt. Rule.

(f) Plantation/Greening Drives:

- Numerous trees and plants are there in the campus to ensure green environment.
- Only natural fertilizers and recycled water is used for plants and trees in the campus.
- Gardens are well maintained ensuring greenery.

(g) Hazardous Waste Management and e-waste Management

- Unusable e-waste like cartridge, spoiled disks, lab waste are properly disposed.
- The condemned batteries are disposed through outside agencies.
- Awareness programmes are initiated for waste management.
- The chemical waste in the Chemistry lab is disposed safely as per the procedure.

(h) e- waste Management

- Used Computer with good quality and low configuration are given to nearby schools.
- Computer peripherals that can be reused are utilized on recommendation of Networking and Hardware section.

7.2 Innovations

The institute dedicates to buildup a very conducive teaching learning and administrative atmosphere. Innovations and new ideas are very much welcomed and incorporated in the system. There is R&D Cell for technical innovations. Students at their departmental level also workout for innovative projects. The management takes care for innovative steps in administration and management through various committees.

7.2.1 Give details of Innovations introduced during last 4 Years, which created positive impact on the functioning of the college.

(a) Innovative Mechanism/Process for internal quality checks

Through the following initiatives the institution promotes an ambience of creativity and innovation in all spheres of academic progression thereby leading to quality enhancement.

Academics/ Teaching-learning

- Teaching – Learning Material, Lecture – Plan, Assignments, Question Banks and reference materials are provided to the students in the beginning of the semester.
- Course material is made available to all the students.
- Close monitoring of course coverage through frequent meeting of HOD with faculty member.
- Course Coordinators are appointed in the beginning of the semester to coordinate syllabus coverage, speed and teaching-learning assistance.
- Regular monitoring of academic activities through College Academic Committee (CAC).
- Library is fully automated. The student entry, searching of books, issuance/submission of books is done through automated OPAC library software.
- E- Journals such as IEEE, Springer are subscribed in library. The faculty members and students access the journals to improve their research performance.
- Value-added courses (Add on) are regularly conducted in all the departments to increase the employability of students.
- Teachers include Bridge–Courses alongwith prescribed syllabus in their teaching plan.
- Internet WiFi facility is also available in campus.
- Biometric attendance system is adopted.
- The institute sponsors technically competent projects of students.
- Addition of latest softwares and equipment is regularly done.
- Experienced academicians and industry experts are invited regularly for delivering guest lectures.
- Teacher Guardian: For every 20 students a separate teacher guardian is assigned who works as friend, philosopher and guide. Necessary guidance on studies and career related issues are resolved by them. Every student can share his/her personal problems, discuss failures and all other issues. A teacher guardian helps understand importance of critical discipline issues, keep record of important facts and analyze overall performance of every student.
- Emphasis is given on Research activities and publication of research papers in different conferences/seminars/journals.
- Incentives are given to those carrying out research and publishing papers in conferences and seminars.
- Periodic collection and analysis of students feedback is carried out in respect of individual subject teachers.
- Review of performance of teachers is done on the basis of feedback and pass percentage of students at University Examination.
- Remedial classes are conducted to make up for weak and slow learners.
- Awards are given to staff members who produce result of 95% and above in University Exam.

- Each student is provided a set of text books at the very beginning of the respective semester so that the students need not put economic burden on their parents. In addition to this 'Book – Bank' facility, students can also get books issued from the library.
- Seminar Halls are provided to each department for a more conducive learning and interaction between faculty and students.
- Students are motivated and sponsored to participate and present papers in National and International Conferences.
- During the first year, the students are trained in communication skills. There is a communication lab setup for this training which is open throughout the working hours with dedicated faculty to help the students.
- Students of second and third year undergo intensive training in enhancing their personality through external agency. They are put into mock interviews, GDs, and presentation on technical and general topics are organized to understand individual needs and development of students.
- Parents are apprised by the TGs and PRC about the performance of the students on regular basis.
- The library has a Reference Section with reading-room facility. This facility is available for students and faculty till 08:00 PM on weekdays.
- Interviews, GDs and presentation on technical and general topics are organized to understand academic development of students.

Administrative System

- The Institute has administrative section that works under Principal and Registrar.
- Anti-ragging Committee is constituted to take immediate action against any case of ragging. The committee also counsels students against ragging.
- Training & Placement Cell takes care of all the aspects related to placement and employability.
- Public Relation Cell (PRC) functions as a channel between institute and the stake holders.
- Institute administration is run through Management, Director, Principal, Dean, HODs and Professors. All decisions are implemented after a consensus is arrived at.
- Research Committee with Director, Principal, and Dean to view all R&D related work at departmental and institutional level.
- An exclusive Exam Section headed by Exam superintendent is constituted.
- The monitoring of the laboratories and equipments is done periodically by lab committees.
- As a part of Lab Audit, some of the practicals are conducted by the concerned teacher in the presence of experienced faculty and HODs.
- Women Grievance Cell looks after gender-sensitive congenial working environment at the institute.
- EPF facility is available for all the faculties and other staff members.

- Group Insurance Scheme is applicable for all staff.
- Lump Sum Medical soft loans as per eligibility of the staff.
- Limited Residential facility is available on campus.

(b) Quality Assurance for Academic Programs

Performance Monitoring

- Based upon the results of Class Tests (CT), Assignments, Presentation and Examination performance of every student is carefully analyzed. Special remedial classes, tutorials are arranged for poor performing students.
- Remedial instructions to the slow learners, repeat instruction sessions for low attendance learner groups are some of the practices adopted.
- Tutorial classes are introduced in the curriculum. In these practice classes, students interact with the concerned teacher to clear their doubts.
- Periodic meetings are conducted by Principal and HODs to review the result of internal and external examination.
- All the teachers prepare a teaching plan for completing the syllabus in stipulated time.
- All the teachers keep subject notes of the subjects they teach, and revise their notes during lean period.
- 'Lab Manuals' are prepared to guide the students for practical learning through experiments.

Administration

- The quality of administration is monitored through various committees like Exam Committee, Finance Committee, Cultural Committee, Women Grievance Cell.
- The HODs look after their respective departmental activities.
- The Governing Body and Governing Board take final decision on major administrative matters.
- Principal monitors teaching-learning process and lab developments.
- The Teacher Guardians guide their respective group of students in the academic activities, extracurricular, sports, cultural and project making by taking personal care. Teacher Guardians closely monitor students' performance and intimate their parents.
- The Registrar administers the supporting administrative affairs of the institute.

Student Evaluation

The performance of the students is evaluated through Class Tests and Assignments.

- In a semester three class tests are taken. The corrected scripts are shown to the students for self evaluation and correction.
- Assignments are designed for the students to encourage academic exploration. In a semester every student has to write two assignments. The checked assignments are graded and returned back.
- Practical Examinations are conducted as per the University norms. Each student is required to undergo practical examinations.
- The evaluation guidelines and procedure for all types of formative and summative assessment are made available to the teachers and students well in advance.

Automation of student Information

- For close monitoring of students attendance, the Institute has developed software which presents comprehensive picture of every individual student's attendance.
- The Institute has developed a computer based system/software that holds semester-wise academic performance of all students and helps projecting comprehensive academic achievement of every student.

(c) Quality Assurance for Administration

- The UG engineering programmes of the institute EEE, MECH, E&I have been accredited for two years in 2013 by NBA after considering the quality of the education imparted and other practices of the Institute and its infrastructure.
- The institute is ISO Certified.
- Reputed companies have been hosting their campus interviews at the Institute.
- Continuous enhancement in Placement Record.
- Improvement in faculty expertise leading to qualified faculty

(d) Innovations in Research and Development and Extension

- Research and Development Cell which comprises of Senior Professors looks after Research and Development activities.
- Faculty members are reasonably engaged in research activities.
- Faculty members are awarded for their research publications every year.
- IEEE and Springer research journals are subscribed to keep the faculty members and students updated with latest trends and technologies.
- Inter departmental research projects are encouraged.
- Departments have been equipped with computers and 24×7 internet facility.

- Financial aid is given to faculty members to attend conferences /seminars/ workshops.
- Project work is mandatory for U.G. students which helps to inculcate research aptitude among the students.
- The Institute organizes National Conference – ‘AICON’ every year to bring together scholars, academicians and industrialists.
- Research aptitude among the students is encouraged and embellished by organizing paper presentation competitions and seminars.

(e) National Events

- Every year the Institute organizes National Conference – ‘AICON’ since 2009 to provide an academic platform for academic interaction among scholars, faculties and peers.
- The Institute organizes National Techno Management Fest – Robotics Festival called as LaQshya since 2009 for engineering students. Students from across the country participate in the event.
- The Institute commemorates the days of National importance with great fervor and ensures participation of the students.

(f) Modern ICT Tools

- The Institute has procured lectures from NPTEL. Students are encouraged to listen to the video lectures of eminent professors available at NPTEL. The lectures of NPTEL are available for access at our Computer Labs.
- The e-library at the Institute contains many reputed technical Journals. The teachers guide the students to access online technical papers on engineering and technology available at our e-library.
- For effective teaching learning process LCD and other teaching devices are used.

(g) Installation of latest Configuration Computers

- Computers and devices with latest configuration are procured to replace the older ones. Majority of computers is P-IV with dual core processors.
- The CRT monitors have been replaced largely by LED/LCD monitors.
- Out dated technical gadgets are discarded or donated.

(h) Engineers’ Day Celebrations

- In memory of one of the greatest engineers Bharat Ratna Sir Vesvesvaraya Engineers’ Day is commemorated every year on 15th September since the

inception of the Institute. The programmes are coordinated by the students which help them to hone their managerial skills.

(i) Students' exposure to Industries

The Institute coordinates frequent visits of the students to different industries related to their branch so that they can observe practical applications of the knowledge they have acquired. Such visits provide more clarity on techniques.

- The Institute has established Industry Institute Partnership Cell (IIPC) that coordinates industrial visits of the students of different departments every semester to relevant industries.
- Experts and Managers from Industries are frequently invited to deliver talks to the students.

(j) Group Insurance for Staff-members

- All employees of the Institute are registered to avail benefit of C.G. state Government's insurance scheme, ESIC (Employee State Insurance Corporation).

(k) Industry Institute Interaction

- Guest lectures by eminent personalities/industrialists are arranged for the benefit of students and staff members.
- Industrial training is arranged for students during summer vacations.
- Students are encouraged to take up industry related projects.
- A Training & Placement Cell is established for providing training and assistance for students.
- The Institute keeps itself in touch with the industrial demands by coordinating interactions through TPC.

(l) Learner Centered Teaching

The primary objective of the teaching-learning process at the Institute is to impart quality education and proper knowledge to the students. A variety of teaching methods are used to achieve this objective. The Institution takes all measures to offer value based education by organizing lectures, talks, workshops and by organizing interactive sessions in selective areas.

- **Lecture Method:**

By this method teacher presents the relevant data and diagrams through PowerPoint/ Chalk board mode. Lectures are made effective by ensuring proper planning. HODs and Course Coordinators Monitor the lectures.

- **Interactive Method:**

Teaching is followed by discussion. The lectures are made interactive by encouraging the students to clear their doubts freely.

- **Project Based Learning:**

Students are asked to prepare projects with a clear concept of the principles learned. In VIII Semester students have to undertake a major technical project which is supervised by a faculty member. The students are made technically competent since Semester-III by encouraging them to take minor projects.

- **Seminar Based Learning:**

Students are asked to come prepared on an assigned topic and present a seminar before the faculty and students. The presentation of the students is assessed by senior faculty members.

- **Computer – Assisted Learning (CAL):**

Computer Assisted Learning is used to handle topics with large and complex data.

(m) Shift from Content to Competency

- The Institute lays emphasis on syllabus related and industry related practicals.
- The Institute organizes talks of expert industrialists and academicians to raise the competency level of the students.
- The curriculum involves minor and major projects compulsory for each student. Apart from this, the students at the Institute work on a small project in every Semester from III Semester onwards.

(n) Innovative Instructional Methods

Teachers use a variety of techniques to address to the individual cognitive needs. Experimental learning, inquiry and discovery problem based learning, collaborative and cooperative learning in groups is encouraged.

(o) E-learning Applications

- Classrooms are in the range of Wi-Fi communication facility of the Institute .
- Video lectures of reputed professors and online journals are available for reference.

(p) Innovative techniques in Assessments

- Assignment, Seminar, Tests are conducted for assessments.
- The competency of the students is judged from Seminar Presentations.

(q) Innovative to be Coherent and Integrated

Orientation programme, Bridge course to help students to integrate academic systems and the external world.

(r) Innovation in Infrastructural Planning:

- A new four- storeyed Air Conditioned Library with modern amenities is under construction and shall be ready by Nov. 2014.
- New class rooms with modern amenities are constructed.

(s) Innovative ‘Higher Focus Course’

- Special training programmes, Computer training, Personality development classes, Spoken English classes, Communication skill classes are organized every year.
- Add on courses are included in the curriculum in every Semester.

(t) Incentives and Reward Schemes

- Every year the University toppers are awarded with cash prizes and certificates.
- Students with hundred percent attendance are felicitated with cash prize and certificate every year by the institute.
- The students who design the best project are awarded with cash prize and certificate.
- Institute provides travel allowance and registration fees to faculty members and students for attending conferences.
- Faculty members who publish research works or undertake projects are awarded every year on Foundation Day (6th September) of the Institute.
- Students and Faculty members are awarded for their exemplary creative contribution.

(u) Innovative Faculty Development Initiatives

- The Institute organizes Faculty Development Programmes on different technical topics to upgrade the knowledge of faculty. Resource persons are acclaimed industrialists, managers and academicians.

Sponsorship to faculty members for higher education:

- M.Tech sponsorship
- Ph.D sponsorship
- The Institute provides special study leave, book grant to the faculty pursuing Ph.D program.

(v) Innovative to be socially responsible

- Student Association activities and all activities like Blood donation camp, visit to orphanages, Helmet Awareness, Meditation Yoga program, Environment activities, are organized to imbibe socially responsible attitude among the students.

Other Innovations that impacted the functioning of the College

- The Institute functions as per its quality document which covers diverse areas.
- The Institute Newsletter, “Sancharika”, brochure etc. are brought out.
- Students exit feedback has helped in understanding the needs of the students.
- Teacher’s Self Appraisal Report, Performance Appraisal of faculty, Result Analysis, Self Appraisal Report of Lab technician and the semester-wise feedback on teachers lead to improvement in teaching learning process.
- College Academic Committee (CAC) whose head is Principal, looks after the improvement in academic affairs. Meetings of CAC are conducted at least once in a semester.
- The College Management Committee (CMC) meets once in 6 months to discuss on Academic, Administrative, maintenance and development activities and communicates its decision to the CAC for follow up action.

Innovative Practices of various Departments

- In addition to the above innovative practices, the respective departments have been adopting various innovative practices which are explained in the evaluation reports.

7.3 Best Practices:

7.3.1 The Institute adopts a number of healthy practices to create a strong academic and congenial working atmosphere. Three Best Practices 'Environment Conservation Initiative', 'Faculty Motivation Sponsorship and Retention' and Personality Development Initiatives' are exemplary.

The Institute acknowledges that the impact of these practices have been very influential and clearly visible. These practices have also cast impact on teaching – learning process.

Best Practice – I

Title of Practice: "Environment Conservation Initiatives"

Goal of the Practice:

Inspire faculty, students and society to use environmental friendly technology and popularize the use of non-conventional energy sources for sustainable development.

The Context:

The Institute realizes that environment is a precious gift to mankind. Being an Institution engaged in nurturing the youth, the Institute shoulders the responsibility of making its students sensitive towards environment and use of non conventional energy sources. The Institute dedicates in creating awareness in the society about environmental conservation too. The Institute has emerged as a focal agency in creating environmental consciousness.

Practice:

The Institute has been dedicating towards environment conservation through some exemplary initiatives.

- The Institute has installed two Solar Power Generation Plants each of 100 Kilowatt capacity. The Power generated through solar energy is used in campus and the remaining is supplied to the Grid.
- The Institute has a Renewable Energy Cell that looks after energy conservation strategies and disseminating the importance of using non-conventional energy sources.
- The Institute has installed a Sewerage Treatment Plant of capacity one lakh litre per day.

- The waste water coming out of institute, hostel premises and mess is treated daily. The treated water is used in construction activities and gardening.
- The Institute has a green cover of about 6 acres. The green cover is kept well maintained throughout the year.
- The Renewable Energy Cell has so far organized 15 Mega Awareness Camps/ Workshops on use of Solar Cookers and Solar driers developed at the Institute.
- This Institute allocates a separate budget for greenery and environmental issues. The budget for green-cover of the Institute for previous fiscal year i.e. (2012-13) was about Rupees Five lakhs.
- The Institute has also manufactured Solar driers for nearby commercial units.
- The Institute has trained rural and urban women on the use of solar energy, solar driers and solar gadgets.

Best Practice- II

Title of Practice: “Personality Development Program for the Students”
(This program has been outsourced)

Goal of the Practice:

- To provide free training to the students to enhance their skills and competency, and make them ready for the selection in core companies and prospective industries.

The Context:

We are in the competitive world. Efficient planning and a different training is required by students to achieve their desired altitudes. Our Institution trains the students from 3rd semester onwards. The focus of the Institute on this training programme has uplifted the status of the students. Consequently, students performance in Interviews, GD and other recruitment tests have shown quantum improvement.

The Practice

The Institution put forth all its effort for bringing out a positive outcome from personality development programme. The Institute has hired an expert agency *BookShelf* that daily conducts classes on Personality Development, Communication Skills, Preparation for Interviews, Aptitude and Reasoning. The agency also counsels students in career planning.

The Process

Some of the noted tasks of this practice are:

- In a week two classes are devoted for Personality Development.
- Regular classes focusing on employability of the students.
- Group Discussions and seminars are arranged for students to eliminate their stage fear.
- Regular monitoring is carried out by agency.
- Regular class test is organized on General Knowledge, Logical Reasoning, Aptitude.
- The Syllabus of the training programme is specifically designed to support the development of more efficient learning strategies, to strengthen personal motivation and commitment to studies and to sustain academic excellence.
- The Institute has a separate budget of about 14 Lakhs per year for this programme.
- Under this program 837 students were personally counseled on one-to-one basis.

Outcome:

The following table provides the details of this practice

Academic Year	Class	No. of classes/ Sessions taken	No. of Students Counseled
2011-2012 (Only January Session)	2nd Yr	27 per section	264
	3rd Yr	27 per section	
2012-2013	2nd Yr	34 per section	204
	3rd Yr	39 per section	
	4th Yr	10 per section	
2013-2014	2nd Yr	29 per section	369
	3rd Yr	26 per section	

Concluding Notes:-

The Institution believes this could be very robust approach for enhancing the students technical and reasoning ability. Personality Development Programme seems to be a way to students career by ensuring their employment in reputed companies. The students have given strong positive response towards this programme so the Institute does continue this programme. The Institution has found Personality Development Programme to be a highly successful way to encourage students preparation before going for

Interview. This also supports students in developing leadership qualities and enhances their personality in all respects.

Best Practice- III

Title of the Practice: “Motivational Extra Academic Support to Academically Weak Students.”

Goal of the Practice:

The main objective of the Institution is to transform the students into well meaning citizens through the committed pattern of Institution based well designed curricular aspects. Extra classes are provided to students who are weak in some particular area.

Context:

The Institute gets a variety of students during admission. While some students who come from rural areas have minimal exposure to technology and modern teaching-learning practice, the others who come from urban background differ a lot in industry exposure and scientific ability. Some of the students do lag behind in communication skills. To address to the vivid academic needs in classroom is a difficult task. The Institute has therefore adopted this practice to bring best out of its students.

The Institute introduced the novel technique of “poor class” and “Question Bank” keeping in view the examination perspective. The students belonging to the Hostel are given extra coaching focusing on their problems and academic needs. Their skills are honed. Keeping in view the patterns of examinations they are given a list of important topics. This has helped them to attain their targets in a better way.

The Practice:

- The students were given extra coaching in all the subjects in general. The students were given extra guidance in the core subjects.
- The Institute ensured that the students are provided with the questions framed by the experienced faculty of the Institute.
- A separate time-table for the evening classes is made.
- The students of the evening classes are provided other basic facilities like breakfast and tea so as to make them feel comfortable and healthy.
- Frequent sessions on yoga, meditation and motivational talks of the peers are organized to keep them motivated.

Evidence of success

The University result of the students of First Year has drastically improved in subjects like English, Physics and Engineering Drawing. The confidence of the students was seen much enhanced towards the end of the First Semester due to this programme.

Annexure - I

5.3.7 Networking with Alumni:

CHHATRAPATI SHIVAJI INSTITUTE OF TECHNOLOGY, DURG

FORMAT III – FEEDBACK FROM ALUMNI

- A. Name : ANISH NAIR
- B. Year of Passing : 2007
- C. Branch : ELECTRTICAL AND ELECTRONICS
- D. Present Address : Korba West Power Company ltd. Chotte Bhandar, PO Badebhandar , Pussaur, Raigarh, (C.G.) 496001
- E. Email id : appunair15@gmail.com
- F. Present Occupation : Sr. Engg in Green Field Project Korba Power Company Ltd. A Subsidiary of Avantha Power Ltd.
- G. Whether undergone higher education : NO
(If yes, please give detail)_____
- H. Please provide your comments on the following:
1. College infrastructure : Good
 2. Effectiveness of Teaching Process : Average
 3. Department Resources : Good
 4. Faculties Helpfulness : Average
 5. Library Facilities : Good
 6. Computing & Internet Facilities : Good
 7. Sports, Extra Curricular activities : Sports(Bad), Good
 8. Personality, Communication Skills / Development Facilities : Average
 9. Placement Cell : Average
 10. Overall rating of the College : Good
- I. Your positive / negative comments : A great institute with omniscient teachers and developing an individual as a complete package.
- Negative : Disciple is a good, but that should not take away an individual's freedom away. A little can be managed, after all ours is an engineering college

ANISH NAIR

Annexure - II

3.6.10 Awards received by the Institution:

CONTRIBUTION

The societal contribution of Mr. Verma as the Chairman of the Institute is visible in the Corporate Social Responsibilities. He started a programme called E-Tap through which every year about six-thousand students of higher secondary classes are educated and counseled about the developments in technical education, admission procedures, and different fields in engineering. Guest and expert talks are organized regularly for the student and faculty every semester to update them on technical developments, industrial trends and corporate expectation. The free coaching classes are conducted every day in the morning hours at the campus for the poor school-going students of the rural area near CSIT. Fee concessional schemes for economically poor students of the college as well as cash awards for academic achievements of students and faculties are presented for the benefit and motivation of students and faculty alike. Awards are given on Foundation Day of the institute as well. The award money goes in lakhs and is exponentially increasing every year. LaQshya a national level mega Techfest is organized annually to encourage technical and co-curricular skills among the engineering students. It also hosts various robotic events which are very popular in the region. Computer Literacy Programme is another such event held for women and the Police department. CSIT has trained many women to use computer as a means to earn a living.

Evaluative Report of the Departments

Evaluative Report of the Department

1. Name of the Department: - Computer Science & Engineering

2. Year of establishment:- 1999

3. Names of programmes offered (UG, PG, M.Phil, Ph.D., Integrated Masters, Integrated Ph.D., D.Sc., D.Litt., etc.) : - UG & PG

4. Name of Interdisciplinary Courses and the departments/units involved:-

S. No.	Semester	Name of Course	Department
1.	1 st	Applied Mathematics I	Mathematics
2.	1 st	Professional Communication in English	Humanities
3.	1 st	Applied Chemistry	Basic Sciences
4.	1 st	Engineering Graphics	Mechanical Engineering
5.	1 st	Elements of Electrical Engineering	Electrical Engineering
6.	2 nd	Applied Mathematics II	Mathematics
7.	2 nd	Environment & Ecology	Civil Engineering
8.	2 nd	Applied Physics	Basic Sciences
9.	2 nd	Basic Civil Engineering	Civil Engineering
10.	2 nd	Fundamentals of Mechanical Engineering	Mechanical Engineering
11.	3 rd	Mathematics III	Mathematics
12.	3 rd	Digital Logic Design	Electronics & Telecommunication
13.	3 rd	Basic Electronics	Electronics & Telecommunication
14.	4 th	Computational Mathematics	Mathematics
15.	4 th	Discrete Mathematics	Mathematics

16.	5 th	Microprocessor & Interface	Electronics & Telecommunication
17.	5 th	Principle of Communication System	Electronics & Telecommunication
18.	7 th	Operational Research	Mathematics

5. Annual/Semester/Choice Based Credit System (Programme Wise):-

- Semester System

6. Participation of the department in the courses offered by other departments: -

S. No.	Name of the Course/ Program	Courses offered by other departments
1	B. E. (Computer Science & Engineering)	Electronics & Telecommunication Engineering
2		Electrical & Electronics Engineering
3		Mechatronics Engineering
4		Information Technology
5		Electronics & Instrumentation

7. Courses in collaboration with other universities, industries, foreign institutions, etc.:-

➤ Nil

8. Details of courses/programmes discontinued, if any, with reason

- Ph.d. programme discontinued
 1. Dr. HR Sharma who was approved guide retired and
 2. Mrs. Sipi Dubey the then professor in CSE who was approved guide left our institute. Since the approved guides were not available in the Dept of CSE, our Ph.D. Program was discontinued.

9. Number of teaching posts:-

Name of post	Sanctioned	Filled and Actual
Professors	2	1
Associate Professors	5	2
Asst. Professors	17	18

10. Faculty profile with name, qualification, designation, area of specialization, experience and research under guidance:-

Presently 21 faculty members are in the department. Faculty profile will be shown on the day of visit.

11. List of senior Visiting Fellows, adjunct faculty, emeritus professors:-

S. No.	Name of expert	Title of Lecture	Year of Visit
1	Mr. Jayant Chaturvedani, (Manager – Operation; Ericsson)	Telecommunication Word, Mobile Network	2014
2	Dr. K. B. Subramaniam (Retd. Principal; Regional Institute of Education, Bhopal)	Nature of Mathematics and Mathematics in Nature	2014
3	Mr. Apoorv Tiwari, (Associate Software Engineer; Accenture)	Cracking Campus Interviews	2014
4	Dr. Ashok Bhansali	Teaching Methodology	2014
5	Dr. Rabindra Ku. Jena	Green Computing	2014
6	Dr. Roheet Bhatnagar	Big Data	2014
7	Dr. Saket Ranjan Praveer(Shankara Group Of Institutions, Bhilai)	Marketing Strategies	2014
8	Mr. Ankit Fadia	Ethical Hacking	2013
9	Mr. M. S. Uday Kiran (HR Manager; HCL) & Mr. Venkat Bhukta (Vertical Head; Idea Cellular Ltd.)	HR Interviews	2013
10	Mr. Ashish Mathur, (Chief General Manager; Escorts Constructions Equipments Ltd.)	Functioning of an Organisation	2013
11	Mr. Rahul Krishna (Manager Recruitment; Espire Infolabs)	Fresher Development - a Taste of Corporate World	2013
12	Mr. Kunjal Jayant Kamdar (Head Recruiter; Quinnox)	Social Media	2013
13	Mr. Ravi Bizoara (Module Lead; Mphasis)	IT Interactive	2013
14	Mr. Abhishek Shrivastava (Assistant Manager; SAIL)	How to Clear SAIL	2013

15	Dr. M. S. Gupta(Gov. Science College, Raipur)	E M Theory	2013
16	Dr. G. K. Goswami(Kalyan P. G College, Bhilai Nagar)	Cosmology	2013
17	Dr. Saket Ranjan Praveer(Shankara Group Of Institutions, Bhilai)	Marketing Methodology	2013
18	Dr.Sudhakar Pandey(NIT,Raipur)	Research methodologies	2013
19	Dr.Keshri Verma(NIT,Raipur)	Innovations in Data mining	2013
20	Dr.Sanjay Kumar(Pt.RSSU,Raipur)	Principles of pipelining	2013
21	Mr. Navneet Dewangan (Principle Infrastructure Engineer; Mphasis Ltd)	My Struggle & Success Story	2012
22	Mr. A. Ganesh Ray (Executive Director; Techlead Solutions Pvt. Ltd.)	Faculty Development Program in MBD	2012
23	Mr. Pankaj P. Kulkarni (Engineer – Structures; Innovent Engg. Solutions Pvt. Ltd.)	ANSYS Mechanical	2012
24	Mr. K. Patel (Advisor – HRD; Jayaswals Neco Industries Ltd.)	Man in the Mirror	2012
25	Mr. Nagaraj Parappagoudar (Solutions Architect; CISCO Systems, Sanjose, CA, USA)	Emerging Technology Areas and Trends in Computer Industry	2012
26	Mr. Anirudh Girey (Sr. Software Engineer; IBM India Limited)	Job Orientation, Career Guidance	2012
27	Dr. Bharti Trivedi (Director; Apex Technology)	Vacuum Systems & Green IT	2012
28	Ms. Tripti Vishwakarma (Consultant Talent Acquisition; Tech Democracy LLC)	Importance of Technical Knowledge	2012
29	Mr. Yashwant Vishwakarma (Consultant; Microsoft)	Career as Database Administrator	2012
30	Mr. Jayant K. Kittur (Asso. Professor; Gogte Institute of Technology)	Career option for Engineering Graduates and Preparation for Campus Placement	2012

31	Mr. Sumit Kumar (SAP Project Manager; Total Environment Building Pvt. Sys.)	ERP Implementation	2012
32	Yashavant Kanetkar & Asang Dani	Work shop on secure programming	2011
33	Mr. Mayank Jha (Manager; IBM)	Knowledge Required for IT Sector	2011
34	Mr. Kaustabh Shukla (Advocate; Corporate Laws)	Corporate Social Responsibilities	2011
35	Mr. Vikram Singh (Regional Manager Corporate & Academic Relation; Aimsvarsity Education Pvt. Ltd.)	Cloud Computing & WEB 2.0	2011
36	Mr. Vaibhav Khare (Director; Gurukul Classes)	Resume Building	2011
37	Mr. Virendra Vishwakarma (Technical Lead; Microsoft)	Latest Trends in IT Sector	2011

12. Percentage of classes taken by temporary faculty – programme-wise information:-

➤ Nil

13. Programme-wise Student Teacher Ratio:-

Year	2 nd year	3 rd year	4 th year	Total	No. of Faculties	Student Teacher Ratio
2010-11	122	77	51	250	15	16
2011-12	115	105	67	287	19	15
2012-13	115	132	93	340	23	15
2013-14	109	112	129	350	24	15

14. Number of academic support staff (technical) and administrative staff:-

Type of staff	Sanctioned	Filled
Support Staff (Technical)	15	15
Administrative Staff	01	01

15. Qualification of teaching faculty with DSc/D.Litt/Ph.D/MPhil/PG. :-

Presently 21 faculty members are in the department. Faculty profile will be shown on the day of visit.

16. Number of faculty with ongoing projects from a) national b) international funding agencies and c) Total grants received.

a) National

S. No	Name of the Faculty	Title of the Project	Funding Agency	Grants received (Rs.)	Duration
1	Nil	Nil	Nil	Nil	Nil

b) International

S. No	Name of the Faculty	Title of the Project	Funding Agency	Grants received (Rs.)	Duration
1	Nil	Nil	Nil	Nil	Nil

17. Departmental projects funded by DST-FIST; UGC-SAP/CAS, DPE; DBT, ICSSR, AICTE, etc.; total grants received.

S. No	Title	Chief coordinator	Amount (Rs)	Ref no
1	MODROB	Dr. Sipi Dubey	1500000/-	Ref no:-200-21/FIN/2001-2002/1016 Date 05/02/10
2	Staff Development Program	Dr. Sipi Dubey	350000/-	Ref:-35/FIN/2001-2002/1066 on 08/02/10

18. Research centre/ facility recognized by University:-

➤ Nil

19. Publications:-

Publication details are available in the Department and will be shown at the time of visit.

20. Area of consultancy and income generated: - Nil

21. Faculty serving in

a) National committees:- Nil

b) International committees: - Nil

c) Editorial Boards

S. No.	Faculty Name	Post Name	Committee
1	Khomlal Sinha	Reviewer	IJERT, Journal
2	Bhupesh Kumar Dewangan	Reviewer	IJERT, Journal

22. Students projects:-

- Percentage of students who have done in-house projects including inter departmental/programme :-100%
- Percentage of students placed for projects in organizations outside the institution i.e. in Research laboratories/industry/other agencies:- Nil

23. Awards/Recognitions received by faculty and students

- No. of recipients of Awards and Recognitions :- 55
- Listing of the Recipients will be shown to the peer team at the time of visit.

24. List of eminent academicians and scientists/visitors to the department:-

S. No	Year	Name of the Academician/ Scientist/ Visitor	Designation	Organization	Purpose of Visit
1	2013-14	Dr.Ashok Bhansali	Assistant Professor	OPJIT, (Raigarh)	Teaching Methodology
2	2013-14	Dr. Rabindra Ku. Jena	Assistant Professor	IIITM, Nagpur	Green Computing
3	2013-14	Dr. Roheet Bhatnagar	Assistant Professor	JIT, Jaipur	Big Data
4	2012-13	Dr.Sudhakar Pandey	Assistant Professor	NIT, Raipur	Research Methodologies

		Dr.Keshri Verma	Assistant Professor	NIT, Raipur	Innovations in Data Mining
		Dr.Sanjay Kumar	Assistant Professor	Pt.RSSU, Raipur	Principles of Pipelining
5	2011-12	Mr. Yashvant Kanetkar & Mr. Asang Dani	Director	KSET Pvt. Ltd., Nagpur,	Workshop

25. Seminars/Conferences/Workshops organized & the source of funding
a) National

S. No.	Year	Name of the Seminar/ Conference/ workshop	Date (s)	Source of Funding
1.	2013-14	AICON'14'	25-26 April'2014'	Chhatrapati Shivaji Institute of Technology, Durg
2	2012-13	AICON'13'	12-13 April'2013'	Chhatrapati Shivaji Institute of Technology, Durg
3	2011-12	AICON'12'	28-29 Jan'2012'	Chhatrapati Shivaji Institute of Technology, Durg
4	2010-11	AICON'11'	20-21 Jan'2011'	Chhatrapati Shivaji Institute of Technology, Durg

b) International : Nil

26.Student profile programme/course wise:

Year	Name of the Course/ Program	Applications Received	Selected (Number Admitted including Lateral Entry)	Enrolled		Pass Percentage
				Male	Female	
2010-14	B. E. (Computer Science &	*	175	105	70	42.88%
2009-13		*	128	78	50	49.20%

2008-12	Engineering)	*	96	63	33	44.44%
2007-11		*	67	41	26	100%

27. Diversity of Students:-

Name of the Course	Academic Year	% of students from the same state	% of students from other state	% of students from abroad
BE(CSE)	2013-2014	98.9	0.01	-
	2012-2013	95.4	0.05	-
	2011-2012	95.9	0.04	-
	2010-2011	98.2	0.01	-

28. How many students have cleared national and state competitive examinations such as NET, SLET, GATE, Civil services, Defense services, etc ?

S. No.	Name of the Course/ Program	Academic Year	No of students				
			GATE	CAT	GRE	MAT	TOFEL
1	B. E. (Computer Science & Engineering)	2013-14	1	-	-	2	-
2		2012-13	1	-	-	-	-
3		2012-13	5	-	-	-	-
4		2010-11	2	-	-	-	-

29. Student progression:-

Student Progression	Year	Against % enrolled	
UG TO PG	2014	Nil	
	2013	03,	3.89%
	2012	02	3.77%
	2011	05	10.41%
	2010	01	2.04%
PG to M. Phil.	-	-	
PG to Ph.D.	-	-	
Employed	2014	89	
	2013	64	
• Campus Selection	2012	02	3.77%
	2011	01	2.08%
	2010	04	8.16%
• Other than Campus recruitment	2014	-	
	2013	-	
	2012	35	66.03%

	2011	28	58.33%
• Entrepreneurship / Self Employment	2010	31	63.26%

30. Details of Infrastructural facilities:-

a) Departmental Library

No. of Titles	No. of Volumes	No. of Computers	No. of E-Journals	No. of Print Journals
198	322	-	-	4

Timing : 9.00 AM to 5.00 PM (Working days only)

b) Internet facilities for Staff & Students:-

- All Computer Labs are having Internet Facility with 10 Mbps speed.

c) Class rooms with ICT facility:-

The Department is having:-

- 8 well equipped Lecture halls, 1 Tutorial room and 1 well equipped Seminar Hall to run the various UG Lectures and conduct seminars.
- All classes are equipped with white and/or chalk boards.
- LCD is primarily used to deliver electronic class notes and to perform demos and presentations.

d) Laboratories

Lab Description	Space/# Students	Curriculum Laboratory
Research & Development Lab	30	M. Tech lab
Programming Lab C/C++	30	Programming with C Lab, Data Structure Lab, Object Oriented Programming Lab
Programming Lab VB/HTML/DHTML	30	Web Technology Lab, Computer Hardware Simulation Lab, Software Techno-logy Lab-1
Computer Center	60	Software Techno-logy Lab-4, Artificial Intelligence Lab, Network Security Lab, Software Technology Lab-5

Advanced Programming Lab	60	Data Base Management System Lab, Computer Network Lab, Computer Graphics Lab, Unix & Shell Programming Lab, Software Technology Lab-2, Soft Computing Lab, Network Programming Lab
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31. Number of students receiving financial assistance from college, university, government or other agencies:-

Agency	No. of students received financial assistance			
	2009-10	2010-11	2011-12	2012-13
College	-	-	-	-
University	-	-	-	-
Government	-	-	-	93
Other agencies	-	-	-	-

32. Details on student enrichment programmes (special lectures /workshops/seminar) with external experts:-

2013 – 14

- Mr. Rahul Krishna (Manager Recruitment; Espire Infolabs) Dated 11/09/2013; Topic - FRESHER DEVELOPMENT - A Taste of Corporate World for 7th Semester CSE.
- Mr. M. S. Uday Kiran (HR Manager; HCL) & Mr. Venkat Bhukta (Vertical Head; Idea Cellular Ltd.) Dated 28/10/2013; Topic – HR Interviews for 7th Semester CSE.
- Mr. Jayant Chaturvedani, (Manager – Operation; Ericsson) Dated 10/02/2014; Topic – Telecommunication Word, Mobile Network for CS & IT Students.

2012 – 13

- Mr. Anirudh Girey (Sr. Software Engineer; IBM India Limited) Dated 11/06/2012; Topic – Job Orientation, Career Guidance for Final Year Students.
- Mr. Nagaraj Parappagoudar (Solutions Architect; CISCO Systems, Sanjose, CA, USA) Dated 21/07/2012; Topic – Emerging Technology Areas and Trends in Computer Industry.

- Mr. Navneet Dewangan (Principle Infrastructure Engineer; Mphasis Ltd) Dated 07/11/2012; Topic – My Struggle & Success Story.
- Mr. Ravi Bizoara (Module Lead; Mphasis) Dated 01/02/2013; Topic – IT Interactive
- Mr. Kunjal Jayant Kamdar (Head Recruiter; Quinnox) Dated 02/02/2013; Topic – Social Media.

2011 – 12

- Mr. Vikram Singh (Regional Manager Corporate & Academic Relation; Aimsvarsity Education Pvt. Ltd.) Dated 26/07/2011 to 28/07/2011; Topic – Cloud Computing & WEB 2.0
- Mr. Vaibhav Khare (Director; Gurukul Classes) Dated 15/09/2011; Topic – Resume Building for 7th Semester CSE.
- Mr. Virendra Vishwakarma (Technical Lead; Microsoft) Dated 30/09/2011; Topic – Latest Trends in IT Sector .
- Mr. Sumit Kumar (SAP Project Manager; Total Environment Building Pvt. Sys.) Dated 14/01/2012; Topic – ERP Implementation for CS & IT 7th Semester Students.
- Mr. Jayant K. Kittur (Asso. Professor; Gogte Institute of Technology) Dated 03/02/2012; Topic – Career option for Engineering Graduates and Preparation for Campus Placement for 6th Semester Students.
- Mr. Yashwant Vishwakarma (Consultant; Microsoft) Dated 15/03/2012; Topic – Career as Database Administrator for Final Year Students.
- Ms. Tripti Vishwakarma (Consultant Talent Acquisition; Tech Democracy LLC) Dated 27/03/2012; Topic – Importance of Technical Knowledge for Final Year Students.
- Dr. Bharti Trivedi (Director; Apex Technology) Dated 17/05/2012; Topic – Vacuum Systems & Green IT for Final Year Students of CSE,

2010 – 11

- Mr. Shashank Pradhan (Manager; Accenture) Dated 14/09/2010; Topic – Adapting Students into Corporate Lifestyle for 7th Semester Students.
- Mr. Tulika Chakraborty (Lead Quality Analyst; IBM) Dated 20/10/2010; Topic – Life Style in Corporate.
- Mrs. Chitra Pathak (Software Engineer; IBM) Dated 05/11/2010; Topic – How to Prepare for IT Sector for 7th Semester & 5th Semester Students (CSE & IT).
- Mr. Mayank Jha (Manager; IBM) Dated 13/01/2011; Topic – Knowledge Required for IT Sector for 7th Semester Students (CSE & IT).
- Mr. Kaustabh Shukla (Advocate; Corporate Laws) Dated 30/06/2011; Topic – Corporate Social Responsibilities for 6th Semester Students.

33. Teaching methods adopted to improve student learning

- Seminars are organized to improve their presentation skill.
- Encouraging Mini Projects from third sem onwards to improve their practical knowledge and building up their confidence.
- Organizing coaching classes for Aptitude and placement tests and personal interview.
- Add-on programs are organized like Advance Java, Matlab, Networking etc to strengthen theoretical learning.
- Coaching Classes for campus placement are organized.

34. Participation in Institutional Social Responsibility (ISR) and Extension activities:-

- On the occasion of Independence Day, a visit to orphanage 'Saraswati Bal Kalyan Ashram' was organized. Dr. N.T. Khobragade (Dean, CSIT) addressed the students. 110 students and various faculties of the college visited the orphanage. The faculties also told the students about their fundamental rights and duties as responsible citizens.



35. SWOC analysis of the department and Future plans?

Strengths:

- Competent and well qualified faculty members.
- High quality infrastructure.
- Working in high impact areas for research in IT, Management and Applied Sciences
- Committed staff and proactive members.
- Adequate proper digitization of records.
- Working with ERP system.
- Student project quality is good.

- Teacher student ratio is good.
- Sufficient emphasis on lab and project work. .
- Good team spirit among faculties.
- Committed technical supporting Staff.
- Department has its student association which regularly conducts programs and events which are student centric.
- Systematic monitoring of teaching learning process.
- Active participation of faculty in research activities.
- Well equipped and specious laboratories.

Weaknesses:

- Lack of good IT /ITES based industries in and around Durg for interaction and Collaboration.
- Placement in private and Government sector is limited.
- Lack of Industrial research and consultancy.

.Opportunities:

- Very good potential in e-governance. Possibility of short term Residential Programmes for faculties, Government employees.
- Availability of qualified, experienced and trained faculty.
- Soft skills and related training by T&P Cell and association with alumni offer the opportunities to raise the employability of the students.
- Scope for Industry Institute Interaction.
- Availability of adequate Journals, Lab facilities for Research.
- Proper connectivity by Road, Rail and Air with major cities in the country.

Constraints:

- Attracting and retaining quality faculty
- Interactions with industries have become a difficult task.
- Insignificant employment potential in the state for CSE students.

Future Plans:

- To achieve 100% result.
- To achieve excellence in providing technical manpower to meet the changes of the times to come, to keep pace with the technological advancements, commensurate with the growth of Information Technology.



- To encourage all faculty to get involved in research program leading to acquiring higher degree.
- To motivate staff towards R & D activities.
- To establish sound industry-institute interaction.

Evaluative Report of the Department

1. **Name of the department :** Electronics & Telecommunication Engineering
2. **Year of Establishment :** 1999
3. **Names of Programmes / Courses offered (UG And PG) :**
 - UG – B.E. (Electronics & Telecommunication Engineering)
 - PG – M.E. (Communication)
4. **Names of Interdisciplinary courses and the departments/units involved:**
 - Computer Science and Engineering
 - Information Technology
 - Mechatronics Engineering
5. **Annual/Semester/Choice based credit system (Program wise):**

B.E. (Electronics & Communication Engineering)
Semester based Credit system.
Electives for Higher semesters.

M.E. (Communication)
Semester based Credit system
Electives for Higher semester.
6. **Participation of the department in the courses offered by other departments:**

S. No	Programme	Courses offered by other departments
1	B.E	Computers Science and Engineering
2		Information Technology
3		Mechatronics Engineering

7. **Courses in collaboration with other universities, industries, foreign institutions, etc.:** Nil
8. **Details of courses/programmes discontinued (if any) with reasons. :** Nil
9. **Number of Teaching posts:**

Designation	Sanctioned	Filled
Professors	2	2

Associate Professors	5	5
Asst. Professors	17	14

10. Faculty profile with name, qualification, designation, specialization:

Presently 21 faculty members are in the department. Faculty profile will be shown on the day of visit.

11. List of senior visiting faculty: Nil

12. Percentage of lectures delivered and practical classes handled (Programme wise) by temporary faculty: Nil

13. Student -Teacher Ratio (programme wise):

UG: (B.E) – 17:1

PG: (M.E) – 12:1

14. Number of academic support staff (technical) and administrative staff; sanctioned and filled:

Staff	Sanctioned	Filled
Academic Support staff (Technical)	10	10
Administrative staff	2	2

15. Qualifications of teaching faculty with DSc/ D.Litt/ Ph.D/ MPhil / PG :

No. of Faculty with Ph.D	No. of Faculty with PG (M.Tech / M.E)	No. of faculty with UG (B.Tech/B.E)
1	15	4

16. Number of faculty with ongoing projects from funding agencies and grants received:

a) National: Nil

b) International: Nil

17. Departmental projects funded by DST - FIST; UGC, DBT, ICSSR, AICTE etc. and total grants received:

S. No.	Name of the project	Funding Agency	Name of the coordinator/s	Grant received	Duration
1	Lab for Printed	AICTE MODROBS	Mr. Mangal Singh	Rs. 13,00,000	2 years

	Circuit Board Design and Centre for Training				
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18. Research Centre /facility recognized by the University: Nil

19. Publications:

a) Publication per faculty (for four years)

Publication details are available in the Department and will be shown at the time of visit.

20. Areas of consultancy and income generated: Nil

21. Faculty as members in

- a) National committees: Nil
- b) International Committees: Nil
- c) Editorial Board: Nil

22. Student projects

- a) Percentage of students who have done in-house projects including inter Departmental/programme: 100%
- b) Percentage of students placed for projects in organizations outside the institution i.e.in Research laboratories/Industry/ other agencies: Nil

23. Awards / Recognitions received by faculty and students: Nil

24. List of eminent academicians and scientists / visitors to the department:

S. No.	Y r.	Name of Academician / Scientist/ Visitor	Designation	Organization	Purpose of visit
1	2011	Mr. Basant Kumar	Coordinator	Prajapati Bramkumari Ishwariya Vishvidyalaya	Guest Lecture
2		Ms.Rupali Behen ji & Mr. Yuvratna Bhai ji	Acharya	Prajapati Bramkumari Ishwariya Vishvidyalaya	Guest Lecture

3	2012	Mr. C. Dinkar	Traffic Education Instructor	Police control Room, Sector 6- Bhilai	Guest Lecture
		Mr.S.R.Bhattacharya	AGM (OHP),	Bhilai Steel Plant, Bhilai	Guest Lecture
5		Dr.Ashok Tripathi & Dr. Anil Karnawat		Raipur, Chhattisgarh	Guest Lecture
6		Mr. Anil Menon	Junior Manager, Rail and Structural Mill,	Bhilai Steel Plant, Bhilai, Chhattisgarh	Guest Lecture
7	2013	Dr. Archana Sharam & Mrs. Chandraprabha Dewangan	Scientist	BARC , Mumbai	Guest Lecture
8		Mr. Neelesh Sawant, Mr. Deepak Doke & Shayali Wagha	Project Incharge	Siemens, Mumbai	Guest Lecture

25. Seminars/ Conferences/Workshops organized & the source of funding

a) National – Since 2009, AICON is conducted by the college and the event is Annual.

b) International – NA

26. Student profile programme/course wise:

Year	Name of the Course/programme (refer question no. 4)	Applications received	Enrolled			
			Selected	*M	*F	Pass %
2010	B.E(ETC)	×	124	80	44	69.71
2011		×	116	62	54	71.94
2012		×	104	54	50	63.84
2013		×	86	44	42	52.38
2010	M.E(ETC)	×	18	8	10	83.3
2011		×	18	8	10	NA
2012		×	18	10	7	NA
2013		×	18	7	11	NA

*M = Male *F = Female

× = Admissions are made by the Directorate of Technical Education.

Basis : Entrance Exam (CGPET) + JEE (Mains) for other state Quota.

27. Diversity of Students

Name of the Course		% of students from the same state	% of students from other States	% of students from abroad
B.E	2009	78	22	Nil
	2010	94	6	Nil
	2011	92	8	Nil
	2012	92	8	Nil
	2013	87	13	Nil
M.E	2010	100	Nil	Nil
	2011	94	6	Nil
	2012	100	Nil	Nil
	2013	100	Nil	Nil

28. How many students have cleared national and state competitive examinations such as NET, SLET, GATE, Civil services, Defense services, etc.:

No. of Students qualified in GATE

Academic Year	ETC
2013-2014	2
2012-2013	7
2011-2012	1
2010-2011	1

No. of Students qualified in GRE/TOFEL/CAT/MAT/PGECET etc

Academic Year	ETC
2013-2014	1
2012-2013	Nil
2011-2012	2
2010-2011	2

29. Student progression

Student Progression	Year	No. of Students
UG TO PG	2014	2
	2013	7
	2012	2
	2011	1
PG to PhD		Nil
Employed	2014	83
• Campus Selection	2013	84
	2012	69
	2011	88
• Other than Campus recruitment	2014	--
	2013	--
	2012	--
	2011	--

30. Details of Infrastructural facilities

a) Library

The following library facilities are available in our department for providing research facility to promote research.

Books: 410 Cost: 98050/-

On working days : 9.00 AM to 5.00 PM

b) Internet facilities for Staff & Students Yes

Computer Lab No.07 and Computer Lab No. 12

c) Class rooms with ICT facility:- Yes, Ramanujan Hall

d) Laboratories

S. No.	Name of Lab	Area (sq. Ft.)	Total Amount (Rs in lakh)
1	Basic /Analog Electronics Lab	792 Sqft	3,00,320/-
2	Electronics Workshop Lab	805 Sqft	9,20,244/-
3	Digital Electronics Circuit lab	805 Sqft	1,93,692/-
4	Microprocessor (8085/8086) Lab	805 Sqft	4,52,892/-
5	Communication System Lab	756 Sqft	15,75,231/-
6	Linear Integrated/ Advanced	805 Sqft	4,32,552/-
7	Microwave Communication Lab	792Sqft	4,05,598/-

8	Microcontroller & Embedded System Lab	805 Sqft	6,37,000/-
9	Optical Communication Lab	756 Sqft	2,32,020/-
10	Computer Lab No. 7	720 Sqft	5,86,000/-
11	Computer Lab No. 12	1368 Sqft	16,45,300/-

31. Number of students receiving financial assistance from college, university, Government or other agencies Yes

Details of financial Assistance given to the students by the State Govt. in year 2010-2011

Scholarship for SC/ST/OBC in BE Course Year 2013-14

S. No	Course	Cast	Total No of Student	Allotted Fund	Total Amount
1.	BE	SC	1	26037/-	26037/-
2.	BE	ST	1	26005/-	26005/-
3.	BE	OBC	16	21545/-	3,44,720/-
Grand Total			18		3,96,762/-

Scholarship for SC/ST/OBC in BE Course Year 2012-13

S. No	Course	Cast	Total No of Student	Allotted Fund	Total Amount
1.	BE	SC	5	23500/-	1,17,500/-
2.	BE	ST	8	23508/-	1,88,064/-
3.	BE	OBC	30	19650/-	5,89,500/-
Grand Total			43		8,95,064/-

Scholarship for SC/ST/OBC in BE Course Year 2011-12

S. No	Course	Cast	Total No of Student	Allotted Fund	Total Amount
1.	BE	SC	4	27396/-	1,09,584/-
2.	BE	ST	5	27742/-	1,38,710/-
3.	BE	OBC	43	22493/-	9,67,199/-
Grand Total			52		12,15,493/-

Scholarship for SC/ST/OBC in BE Course Year 2010-11

S. No	Course	Cast	Total No of Student	Allotted Fund	Total Amount
1.	BE	SC	2	21160/-	42,320/-
2.	BE	ST	5	20453/-	1,02,265/-
3.	BE	OBC	49	17997/-	8,81,853/-
Grand Total			56		10,26,438/-

32. Details on student enrichment programmes (special lectures / workshops /Seminar) with external experts: Nil

33. Teaching methods adopted to improve student learning

- All classes are equipped with chalk boards.
- LCD projectors are primarily used to deliver electronic class notes and to perform demos and presentations as and when required.

34. Participation in Institutional Social Responsibility (ISR) and Extension activities:

Year	Name of Activity	Location of Activity	Date	No. of Participants
Academic year 2013 – 2014	Prayas'13, Visit to Deaf & Dumb School	Supela, Bhilai	15/08/2013	130
Academic year 2012-2013	Old Age Home Visit	Chandkhuri, Durg	28/07/2012	130
	Prayas'12 , Visit to Deaf & Dumb School	Supela, Bhilai	15/08/2012	120
Academic year 2011-2012	Old Age Home Visit	Chandkhuri, Durg	15/08/2011	150
	Prayas'11 , Visit to Deaf & Dumb School	Supela, Bhilai	21/08/2011	225
Academic year 2011-2012	Prayas'10 , Visit to Deaf & Dumb School	Supela, Bhilai	15/08/2010	150

35. SWOC analysis of the department and Future plans
Strengths

- Department had NBA accreditation and ISO certification.
- The department has very active and very supportive alumni.
- The retention of the faculties in the department is good.
- Maximum faculties in the department are well qualified and experienced.
- Department has its student association that regularly conducts the programs and events which are student centric.

- Excellent geographic location to provide service locally and regionally.
- Well equipped and spacious laboratories.
- Dedicated and experienced support staff.
- Faculties are involved in their professional organizations and research activities.
- The department is strongly student centered and focused. Faculties are supportive to each other.

Weaknesses

- Too few summer classes offered at the graduate level.
- Lack of off-campus and alternative programming.
- First year result is not as good as expected.
- Lack of software industries in nearby places.
- Poor communication skills of the students at entry level due to rural background. This becomes a hindrance in getting proper placement.

Opportunities

- Scope for Industry Institute Interaction.
- Availability of adequate journals, Lab. Facilities for research.
- Proper connectivity by road, rail and air with important cities of India.
- Availability of qualified, experienced and trained faculty.
- Conduction of national conferences for students/ staff in thrust areas of research.

Constraints

- Lack of good communication skill in English. This becomes a barrier for students in getting selected in campus placement.
- Lack of Electronic Industries in the state.
- Competition from other institutes.
- Quality of intake is poor, especially in rural engineering colleges. This may be due to proliferation of engineering colleges.

Future Plans

- To establish research oriented laboratories.
- To achieve 95% and above result.
- To achieve 95% and above placement for eligible candidates.
- To get recognition as a research centre.
- To conduct Student Development Programme in the emerging areas.
- To conduct faculty development programmes in the latest technologies and teaching methodologies.

Evaluative Report of the Department

1. **Name of the department** : Mechanical Engineering
2. **Year of Establishment** : 1999
3. **Names of Programmes /Courses offered:** UG: B.E (Mechanical Engineering) & PG: M.Tech (Industrial Engineering & Management)
4. **Names of Interdisciplinary courses and the departments/units involved:** Nil
5. **Annual/ semester/choice based credit system (programme wise) :** B.E (Mechanical Engineering) Semester based credit system – Electives in Higher Semesters

6. Participation of the department in the courses offered by other departments:

S.No.	Programme	Courses offered by other departments
1	BE	Electronics & instrumentation Engg.
2	BE	Information Technology
3	BE	Mechatronics Engineering
4	BE	Electronics & Electrical Engineering
5	BE	Computer Science & Engineering
6	BE	Electronics & Telecommunication Engg.

8. Courses in collaboration with other universities, industries, foreign institutions, etc. : Nil

8. Details of courses/programmes discontinued (if any) with reasons: Nil

9. Number of Teaching posts:

	Sanctioned	Filled
Professors	2	4
Associate Professors	4	4
Asst. Professors	19	19
Total		27

10. Faculty profile with name, qualification, designation, specialization, (D.Sc. / D.Litt. /Ph.D. / M. Phil. etc.)

Presently 27 faculty members are in the department. Faculty profile will be shown on the day of visit.

11. List of senior visiting faculty: 04

- Dr.S.Sanyal (NIT Raipur)
- Dr.Nitin Jain (NIT Raipur)
- Dr.S. Mahapatra (NIT Rourkela)
- Dr. Baiju Tharkan (Parthivi College Bhilai)

12. Percentage of lectures delivered and practical classes handled (programme-wise) by temporary faculty: Nil

13. Student -Teacher Ratio (Programme-wise): UG: B.E –15: 1, PG: M.Tech.- 12 : 1

14. Number of academic support staff (technical) and administrative staff; sanctioned and filled:

	Sanctioned	Filled
Academic support staff (Technical)	10	13
Administrative staff	2	2

15. Qualifications of teaching faculty with DSc/ D.Litt/ Ph.D/ MPhil / PG

No. of Faculty with D.Sc/ D.Litt	No. of Faculty with Ph.D	No. of Faculty with M.Phil	No. of Faculty with PG (M. Tech / MA/ M.Com / M. Sc/ MBA / MCA)	No. of faculty With UG (B. Tech)
-	4	-	16	7

16. Number of faculty with ongoing projects from a) National b) International funding agencies and grants received:

- a) National : 07
b) International : Nil

S. N o.	Nature of the project	Dura tion	Title of the project	Name of the funding agency	Name of facult y	Total grant received till date	Project ID
1	R& D project	24/02 /2010 to Sept./ 2014	Development & Popularisation of Low cost Solar Cooker & Low cost Solar Driers in Rural Areas	NABARD	Prof. N.H. Kela	300000/-	NB(CG)/DPD-NFS/5438/RIF-Innovation/RPR10010/CSIT/2009-10
2	R& D project	2013 -14	Cost effective fully Automatic Power wheel chair for old age & disabled	Instituti on of Enginee rs (India)	Prof. Sanjay Singh	30000/-	RDUG2014045
3	R& D project	20/10 /2009 to 13/08 /2013	Indirect type Solar driers for processing food products under TePP	Ministry of Science & Technolog y Dept, of Scientific & Industrial Research	Prof. N.H. Kela	37000/-	DSIR/T ePP/TM C-CSV TU -1/09/96
4	R& D project	20/04 /2012	Vibration Analysis of Cracked Rod with Circumferentia l Crack	Instituti on of Enginee rs (India)	Prof. Sanjay Singh	75000/-	RDUG2012010
5	R& D project	2012 -13	Hybrid Limbed Exploration Vehicle	Instituti on of Enginee rs (India)	Prof. Sanjay Singh	50000/-	RDUG2013018

6	R&D project	2012-13	Cost effective Micro Propulsion through indigenous Development of Valve less Pulsejet Systems	Institution of Engineers (India)	Prof. Sanjay Singh	50000/-	RDUG2 013096
7	Review of drawing of SEMI BED TRUCK TRAILER	25/06/2011	Review of drawing of SEMI BED TRUCK TRAILER	Khalsa Engg. & Foundary Works	Mr. A. P. Kumbhare	20000/-	

17. Departmental projects funded by DST - FIST; UGC, DBT, ICSSR, etc. And total grants received:

S. No.	Nature of the project	Duration	Title of the project	Name of the funding agency	Name of faculty	Total grant received till date	Project ID
1	MODROBS	2013-14	Systems Lab	AICTE	Dr. Anurag Verma	10,68,349.93/-	9-38/RIFD/MODROB/POLICY-1/2013-14(PVT)
2	RPS	23/03/2012 to 22/03/2014 (Two Years)	Advanced Mechanism Design Centre	AICTE	Dr. Anurag Verma	700000/-	8023/RID/RPS-21//2011-12/

3	MODROBS	4/06/2012 (1 Year)	Development of Centre of Excellence-Product Development Centre	AICTE	Mr. S. K. Sharma	920000/-	8024/RIF D/MOD-29(pvt)/policy-IV/2011-12
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18. Research Centre /facility recognized by the University: Nil

19. Publications:

Publication details are available in the Department that will be shown at the time of visit.

20. Areas of consultancy and income generated:

Design & Material testing, Consultancy Amount: Various lab equipments are designed and fabricated at CSIT & supplied to nearby Engineering Colleges; The Details area as follows:-

S. No.	Year	Order Form	Income Generated
1.	2011-12	Ashoka Institute of Technology & Management, Rajnandgaon	1,60,000/-
2.	2011-12	Chhattisgarh Engineering College, Durg	3,86,000/-
3.	2012-13	Rungta Group of Colleges (RSRRCET) Kohka-Kurud, Bhilai (C.G.)	2,50,000/-
4.	2012-13	Rungta Group of Colleges (GDR CET) Kohka-Kurud, Bhilai (C.G.)	2,00,000/-
5.	2012-13	Garv Institute of mgmt. & Tech. (GIMT) Purai, Bhilai-Durg (C.G.)	4,55,000/-

21. Faculty as members in

(a) National committees:

Dr.Mahesh B. Parappagoudar

- National advisory committee member (ICARET-2013) IEEE International Conference on advanced research in Engineering and Technology Feb. 8&9 2013 K L University, Vaddeswaram A.P. India
- National advisory committee member (ICARET-2012) IEEE International Conference on advanced research in Engineering and Technology Dec. 7&8 2012 Vijayawada A.P. India

Prof. Santosh Kumar Sharma

- Coordinator IIIE ,CSIT Chapter
- Member Board of Studies , Mechanical Engineering Branch CSVTU Bhilai, session since 2013-14
- Member Syllabus Review Committee, Mechanical Engineering Branch, CSVTU Bhilai, A.Y. 2013-14
- Member Examination committee , Mechanical Engineering Branch, CSVTU Bhilai,

Mr. Jitendra Kumar Chaturvedani

- Member State Innovation Council, CSVTU Bhilai, C.G. India.

(b) International Committees

Dr. Mahesh B. Parappagoudar

- International Technical Committee Member, ICIT 2014 2nd Science One International Conference on Information Technology, Dubai, UAE, January 21-23, 2014.
- International Scientific Committee Member, ICMAS 2013, 22nd International Conference on Manufacturing Systems- ICMAS-2013, Nov 14-15,2013, University Polithnica of Bucharest, Romania.
- Technical Committee Member, ICGTIS 2013, International Conference on Global Innovations in Technology and Sciences, 4-6 April,2013, SAINTGITS College of Engineering, Kerala, India.

(c) Editorial Boards

Dr. Mahesh B. Parappagoudar

- Editorial Board Member, CSVTU Research Journal.

22. Student projects

- a. Percentage of students who have done in-house projects including inter departmental /programme: **90%**
- b. Percentage of students placed for projects in organizations outside the institution i.e.in Research laboratories/Industry/other agencies: **10%**

23. Awards/ Recognitions received by faculty and students:

(a) Faculty

Dr. Mahesh B. Parappagoudar Bibliography (distinguished personality) is published in 30th Edition of Marquis Who's Who in the World 2013.

Recognition received by the Students:

S. No	Name of the Student	Year	Program	Award
1	Mr.Mintu P. Gupta	2012-13	B.E	Fund for Project from CSIT Management
	Mr. Rahul Kurmi			
	Mr.Sourav Lahiri			
	Mr.Sumit Kr. Sahu			
	Mr.Pauras Vental			
	Mr.Nagendra Deshmukh			
	Mr.P.Sandeep			
	Mr.Praveen Kr. Sahu			
2	Mr.Aman Wani	2012-13	B.E	Fund for Project from CSIT Management
	Mr.Atul Tiwari			
	Mr.Chandrakant			
	Mr.D. Kumar Swami			
3	Mr.A.V.Anurag	2012-13	B.E	Fund for Project from CSIT Management
	Mr.Akash Devan k.			
	Mr. Abhishek Mangalgiri			
	Mr.Sameer Gupta			

**24. List of eminent academicians and scientists/ visitors to the departments
2010-11**

S. No.	Name of program	Date of program	Resource persons	Industry
1.	Introduction about Machine Engg.	01.10.10	Mr. Vikrant Naramdeo	The Great Eastern Shipping Co. Ltd.
2.	Leadership	7.03.11	Mr. Bhanu Murthy	HRD BSP.
3.	Robotics	28.03.11	Mr. V. Pandu Ranga	HSMIC College of Tech. Kochiguda, Hyderabad.
4.	Pro – E Advance Machines Assembly, Mold, 2 D Drawing	25.06.11	Mr. Loknath	CADD Centre, Pro – E Engineer, Bhilai power house (CH)
5.	Ansys & Basics of FEA	26.06.11	Mr. Manish Mudotiya	CADD Centre Traning services Pvt. Ltd., Indore (MP)

6.	Corporate Social Responsibilities	30.06.11	Mr. Naustubh Shukla	Kumar Law Firm, New Delhi
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2011-12

S. No.	Name of program	Date of program	Resource persons	Industry
1.	Dining Etiquette & Table Manners	5.07.11	Mr. Atul Kumar Mishra	VEL Global, Bangalore
2.	Entrepreneurship & Fiber Reinforced plastic	09.07.11	Mr. Vijay Kumar	M/s Auto Fibre Craft, Jamshedpur
3.	AMCAT Assessment	22.07.11	Mr. Aashish Pratap Singh	Aspiring Minds., Gurgaon Haryana
4.	Water Requirement & its uses in Bhilai Steel Plant	01.08.11	Mr. J. P. Singh	Water Management Dept. BSP
5.	Future in power sector	09.01.12	Mr. Aashish Dewangan	Jindal Power Limited Angul (Orissa)
6.	Career option for engineering graduation & preparabon for campus placement	03.02.12	Mr. Jayant Kumart Kittur	Gogte Institute of Technology, Udyambag (Narnatana)
7.	Vacuum system & Green IT	17.05.12	Mr. Ajay Trivedi & Dr. Bharti Trivedi	Apex Technology, vadodara

2012-13

S. No.	Name of program	Date of program	Resource Persons	Industry
1.	The Ethics of students life	18.07.12	Mr. K. Bhanu Murthy	HR, Bhilai Steel Plant
2.	Man in the Mirror for Final Year Mechanical Student.	09/08/2012	Mr. K. Patel	HRD, Jayaswals Neco Industries Ltd.

3.	How to Clear SAIL	08/01/2013	Mr. Abhishek Shrivastava	Assistant Manager; SAIL
4.	Basics of Cellular Communication	12/01/2013	Prof. Mangal Singh	HOD ETC Dept. CSIT
5.	Solar Energy & Power Consumption in PV modules	29/01/2013	Mr. Sanjay Dehnan/ Mr. N. Bala	Lancho Solar Pvt Ltd Rajnandgaon (C G)
6.	Applications & scopes of CAD software's	21/02/2013	Mr. Jayant Waghmare	CAD Machine Design Hirga Road, Nagpur

2013-14

S. No.	Name of program	Date of program	Resource persons	Industry
1.	Mechanical Maintenance and its type	11/09/2013	Mr. H. K. Shetty	Ex GM BSP Bhilai
2.	FUNCTIONING OF AN ORGANISATION	21/09/2013	Mr. Ashish Mathur,	Chief General Manager; Escorts Constructions Equipments Ltd.)
3.	Plate Mill Manufacturing Process	08/10/2013	Mr. S. R. Babu	DGM Plate Mill BSP Bhilai
4.	Cracking Campus Interviews	09/01/2014	Mr. Apoorv Tiwari	Associate Software Engineer; Accenture
5.	Types of Gears and their manufacturing	25/02/2014	Mr. P.C. Lalwani	MD Gearcon Engineering, Rasmada, Durg

25. Seminars/ Conferences/Workshops organized & the source of funding

(a) National

S. No.	Name of Faculty	Program	Title	Organizing Institute	Source of Funding
1	Dr. Anurag Verma	FDP-2014	Analysis of Hydraulics & pneumatics using	CSIT Durg	AICTE

			Fluidsim software		
2	Mr.S.K. Sharma	AICON-2014	Intelligence System in Mechanical & Mechatronics Engineering	CSIT Durg	CSIT Durg
3	Mr. S.K. Sharma	FDP-2013	FEA & CFD using ANSYS 14 & ABAQUS	CSIT Durg	AICTE
4	Mr.S.K. Sharma	AICON-2013	Global Innovation for Technological Development in Mechanical & Mechatronics Engineering	CSIT Durg	CSIT Durg
5	Dr. Anurag Verma	FDP-2013	Analysis using ADAMS software	CSIT Durg	CSIT Durg
6	Dr. Anurag Verma	FDP-2012	Analysis using ANSYS-14 software	CSIT Durg	CSIT Durg
7	Mr.S.K. Sharma	AICON-2012	Green Technology for Sustainable Development in Mechanical & Mechatronics Engineering	CSIT Durg	CSIT Durg
8	Mr.D.R. Nishad	FUTEC H Green-2012	State Level Paper Presentation for students	CSIT Durg	CSIT Durg
9	Mr.S.K. Sharma	AICON-2011	Latest Development in Engineering and Science in Mechanical & Mechatronics Engineering	CSIT Durg	CSIT Durg

(b) International: Nil

26. Student profile programme/course wise:

Year	Name of the Course / Program	Selected (Number admitted including Lateral Entry)	Enrolled		Pass%
			Male	Female	
2013-14	BE (Mechanical)	120+9*	128	1	75.18
	ME (Industrial Engg. & Management)	16+01 [#]	16	1	Awaited
2012-13	BE (Mechanical)	120+9*+4**	133	0	74.43
	ME (Industrial Engg. & Management)	06+01 [#]	7	-	-
2011-	BE (Mechanical)	120+12* +6**	136	2	65.21

12	ME (Industrial Engg. & Management)	10+02 [#]	10	1	60
2010-11	BE (Mechanical)	120+12*+6**	136	2	67.39
	ME (Industrial Engg. & Management)	07+02 [#]	9	-	85.7

* Fee waiver

**Lateral Entry

Dropped

27. Diversity of Students

Year of Admission	Name of the Course	% students from the same State	% of students from other States	% of students from abroad
2010-11	BE (Mechanical)	94.36%	5.64	0
2011-12		90.83	9.17	0
2012-13		91.27	8.73	0
2013-14		96.12	3.88	0

28. How many students have cleared national and state competitive examinations such as NET, SLET, GATE etc.

Name of the Competitive examination	No. of students Qualified			
	2010-11	2011-12	2012-13	2013-14
GATE	10	07	02	05
Civil services	-	-	-	-
Defense services	-	-	-	-
SLET	-	-	-	-
NET	-	-	-	-
Public Sectors	-	-	01	-
CAT/GRE/MAT/TOEFL	-	-	03	02

29. Student progression

Student Progression	No. of students Selected			
	2010-11	2011-12	2012-13	2013-14
UG to PG	11	08	05	02
PG to M.Phil.	-	-	-	-
PG to Ph.D.	-	-	-	-

Employed				
• Campus selection	46	77	98	95
• Other than campus recruitment	27	10	06	-

30. Details of Infrastructural facilities

(a) Library : Area - 1912 sq.mts.

No. of Titles	No. of Volumes	No. of Computers	No. of E-Journals	No. of Print Journals
6842	58761	10	194	80

(b) Internet facilities for Staff & Students:

- 10 Mbps broad band leased line from BSNL

(c) Laboratories

S. No.	Name of the laboratory	Area (Sq. Mts)	Total Investment (Rs.)
1	Engineering Mechanics Lab	107	39404/-
2	Fluid Mechanics & Machinery Lab	120	1142125/-
3	Dynamics & Kinematics of Machine Lab	86.4	527789/-
4	Material Testing Lab	89.76	542150/-
5	Mechanical Measurement &	112	355716/-
6	Thermodynamics & Energy Conversion System Lab	80	80370/-
7	Heat & Mass Transfer Lab	76	346672/-
8	Refrigeration & Air Conditioning	90.83	269794/-
9	Internal Combustion Engine Lab	104	479937/-
10	Automobile	112	69792/-
11	Workshop	442.47	1149341/-
12	CAM Lab	47.03	1907152/-
13	CAD Lab	83.61	1568844/-
14	Robotics Lab	81.29	15084/-

31. Number of students receiving financial assistance from college, university, government or other agencies:

Agency	No. of students received financial assistance
--------	---

	2010-11	2011-12	2012-13	2013-14
College	-	-	16	-
University	-	-	-	-
Government	-	-	-	136
Other agencies	-	-	-	-

32. Details on student enrichment programmes (special lectures / workshops / Seminar) with external experts:

Year	Topics	Special Lectures / Workshop / Seminar	Date	No. of Students	Duration
2013	CREO 2.0	Workshop	21-01-2013	67	6 month
	CREO 2.0	Workshop	21-08-2013	33	1 month
2014	HIGH END INTEGRATED DESIGN SOFTWARE	Special Lectures	24-02-2014	49	1 day
		Special Lectures	25-02-2014	50	1 day
		Special Lectures	26-02-2014	14	1 day
		Special Lectures	05-03-2014	53	1 day
		Special Lectures	06-03-2014	55	1 day
		Special Lectures	21-03-2014	39	1 day

33. Teaching methods adopted to improve student learning

- For effective presentation/teaching, we use audio visual aids like power point, charts, models etc.
- Advising students to participate in group discussions, technical quizzes, and educational games activities to develop analytical and problem solving abilities.
- Prepare them for presenting technical papers in technical conferences, seminars and paper contests organized by reputed institutions.
- Grievances and redressal mechanism on Teaching – Learning process by taking feedback from the students.
- Motivating to access latest online journals, reference material and help them to understand the emerging trends in the field of Mechanical Engineering.

- Pre-placement training programs to students by external & internal experts to prepare them for campus recruitment.

34. Participation in Institutional Social Responsibility (ISR) and Extension activities

S. No.	Name	Place of Camp	Organiser/ Inviter	Date from	To	Total no of Days
1	Training of Self Help Groups & Farmers	Balod	NGO MP Vigyan Sabha Balod & CSIT	14-05-10	-	One
2	Main Market for Farmer	Village-Latabod	NGO MP Vigyan Sabha Balod	17-05-10	-	One
3	Market of Karhi Bhadar for Farmer	Village-Karhibhadar	NGO MP Vigyan Sabha Balod	24-05-10	-	One
4	Market of Mulley Guda for Farmers	Village-Mulleyguda	NGO MP Vigyan Sabha Balod	22-06-10	-	One
5	C.G. Rajyotsava for Visitors	Science College Ground Raipur	CSIT	26-10-10	01-11-10	seven Days
6	Vegetable grower students and principal of school	Krishi Vigyan Kendra, Durg	Krishi Vigyan Kendra, Durg	29-11-11	-	One
7	Swadeshi Jagran Manch Rajnandgaon	Rajnandgaon	Swadeshi Jagran Manch Rajnandgaon	12-04-12	-	One
8	National Science Day Celebration 2012	Govt V.Y.T. PG	Secretary Science club govt	18-05-12	-	One

		Autonomous College, Durg	college, Durg			
9	SHGS, Farmer Club Individual Entrepreneur	Bemetara	Through Krishi Vigyan Kendra, Durg	16-05-12	-	One
10	SHGS, Farmer Club Individual Entrepreneur	Balod	Through Krishi Vigyan Kendra, Durg	17-05-12	-	One
11	Students of Central College of Engg. & Management	Kabir Nagar, Raipur	CCEM Raipur	09-02-13	-	One
12	National Conference on Expanding Horizons of National Communication (Exhibition & Lecture)	Agriculture University, Raipur	CG Cost Raipur	05-01-13	06-01-13	Two
13	Training of SHG Jan kalyan Samiti Gundardehi	CSIT, Durg	CSIT, Durg	14-05-14	-	One
14	Akasi Kisan Mela	Thanaud	Agriculture Dept.	16-02-13	17-02-13	Two
15	Maheshwari Yuva Mahila Mandal Raipur	Virndavan Hall Raipur	Maheshwari Yuva Mahila Mandal Raipur	06/03/2014	-	One

35. SWOC analysis of the department and Future plans

1. Strengths:

- Reaching expected outcome by imparting quality education and securing good academic results.
- Continuous improvement in teachers performance.
 - Promoting Participation in knowledge improvement program.
 - Up graduation of Qualification
 - Incentive for good Academic Performance
- Exposure of students to practical pursuit by organizing industrial visits, and working on good projects.
- Providing consultancy services to the technical institutes and nearby industries in the state.
- Students' active participation in model making and paper presentations.
- Qualified faculty & well equipped laboratories.
- High-End software packages are made available for the students and faculty to carry out their projects and research work.
- Receiving grants from AICTE under MODROBS, RPS (Research Promotion Scheme) and FDP schemes.
- Set up two centres of excellence:
 - Advanced Mechanism Design Centre
 - System lab

2. Weaknesses:

- Mismatch between industrial requirement and curriculum.
- Need to strengthen R&D activities.
- Liaison with alumni network is yet to be developed.

3. Opportunities:

- Growing demand for Mechanical Engineers because of industrial growth in nearby areas and overall pace of growth in infrastructural projects in the country.

4. Constraints:

- Decrease in quality of admissions due to mushrooming of engineering colleges and deemed Universities.
- It is very difficult to motivate students with rural background towards academics and soft skills.

Evaluation Reports of the Department

1. **Name Of the department:** - Department of Electrical and Electronics Engineering.
2. **Year of Establishment:** -2002.
3. **Name of the Programmes / Courses offered:** -BE. Electrical and Electronics Engineering.
4. **Names of Interdisciplinary courses and the departments/units involved:** -

S. No.	Semester	Interdisciplinary Courses	Departments Involved
1	BE I	Professional Communication in English (Theory and Lab)	Humanities
2	BE I	Engineering Physics (Theory and Lab)	Science
3	BE I	Engineering Chemistry (Theory and Lab)	Science
4	BE I	Engineering Graphics (Theory and Lab)	Mechanical
5	BE I	Fundamental of Mechanical Engineering (Theory and Lab)	Mechanical
6	BE I	Environment and Ecology	Science
7	BE I	Basic Civil Engineering	Civil
8	BE I	Workshop Practice	Mechanical
9	BE I	Mathematics-I & II	Maths
10	BE III	Mathematics-III	Maths
11	BE III	Numerical Analysis	Maths
12	BE IV	Programming With C Lab	CSE
13	BE IV	Analog Electronics Lab	ET&T
14	BE V	Digital Electronics and Logic Circuit Lab	ET&T
15	BE V	Linear Integrated Circuit Lab	ET&T
16	BE V	Linear Control System Lab	E&I
17	BE VI	Microprocessor and Interfacing Lab (8085)	ET&T

18	BE VII	Advanced Microprocessor and Interfacing Lab(8086)	ET&T
19	BE VII	Microcontroller Lab (8051)	ET&T
20	BE VII	Advanced Electronics Circuit Lab	ET&T

5. **Annual/semester/choice based credit system (Programme wise): -**

- BE Electrical and Electronics Engineering : Semester system.

6. **Participation of the department in the courses offered by other departments:**

S. No	Programme	Courses offered by other departments
1	All Branches of BE 1 st Year	Elements of Electrical Engineering (Theory)
2	All Branches of BE 1 st Year	Elements of Electrical Engineering Lab
3	BE III Electronics and Instrumentation Engineering	Electrical Machines (Theory)
4	BE V Electronics and Instrumentation Engineering	Industrial Power Electronics Lab
5	BE III Electronics and Instrumentation Engineering	Electrical and Electronics Measurement lab

7. **Courses in collaboration with other universities, industries, foreign institution, etc.: Nil**

8. **Details of courses/ programmes discontinued (if any) with reason:Nil**

9. **Number of Teaching posts: -**

	Sanctioned	Filled
Professors	1	2
Associate Professors	4	3
Asst. Professors	11	12

10. **Faculty profile with name, qualification, designation, specialization, (D.Sc./ D.Litt./ Ph.D./ M. Phil. Etc.) :-**

Presently 17 faculty members are in the department. Faculty profile will be shown on the day of visit.

11. List of senior visiting faculty: -

- Dr.A.S.Thoke, Professor and HoD NIT Raipur.
- Dr.Kavita Thakur, Porfessor School of Studies in Electronics Pt.R.S.S.U. Raipur.
- Dr.R.N.Patel, Professor, SSTC, Bhilai.

12. Percentage of lectures delivered and practical classes handled (programme wise) by temporary faculty: -

➤ Nil

13. Student-Teacher Ratio (programme wise): -

Name	Student Teacher Ratio
BE(Electrical and Electronics Engineering)	14:1

14. Number of academic support staff (technical) and administrative staff; sanctioned and filled: -

Category	Sanctioned	Filled
Academic support staff (technical)	7	7
Administrative staff	1	1

15. Qualifications of teaching faculty with DSc / D. Litt / Ph.D. / MPhil / PG: -

No. of faculty with Ph.D.	No. of faculty with PG (M.Tech/MA/M.Com/M.Sc/MBA/MCA)	No. of faculty with UG (BE)
2	8	7

16. Number of faculty with ongoing projects from a) National b) International funding agencies and grants received:-

Project title	Funded By	Total Grant Received (Rupees)	Faculty	Status of the Project
MODROBS PSPS lab	AICTE	1220000/-	Dr.V.K.Chandra	In progress

PSPS: Power System Protection and Switch Gear Lab.

17. Departmental projects funded by DST-FIST; UGC, DBT, ICSSR, etc. and total grants received: -

Project title	Funded By	Total Grant Receive (Rupees)	Faculty	Status of the Project
Faculty Development Program	AICTE	490000/-	Dr.V.K.Chandra	Complete
MODROBS PSPS Lab	AICTE	1220000/-	Dr.V.K.Chandra	In progress

18. Research Centre / facility recognized by the University: -

➤ Nil

19. Publication: -

Publication details are available in the Department that will be shown at the time of visit.

20. Areas of consultancy and income generated: Nil

21. Faculty as member in

a) National committees b) International Committees c) Editorial Boards...

S. No.	Name of the Faculty	Members
1	Dr. V.K. Chandra	Member of Board of Studies in University Member of Syllabus Review Committee in University
2	Mr. Ashraf Jafri	Member of Expert Committee in University Member of State Innovative Council
3	Mrs.Padmini Sharma	Member of Syllabus Review Committee in University Member of Board of Studies

22. Student Projects

a) Percentage of student who have done in-house projects including inter departmental / programme: -

S. No.	Academic Year	Percentage of students who have done in-house projects
1	2010-11	100
2	2011-12	100

3	2012-13	100
4	2013-14	100

- b) Percentage of students placed for projects in organizations outside the institution i.e. in Research laboratories/ Industry / other agencies: -

➤ Nil

23. Awards / Recognition received by faculty and students: -

I. Faculty activities/ achievements

- Many of the faculty members have energetically participated in various national, international conferences.

- a) Dr. V. K. Chandra has been Awarded “**Young Scientist Award**” in 5th Chhattisgarh Young Scientist Congress 2007 in Engineering Science by Chhattisgarh Council of Science & Technology, Raipur (CG)

- b) Research publication of the members of faculty are given below:

S. No.	Academic Year	Number of Publication
1	2013-14	12
2	2012-13	07
3	2011-12	12
4	2010-11	02

*Details will be shown to the peer team during its visit

- c) Research publication in conferences of the members of faculty are given below:

S.	Academic Year	Number of Publication
1	2013-14	17
2	2012-13	00
3	2011-12	08
4	2010-11	07

*Details will be shown to the peer team during its visit

- d) Details of conference seminar/workshops attended by faculty

S. No.	Academic Year	Number of Workshop/Seminar Attended
1	2013-14	03
2	2012-13	02
3	2011-12	06
4	2010-11	03

*Details will be shown to the peer team during its visit

II. Student activities/ achievements:

- Many of the students have participated in various technical events; national conferences held at various engineering colleges and won prizes.
- University Merit List rank holder (2009-10)
 - Ankita Sinha
 - Ishwari Prasad Sahu

a) Details of values added courses/ seminar / symposium/ workshops organised for student:

The Entrepreneurship Development cell has organised many programs for the improvement of the skill of Entrepreneurship among the students. The students of Electrical & Electronics Engineering Branch have attended the program. The details are given below:-

S. No.	Workshop	Duration/Date	Total No. of Participants	Participants from EEE Branch
1	PCB Designing	15 Feb 2010 – 6 March 2010 (10)	31	22
2	Hardware Assembling	15 th Feb to 6 th Mar 2010	37	02
3	Mobile Repairing	8 th to 10 th March And 18 th to 25 th March	41	03
4	Computer Networking	30 th Aug to 8 th Aug 2010	19	04
5	Air Conditioning &	11 th Aug to 28 th August 2010	36	02

b) The details of student's achievements at various places during four academic years.

S. No.	Title of the paper	Student (s) Name	Class	Name of the Event and Venue	Date (s)	Award/ Reward
1.	Vocational training program presentation	Bharti Pandey	5th	Vocational training program presentation, CSVTU, Bhilai	Dec 2007	1 st
2.	Vocational training	Prankur Chaturvedi	5th	Vocational training	12 th Jan 2010	1 st

	program presentation			program presentation, CSVTU, Bhilai		
3.	Tech Yantra	Shivani Palia Yugal Kishore Trilok Deshmukh Prashant Verma	6th	'Di – Tech Fest 2008', DIMAT, Raipur	16 -17 Jan 2008	2nd
4.	TE – NET Competition	Vimlesh Singh Abhishek Pandey	6th	Gajajyoti – 08, JITM	22 – 25 Feb 2008	3rd
5.	State Level Model Competition	Yugal Kishore Trilok Deshmukh Prashant Verma	6th	National Science Day, CSIT, Durg	28 Feb 2008	3rd
6	PICASO	Chaman yadav Akhilesh Tiwari Chandrakant Verma Nidhi Nirmalkar	8 th	Laqshya 12, CSIT, Durg	3 -5 March 2012	Third
7	Inter college football competition	Vinu Sayani	6 th	Inter college football competition 2012	15 th March 2012	Second
8	Rangoli Competition	Mousmi Sonkamle	3 rd	CSVТУ Youth Fest 2012	24 th April 2012	Second
9	Essay Competition	Komal Alwa	5 th	Essay And Poster Competition Conducted By Ecc, Csit Durg	10 th August 2011	Second

10	Dance Competition Held by MEA	Swati Chandrakar	4 th	Make your move	11 Feb 2012	Second
11	Vollyboll Competition Held By EI Department	Monika chandrakar	4 th	CSIT DURG 2011	28 th -3 rd March, 2011	First
12	Talent Zone (Greeting Card Design)	Monika chandrakar	4 th	Laqshya 11, CSIT, Durg	10-12 March, 2011	First
13	Wrestlemania	Bhenuka khare	4 th	Laqshya 11, CSIT, Durg	10-12 March, 2011	Second
14	Wrestlemania	Bhenuka khare	6 th	Laqshya 11, CSIT, Durg	10-12 March, 2011	Third
15	Talent Zone (Greeting and logo)	Bhenuka khare	4 th	Laqshya 11, CSIT, Durg	10-12 March, 2011	Second
16	Inter college football competition	Vinu Sayani	4 th	Inter college football competition	12 th March 2011	First
17	Rangoli Competition	Mousmi Sonkamle	1 st	CSVТУ Youth Fest 2011	30 th April 2011	Second
18	Slow Bicycle Race Competition Held by E and I Department	Pankaj Kumar Sahu	4 th	CSIT DURG 2011	8 th Oct 2011	Winner
19	Tornixx	Sourabh Brigwani	2 nd	Laqshya 11, CSIT, Durg	10-12 Mar. 2011	Second

24. List of eminent academic and scientists/visitors to the department.

- Dr.A.S.Thoke, Professor and HoD NIT Raipur.
- Dr.Kavita Thakur, Porfessor School of Studies in Electronics, Pt.R.S.S.U. Raipur.
- Dr.R.N.Patel, Professor, SSTC, Bhilai.

25. Seminar/ Conference/Workshops organized and the source of funding.

a. National

S. No.	Title of Seminar, Conference, Workshops	Duration	Organizer
1	An FDP on “Advancement In Power System And Power Electronics”.	29 th to 11 th May 2013	Sponsored by AICTE
2	Three-Day Workshop On “Energy Audit”	22 nd to 24 th Feb 2013	Sponsorship Of Chhattisgarh State Renewable Energy Development Agency (CREDA), Raipur
3	MI-POWER software	1 st May to 3 rd May 2014	Power Research and Development Consultant Private limited

b. International:

➤ Nil

26. Student profile programme/course wise:

Name of the course/ programme (refer question no.04)	Academic Year	Applications received	Selected	Enrolled		Pass Percentage
				*M	*F	
BE (EEE)	2013-14	Admissions are done through CGPET	114	94	20	69
	2012-13		64	49	15	75
	2011-12		63	50	13	60
	2010-11		67	41	26	65

*M=Male *F=Female

27. Diversity of Student:-

Name of the Course	Academic Year	% of Students from the same state	% of Students from the other state	% of Students from the abroad
B.E. Electrical and Electronics Engineering	2013-14	98.40%	1.60%	0
	2012-13	91%	9%	0
	2011-12	89%	11%	0
	2010-11	96.70%	3.30%	0

28. How many students have cleared national and state competitive examinations such as NET, SLET, GATE, Civil service, Defence services, etc.?

Name of the Course	Academic Year	Number of Students					
		GATE	GRE	CAT	GMAT	TOFEL	CIVIL SERVICE
B.E. Electrical and Electronics Engineering	2013-14	0					
	2012-13	8					
	2011-12	2					
	2010-11	1	1				

29. Student Progression

Student Progression	Against % enrolled			
	2013-14	2012-13	2011-12	2010-11
UG to PG	-	1%	7%	25%
PG to M.Phil	-	-	-	-
PG to Ph.D.	-	-	-	-
Ph.D. to Post-Doctoral	-	-	-	-
Employed				
• Campus Selection	72%	47%	55%	78%
• Other than Campus recruitment	-	1	-	1
Entrepreneurship / Self – employment	-	-	-	1

30. Details of Infrastructural facilities

a) Library

Room Description	Usage	Capacity/area	Room Equipped with
Room No. 211	Departmental Library	124.38 m ²	Books, Magazines, Project Report, Training Report, 5Almira, 1 Table, Chair, Record of Sancharika.

b) Internet facilities for Staff & Students

- Faculty can access the Internet with 12Mbps from Power System Simulation lab.
- Students can also access the internet from Power System Simulation Lab.

- Wi-Fi Facility is available.

c) Class rooms with ICT facility

Room Description	Usage	Capacity/area	Room Equipped with
Class Room No: 185	Class Room for 1 st year	64/85 Sq.m.	Glass board, OHP with White Screen Table for faculty, Students Benches, Chalks, Dusters, Fans, Tube light and Dust Bin
Class Room No: 220	Class Room for 2 nd year,	66/85 Sq.m.	Glass board, OHP with White Screen Table for faculty, Students Benches, Chalks, Dusters, Fans, Tube light and Dust Bin
Class Room No: 342	Class Room for 2 nd year,	64/80.29 Sq.m.	Glass board, OHP with White Screen Table for faculty, Students Benches, Chalks, Dusters, Fans, Tube light and Dust Bin
Class Room No: 328	Class Room for 3 rd year	66/85 Sq.m.	Glass board, OHP with White Screen Table for faculty, Students Benches, Chalks, Dusters, Fans, Tube light and Dust Bin
Class Room No: 205	Class Room for 4 th year	64/85 Sq.m.	Glass board, OHP with White Screen Table for faculty, Students Benches, Chalks, Dusters, Fans, Tube light and Dust Bin
Class Room No:212	Raman Hall (Seminar Hall)	150/180 Sq.m.	Overhead projector, capacity 200.

d) Laboratories

S. No.	Name of Laboratory	Room No.	Space/No. of Students
1	Basic Electrical Engineering Laboratory	111	85.67 m ² /30
2	Basic Electrical Machines Laboratory	110	96.49 m ² /30
3	Electric Circuit Laboratory	208	124.38m ² /30
4	Measurement & Instruments Laboratory	208	124.38m ² /30
5	Rotating Electrical Machine Laboratory	108	90.83 m ² /30

6	Power System Simulation Laboratory	232	66.88 m ² /30
7	Industrial & Power Electronics Laboratory	107	88.88 m ² /30
8	Power System Protection & Switch Gear Laboratory	112	74.24 m ² /30
9	Utilization of Electrical Energy Laboratory	112	74.24 m ² /30
10	High Voltage Engineering Laboratory	107	88.88 m ² /30

31. Number of students receiving financial assistance from college, university, government or other agencies.

Criteria	Number of students receiving Financial Assistance			
	2010-2011	2011-2012	2012-2013	2013-2014
University	2	2	--	2
Government	--	--	--	--
Other Agencies	--	--	--	--

32. Details on student enrichment programmes (special lectures / workshops / seminar /) with external expert

Module Description	Developed / organized by	Duration	Resource Persons	Target Audience	Usages and Citation etc.
3 days' workshop on "Entrepreneurship development"	Mr. N.H. Kela Coordinator of ED Cell CSIT, Durg.	03 days	Mr. G.S. Day Ex. Bank dist coordinator Mr. K.C. Thomas P.O. CITCON Mr. P.K. Nimonkar Project head	IV year Students	Participants can the knowledge about Entrepreneurship encourage them to make their future as
Four day workshop on "MATLAB"	Mrs. Padmini Sharma	04 Day	Mr. O.P.Yadav Asst. Prof. ,Dept. Of Electronics & Tele comm.	III year Students	Participants should familiar with the software to use in the field of Electrical engineering
One day lecture on	TNP Departmen	01 Day	Mr. Kaustabh Shukla, Corporate	VI Sem	Participants can get

Corporate Social Responsibilities	t		law advocate & legal consultant	Students	knowledge about Corporate social responsibilities
One day lecture on Security Programming	TNP Department	01 Day	Mr. Yashwant Kanetkar, Director – SCET, Nagpur	III & IV year Student	Participants can get knowledge of security system
One day lecture on Description about CSEB work	SAEEE	01 Day	Mr. Rajanikant Awasthi	III year Student	Participants can get knowledge about CSEB work culture
One day lecture on Stress Management	SAEEE	01 Day	Mr. Sayed Gafoor Ali	III year Student	Participants can get knowledge about how to release stress from student

33. Teaching methods adopted to improve student learning:-

- By giving assignments to the students.
- Through audio visual lectures.
- Students enrichment programmes through different seminars and workshops.
- By relating the theory with the practical.

34. Participation in Institutional Social Responsibility (ISR) and Extension activities

S. No.	Academic Year	Name of the Event	Organized Under	Date	No. of Participants
1	2013-14	SNEH –SAMPADA	SAEEE	15AUG. , 2013	45
2	2012-13	VISIT TO OLD AGE HOME	SAEEE	15 th Aug. 2012	46
		JEEVAN AADHAR(BLOOD DONATION CAMP) AND “BLOOD TEST	SAEEE	23 rd feb. 2012	266

		CAMP			
3	2011-12	SNEH –SAMPADA SAEEE ASSOCIATION	SAEEE	15AUG. , 2011	40

35. SWOC analysis of the department and future plans

STRENGTHS

- Well qualified faculty members in the department with good retention.
- Accredited by National Board of Accreditation in February 2013.
- Well-equipped laboratories.
- Well stocked Departmental Library.
- Motivated students with confidence.
- Online journals subscription.
- Counseling and Remedial classes for weak students.
- AICTE grants received for modernisation and removal of obsolescence of lab (MODROBS) and for Faculty Development Program(FDP).
- Continuous improvements in performance of both the faculty and students.
- Systematic monitoring of teaching-learning process.

WEAKNESSES

- Research & Development work needs to be strengthened.
- Industry institute interaction needs to be improved.
- No Consultancy work is carried out.

OPPORTUNITIES

- There is tremendous growth proposed for power industry at the national level. Therefore electrical engineering graduates will have very good job opportunities.
- Faculty pursuit for Ph.D with local universities.
- To obtain sponsored projects from AICTE and other organisations.

CONSTRAINTS

- Decrease in quality of admissions.
- Declining interest in engineering education.
- Poor communication skill of students.
- Increase in number of engineering colleges in the state.
- No significant employment potential in the state for EEE students.

FUTURE PLANS

- To Start post graduate course.
- To start research centre.
- To start consultancy services.
- To obtain reaccreditation.
- To ensure better placement of students.

Evaluative Report of the Department

1. **Name of the Department :** Electronics & Instrumentation Engineering
2. **Year of Establishment :** 2005
3. **Names of Programmes/ Courses offered : (UG) :** B.E. - Electronics & Instrumentation Engineering
4. **Names of Interdisciplinary courses and the departments/units involved :**

S. No.	Semester	Name of Course	Department
1.	1 st	Applied Mathematics – I	Mathematics
2.	1 st	Professional Communication in English	Humanities
3.	1 st	Applied Chemistry	Basic Sciences
4.	1 st	Engineering Graphics	Mechanical Engineering
5.	1 st	Elements of Electrical Engineering	Electrical & Electronics Engineering
6.	2 nd	Applied Mathematics – II	Mathematics
7.	2 nd	Environment & Ecology	Basic Sciences
8.	2 nd	Applied Physics	Basic Sciences
9.	2 nd	Basic Civil Engineering	Basic Sciences
10.	2 nd	Fundamentals of Mechanical Engineering	Mechanical Engineering
11.	3 rd	Mathematics – III	Mathematics
12.	3 rd	Programming with 'C'	Computer Science
13.	4 th	Mathematics – IV	Mathematics
14.	4 th	Electrical Machines	Electrical & Electronics Engineering

5. **Annual/ semester/choice based credit system (programme wise)**

B.E - (Electronics & Instrumentation Engineering) –

- Semester based credit system.
- Electives in higher semesters.

6. **Participation of the department in the courses offered by other departments**

S. No.	Programme	Courses offered by other departments
1.	B.E.	Electronics & Telecommunication Engineering
2.		Mechatronics Engineering

7. Courses in collaboration with other universities, industries, foreign institutions, etc.: Nil

8. Details of courses / programmes discontinued (if any) with reasons : Nil

9. Number of Teaching posts

S. No.	Position	Sanctioned	Filled
1.	Professor	1	1
2.	Associate Professor	2	1
3.	Sr. Assistant Professor	1	1
4.	Assistant Professor	8	7

10. Faculty profile with name, qualification, designation, specialization, (D.Sc./D.Litt. /Ph.D. / M. Phil. etc.),

Presently 10 faculty members are in the department. Faculty profile will be shown on the day of visit.

11. List of senior visiting faculty :

Session 2013 - 2014		
S. No.	Name of The Faculty	Designation
1.	Dr. R. N. Patel	Professor & Head EEE, SSCET Bhilai
2.	Dr. Rajesh Pandey	Principal, Shri Ramdeobaba College Of Engineering & Management, Nagpur
3.	Dr. Kavita Thakur	Professor in Electronics, Pt. R.S.U Raipur
Session 2012 - 2013		
4.	Dr. Neeta Tripathi	Principal SSITM Bhilai
5.	Dr. S. K. Patra	Professor & Dean , NIT Rourkela
6.	Dr. Kavita Thakur	SOS Professor in Electronics, Pt. R.S.U Raipur

7.	Dr. M. K. Kowar	Director BIT Durg
Session 2011 - 2012		
8.	Dr. K. Vishwanath	HOD (E&T) Tumkur, Karnataka
9.	Dr. Neeta Tripathi	Principal, SSITM Bhilai
10.	Dr. Kavita Thakur	Assistant Professor , Pt, R.S.U Raipur
Session 2010 - 2011		
11.	Dr. B.C. Mal	Vice Chancellor CSVTU
12.	Dr. Narendra Nath	Assistant Professor, NIT Suratkal
13.	Dr. J. Shiv Kumar	Assistant Professor, Bapuji College Of Engineering, Karnataka
14.	Dr. G. L. Dutta	Vice Chancellor, K.L University Vijayawada

12. Percentage of lectures delivered and practical classes handled (programme wise) by temporary faculty : Nil

13. Student - Teacher Ratio (Programme wise) : UG - 17:1

14. Number of academic support staff (technical) and administrative staff; sanctioned and filled

Type of staff	Sanctioned	Filled
Support Staff (Technical)	04	04
Administrative Staff	01	01

15. Qualifications of teaching faculty with DSc/ D.Litt/ Ph.D/ MPhil/PG.

No. of Faculty with D.Sc / D. Litt	No. of Faculty with Ph.D	No. of Faculty with M.Phil	No. of Faculty with PG (M.Tech / MA/ M.Com / M.Sc/ MBA / MCA)	No. of faculty with UG (B.Tech)
0	00	00	09	01

**16. Number of faculty with ongoing projects from a) National
b) International funding agencies and grants received :**

a) National:

S. No	Name of the Faculty	Title of the Project	Funding Agency	Grants received (Rs.)	Duration
1	Nil	Nil	Nil	Nil	Nil

b) International: Nil

17. Departmental projects funded by DST-FIST; UGC, DBT, ICSSR, etc. and total grants received

S. No.	Title of the project	Functioning Agency	Name of the Coordinator	Grants Received (Rs)	Duration
	Nil	Nil	Nil	Nil	Nil

18. Research Centre /facility recognized by the University: Nil

19. Publications (last 4 Years) :

Publication details are available in the Department that will be shown at the time of visit.

20. Areas of consultancy and income generated : Nil

21. Faculty as members in

a) National Committees:

- Mrs. Archana Tiwari is the Chairman of Board of Studies, Department of Electronics & Instrumentation Engineering, CSVTU, Bhilai, Chhattisgarh for the duration 2010-13 & 2014-17.
- Mrs. Archana Tiwari is Member of Subject Review Committee, Teachers Recruitment Committee and Examination Committee in university, CSVTU, Bhilai.
- Mrs. Archana Tiwari was Convener of National Level Conference (AICON) organized at CSIT Durg since last six years i.e. from 2009 to 2014.
- Mrs. Archana Tiwari has Lifetime Membership of Indian Society for Technical Education (ISTE).

- Mrs. Archana Tiwari has Lifetime Associate Membership of Institution of Electronics and Telecommunication Engineers (IETE).
- Mr. Rajesh Kumar Gupta is Member of Board of Studies, Department of Electronics & Instrumentation Engineering, CSVTU, Bhilai, Chhattisgarh.
- Mr. Rajesh Kumar Gupta is Member of Subject Review Committee in university, CSVTU, Bhilai.
- Mr. Rajesh Kumar Gupta has Lifetime Associate Membership of Institution of Electronics and Telecommunication Engineers (IETE).
- Mr. Rajesh Kumar Gupta has Lifetime Membership of Indian Society for Technical Education (ISTE).
- Mr. Prabhas Kumar Gupta is member of subject review committee in university, CSVTU, Bhilai.
- Mr. Prabhas Kumar Gupta has Lifetime Membership of Indian Society for Technical Education (ISTE).
- Mr. Prabhas Kumar Gupta has the Membership of Instrument Society of India (ISOI).
- Mr. Shailendra Kumar Dewangan is member of Board of Studies, Department of Electronics & Instrumentation Engineering, CSVTU, Bhilai, Chhattisgarh.
- Mr. Shailendra Kumar Dewangan is member of subject review committee in university, CSVTU, Bhilai.
- Mr. Shailendra Kumar Dewangan has Lifetime Associate Membership of Institution of Electronics and Telecommunication Engineers (IETE).
- Mr. Shailendra Kumar Dewangan has Lifetime Membership of Indian Society for Technical Education (ISTE).
- Mrs. Remya Ani has the Membership of Instrument Society of India (ISOI).
- Mr. Umesh Kumar Sahu has Lifetime Associate Membership of Institution of Electronics and Telecommunication Engineers (IETE).
- Mrs. Urmila S. Soni has Lifetime Associate Membership of Institution of Electronics and Telecommunication Engineers (IETE).
- Mr. Akash Kumar Mandal has Lifetime Associate Membership of Institution of Electronics and Telecommunication Engineers (IETE).
- Mr. Akash Kumar Mandal has Lifetime Membership of Indian Society for Technical Education (ISTE).
- Ms. Roma Singhal has the Membership of Instrument Society of India (ISOI).

b) International Committees: Nil

c) Editorial Boards:

- Mrs. Archana Tiwari had been Reviewer for IEEE Conference twice.
- Mrs. Archana Tiwari is the Reviewer of Journal of Information Security

(ISSN No. Print: 2153-1234 & Online: 2153-1242).

22. Student Projects:

a) Percentage of students who have done in-house projects including interdepartmental/ programme : **100%**

b) Percentage of students placed for projects in organizations outside the institution i.e. in research laboratories/industry/other agencies : Nil

23. Awards / Recognitions received by faculty and students:

a) Faculty

The listing of awards and certificates of appreciation issued to various members of faculty shall be shown to the peer team at the time of visit.

b) Students

- P. Sneha won 1st position in Group Song in Youth Festival 2014 of CSVTU, Bhilai in the academic year 2013-14.
- Kedar Rajpurohit won 3rd position in Technical Model in Youth Festival 2014 of CSVTU, Bhilai in the academic year 2013-14.
- Nikita Swain won 1st position in Solo Song in Youth Festival 2013 of CSVTU, Bhilai in the academic year 2012-13.
- P. Sneha won 2nd position in Group Song in Youth Festival 2013 of CSVTU, Bhilai in the academic year 2012-13.
- Dimple Verma won 2nd position in Group Dance in Youth Festival 2013 of CSVTU, Bhilai in the academic year 2012-13.
- Sheikh Rafik Manihar published paper in IJS Volume 3, Issue 5, May 2012 ER IJSER International Journal of Scientific and Engineering Research (IJSER) - (ISSN 2229-5518) in the academic year 2011-2012.
- Sheikh Rafik Manihar published paper 1st Global Conference for Academic research Kuala Lumpur, Malasiya in the academic year 2011-2012 on 8th April.
- Sheikh Rafik Manihar and Komal Prasad Dewangan paper published in Cognizance 2012 "The Design and Construction of Low Cost Propeller Led Display" IIT Roorkee First Position in the academic year 2011- 2012 on 23-25 March.
- Sheikh Rafik Manihar published Paper in Global Journal of Research in Engg Global Journal, Indore (IJRE Volume 12 Issue 4) in the academic year 2011- 2012 on 1 March.
- Daksha Biyani won 1st position in Extempore in Youth Festival 2012 of CSVTU, Bhilai in the academic year 2011-12.
- Daksha Biyani won 2nd position in Speech in Youth Festival 2012 of CSVTU, Bhilai in the academic year 2011-12.

- Chaithra, Ekta & Udipt won 2nd position in Group Song in Youth Festival 2012 of CSVTU, Bhilai in the academic year 2011-12.
- Komal Prasad Dewangen won 1nd prize in Robowar, event in Laqshya - 2012 in the academic year 2011-2012.
- Sheikh Rafik Manihar won 1nd prize in Robowar, event in Laqshya - 2012 in the academic year 2011-2012.
- Akansha Sahu participated in Circuit mania event in Laqshya - 2012 in the academic year 2011-2012.
- Akansha Sahu participated in Ad- Mad Show event in Laqshya - 2012 in the academic year 2011-2012.
- Chitra Rekha Yadav participated in Circuit mania event in Laqshya - 2012 in the academic year 2011-2012.
- Chitra Rekha Yadav participated in Nirman event in Laqshya - 2012 in the academic year 2011-2012.
- Khushbu Sahu participated in Nirman event in Laqshya - 2012 in the academic year 2011- 2012.
- Nisha Chandrakar won 2nd prize in Circuit mania event in Laqshya - 2012 in the academic year 2011-2012.
- Abdul Jahid won 2nd prize in Trumble the Genre event in Laqshya - 2012 in the academic year 2011-2012.
- Abdul Jahid participated in Robo Soccer League event in Laqshya - 2012 in the academic year 2011-2012.
- Abdul Jahid participated in Deep Blue event in Laqshya - 2012 in the academic year 2011-2012.
- R. Nitin participated in Deep Blue event in Laqshya - 2012 in the academic year 2011-2012.
- Ankita Agarwal participated in Nirman event in Laqshya - 2012 in the academic year 2011-2012.
- Akanchha Mishra participated in Nirman event in Laqshya - 2012 in the academic year 2011-2012.
- Ankita Agarwal participated in Xxcelerate event in Laqshya - 2012 in the academic year 2011- 2012.
- Akanchha Mishra participated in Xxcelerate event in Laqshya - 2012 in the academic year 2011-2012.
- Chhatrapal Singh Kaushik participated in Xxcelerate event in Laqshya - 2012 in the academic year 2011-2012.
- Ankita Agarwal participated in Su- Do- Ku event in Laqshya - 2012 in the academic year 2011-2012.
- Chhatrapal Singh Kaushik participated in Su- Do- Ku event in Laqshya - 2012 in the academic year 2011- 2012.
- Akanchha Mishra participated in Su- Do- Ku event in Laqshya - 2012 in the academic year 2011- 2012.
- Kanha Gandhi Participated in Ad- Mad Show event in Laqshya - 2012 in

the academic year 2011- 2012.

- Kanha Gandhi participated in NFS event in Laqshya - 2012 in the academic year 2011- 2012.
- Kanha Gandhi won 1st prize in Dummy market event in Laqshya - 2012 in the academic year 2011- 2012.
- Dimple Verma participated in Nirman event in Laqshya - 2012 in the academic year 2011- 2012.
- Jyoti Chandravanshi participated in Nirman event in Laqshya - 2012 in the academic year 2011- 2012.
- Jyoti Chandravanshi participated in Ad- Mad Show event in Laqshya - 2012 in the academic year 2011- 2012.
- Kundan Singh participated in Circuit mania event in Laqshya - 2012 in the academic year 2011- 2012.
- Kundan Singh participated in Xxcelerate event in Laqshya - 2012 in the academic year 2011- 2012.
- B. Dinesh Kumar participated in Circuit mania event in Laqshya - 2012 in the academic year 2011- 2012.
- Daksha Biyani published paper "Role Of speaker Recognition System" CCET, Bhilai in the academic year 2011-2012 on 1-2 Mar.
- Khushbu Sahu paper presentation Prabhat 2012 SSCET, Bhilai in the academic year 18-19 Feb 2012.
- Kaushal Prajapati paper presentation Prabhat 2012 SPARKS, SSCT, Bhilai in the academic year 2011-2012 on 18-19 Feb.
- B. Dinesh Kumar Tech Patra 11-12 COMET, CSIT Durg in the academic year 2011-2012 on 18 Feb.
- Hem Hirwani participated in CITCON 2012 EDC, CSIT Durg in the academic year 2011-2012 on 18 Feb.
- Ankita Agrawal participated in OJAS Sliver Edition BIT, DURG in the academic year 2011-2012 on 16-18 Feb.
- Kundan Singh participated in OJAS Sliver Edition BIT, DURG in the academic year 2011-2012 on 16-18 Feb.
- Akanchha Mishra participated in OJAS Sliver Edition BIT, DURG in the academic year 2011-2012 on 16-18 Feb.
- Kanha Gandhi participated in OJAS Sliver Edition BIT, DURG in the academic year 2011-2012 on 16-18 Feb.
- G.Shashank participated in OJAS Sliver Edition BIT, DURG in the academic year 2011-2012 on 16-18 Feb.
- Sheikh Rafik Manihar participated in Robotics workshop AITM, Rajnandgaon in the academic year 2011- 2012 on 10 Feb.
- Khushbu Sahu participated in Robotics workshop AITM, Rajnandgaon in the academic year 10 Feb 2012.
- Nish Chandrakar participated in Robotics workshop AITM, Rajnandgaon in the academic year 2011-2012 on 10 Feb.

- Chitra Rekha yadav participated in Robotics workshop AITM, Rajnandgaon in the academic year 2011-2012 on 10 Feb.
- Akansha Sahu participated in Robotics workshop AITM, Rajnandgaon in the academic year 2011-2012 on 10 Feb.
- Kundan Singh won 1st prize in Volleyer's Campania ELITE, CSIT Durg in the academic year 2011-2012 on 10 Feb.
- Saraswati Sharma won 2nd prize in Volleyer's Campania ELITE, CSIT Durg in the academic year 2011-2012 on 10 Feb.
- Charanpreet Singh & Shilpi Pauranik won 2nd position in Technical Model Competition in Youth Festival 2011 of CSVTU Bhilai in the academic year 2010-11.
- S. Vinita & Manisha Pradhan won 2nd position in 4x100 Girl's Relay Race in Youth Festival 2011 of CSVTU Bhilai in the academic year 2010-11.
- S. Vinita, Manisha Pradhan, Shakti Chandra & Dipti Bhawsar won 2nd position in Volleyball Girl's in Youth Festival 2011 of CSVTU Bhilai in the academic year 2010-11.
- Prashant Sahu won 2nd position in Football in Youth Festival 2011 of CSVTU Bhilai in the academic year 2010-11.

24. List of eminent academicians and scientists / visitors to the department.

S. No	Year	Name of the Academician/ Scientist/ Visitor	Designation	Organization	Purpose of Visit
1	2013-14	Mr. L. Umakant, Mr. M. Venkatesh Babu	General Manager, Assistant General Manager	Bhilai Steel Plant, Bhilai	Guest Lecture on "Role of Instrumentation Engineering in Industries"
2	2012-13	Mr. J.P.S. Chouhan	Sr. Manager (C&IT)	Bhilai Steel Plant, Bhilai	Manufacturing Execution System (MES) Implementation in BSP
		Mr. Siddharth Bardiya & Mr. Sumit Singh Thakur	Jr. Manager	Neco Jaiswal Industry Ltd. Raipur	Industrial Automation

		Dr. Sameer Bardia	Dentist		Dentistry & Personality
1	2011-12	Mr. Yashvant Kanetkar & Mr. Asang Dani	Director	KSET Pvt. Ltd., Nagpur,	Workshop
2		Shri Vyomapada Dasa, Shri Gopa Vrindapada, dasa & Shri Sumadura Krishna Dasa	Head	Akshya Patra Foundation, Bhilai	Guest Lecture on "IQ, EQ & SQ in Management & Existence of God is it Reality"
3	2011-12	Mr. Sudipta Garnayak	Faculty	ENTEC Power & Automation Training Institute, Raipur	Workshop on "Industrial Automation with Electrical Applications"
1	2010-11	Mr. Naveen Verma	Associate Financial Planning Manager	ICICI Prudential, Bangalore	Guest Lecture on "Management Scenario of Electronics in Marketing"
2		Mr. Pratish Sharma	AGM, ERP deptt.	Bhilai Steel Plant, Bhilai	Guest Lecture on "Entrepreneurship Resource & Planning"

25. Seminars/ Conferences/Workshops organized & the source of funding

a) National

S. No.	Year	Name of the Seminar / Conference / workshop	Date (s)	Source of Funding
1.	2012-13	Eureka	22.02.2013	Indian Society for Technical Education (ISTE)

b) International: Nil

26. Student profile programme /course wise:

Year	Name of the Course/ Program	Applications Received	Selected (Number Admitted including Lateral Entry)	Enrolled		Pass percentage
				Male	Female	
2010-14	B. E.	*	57	34	23	73.68%
2009-13	(Electronics & Instrumentation)	*	65	43	22	63.07%
2008-12		*	63	40	23	61.9%
2007-11		*	69	40	29	65.21%

*Admissions are through CGPET/ JEE and as per the rules of CG state government.

27. Diversity of Students:

Year of Admission	Name of the Course	% students from the same State	% of students from other States	% of students from abroad
2013-14	B. E. (Electronics & Instrumentation)	95.65%	4.35%	-
2012-13		100%	-	-
2011-12		98.24%	1.76%	-
2010-11		100%	-	-

28. How many students have cleared national and state competitive examinations such as NET, SLET, GATE, Civil services, Defense services, etc. ?

Students clearing GATE exam: 15

29. Student progression:

Student Progression	Against % enrolled
UG to PG	35
PG to M. Phil.	N.A.
PG to Ph.D.	1
Ph.D. to Post-Doctoral	N.A.
Employed • Campus selection • Other than campus recruitment	72

Entrepreneurship / Self-employment	450
------------------------------------	-----

30. Details of Infrastructural facilities

a) Departmental Library : Plinth Area – 33.582 sq.mts.

No. of Titles	No. of Volumes	No. of Computers	No. of E-Journals	No. of Print Journals
198	224	1	-	8

b) Internet facilities for Staff & Students:

- Computer Lab – VIII with 30 computers having internet connection.

c) Class rooms with ICT facility:

- All classes are equipped with white and/or chalk boards.
- OHP's and LCD's are primarily used to deliver electronic class notes and to perform in-class demos and presentations.

d) Laboratories: 8

S. No.	Name of Laboratory	Area (sq.m)	Total Amount (in Rs.)
1	Sensor & Transducer Laboratory	78.0385 m ²	2,08,018/-
2	Automatic Control System Laboratory	78.0385 m ²	1,80,500/-
3	Industrial Instrumentation Laboratory	78.0385 m ²	2,08,018/-
4	Analytical Instrumentation Laboratory	78.0385 m ²	78,724/-
5	Instrumentation System Design Lab	78.0385 m ²	6,12,750/-
6	PC Based Instrumentation Laboratory	78.0385 m ²	3,11,750/-
7	Basic Electronics Laboratory	78.0385 m ²	4,53,540/-
8	Computer Lab - VIII	80 m ²	13,53,500/-

31. Number of students receiving financial assistance from college, university, government or other agencies:

Agency	No. of students received financial assistance			
	2009-10	2010-11	2011-12	2012-13
College	-	-	-	-

University	-	-	-	-
Government	-	-	-	-
Other agencies	-	-	-	-

32. Details on student enrichment programmes (special lectures / workshops /seminar) with external experts:

Yr.	Title of the Program	Special Lectures/ workshop/seminar	Date(s)	No. Benefited
2009-10	LABVIEW Programming Basics	Workshop	10/08/09 to 14/08/09	61
	Personality Development	Special Lectures	17/02/10	174
	Industrial Automation & SCADA	Special Lectures	29/01/10	184
2010-11	Management Scenario of Electronics in Marketing	Special Lectures	1/2/2011	78
	Entrepreneurship Resource & Planning	Special Lectures	28/09/10	97
2011-12	ENTEC Power Automation training Institute	Workshop	13/09/2011 to 14/09/2011	61
	Workshop on Secure Programming	Workshop	28/08/11 to 29/08/11	60
	Secret of success by Akshay Patra Foundation	Special Lectures	21/02/12 to 23/02/12	151
	IQ,EQ & SQ in Management & Existance of God, is it a Reality ?	Special Lectures	22/07/11	107
Yr.	Title of the Program	Special Lectures/ workshop/seminar	Date(s)	No. Benefited
2012-13	Workshop on PLC	Workshop	11/03/13 to 15/03/13 (5 Day).	70

	Industrial Instrumentation & Automation	Special Lectures	31/01/2013	120
	Manufacturing Execution System (MES)	Special Lectures	30/08/2013	125

33. Teaching methods adopted to improve student learning

- Seminars are organized to improve their presentation skill.
- Encouraging Mini Projects in each semester from third semester onwards to improve practical knowledge of students and build up their confidence.
- Organizing coaching classes for Aptitude and placement tests and personal interview.
- Add-on program is organized like MATLAB, Simulink, PLC training etc to strengthen theoretical learning.
- Organizing Coaching Classes for facing Technical Interview during campus placement.

34. Participation in Institutional Social Responsibility (ISR) and Extension activities

a) NSS activities:

Yr.	Name of the Activity	Venue	Date	No. of Students Participated
2012-13	Old Age Home	Vriddhashram, Shivaji Nagar, Balod Road, Durg (C.G)	6/10/2012	150
	Tree Plantation	CSIT Campus	10/8/2012	15
2011-12	Tree Plantation	CSIT Campus	7/7/2011	10
	Blood Donation Camp	CSIT Campus	29/07/2011	15

35. SWOC analysis of the department and Future plans

Strengths

- The Department is accredited by NBA-AICTE.
- Student project quality is good.
- Teacher student ratio is good.
- Sufficient emphasis on lab and project work.
- Experienced and qualified faculties.

- Good team spirit among faculties.
- Committed technical supporting Staff.
- Department has its student association that regularly conducts programs and events which are student centric.
- Systematic monitoring of teaching learning process.
- Conducive atmosphere for the all round development of students.
- Active participation of faculty in research activities
- Well equipped and spacious laboratories.
- Good interaction with professional bodies.

Weaknesses

- First year result is poor.
- Enhancement of faculty qualification and retention need improvement.
- Poor communication skill of students at entry level due to rural background. This becomes a hindrance in getting proper placement.
- Lack of software industries in nearby places.
- Research initiatives need to be increased.

Opportunities:

- The Department is accredited by NBA hence eligible for grants from AICTE and other public sector agencies.
- Availability of qualified, experienced and trained faculty.
- Soft skills and related training by T&P Cell and association with alumni lead to the opportunities to raise the employability of the students.
- Scope for Industry Institute Interaction.
- Availability of adequate Journals, Lab facilities for Research.
- Proper connectivity by road, rail and air with important cities of India.
- Staff development programmes.
- Conduction of national conferences for Students / Staff in thrust areas of research.

Constraints:

- Lack of good communication skill in English language.
- Competition from other institutions
- Lack of stable human resource
- Development of application of electronics to cater to the needs of common man in the field of agriculture, health, education etc.
- Quality of intake is poor, especially in rural engineering colleges. This may be due to proliferation of engineering colleges.

Future Plans:

- To establish research oriented laboratories.



- To attain a level of five year accreditation by NBA.
- To achieve 95% and above result.
- To achieve 95% placements for eligible candidates through campus selections.
- To enhance Industry Institute Interaction to attract leading MNCs to conduct campus recruitment.
- Up gradation of faculty qualifications.

Evaluative Report of the Department

1. **Name of the Department :** Information Technology
2. **Year of Establishment :** 2008
3. **Names of Programmes / Courses offered (UG, PG, M.Phil., Ph.D., Integrated Masters; Integrated Ph.D., etc.) :**UG
4. **Names of Interdisciplinary courses and the departments/units involved**

S. No.	Semester	Subject	Departments Involved
1.	1st	Language (Professional Communication in English)	Humanities
2.	1st	Applied Chemistry	Basic Sciences
3.	1st	Applied Physics-I	Basic Sciences
4.	1st	Applied Maths-I	Mathematics
5.	1st	Elements of Electrical Engineering	Electrical and Electronics Engineering
6.	1st	Engineering. Graphics	Mechanical Engineering
7.	2nd	Engineering. Mechanics	Mechanical Engineering
8.	2nd	Environment & Ecology	Humanities
9.	2nd	Applied Physics –II	Basic Sciences
10.	2nd	Applied Maths -II	Mathematics
11.	2nd	Fundamental of Mechanical Engineering.	Mechanical Engineering
12.	2nd	Basic Civil Engineering.	Civil Engineering
13.	3rd	Mathematics-III	Mathematics Department
14.	3rd	Discrete Structures	Mathematics Department
15.	3rd	Digital Electronics and Logic Design	Electronics and Telecommunication
16.	4th	Computational Mathematics	Mathematics Department
17.	4th	Analog Electronics Circuits	Electronics and Telecommunication
18.	4th	Telecom Switching and Computer Networks	Electronics and Telecommunication
19.	5th	Principal of Communication System	Electronics and Telecommunication
20.	5th	Microprocessor and Interfaces	Electronics and Telecommunication

21.	6th	Information Theory and Coding	Electronics and Telecommunication
22.	6th	Cellular and Mobile Computing	Electronics and Telecommunication

5. **Annual/ semester/choice based credit system (programme wise)**

B.E. - (Information Technology) - Semester system

6. **Participation of the department in the courses offered by other departments: Nil**

7. **Courses in collaboration with other universities, industries, foreign institutions, etc.: Nil**

8. **Details of courses / programmes discontinued (if any) with reasons : Nil**

9. **Number of Teaching posts**

Designation	Sanctioned	Filled
Professors	01	01
Associate Professors	02	00
Asst. Professors	09	05

10. **Faculty profile with name, qualification, designation, specialization, (D.Sc./D.Litt./Ph.D./M.Phil. etc.,)**

Presently 06 faculty members are in the department. Faculty profile will be shown on the day of visit.

11. **List of senior visiting faculty :**

S. No.	Name of expert	Title of Lecture	Year of Visit
1	Mr. Jayant Chaturvedani, (Manager – Operation; Ericsson)	Telecommunication Word, Mobile Network	2014
2	Dr. K. B. Subramaniam (Retd. Principal; Regional Institute of Education, Bhopal)	Nature of Mathematics and Mathematics in Nature	2014
3	Mr. Apoorv Tiwari, (Associate Software Engineer; Accenture)	Cracking Campus Interviews	2014
5	Dr. Rabindra Ku. Jena	Green Computing	2014
6	Dr. Roheet Bhatnagar	Big Data	2014

8	Mr. Ankit Fadia	Ethical Hacking	2013
9	Mr. M. S. Uday Kiran (HR Manager; HCL) & Mr. Venkat Bhukta (Vertical Head; Idea Cellular Ltd.)	HR Interviews	2013
10	Mr. Ashish Mathur, (Chief General Manager; Escorts Constructions Equipments Ltd.)	Functioning of an Organisation	2013
11	Mr. Rahul Krishna (Manager Recruitment; Espire Infolabs)	Fresher Development - a Taste of Corporate World	2013
12	Mr. Kunjal Jayant Kamdar (Head Recruiter; Quinnox)	Social Media	2013
13	Mr. Ravi Bizoara (Module Lead; Mphasis)	IT Interactive	2013
14	Mr. Abhishek Shrivastava (Assistant Manager; SAIL)	How to Clear SAIL	2013
18	Dr. Sudhakar Pandey (NIT, Raipur)	Research methodologies	2013
19	Dr. Keshri Verma (NIT, Raipur)	Innovations in Data mining	2013
20	Dr. Sanjay Kumar (Pt. RSSU, Raipur)	Principles of pipelining	2013
21	Mr. Navneet Dewangan (Principle Infrastructure Engineer; Mphasis Ltd)	My Struggle & Success Story	2012
22	Mr. A. Ganesh Ray (Executive Director; Techlead Solutions Pvt. Ltd.)	Faculty Development Program in MBD	2012
24	Mr. K. Patel (Advisor – HRD; Jayaswals Neco Industries Ltd.)	Man in the Mirror	2012
25	Mr. Nagaraj Parappagoudar (Solutions Architect; CISCO Systems, Sanjose, CA, USA)	Emerging Technology Areas and Trends in Computer Industry	2012
26	Mr. Anirudh Girey (Sr. Software Engineer; IBM India Limited)	Job Orientation, Career Guidance	2012
27	Dr. Bharti Trivedi (Director; Apex Technology)	Vacuum Systems & Green IT	2012
28	Ms. Tripti Vishwakarma (Consultant Talent Acquisition; Tech Democracy LLC)	Importance of Technical Knowledge	2012
29	Mr. Yashwant Vishwakarma (Consultant; Microsoft)	Career as Database Administrator	2012

30	Mr. Jayant K. Kittur (Asso. Professor; Gogte Institute of Technology)	Career Option for Engineering Graduates and Preparation for Campus Placement	2012
31	Mr. Sumit Kumar (SAP Project Manager; Total Environment Building Pvt. Sys.)	ERP Implementation	2012
32	Yashavant Kanetkar & Asang Dani	Workshop on secure programming	2011
33	Mr. Mayank Jha (Manager; IBM)	Knowledge Required for IT Sector	2011
34	Mr. Kaustabh Shukla (Advocate; Corporate Laws)	Corporate Social Responsibilities	2011
35	Mr. Vikram Singh (Regional Manager Corporate & Academic Relation; Aimsvarsity Education Pvt. Ltd.)	Cloud Computing & WEB 2.0	2011
36	Mr. Vaibhav Khare (Director; Gurukul Classes)	Resume Building	2011
37	Mr. Virendra Vishwakarma (Technical Lead; Microsoft)	Latest Trends in IT Sector	2011
38	Mrs. Chitra Pathak (software engineer; IBM)	How to prepare for IT sector	2010
39	Ms. Tulika Chakraborty (Lead Quality Analyst; IBM)	Life Style in Corporate World	2010
40.	Mr. Shashank Pradhan (Manager; Accenture)	Adapting Students into Corporate Life Style	2010

12. Percentage of lectures delivered and practical classes handled (programme wise) by temporary faculty : Nil

13. Student -Teacher Ratio (programme wise)

Student -Teacher Ratio (programme wise) Year	II Year	III year	IV year	Total	No. of Faculties	Student Teacher Ratio
2010-2011	38	16	NIL	54	4	14:1
2011-2012	25	39	13	77	6	13:1
2012-2013	42	33	42	117	9	13:1
2013-2014	25	44	28	97	9	11:1

14. Number of academic support staff (technical) and administrative staff; sanctioned and filled

Type of staff	Sanctioned	Filled
Support Staff (technical)	Nil	Nil
Administrative staff	01	01

15. Qualifications of teaching faculty with DSc/ D.Litt/ Ph.D/ MPhil/PG.

Presently 05 faculty members are in the Department. Faculty profile will be shown on the day of visit.

16. Number of faculty with ongoing projects from a) National b) International funding agencies and grants received : Nil

17. Departmental projects funded by DST-FIST; UGC, DBT, ICSSR, etc. and total grants received : Nil

18. Research Centre /facility recognized by the University :

1. Library Facilities

➤ **E-Journals:**

The Institute has a collection of 145-IEEE, 49-SPRINGER journals.

➤ **Print Journals:**

There are about 80 print Journals in the library of different categories.

➤ **E-Books:**

Total 35 e-books are available in the library.

➤ **No.of Computers:**

Currently there are 20 Systems with Broadband Connectivity of 10 Mbps BSNL Leased Line

➤ **Library Server:**

Servers in use are HP Server Proliant, Intel Xeon Processors with a capacity of 2Gb RAM and 146Gb HDD.

➤ **Facility:**

The institute provides discussion rooms, Video Projection Room etc.

➤ **E-Learning Material:**

There are mainly 8000Hrs of NPTEL Video Courses on 200 subjects and Lecture materials from IIT,MIT Stanford and Harvard Universities

19. Publications (last 4 Years) :

Publication details are available in the Department that will be shown at time of visit.

20. Areas of consultancy and income generated :Nil

21. Faculty as members in

- a) National committees: Nil
- b) International Committees : Nil
- c) Editorial Boards : Nil

22. Student projects

- a. Percentage of students who have done in-house projects including interdepartmental/programme : 100%
- b. Percentage of students placed for projects in organizations outside the institution i.e. in Research laboratories/Industry/other agencies: Nil

23. Awards / Recognitions received by faculty and students:

- a) Faculty : 1
- b) Student :10

24. List of eminent academicians and scientists/ visitors to the department:

S. No	Yr	Name of the Academician/ Scientist/ Visitor	Designation	Organization	Purpose of Visit
1	2013-14	Dr.Ashok Bhansali	Assistant Professor	OPJIT, Raigarh)	Teaching Methodology
2		Dr. Rabindra Ku. Jena	Assistant Professor	IIITM, Nagpur	Green Computing
3		Dr. Roheet Bhatnagar	Assistant Professor	JIT,Jaipur	Big Data
1	2012-13	Dr.Sudhakar Pandey	Assistant Professor	NIT,Raipur	Research methodologies
		Dr.Keshri Verma	Assistant Professor	NIT,Raipur	Innovations in Data mining
		Dr.Sanjay Kumar	Assistant Professor	Pt.RSSU, Raipur	Principles of pipelining
1	2011-12	Mr. Yashvant Kanetkar & Mr. Asang Dani	Director	KSET Pvt. Ltd., Nagpur,	Workshop

25. Seminars/ Conferences/Workshops organized & the source of funding

a) National

S. No.	Yr	Name of Seminars/ Conferences/Workshops	Date
1.	2010-11	Seminar on “C” by Mr. Yashavant Kanetkar	10 th Feb 2011
		All India Conference on Modern Trends in Computer Science Engineering and Information Technology	28 th to 29 th Jan 2011
2.	2011-12	Seminar on “Secure Programming”	28 th & 29 th August 2011
		Workshop on “Etrix”	8 th Oct 2011
		Workshop on “Cloud Computing”	26 th -28 th July 2011
		All India Conference on Green in Computer science Engineering and Information Technology	20 th to 21 st Jan 2012
3.	2012-2013	Workshop on “Ethical Hacking” By Ankit Fadia	19 th -20 th Feb 2013
		All India Conference on Global Innovations in Computer science Engineering and Information Technology	12 th to 13 th April 2013
		Workshop on “RC Aircraft & Manufacturing	25 th to 26 th July 2012
4.	2013-2014	All India Conference on Intelligent Systems in Computer science Engineering and Information Technology	25 th to 26 th April 2014

b) International : Nil

26. Student profile programme/course wise:

Year	Name of the Course / Program	Applications received	Selected (Number Admitted including Lateral Entry)	Enrolled		Pass Percentage
				Male	Female	
2008-12	BE(IT)	*	39	30	9	44.73
2009-13	BE(IT)	*	38	12	26	71.79
2010-14	BE(IT)	*	26	9	17	50.00

27. Diversity of Students

Year of Admission	Name of the Course	% students from the same State	% of students from other States	% of students from abroad
2010	B.E.(IT)	97%	3%	Nil
2011	B.E.(IT)	100%	Nil	Nil
2012	B.E.(IT)	100%	Nil	Nil
2013	B.E.(IT)	100%	Nil	Nil

28. How many students have cleared national and state competitive examinations such as NET, SLET, GATE, Civil services, Defense services, etc. (APSEB,POWER GRID,NTPC,BHEL,RRB.... ?

Name of the Competitive qualified examination	No. of students qualified			
	2010-11	2011-12	2012-13	2013-14
GATE	-	-	1	-

29. Student progression

Student Progression	Year	Against % Enrolled
UG to PG	2013-14	Nil
	2012-13	2.38%
	2011-12	16%
	2010-11	NA
PG to M.Phil.	-	-
PG to Ph.D.-	-	-
Employed Campus Selection Other than Campus Recruitment	2013-14	11%
	2012-13	48%
	2011-12	7.6%

	2010-11	NA
Entrepreneurship/Self-employment	2013-14	Nil
	2012-13	Nil
	2011-12	7.6%
	2010-11	NA

30. Details of Infrastructural facilities

a) Library : Plinth Area –861.11sq. ft.

No. of Titles	No. of Volumes	No. of Computers	No. of E-Journals	No. of Print Journals
135	194	NIL	NIL	NIL

b) Internet facilities for Staff & Students:

- 10 Mbps Broadband Leased line from BSNL
- Wi-Fi enabled Campus

c) Class rooms with ICT facility : LCD projectors are used in class rooms and in laboratory

d) Laboratories (on shared basis)

S. No.	Name of the laboratory	Space# students	Curriculum Laboratory
1	Programming C/C++ Lab	30	Programming with C Lab, Data Structure Lab, Object oriented Programming Lab
2.	Programming Lab HTML/DHTML	30	Web Technology Lab, Computer Hardware simulation Lab
3.	Computer Center	60	Software Technology Lab
4.	Advanced Programming Lab	60	Database Management System Lab, Computer Network Lab,
5.	Microprocessor and Interfacing Lab	30	Microprocessor and Interfacing Lab
6.	Digital Electronics and Logic Design Lab	30	Digital Circuits Lab
7.	Basic Electronics	30	Basic Electronic Circuits Lab

31. Number of students receiving financial assistance from college, university, government or other agencies : 2

32. Details on student enrichment programmes (special lectures / workshops /seminar) with external experts:

2013 – 14

- Mr. Rahul Krishna (Manager Recruitment; Espire Infolabs) Dated 11/09/2013; Topic - FRESHER DEVELOPMENT - A Taste of Corporate World.
- Mr. M. S. Uday Kiran (HR Manager; HCL) & Mr. Venkat Bhukta (Vertical Head; Idea Cellular Ltd.) Dated 28/10/2013; Topic – HR Interviews.
- Mr. Jayant Chaturvedani, (Manager – Operation; Ericsson) Dated 10/02/2014; Topic – Telecommunication Word, Mobile Network for IT Students.

2012 – 13

- Mr. Anirudh Girey (Sr. Software Engineer; IBM India Limited) Dated 11/06/2012; Topic – Job Orientation, Career Guidance for Final Year Students.
- Mr. Nagaraj Parappagoudar (Solutions Architect; CISCO Systems, Sanjose, CA, USA) Dated 21/07/2012; Topic – Emerging Technology Areas and Trends in Computer Industry.
- Mr. Navneet Dewangan (Principle Infrastructure Engineer; Mphasis Ltd) Dated 07/11/2012; Topic – My Struggle & Success Story.
- Mr. Abhishek Shrivastava (Assistant Manager; Sail) Dated 08/01/2013; Topic-How to Clear SAIL for 6th Semester students.
- Mr. Ravi Bizoara (Module Lead; Mphasis) Dated 01/02/2013; Topic – IT Interactive
- Mr. Kunjal Jayant Kamdar (Head Recruiter; Quinnox) Dated 02/02/2013; Topic – Social Media.

2011 – 12

- Mr. Vikram Singh (Regional Manager Corporate & Academic Relation; Aimsvarsity Education Pvt. Ltd.) Dated 26/07/2011 to 28/07/2011; Topic – Cloud Computing & WEB 2.0
- Mr. Vaibhav Khare (Director; Gurukul Classes) Dated 15/09/2011; Topic – Resume Building for 7th Semester CSE.
- Mr. Virendra Vishwakarma (Technical Lead; Microsoft) Dated 30/09/2011; Topic – Latest Trends in IT Sector .
- Mr. Sumit Kumar (SAP Project Manager; Total Environment Building Pvt. Sys.) Dated 14/01/2012; Topic – ERP Implementation for CS & IT 7th Semester Students.
- Mr. Jayant K. Kittur (Asso. Professor; Gogte Institute of Technology) Dated 03/02/2012; Topic – Career option for Engineering Graduates and Preparation for Campus Placement for 6th Semester Students.
- Mr. Yashwant Vishwakarma (Consultant; Microsoft) Dated 15/03/2012; Topic – Career as Database Administrator for Final Year Students.

- Ms. Tripti Vishwakarma (Consultant Talent Acquisition; Tech Democracy LLC) Dated 27/03/2012; Topic – Importance of Technical Knowledge for Final Year Students.
- Dr. Bharti Trivedi (Director; Apex Technology) Dated 17/05/2012; Topic – Vacuum Systems & Green IT for Final Year Students of CSE,

2010 – 11

- Mr. Shashank Pradhan (Manager; Accenture) Dated 14/09/2010; Topic – Adapting Students into Corporate Lifestyle for 7th Semester Students.
- Mr. Tulika Chakraborty (Lead Quality Analyst; IBM) Dated 20/10/2010; Topic – Life Style in Corporate.
- Mrs. Chitra Pathak (Software Engineer; IBM) Dated 05/11/2010; Topic – How to Prepare for IT Sector for 7th Semester & 5th Semester Students (CSE & IT).
- Mr. Mayank Jha (Manager; IBM) Dated 13/01/2011; Topic – Knowledge Required for IT Sector for 7th Semester Students (CSE & IT).
- Mr. Kaustabh Shukla (Advocate; Corporate Laws) Dated 30/06/2011; Topic – Corporate Social Responsibilities for 6th Semester Students.

33. Teaching methods adopted to improve student learning

- Guest lectures by Academicians & Industrial Experts.
- Distribution of Course Handouts.
- Maintenance of Course and Subject files.
- University results are critically analyzed and discussed in the faculty meeting.
- Special classes are conducted for slow learners for improving their academic performance.
- Feedback on classroom teaching is taken twice in a semester.
- Department conducts staff meetings for review of academics and other related works.

34. Participation in Institutional Social Responsibility (ISR) and Extension activities: NSS, blood donation.

- CSIT in association with the Chhattisgarh Police organized a **Helmet Awareness Drive** on 7th Jan 2013. The drive was inaugurated by Mr. O.P. Pal, IPS, Senior Superintendent of Police, Durg in the CSIT Campus. Mr. Dinakaran, Traffic Instructor, Chhattisgarh Police addressed students on the importance of road safety and the usage of helmet.



c) Training imparted to the local public : Nil

35. SWOC analysis of the department and Future plans

Strengths:

- Regular Emphasis is given on strong academics along with overall development of students.
- Continuous upgrading of infrastructure facilities.
- Continuous improvement in the performance of teachers.
- Conduction of Guest lectures by the Software Company Experts
- Regular Training programs for preparation of competitive examinations.
- Extra curricular and co-curricular activities for overall development of students.
- Well equipped labs with modern equipment.

Weaknesses:

- R & D work to be strengthened
- Institute & Industry Interaction must be improved.

Opportunities:

- Adequate Journal and Lab Facility for research work
- Facilities are provided to attend National and International Seminar and Conference.
- 'IT' park may be established in C.G. in near future.

Constraints:

1. The branch of Information Technology all over the country has started losing its popularity.
2. The branch of Information Technology does not attract students. We are on the verge of its closure.



Future Plans:-

Long term Goals:

- To start P.G. program.
- To undertake research projects and Consultancy works.
- To strengthen the industry institute interaction.

Short term goals

- To prepare the students to keep pace with the latest technological advancements especially in the field of Information Technology and to combat the challenges being faced by the Industry.
- To maintain the development of resources in the Laboratories so that students are in positive learning environment.

Evaluative Report of the Department

1. **Name of the department :** Mechatronics Engineering
2. **Year of Establishment :** 2009
3. **Names of Programs/Courses offered :** UG: B.E. (Mechatronics Engineering)
4. **Names of Interdisciplinary courses and the departments/units involved:** Nil
5. **Annual/ semester/choice based credit system (programme wise)**
B.E. (Mechatronics Engineering) - Semester based credit System – Electives in Higher Semesters
6. **Participation of the department in the courses offered by other departments**

S. No	Program	Courses offered by other departments
1	BE	Electronics & Communication Engineering
2		Computer Science and Engineering
3		Mechanical Engineering
4		Electronics & Instrumentation

7. **Courses in collaboration with other universities, industries, foreign institutions, etc.:** Nil
8. **Details of courses / programmes discontinued (if any) with reasons:** Nil
9. **Number of teaching posts:**

	Sanctioned	Filled
Professors	1	2
Associate Professors	1	1
Assistant Professors	6	6

10. **Faculty profile with name, qualification, designation, specialization, (D.Sc./D.Litt. /Ph.D. / M. Phil. etc.,)**

Presently 09 faculty members are in the department. Faculty profile will be shown on the day of visit.

11. **List of senior visiting faculty:** Nil

12. **Percentage of lectures delivered and practical classes handled (programme wise) by temporary faculty:** Nil

13. **Student -Teacher Ratio (Programme wise):** 20: 1

14. Number of academic support staff (technical) and administrative staff; sanctioned and filled

	Sanctioned	Filled
Academic support staff (Technical)	1	1
Administrative staff	1	1

15. Qualifications of teaching faculty with DSc/ D.Litt/ Ph.D/ MPhil/PG.

No. of Faculty with D.Sc / D. Litt	No. of Faculty with Ph.D	No. of Faculty with M.Phil	No. of Faculty with PG (M.Tech / MA/ M.Com / M.Sc/ MBA / MCA)	No. of faculty with UG (B.Tech)
---	-	---	6	3

16. Number of faculty with ongoing projects from a) National b) International funding agencies and grants received:

(a) National:

S. No	Name of the Faculty	Title of the Project	Funding Agency	Grants received (Rs.)	Duration
1	Chandrakant Badgaiyan	To fabricate a cost effective fully automatic power wheel chair for old aged and disabled	Institution of Engineer (India)	30,000/-	2013-14

(b) InterNational: Nil

17. Departmental projects funded by DST-FIST; UGC, DBT, ICSSR, etc. and total grants received: Nil

18. Research Centre /facility recognized by the University: Nil

19. Publications:

Publication details are available in the Department, and will be shown at the time of visit.

20. Areas of consultancy and income generated: Nil

21. Faculty as members in

a) National committees:

Prof Rajesh Kumar

- Member, Academic Council, CSVTU Bhilai
- Chairman, Board of Studies, CSVTU Bhilai
- Member, Executive Council, Mechantronics Engg., CSVTU Bhilai
- Member, Syllabus Review Committee, Mechatronics Engg., CSVTU Bhilai
- Observer, University Examination, CSVTU Bhilai

b) International Committees: Nil

c) Editorial Boards:

- Mr. Chandrakant Badgaiyan Asst.Professor as Advisory Board Member of International Journal “Journal of Harmonized Research”.
- Mr. Chandrakant Badgaiyan Asst.Professor as Reviewer Member of “International Journal of Advanced Information and Communication Technology (IJAICT)”
- Mr. Chandrakant Badgaiyan Asst.Professor Reviewer Member of “International Journal of Scientific & Engineering Research (IJSER)” ISSN 2229-5518.
- Mr. Chandrakant Badgaiyan Asst.Professor Reviewer Member of “International Journal of Engineering Research and Sports Science (IJERSS)” ISSN 2348 – 2400(Print), ISSN 2348 –1404(Online)

22. Student projects

a) Percentage of students who have done in-house projects including inter Departmental Programme: **100%**

b) Percentage of students placed for projects in organizations outside the institution i.e. in Research laboratories/Industry/other agencies:-0%

23. Awards/ Recognitions received by faculty and students

a) Faculty: Nil

b) Students

- Jaikishan Dewangan (3036709028) received Gold Medal from CSVTU, Durg for his Outstanding Academic performance as a topper in Mechatronics Engineering branch of all affiliated colleges in the University region for 2009-13 passed out batch.

24. List of eminent academicians and scientists/ visitors to the department

S.	Yr.	Name of the academician / scientist	Designation	Organization	Purpose of Visit
1	2013	Mr. H K Sethi,	Ex GM	Bhilai Steel Plant Bhilai	Guest Lecture
2	2013	Mr. S R Babu,	DGM	BSP	Guest Lecture
3	2014	Mr. P C Lalwani	Engineer	Gearcon. Rajnandgaon	Guest Lecture
4	2014	Mrs. Pragya Yadav		B& R Automation Pune	Guest Lecture

25. Seminars/ Conferences/Workshops organized & the source of funding

a) National :

S. No	Yr	Name of the Seminar /Conference / Workshop	Date(s)	Source of funding	Amount spent (Rs.)
	2013-14	AICON' 13	25-26 April 2014	Self	
		Workshop on "Mobile Application development on Android Platform for Beginners"	11 March 2014		
		Add On program on "Matlab and Simulink"	24-26 Feb 2014		
		Webinar on "Industrial Automation and Special purpose Machine Building"	20 th Feb 2014		
		Add On Program on "PCB Circuit Designing"	31 July to 10 th Aug 2014		
		Add On Program on "Basics of Matlab Programming and Digital Image Processing"	5 th Aug – 10 th Aug 2013		

	2012-13	AICON' 12	12-13 April 2013	Self	
		Add On Program on "PCB Circuit Designing" for 5 th sem	8 th to 17 th Oct 2012		
		Add On Program on "PCB Circuit Designing" for 4 th sem	22 nd to 25 th Feb 2012S		
	2011-12	AICON' 11	20-21 Jan 2012	Self	
	2010-11	AICON' 10	28-29 Jan 2011	Self	

b) International: Nil

26. Student profile programme/course wise:

Year	Name of the Course/program	Applica tions received	Selected (Number Admitted including Lateral Entry)	Enrolled		Pass percen tage
				Male	Female	
2009-13	BE(Mex)	-	65	60	5	58.73
2010-14		-	61	57	4	21.67

27. Diversity of Students

28. How many students have cleared national and state competitive examinations such as NET, SLET, GATE, Civil services, Defense services, etc.?

Name of the Competitive qualified examination	No. of students qualified			
	2010-11	2011-12	2012-13	2013-14
GATE	-	-	1	-
Banking Sector	-	-	-	-
CAT	-	-	2	-

NET	-	-	-	-
GRE	-	-	-	-
Public Sector	-	-	1	1

29. Student progression

Student progression	Against % enrolled			
	2010-11	2011-12	2012-13	2013-14
UG to PG	Nil	Nil	Nil	6
PG to M.Phil.	Nil	Nil	Nil	Nil
PG to Ph.D.	Nil	Nil	Nil	Nil
Ph.D. to Post-Doctoral	Nil	Nil	Nil	Nil
Employed	Nil	Nil	Nil	40.98%
• Campus selection				
• Other than campus recruitment				
Entrepreneurship/Self-employment	Nil	Nil	Nil	Nil

30. Details of Infrastructural facilities

a) Library: Plinth Area: 325.00 Sqfeet (Approx) =30Sq m

No.of Titles	No.of Volumes	No.of Computers	No.of E-Journals	No.of Print Journals
-	165	Nil	Nil	Nil

b) Internet facilities for Staff & Students:

- 10 Mbps broad band leased line from BSNL

c) Class rooms with ICT facility: 01

d) Laboratories

S. No	Name of the laboratory	Area (Sq.Mts)	Total Investment(Rs.)
1	Systems	76	Nil

31. Number of students receiving financial assistance from college, university, government or other agencies:

Agency	No. of students received financial assistance				
	2010-14	2009-13	2008-12	2007-11	2006-10

College	-	-	-	-	-
University	-	-	-	-	-
Government	50	-	-	-	-
Agencies	-	-	-	-	-

32. Details on student enrichment programmes (special lectures / workshops / seminar) with external experts:

Year	Title of the Programme	Special lectures / workshops / seminar	Date(s)
2013-14	Guest lecture	Automation & various technologies used in Industries	25/02/14
	Guest lecture	Gear Manufacturing Process	14/03/14
	Guest lecture	Industrial requirements in the Fields of Automation & Drives	23/10/13
	Guest lecture	Mechanical Maintenance & its Types	11/09/13
	Guest lecture	Plate Mill Manufacturing Process & Safety	08/10/13
	Workshop on	Mobile Application development on Android Platform for Beginners	11/03/14
	Add On Program	Matlab and Simulink	24-26 Feb 2014
	Webinar	Industrial Automation and Special Purpose Machine Building	20/02/14
	Add On Program	PCB Circuit Designing	31 July to 10 th Aug 2014
	Add On Program	Basics of Matlab Programming and Digital Image Processing	5 th Aug – 10 th Aug 2013
2012-13	Guest lecture	Solar Energy Power Generation using PV modules	29/01/13
	Guest lecture	National Seminar on Energy Technologies of Wireless communication	19/01/13
	Add On Program	PCB Circuit Designing	8 th to 17 th Oct 2012
	Add On Program	PCB Circuit Designing	22 nd to 25 th Feb 2012
2011-2012	Guest lecture	Operation & Optimization in a Steel Plant Rolling Mill	24/9/11
	Guest lecture	Seminar on Industrial Automation	24/8/11
2010-11	Nil		

2009-10	Nil
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33. Teaching methods adopted to improve student learning

- Guest lectures by Academicians & Industrial Experts.
- Distribution of Course Handouts.
- Maintenance of course and Subject files.
- University results are critically analyzed and discussed at the faculty meeting. Follow-up actions are initiated.
- Special classes are being conducted for slow learners for improving their academic performance.
- Feedback on classroom teaching is being taken twice in a semester.
- Department conducts staff meetings for review of academics and other related works
- Organizing Seminars, Technical Quizzes, and Model making contests.
- Development of student support material for poor learners and enthusiastic learners.
- Encouraging Mini Projects as a part of teaching by the method of project.
- Support of Add-on-courses to reinforce theoretical learning.

34. Participation in Institutional Social Responsibility (ISR) and Extension activities

a) NSS activities:

Year	Name of the activity	Venue	Date(s)	No. of students participated
2013-14				

b) Training imparted to the local public: Nil

35. SWOC analysis of the department and Future plans

Strengths

- Qualified and experienced faculty with good retention
- Active participation of faculty in research activities
- Systematic monitoring of teaching-learning process.
- Active participation of alumni for career guidance and improvement
- Continuous upgrading of infrastructure facilities and Teaching and learning Methodologies.
- Conduction of Guest lectures by Industry Experts
- Extracurricular and co-curricular activities for overall development of students.

- Exposure of students to practical pursuit.

Weaknesses

- Lack of Industrial research and consultancy.
- R&D work to be strengthened

Opportunities

- Continuous faculty development programmes at Institute.
- Availability of e-Journals as well as print Journals.
- Well Equipped Laboratories
- By conducting National conferences for Students / Staff in thrust areas of research.

Challenges:-

- To keep up with the expectations of the society.
- To be the forerunner in multidisciplinary technology.

Future Plans**Long term goal**

- To start P.G. program in the Department.
- To undertake research projects and Consultancy works.
- To strengthen the industry - institute interaction.

Short term goals

- Applying for external funded research projects with collaboration of government and private organizations to maintain the development of resources in the Laboratories so that students are in positive learning environment.
- To enhance Industry Institute Interaction and to attract leading MNCs to conduct campus recruitment.

Forwarded message -----

From: "East & North East Region, NAAC" <naac.ner@gmail.com>

Date: 12 Jun 2014 15:51

Subject: loi

To: <registrar@csitdurg.in>, <principal@csitdurg.in>

Cc:

NAAC/E&NE/BSM/CYCLE-1/SSR/ CHCOGN21243/2014

12th June,2014

The Principal

CHHATRAPATI SHIVAJI INSTITUTE OF TECHNOLOGY

DURG-BHILAI NAGAR – 491001

Chhattisgarh

**Dear Sir/Madam,
Greetings from NAAC!**

This has reference to your online submission of Letter of Intent (LOI) on 16th May , 2014 expressing your intention to go in for Assessment and Accreditation by NAAC.

On scrutiny of your LOI and finding your institution eligible for accreditation (henceforth referred to as 1st Cycle accreditation), you are advised to proceed further for submission of Self-Study Report (SSR). The format for submission of SSR is same for all cycles of accreditation. You may download and use the format "Affiliated / Constituent Colleges" from our website: http://www.naac.gov.in/manuals_ass_accrd.asp.

Please note that the new process for Assessment and Accreditation requires uploading of the SSR on your institution website one month prior to submission to NAAC. Once the SSR are uploaded on the institution website you are requested to keep the concerned NAAC officer informed. Kindly, note that the uploaded SSR should not be password protected and accessible online to all stakeholders till the completion of assessment and accreditation process by NAAC.

Kindly submit within **Six months** of the acceptance of the LOI and one month after uploading the SSR on institutional website (one month period is part of the six month deadline), five hard copies and one soft copy (cd or pendrive) of the SSR complete in all respects duly filled in certificate of compliance in the enclosed format along with requisite fee as applicable kindly check the fee details in the link <http://www.naac.gov.in/fees.html>

For any further clarification kindly, log on to NAAC website www.naac.gov.in or contact the undersigned.

Thanking you,

Yours sincerely,
(B.S. Madhukar)